

The global energy race

Energy in the age of fragmentation

AI, electrification and energy security are driving energy demand faster than supply and infrastructure can respond, creating a more inflationary, capital-intensive and strategically important energy system than markets expect.

The outlook for the global energy sector is the most attractive it has been for two decades. The energy transition increasingly looks like a process of energy addition rather than substitution, with demand for oil, gas, electricity and renewable power rising simultaneously. Population growth, economic development, electrification, AI and digital infrastructure are driving energy demand higher, while geopolitical fragmentation has increased the value of reliable and secure supply. Following a decade of underinvestment in parts of the sector, supply and infrastructure are struggling to keep pace. The result is a more capital-intensive, strategically important and potentially more inflationary energy system than markets currently assume. We estimate annual energy investment requirements could reach approximately \$3.6trn in 2027, making energy one of the largest capital-allocation opportunities of the coming decade.

Energy is a growth sector: Often described as "old economy" by wider markets, we take the opposite view. We see the broad energy ecosystem as one of the most attractive structural growth opportunities globally and strategically more important than the market perceives. Rather than entering terminal decline, energy demand is being supported by rising electricity consumption, industrial development, digital infrastructure and growing emerging-market energy needs. Demand growth in 2025 exceeded the 10-year average and we expect above-trend growth to persist for at least the next five years. Meeting this demand will require contributions from all forms of energy. While renewable energy will play an increasingly

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important role in reducing emissions, conventional fuels remain essential for reliability, affordability and security of supply. The result is not a simple replacement of one energy source by another, but a more diversified and capital-intensive system: an era of energy addition, rather than substitution.

Our new proprietary global energy model supports this conclusion: We forecast energy demand growth of **1.9% CAGR from 2025-2050E**. We expect energy demand from data centres to drive **~32 quad BTU of incremental energy needs by 2040** (ie, over 600GW of data centre capacity), or ~3.4% of total energy demand (this is roughly equivalent to Russia's total energy use in 2025). We expect a steady increase in global EV fleet penetration, with EV mix as a percentage of the overall fleet growing at ~15% CAGR between 2020 and 2050, reaching nearly ~60% of the fleet in 2050. This will leave transportation as ~14% of global energy use by 2050, down from 17.5% in 2025.

Global race for capital: Meeting energy demand will require one of the largest investment cycles in modern industrial history. Capital will be needed not only to maintain existing production capacity, but also to expand generation, transmission, storage and supporting infrastructure as governments and corporations prioritise energy security. Across both conventional and renewable energy markets, project pipelines continue to grow, creating opportunities for equipment suppliers, engineering contractors and infrastructure developers. We estimate annual capital expenditure across energy supply, power systems and end-use infrastructure will reach **\$3.6trn in 2027**, growing more than **5% per year**. This is more than **three times** the capital currently required for the planned AI infrastructure build-out, highlighting the scale of the opportunity and making access to capital and balance-sheet strength increasingly important competitive advantages.

Increased risk of higher commodity-driven inflation: Previous periods of energy underinvestment were typically followed by supply responses. This cycle is different. Electrification, AI and energy security are creating new, long-duration sources of demand, while supply constraints, permitting challenges and infrastructure bottlenecks are limiting the pace at which new capacity can be brought online. The resulting imbalance increases the risk of structurally tighter energy markets, higher commodity price volatility and more persistent inflation than markets currently expect.

Energy security increasingly influences policy decisions: Recent geopolitical developments have reinforced the strategic importance of energy security for governments worldwide. Ensuring access to reliable, affordable and secure energy supplies has become a central policy objective, driving support for domestic production, infrastructure investment and supply-chain diversification. As a result, energy security is increasingly shaping both policy and capital allocation decisions, providing additional support for investment across the energy sector. We outline ten geo-economic trends that we believe will shape global energy markets over the coming decade.

Wide-ranging investment opportunities across the value chain: The convergence of rising energy demand, substantial investment requirements, supportive pricing dynamics and increasing energy-security priorities is creating opportunities across multiple segments of the sector and across asset classes. Upstream producers, energy services companies, infrastructure owners, power utilities and technology providers enabling system expansion all stand to benefit from the evolving landscape. Given the scale of investment required and the long duration of many energy projects, companies with strategic assets, strong balance sheets and access to capital appear particularly well positioned. We identify opportunities across the energy value chain, including integrated energy companies, LNG, pipelines, power infrastructure, utilities, energy services and companies enabling system expansion. We outline our preferred equity and

credit exposures in Figures 60-64. Across our equity coverage, we see significant scope for earnings upgrades and valuation re-rating. We identify our preferred OW stocks across the global energy complex and note that on average, our 2028 EPS forecasts are **11% above consensus**, while our price targets imply **30% of potential upside**. We are most optimistic on 2028e earnings upgrades for our preferred OW stocks in the **European Oil Services (33%)**, **European O&G (28%)** and **US Oil Services (28%)** sectors and see the greatest re-rating potential upside in the next 12 months for our preferred OW stocks in **North America Clean Tech**, **European Oil Services** and **European Utilities**.

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Foreword

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The global energy race

AI's speed with which it spreads through the world economy, the use cases it generates in industry after industry, its effect on the labor market, the societal implication, etc: this will be the defining macro theme for the next decade. The global energy race, we believe, will be a close second.

In 2025, every major energy source – oil, natural gas, coal, solar, wind, nuclear – notched new highs. All of them. Oil consumption rose to a record. Natural gas hit a record. Coal hit a record. And renewable capacity additions were the largest in history, with solar photovoltaic systems (PV) alone meeting more than a quarter of global demand growth for the first time. This is not what the consensus predicted. A decade ago, the prevailing narrative was sequential. Renewables would rise, but fossil fuels were supposed to *decline*. The energy system would transition from one to the other, in an orderly fashion. The reality has been messier, more expensive, and hence more consequential for capital markets.

The world is not substituting one energy system for another. It is running both simultaneously, and it does not have enough capital deployed in either. That is the central argument of this report. Energy demand is rising at the same time as geopolitical fragmentation is increasing the premium on energy security. The result is a growing race for capital across resources, infrastructure, and power systems – a race that the current level of investment is not winning. Not only has the energy transition not stopped, but the next phase will require significantly more investment across a far broader set of assets than the market currently appreciates. That includes not only renewables and electrification, but also oil and gas exploration, LNG export capacity, power transmission and storage networks, refining resilience, energy services, and other critical infrastructure that keeps the system functioning while it transforms.

The implications for equity, credit, and commodity markets are substantial. The sector-by-sector analysis that follows in subsequent sections, written by our analysts across the energy complex, identifies which parts of the value chain are best positioned to attract this capital and generate returns from it. This overview sets the stage.

The capital misallocation

The energy transition was always going to be expensive. What was underestimated was how long the existing system would need to keep running at full capacity while the new one was being built. Between 2015 and 2022, a combination of ESG-driven capital reallocation, shareholder pressure for returns over growth, and policy signals favouring clean energy led to a sustained reduction in upstream oil and gas investment. Global upstream capital expenditure is still roughly 45% below peak levels. The industry was told – by investors, regulators, and in some cases its own management teams – that demand was peaking and that new supply investment was stranded capital. The market acted accordingly. But demand did not peak.

We expect underlying oil demand growth at over 6 million barrels per day in 2025-30, driven overwhelmingly by non-OECD economies. At the same time, the industry must

replace more than 7 million barrels per day of output annually just to *maintain* existing supply, before any demand growth is accounted for. It is not investing enough to do even that.

Meanwhile, clean energy investment has surged, reaching a record \$2.3 trillion in 2025, more than double the \$1 trillion flowing into fossil fuels. This is a remarkable achievement. But it has not been sufficient to displace the fossil fuel system it was intended to replace.

The arithmetic is unforgiving: global energy demand grew 1.3% in 2025, and while low-emissions sources contributed nearly 60% of that, the remaining 40% still required more oil, more gas, and more coal. The world added clean energy at an impressive pace. It just was not fast enough to allow the old system to shrink. The result is a system that is underinvested on both sides. Traditional energy has been starved of the capital needed to maintain supply. Clean energy, while attracting record investment, remains geographically concentrated – roughly 75% of global clean energy investment flows to a small number of markets – and has not yet built the grid infrastructure, storage capacity, or baseload reliability needed to serve as a full replacement. *The capital gap exists everywhere one looks.*

And now come the new demand drivers

What makes the current moment different from previous cycles of energy underinvestment is that demand is being driven by forces that are structural rather than cyclical. Three in particular deserve attention.

1. Electrification is accelerating globally

Global electricity demand is forecast to grow 3.7% in 2026, more than twice the rate of total energy demand. The IEA projects this pace will continue through 2027 and beyond, driven by electric vehicles, heat pumps, air conditioning, and industrial electrification, particularly in emerging economies where appliance ownership and cooling demand are rising rapidly. Electrification is the mechanism through which de-carbonization is supposed to work: move end-uses from direct fossil fuel combustion to the grid, then clean the grid. The problem is that the first half of that equation is proceeding faster than the second. Countries are electrifying their economies before the grid – and the generation capacity behind it – is ready.

2. AI and data centres are driving a step-change in electricity demand

Global data centre electricity consumption is forecast to reach 565 terawatt-hours in 2026, up 26% from 447 TWh in 2025. Related power demand is expected to reach 132 gigawatts this year and could hit 290 gigawatts by 2030. AI-optimized servers now account for 31% of total data centre power consumption, up from roughly 20% a year ago. The IEA projects that data centre electricity consumption could double to 945 TWh by 2030, approximately 3% of total global electricity demand. This is not a rounding error. It is a new structural source of baseload electricity demand that did not exist at scale five years ago; that is growing at 25-30% annually; and that requires the kind of reliable, continuous power supply that intermittent renewables alone cannot yet provide. It is showing up in the inflation data – AI-related electricity costs are a measurable contributor to the rise in core PCE – and it is reshaping utility planning, grid investment, and power procurement across every major economy.

3. Geopolitical shocks have repriced energy security

The war in Ukraine, the conflict with Iran, and the broader fragmentation of the global trading system have fundamentally altered how governments think about

energy supply. The assumption that energy could be sourced from the cheapest global supplier, transported through open sea lanes, and priced in transparent commodity markets has been challenged in ways that are not reversible. Europe's forced pivot away from Russian pipeline gas replacing it with LNG from the United States, Qatar, and a patchwork of alternative suppliers – was the most dramatic example, but it is not the only one. Asia's LNG importers scrambled to diversify supply chains after the Iran conflict threatened critical transit routes. Countries across the Middle East, Africa, and Latin America are positioning themselves as alternative suppliers and competing to attract international energy capital.

The result is a global energy system that is less integrated, more regionalised, and more expensive to operate, requiring significantly more redundant infrastructure than the pre-2022 model assumed.

Energy security is reshaping capital allocation

The geopolitical dimension deserves particular emphasis because it changes the investment calculus in ways that standard energy models do not capture. *When energy security is taken for granted, as it largely was in the two decades before 2022, capital allocation optimises for cost.* The cheapest source of supply wins. That logic favoured concentrated production (OPEC, Russia), long global supply chains (LNG tankers, oil tankers through the Strait of Hormuz), and minimal investment in redundancy. It was efficient. It was also fragile. The system that is emerging is different.

Countries are building strategic reserves, contracting for long-term LNG supply, investing in domestic production capacity, and – critically – diversifying not just their energy sources, but also their energy suppliers. This is not cost-optimised. It is resilience-optimised. And resilience requires capital. More LNG regasification terminals. More pipeline interconnectors. More strategic storage. More refining capacity distributed across regions, rather than concentrated in the lowest-cost locations. Global refining is a case in point. Permanent plant closures and war-related damage have cut global refinery output by an estimated 4.5 million barrels per day in the second quarter of 2026. Crack spreads – the margin between crude oil and refined products – have reached record levels, with the US 3-2-1 crack spread hitting \$64 per barrel and global refining margins touching \$59. These are not normal numbers. They reflect a structural shortage of refining capacity that years of underinvestment and geopolitical disruption have created. Even if both the Ukraine and Iran conflicts were to end tomorrow, it would take years to rebuild the refining capacity that has been lost. The market is telling us, in the clearest possible terms, that the downstream infrastructure is insufficient. The same logic applies across the energy value chain.

Grid investment needs to grow more than 50% by 2030, from roughly \$400 billion annually to over \$600 billion, just to meet national climate targets, according to the IEA. More than 2,500 gigawatts of renewable energy, storage, and large-load projects (including data centres) are currently stalled in grid connection queues worldwide. The bottleneck is not generation. It is the wires, transformers, substations, and interconnectors that move electricity from where it is produced to where it is consumed. This infrastructure takes years to permit and build, has been chronically underfunded, and is now the binding constraint on both the energy transition and the AI build-out.

Energy addition, not substitution, is the present – and the future

The data support a clear conclusion: the world is in an era of energy addition, not substitution. All major energy sources are growing. The IEA's own Global Energy Review 2026 confirms that oil, natural gas, and coal each reached record consumption levels in 2025. Renewables are growing faster – solar PV contributed the largest single share of demand growth for the first time – but they are growing *in addition to* fossil fuels, not instead of them. This is likely to persist for several years. Non-OECD energy demand is being driven by industrialisation, urbanisation, and rising living standards, forces that are not discretionary and that require energy in all its forms. India's electricity demand is growing at 6-7% per year. Southeast Asia is building coal plants alongside solar farms. The Middle East is investing in renewable capacity and upstream oil production simultaneously.

The assumption that the world would transition linearly from fossil fuels to renewables was always an abstraction. The reality is a system that needs more of everything, for longer, than the models assumed. This is not an argument against the energy transition. It is an argument for realism about its pace and its capital requirements.

The transition is happening: renewable capacity additions, EV adoption rates, and clean energy investment are all at record levels. But it is happening inside a system where total energy demand continues to grow, where fossil fuel infrastructure is ageing and underinvested, where geopolitical risk demands redundancy, and where entirely new sources of demand (AI, data centres, electrification) are adding load faster than the grid can absorb it.

The net result is that investment is required across resources and infrastructure – traditional and renewable, upstream and downstream, generation and transmission – to maintain a secure, affordable, and resilient energy system. The capital requirements are larger, more diverse, and more sustained than the market is currently pricing.

Opportunities across sectors and markets

The remainder of this report moves from the macro to the specific. Our sector analysts across equities and credit have identified the parts of the energy value chain where this capital race creates the most compelling investment opportunities – and the most significant risks.

The analysis spans upstream exploration and production, LNG infrastructure, power generation and transmission, refining and petrochemicals, energy services, and the enabling technologies that sit at the intersection of the old energy system and the new.

The macro framing is simple. Energy demand is growing. Energy security is being repriced. The capital deployed across the system is insufficient. And the next phase of the energy transition, whether measured in barrels, megawatts, or miles of transmission line, will require more investment, across more assets, for longer than the market currently expects. The question for investors is not whether this capital will be deployed. It is who deploys it, where, and at what return.

Quick takes - Macro, Thematic, Equity, Credit and Strategy

FIGURE 1.

Asset Class	Sector	Quick take	Lead analyst
Macro	Executive Summary	Energy addition, not substitution: The world is not simply replacing one energy source with another. Demand growth means almost every part of the energy system needs more capital, creating a larger and more complex energy ecosystem.	Ajay Rajadhyaksha
	Capital	Capex across the energy sector, spanning oil and gas, pipelines, refineries, LNG, grids, power generation, renewables and electrification, is set to rise from \$2.7tn in 2015 to \$3.4tn in 2026 (IEA). Our spending forecasts are based on the gap between the supply capacity delivered by current investment levels over the past decade and the new capacity required by our demand outlook. Put simply, above-trend energy demand growth (1.9% CAGR 2026-2050e) will require above-trend growth in spending.	Team view
	Geo-economics	Energy is increasingly strategic. The US and China are best positioned through energy resources, infrastructure and supply chains, while Europe faces major investment needs. The Middle East is tilting toward Asia while Africa, Latam and parts of Asia should benefit from rising demand. Australia is set to be a key geo-economic player being a capital-rich advanced economy blessed with an abundance of renewable, fossil, nuclear fuel deposits.	Christian Keller
	Inflation	Markets may be underestimating inflation risk: energy scarcity, commodity tightness, geopolitical fragmentation could prove more inflationary than consensus expects, challenging the prevailing disinflation narrative.	Jon Hill, Pooja Sriram
	Commodities	Commodity markets remain structurally tight as inventories stay low and investment remains constrained. We continue to see upside risks to energy prices relative to market expectations. Energy price volatility is likely to remain a defining feature of this cycle. We recommend long Brent call spreads in the futures space.	Amarpreet Singh
	FX	We believe energy security and data sovereignty will remain key drivers of global capital flows, reinforcing US exceptionalism and supporting a structurally stronger US dollar. At the same time, a growing divide between energy and digital "haves" and "have-nots" is likely to favour energy and commodity exporters, while weighing on the competitiveness and currencies of more import-dependent economies.	Themos Fiotakis, Lefteris Farmakis
Thematic	Energy Transition	Energy security and competitiveness are emerging as transition drivers on par with sustainability. Rapidly rising energy transition capex is still insufficient to meet global needs. In the new energy equation, reliability is attracting a premium - we see nuclear fusion as a future technology with the potential to revolutionise the energy system and potentially solve the energy trilemma in the second half of the century.	Maggie O'Neal
	US Public Policy	US Energy, industrial and national-security objectives are becoming more closely linked. US policy support may help accelerate investment, but geopolitical risks remain a key source of market volatility.	Michael McLean
Equities	US Oil Services	We view the current environment as the best setup for the Energy Services sector in 20 years. Following a decade of underinvestment in oil & gas, we expect a rebound in the years ahead as energy security priorities/concerns come to the forefront. Energy security, offshore growth and AI power needs extend the cycle. OW: HAL, FTI, SEI. UW: NOV.	J. David Anderson
	Europe Oil Services	As Europe seeks a greater diversification of its energy supply by both geography and source, European Energy Services will become a key enabler, serving as a principal conduit through which conventional, unconventional and renewables are delivered to market. The set-up, in our view, is akin to the mid-2000s. OW: Technip Energies, Técnicas, TGS, Tenaris, Vallourec.	Mick Pickup
	US O&G	A mature U.S. shale sector does not imply declining production, but it does imply greater scarcity value for future growth, higher strategic importance for secure molecules, and increasing competition for access to long-duration resources. We believe investors continue to underappreciate this shift. OW: XOM, COP, OXY, EQT. UW: CNX.	Betty Jiang
	Europe O&G	Following a period when capital allocation increasingly focused on renewables and low-carbon opportunities, we are seeing a renewed emphasis on upstream oil and gas. Companies that combine strong balance sheets, asset scale and capital-allocation flexibility will be best placed to fund this next phase of energy development. OW: BP, TotalEnergies, Eni. UW: OMV.	Lydia Rainforth
	North America Midstream Refining	We expect global demand for US NGLs, LNG, and refined product exports to remain structurally higher going forward, giving rise to contracting tailwinds and incremental expansion opportunities for our midstream operators with exposure to these assets. OW: ET, PPL, VLO. UW: PAA.	Theresa Chen
	US Utilities	The US utility sector is experiencing its biggest demand shift since air conditioning, driven by AI and data centres. AI and electrification make power, grids and flexibility scarce assets. Higher load growth translates into the higher capital requirements and, therefore, greater financing needs. This creates higher equity needs to maintain balance sheet health and credit ratings. OW: PPL, Entergy, Xcel.	Nicholas Campanella

Asset Class	Sector	Quick take	Lead analyst
Credit	Europe Utilities	Europe has reached a power-demand inflection point, increasing the value of grids, networks and flexible generation. OW: Enel, Centrica, Solaria, United Utilities, RWE, SSE, Drax, Engie.	Dominic Nash
	North America Clean Tech	The energy infrastructure space had been in a state of disinvestment for the past few decades, as electricity requirements stalled despite continued economic and population growth. The hyperscaler power shock, combined with increasingly insular energy policies, has re-accelerated investment in clean tech. AI is accelerating demand; speed-to-power themes favour nuclear, fuel cells and solar. OW: OKLO, BE, FSLR. EW: GNRC.	Christine Cho
	Europe O&G	We see three themes shaping European energy credit performance: commodity volatility is back but balance sheets are stronger; integrated models continue to provide superior earnings resilience; and capital discipline is increasingly supporting credit quality through the cycle. OW BP, TotalEnergies.	Franck Bataille
	US Energy	LNG, deepwater and gas infrastructure are entering a new investment cycle. We also think that working capital needs could structurally increase in a more fragmented environment, creating an opportunity for capital providers to step in and help bridge short-term liquidity needs. Buy RIG 2031s/2038s and FLQIII 5.55s or FLIQII 5.92s.	Harry Mateer, Paul Chambers
	US & Latam Utilities	The energy buildout supports utilities, midstream infrastructure and power-linked credits. Favour issuers with strategic assets, scale and funding flexibility. OW: Vistra. UW: NextEra.	Srinjoy Banerjee, Stefan Fenistyk
	Europe Utilities	Europe is entering a utility investment super-cycle. The focus is shifting from adding renewables to funding grids, baseload power and energy security while preserving balance-sheet strength. OW: EDF, TenneT. UW: Naturgy.	Franck Bataille
	APAC Energy	Asia drives global demand growth, with LNG, electrification and storage key themes. We expect Asian countries to diversify their energy sources by: 1) reducing their dependence on energy products from the Middle East by diversifying purchases to other countries. (US, Australia, Africa and Singapore to benefit), 2) extending the useful lives of existing coal-fired plants and/or building new ones; and 3) increasing investments in renewable energy and nuclear power plants. OW: Santos, LGES. UW: KORGAS, KORELE.	Jit Ming Tan, Avanti Save
	EEMEA Corporates	Energy security is driving investment in LNG, domestic power and renewables. Buy AYDEMT 2030s, AZURGY 2030s and KOS GTA 11.25% 2031s.	Stella Cridge, Evgeniia Bystrova
	US	The broad energy ecosystem is in the midst of a growth phase: accelerating electrification, AI and data center expansion, and optimization for resilience instead of cost are likely to exacerbate energy bottlenecks and necessitate substantial investment. However, we see more attractive opportunities in select stocks and sub-industries than in a broad allocation to the Energy sector. We remain constructive on Utilities, Industrials and Tech, where exposure to power transmission and generation, grid modernization, infrastructure build-out, electrification, power management and battery technologies offer a more widespread opportunity set.	Venu Krishna
Equity Strategy	Europe	Energy has returned to focus for European investors as energy security has evolved from a policy objective into an economic imperative and a core pillar of European strategic autonomy. Despite strong year-to-date performance, the sector continues to trade at a discount to history. We favour Oil Services and Utilities (OW) and remain MW on Integrated Energy, while seeing medium-term re-rating potential given still-depressed investor ownership across both Energy and Utilities.	Emmanuel Cau
Credit Strategy	US	Current valuations leave little room for forgiveness, especially for the Energy sector. IG Energy currently trades at its richest level versus IG Industrials in more than a decade, with the spread ratio sitting just below 1.0x. That richness was earned through years of balance sheet repair, capital discipline, limited issuance and higher energy prices recently. The challenge is that a significant buildout of LNG, gas infrastructure, and power related investment points to a rising cadence of debt issuance ahead. Utilities face an even steeper financing curve, which argues for continued relative underperformance there. Strong fundamentals can support tight spreads, yet history suggests sectors embarking on large capital cycles tend to underperform the rest of the index over time.	Dominique Toublan
	Europe	We are Overweight the Energy and Utility sector for 3 core reasons. 1) fundamentals are generally strong. Capex build-outs will be disciplined with the supply cycle supported by government (government liquidity support or ownership stakes) "Nationalisation" is a scary word for an equity investor, but absent haircuts to bonds, there is usually upside for a credit investor. 2) The sectors will likely maintain their non-cyclical attributes, with fundamentals less affected by any growth downside, unlike, say, hyperscalers. 3) Valuation, unlike the setup on the \$IG side, relative to the broader market are either mid-range for Energy or fairly wide for Utilities.	Soren Willemann

Source: Barclays Research

Global energy model

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Energy outpacing resulting emissions through 2050

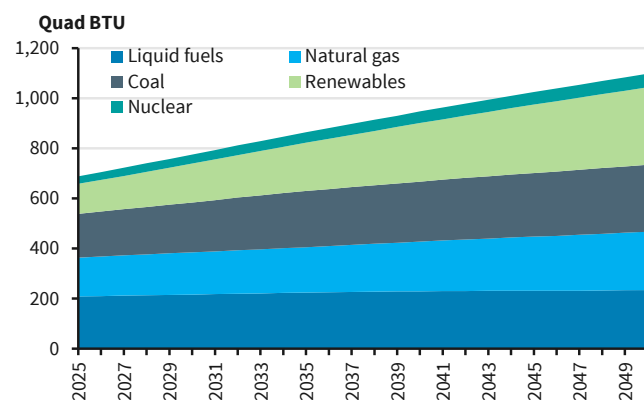
- Rising energy use globally.** In our base case, global energy use will increase ~1.9% CAGR over 2025-50, reflective of population growth (+0.7% CAGR), rising real GDP (+2.5% CAGR) and real income growth (+1.8% CAGR). We also expect the global share of electric vehicles to rise to more than 50% by 2050 and new data centres to reach over 600GW, the bulk of which (400GW) we expect to be in the US. At the same time, we expect carbon emissions growth (+1.3% CAGR) to be below the growth rate of global energy use due to both greater electrification and the growing share of renewables (+15-20pts to ~50%) in the global energy mix. Our base case is for a more moderate increase in CO2 emissions per capita (+0.6% CAGR) through 2050.

FIGURE 2. Energy Demand Model Outputs

Total Energy Use by segment (quad BTU)	2025	2030	2035	2040	2050	CAGR	Absolute Growth
Residential	117.0	126.4	136.1	146.0	171.1	1.5%	46.3%
Data Center	0.6	10.4	20.7	32.1	31.6	16.9%	4896.7%
Other Commercial	74.4	82.3	90.4	97.4	112.8	1.7%	51.7%
EV	2.0	5.5	12.0	22.8	55.3	14.3%	2721.5%
Traditional Vehicles	90.1	87.7	84.1	77.8	59.3	-1.7%	-34.2%
Other Transport	28.2	30.3	32.5	34.5	38.8	1.3%	37.9%
Industrial	376.1	432.6	488.3	536.6	629.3	2.1%	67.3%
Total	688.3	775.2	863.9	947.3	1098.2	1.9%	59.5%

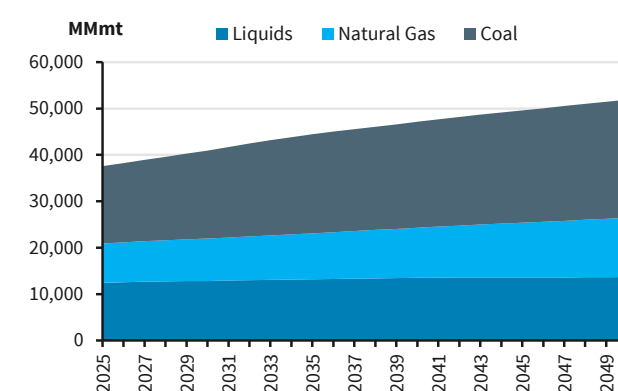
Source: EIA, Barclays Research

FIGURE 3. Energy use by fuel source



Source: EIA, Barclays Research

FIGURE 4. CO2 emissions from energy usage by fuel

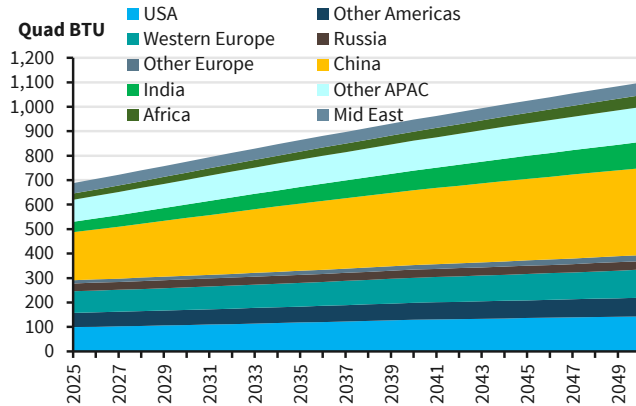


Source: EIA, Barclays Research

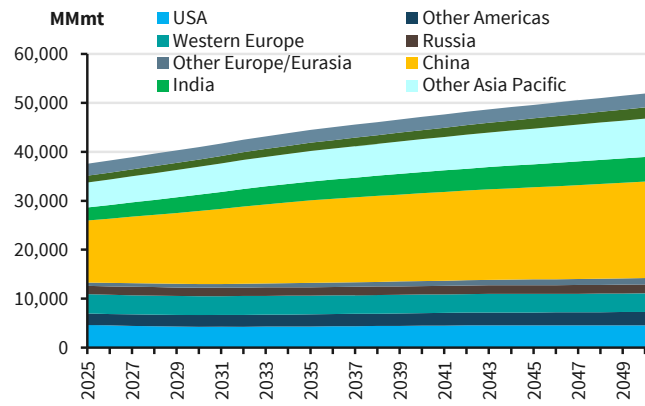
- Emerging markets (eg, India/ China) to drive emissions on higher energy use.** On a regional level, our base case is for emerging markets to be the key driver of global energy use, with India (+3.8% CAGR) showing the strongest growth. Our base case has China, the largest energy consumer today, growing at a slightly slower pace (+2.4% CAGR) than some emerging economies, but still outpacing the US (+1.5%). We expect China's energy usage to be heavily

driven by energy efficiency in industrial and commercial sectors (where we eased the EIA's assumption of significant energy efficiency gains). For CO₂ emissions, we see India (+2.6% CAGR) registering the largest growth in emissions, outpacing both China (2.0% CAGR) and the US (showing about flat).

FIGURE 5. Energy use by region



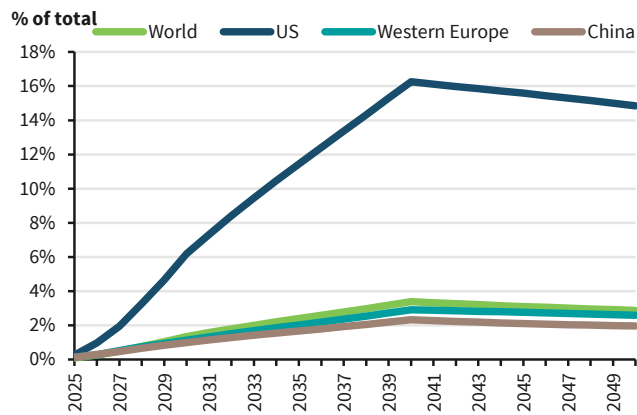
Source: EIA, Barclays Research

FIGURE 6. CO₂ emissions by region

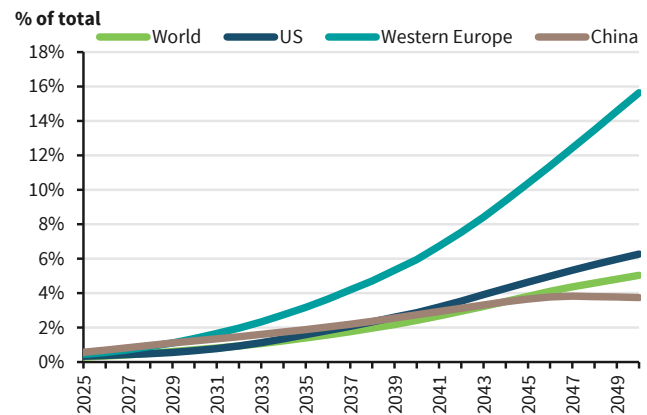
Source: EIA, Barclays Research

- Over the past three years, the construction of data centres has resulted in net increases in expected energy demand:** Our energy model utilizes the [EIA's International Energy Outlook 2023](#) reference scenario, adapted to Barclays house views, particularly for data centres and EVs, where we see key deviations today in terms of the outlook versus 2023¹. Our base case calculates 32 quad Btu of new data centre energy demand by 2040. For reference, we expect ~950 quad Btu energy usage globally by 2040 (data centres will be ~12% of the increase). Past 2040, we do not model, in aggregate, more data centre additions, given lower visibility.
- As global EV penetration increases, transportation will decline as a percentage of total energy use:** With higher EV penetration (over 50% of global vehicles by 2050), we expect transportation to ease as a percentage of global energy use, to about 14% by 2050 versus 17.5% in 2025.
- In absolute terms, data centres will need more energy than entire regions:** By 2040, we estimate more than 600GW of data centre capacity will require ~32 quad Btu of energy (electricity and conversion losses from producing electricity). For context, this is roughly equivalent to 2025 energy use in Russia and slightly lower than in India.

¹ We also dampen the EIA's more significant assumptions around energy efficiency improvements for commercial and industrial sectors, particularly for developed economies

FIGURE 7. Data Centre Energy Usage as a % of Total Energy

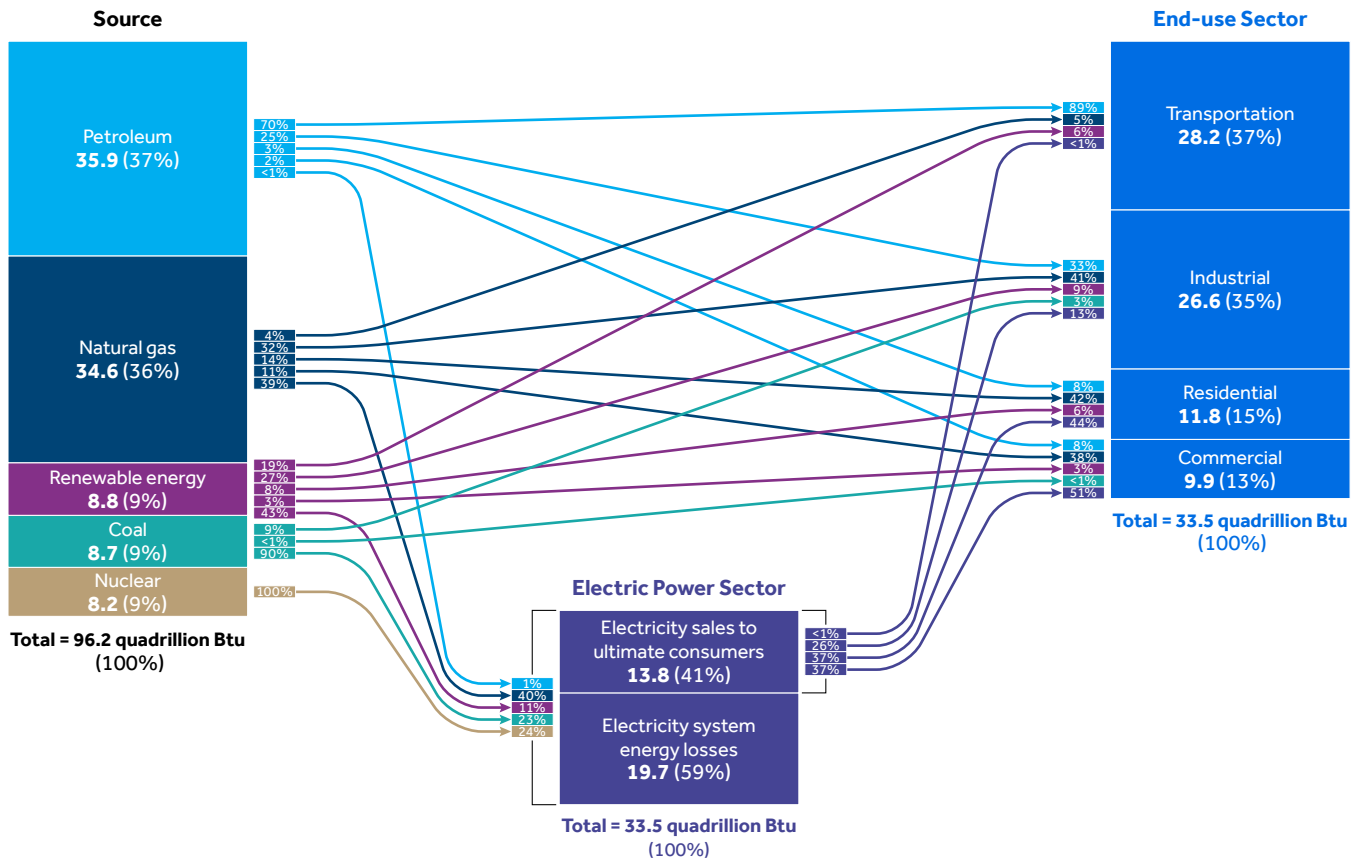
Source: EIA, Barclays Research

FIGURE 8. EV Energy Usage as a % of Total Energy

Source: EIA, Barclays Research

- Model methodology:** The end users of energy are categorized into four principal sectors: transportation, industrial, residential and commercial. Fossil fuels (petroleum, natural gas and coal) are the key primary energy sources, along with nuclear and renewables as non-CO2 emitting sources. Electricity generation is interesting in that it is both a user of energy sources and a provider of energy to end-use sectors. The process of generating electricity, however, results in material electrical system losses (~60%). [Figure 9](#) shows the EIA's breakdown of US energy consumption by source and sector in 2025.

FIGURE 9. US 2025 energy usage



Source: EIA

Exploring our data centre and EV growth forecasts

Data centres

- We expect data centre capacity growth to accelerate through 2026-2027e at a 35% annual rate before plateauing in 2028e.
- The US is expected to be the primary driver of global data centre capacity growth, accounting for ~60% of installed IT capacity by 2029, according to our estimates.
- Outside of China, we expect hyperscalers to account for the largest share of data centre build out, contributing ~60% of total new capacity additions over our forecast period.
- We see lower completion rates for non-hyperscaler projects due to limited visibility on phasing, permitting status, and delivery times.

A historic build-out

We estimate that data centre installed capacity growth will average >30% p.a. in 2026-27, before moderating to c.20% in 2028-29 (new capacity additions could plateau in 2028). The US is expected to be the largest contributor, with installed IT capacity rising from 34GW in 2025 to over 110GW by 2029, implying >40% growth in the first two years. This represents a very significant acceleration in US build-out relative to other regions. For the rest of the world, we assume a more moderate growth trajectory, reflecting slower demand progression and infrastructure deployment outside of the US. We assume non-US markets (ex-China) will grow slower, while China is expected to pick up pace in the coming years.

Global installed data centre IT capacity reached ~70GW in 2025. We estimate that capacity additions could increase by about 25GW in 2026 and >30GW in 2027, with the US accounting for the largest share of incremental capacity, contributing ~15GW and >21GW, respectively. Beyond 2027, we expect the pace of additions to plateau, with the US adding a further 20GW+ per annum, reflecting a transition from the acceleration phase to a more sustained growth trajectory as indicated above.

We expect hyperscalers to account for the largest share of new capacity additions, representing c60% of total additions over our forecast period (excluding China). After analysing close to 500 projects, we conclude that a significant proportion of those announced by non-hyperscalers could face clear execution bottlenecks and might not be realised at all. As a result, we believe realised capacity additions are likely to be more heavily skewed towards hyperscalers than announced project pipelines alone would suggest.

Upside risk

While current forecasts already imply substantial growth in AI infrastructure, we believe the debate is increasingly shifting from whether demand will emerge to how large and durable that demand ultimately becomes. Several factors could drive data centre demand materially above expectations, including broader enterprise AI adoption, continued AI scaling, increasing inference intensity, ongoing cloud growth, and the emergence of sovereign AI as a strategic priority.

AI adoption may still be in its early stages. Despite concerns about efficiency gains and AI commoditization following the DeepSeek release in early 2025, hyperscalers have continued to raise capital spending plans, suggesting confidence in AI-driven returns and sustained demand growth. Data centre fundamentals also remain supportive, with low vacancy rates and strong pricing power. Hyperscaler electricity consumption has increased more than 25% annually for eight consecutive years, largely reflecting continued cloud expansion, which provides a useful precedent, given public cloud continues to grow more than 15 years after its commercial emergence. Most AI spending today remains concentrated among hyperscalers and model developers, while many enterprises are still in the experimentation and pilot phase. As AI copilots, agents, workflow automation, and custom models move into broader production deployment, infrastructure demand could expand well beyond current forecasts.

AI scaling laws continue to support higher compute intensity. Achieving more capable foundation models still requires significant increases in training compute, while the industry's transition toward reasoning models, agentic AI, and physical AI is making inference workloads increasingly compute-intensive.

Sovereign AI represents an additional demand vector. Governments, enterprises, and national champions are increasingly investing in domestic AI infrastructure to support economic competitiveness, national security, and technological leadership, potentially creating a second wave of demand beyond the major hyperscalers.

Downside risks

Despite strong demand signals, the pace and magnitude of future infrastructure growth remain subject to several important risks. While demand for AI compute appears robust, AI efficiency gains, infrastructure bottlenecks, regulatory challenges, workforce constraints, and potential shifts in capital allocation could all result in slower growth than current expectations imply.

AI efficiency gains. Improvements in model architectures, algorithms, software optimization, and compute hardware could reduce the compute and power required per unit of AI output. While lower costs often stimulate additional demand (Jevons Paradox), efficiency gains may outpace workload growth, reducing AI's infrastructure intensity. Better token routing, smaller

domain-specific models, inference optimization, and the growing availability of high-performing open-weight models may lessen the need to train frontier models repeatedly, enabling greater reuse, fine-tuning, and distillation. As a result, AI capabilities could improve while requiring less compute, tempering growth in data centre, power, and capital spending.

Power remains the primary constraint. Grid infrastructure was not designed for individual customers requiring hundreds of megawatts of capacity. Transmission congestion, substation limitations, lengthy interconnection queues, and long lead times for power equipment could delay or limit development, regardless of demand.

Labor shortages could slow deployment. Data centres and supporting power infrastructure require highly specialized workers, including electricians, engineers, mechanics, and pipefitters. Labor constraints are particularly acute in major development hubs where projects often compete for the same limited workforce.

Political and permitting resistance is rising. As AI infrastructure expands, local opposition related to electricity costs, grid reliability, water consumption, emissions, noise, and land-use impacts is becoming more common. While national policymakers broadly support AI leadership, development approvals remain governed primarily at the local level, creating growing permitting uncertainty and geographic limitations.

Capital discipline could emerge as a constraint. Current spending levels reflect strong confidence in AI monetization, but if returns on investment prove slower than expected, capital allocation priorities could shift. Higher financing costs, weaker enterprise AI adoption, or concerns about AI utilization rates could lead hyperscalers, enterprises, and investors to moderate infrastructure spending plans, reducing the pace of capacity additions.

Electric vehicle growth forecasts

- **We expect a steady increase in global EV fleet penetration, with EV mix as a % of the overall fleet growing at a ~15% CAGR between 2020 and 2050, reaching nearly ~60% of the fleet in 2050.**
- **China will be the primary driver of EV fleet growth, with its EV fleet mix exceeding 90% by 2050. Western Europe's ~70% EV fleet mix will also be supportive, while the US's just over ~50% mix will remain a laggard.**
- **EV uptake in most regions will be driven largely by regulatory policy, with key differences between countries with EV targets/mandates and those without.**

Unpacking our EV fleet build assumptions

Our model is based largely on forecasts from BNEF's 2026 Electric Vehicle Outlook. Unpacking key model assumptions:

- **China:** Among the major markets, China will lead the EV fleet transition by a notable margin. Growth in the EV fleet mix is driven by high EV penetration of new vehicle sales (~64% in 2026, rising to more than 90% by the mid-2040s), as well as ongoing government incentives to encourage the transition from ICE to EVs (ie, "scrappage" programs that have been in place since 2024).
- **Europe:** Growth in the EV fleet mix will likely be driven by regulatory policy, as the EU continues to maintain its 2035 targets. EV penetration is expected to see a sharp uptick ahead of this, increasing from ~30% in 2025 to ~75% in 2035.
- **US:** The EV fleet mix will lag due to slower adoption of EVs, with 2035 EV penetration on new sales of ~38%, roughly half the level expected in China/EU. The expected slower US uptake

reflects the current lack of regulatory support for EV adoption, as well as the high mix of large pickup trucks/large SUVs in the fleet, as these segments will be the slowest to electrify.

- **RoW:** We see RoW EV uptake as somewhat contingent on the extent to which Chinese OEMs continue to export EVs into these markets. The EV fleet transition will be more mixed across countries, with East Asia and Southeast Asia likely leading EV adoption.

Upside risk

Potential US outperformance is a source of upside risk, as current expectations are for the US to be a clear laggard in EV uptake and EV fleet penetration. While it is clear that in the near to mid-term the US will lag other major markets such as China and Europe, stronger-than-expected-adoption over the next decade could result in a higher US EV fleet mix and, correspondingly, a higher global EV fleet mix, given that the US is the third largest market.

Downside risks

Regulatory risk is the key potential headwind we see, as current EV forecasts assume that regulations in many markets will continue to encourage EV uptake. The US is a clear example of this, as out-year EV forecasts have been cut significantly over the past ~2 years following the rollback of auto regulations and incentives that had previously supported EV adoption. We view EU regulations as a key area to watch, as EV targets have become an increasingly contested issue.

China weakness is another headwind to consider. China has an outsized effect on global EV adoption, as its domestic market leads the major ones in the EV transition and its OEMs export increasing volumes of EVs to outside markets, supporting Europe and RoW EV adoption. A downturn in either of these trends would likely slow global EV fleet growth.

Part 1: Macro and Thematic views

The changing geo-economics of energy

Christian Keller⁽ⁱⁱ⁾ Barclays, UK

New relevance for an old question

Geography has always mattered for economies. In an increasingly integrated global economy, the trade of primary energy sources such as oil, coal and gas was driven largely by efficiency considerations, with a clear pattern between exporters and importers. Rising geopolitical fragmentation and technological advances have started to change this. Russia's war on Ukraine, disruption in the Strait of Hormuz, the US shale revolution, Europe's nuclear exit and China's mass production of low-cost green technologies have all reshaped the global energy map.

Three challenges: molecules, electrons, science

1. Securing the flow of molecules

In a world that still needs hydrocarbons, the traditional quest to secure the flow of oil and natural gas remains. Rising geopolitical conflict has led energy importers to diversify supply chains, infrastructure and storage. The implications are the need for more investment in storage capacity, pipelines, refining, transport infrastructure and production outside traditional conflict zones and choke points.

2. Growing the flow of electrons

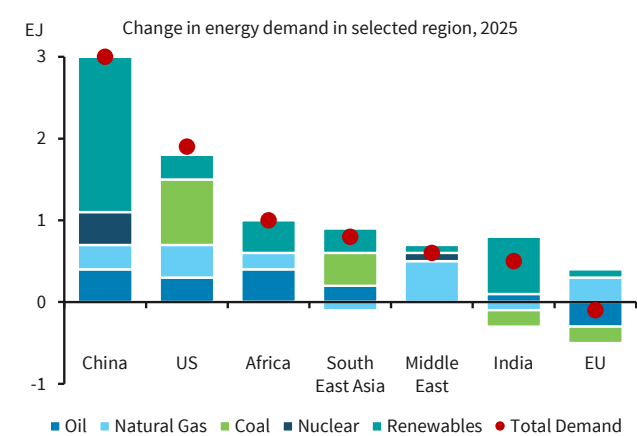
With surging electricity demand, including from AI-related data centres, the key challenge is providing a sufficient and stable flow of electricity. This requires investment in generation, grid capacity, storage and baseload power.

Electrification broadens the range of energy sources, but also creates new dependencies. Reliance shifts from hydrocarbon exporters to suppliers of green technology and to those who extract and refine critical minerals. China occupies a central role across many of these supply chains.

3. Searching for new ways to power

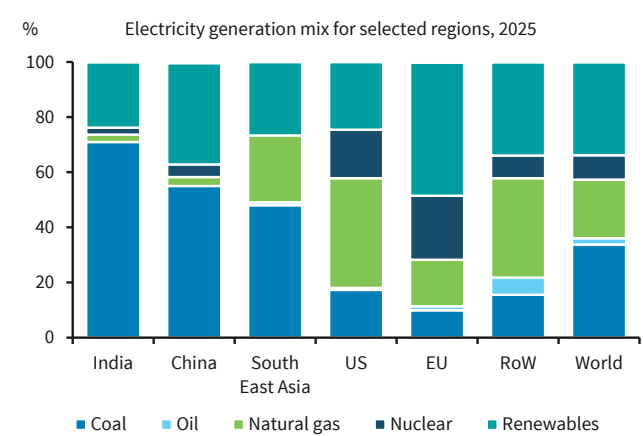
Advancing existing technologies and pursuing breakthrough technologies will be crucial. This includes improvements in solar and nuclear technologies, as well as the search for commercially viable fusion power. Scientific capabilities and access to capital for large-scale, long-term investment will be key.

FIGURE 10. Change in energy demand in selected regions, 2025



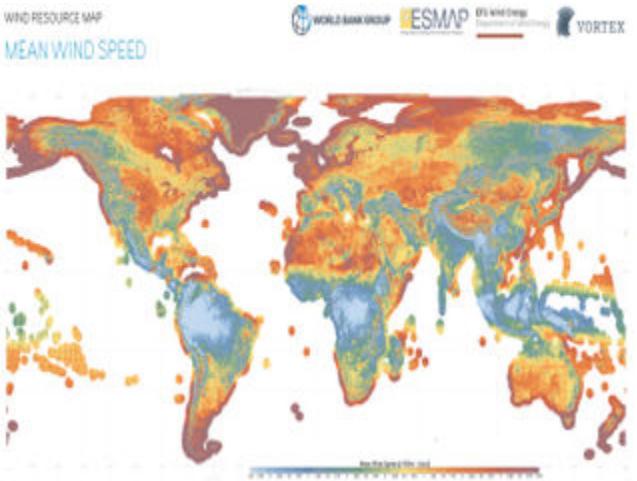
Source: IEA, Barclays Research

FIGURE 11. Electricity generation mix for selected regions, 2025



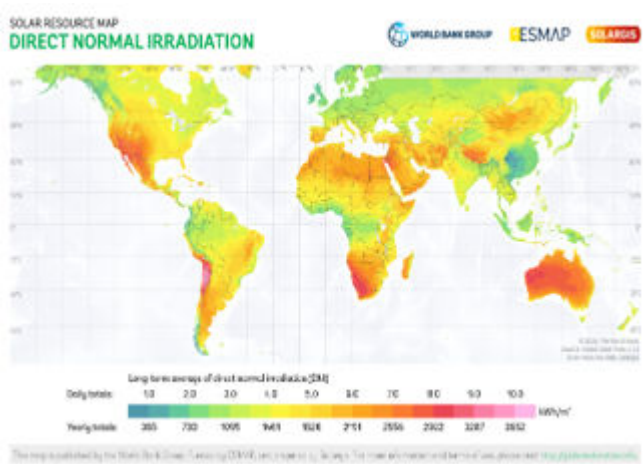
Note: 'Renewables' includes solar PV, wind, hydropower, geothermal, bioenergy and waste, concentrating solar power (CSP), and marine.
Source: IEA, Barclays Research

FIGURE 12. Mean wind speed



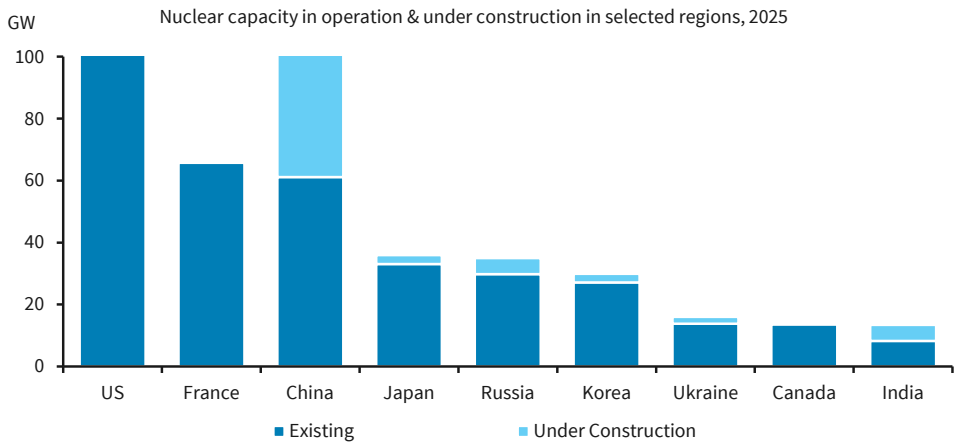
Source: World Bank, ESMAP

FIGURE 13. Direct Normal Radiation



Source: World Bank, ESMAP

FIGURE 14. Nuclear capacity in operation & under construction in selected regions, 2025



Source: IEA, Barclays Research

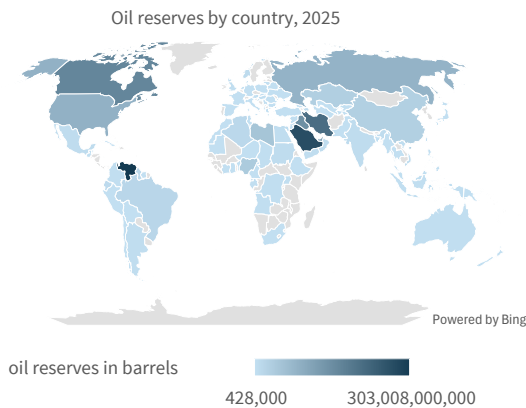
New alliances and new dependencies

Diversifying drilling and distillation

Securing oil supplies in a geopolitically uncertain world supports the diversification of production, refining and transport infrastructure beyond traditional producing regions.

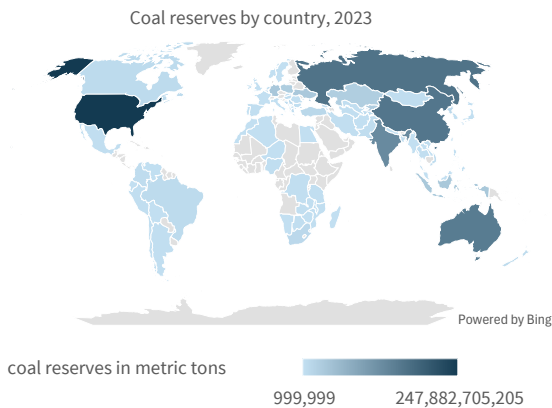
The energy transition also creates a growing focus on critical minerals. Reserves and refining capacity are often concentrated in a small number of countries, with China playing a particularly important role. Access to critical minerals is therefore likely to become an increasingly important factor in trade negotiations, investment flows and economic statecraft.

FIGURE 15. Oil reserves by country, 2025



Source: Worldometers, Barclays Research

FIGURE 16. Coal reserves by country, 2023



Source: Worldometers, Barclays Research

Negotiating critical minerals supply chains

Electricity infrastructure requires critical minerals, the reserves and refining capacity for which are often concentrated in a small number of countries. China plays a particularly important role across many critical mineral supply chains, including graphite, gallium and rare earths, and has used this position to gain strategic leverage. As a result, access to critical minerals is likely to become an increasingly important factor in trade negotiations, geopolitical relationships and industrial policy. Beyond resources and refining capacity, securing the necessary capital to develop alternative supply chains will also be critical, with economic statecraft increasingly

centred on a mix of export controls, investment restrictions, access to foreign capital and sanctions.

Bonding over grids and infrastructure

The need to build electricity grids, pipelines and storage infrastructure requires greater international cooperation. Electricity grids will become increasingly important both within and across borders. For Europe in particular, the challenge is not only generating power, but also transmitting it efficiently across the continent.

Allocating production according to energy abundance

Cost always matters. Energy-intensive industries tend to move to locations where energy is abundant and inexpensive. As a result, energy availability is likely to become a more important driver of industrial location decisions. Regions with access to low-cost energy and reliable infrastructure should become increasingly attractive destinations for investment.

This trend of re-allocating industries could extend further to manufacturing and other industries within the EU. With northern Europe cut off from Russian gas, southern regions such as Spain are able to offer a cheaper energy mix of renewables and pipelined gas from North Africa. This not only makes existing Spanish industries more competitive, but also has led to energy-intensive investments, including data centres, to be directed towards Spain.

AI services as an 'energy export'

In a manufacturing-dominated world, cheap labour has typically been a crucial factor in the allocation of production, but as value creation shifts towards digital and AI-enabled services, it becomes less important, while energy abundance becomes more important. The electricity required to power data centres means that energy-rich regions may increasingly export computing capacity and AI services in much the same way that manufacturing exporters effectively exported cheap labour in previous decades.

Trends for the next ten years

- Energy abundance will gain importance as a determinant of economic activity and international trade, while abundant cheap labour becomes less significant.
- Electricity generation is key, which widens the energy sources to renewable (wind, solar, geothermal, etc.) and brings a refocus on nuclear. This redraws the global map of where power can be generated cheaply.
- Electrification and renewable energy shifts vulnerabilities 'upstream' (grid infrastructure, green tech access), while also creating new downstream dependencies for critical minerals, which have a different geography than the traditional fossil fuel dependence. This requires large investments, favouring regions that are capital rich and have abundance of potential energy sources (eg, US, China). For capital-rich/energy-poor ones (eg, Europe, Japan), it may require increasing foreign investments to capital-poor/energy-abundant areas (Africa, LatAm) to secure energy sources and critical minerals.
- Similarly, capital-rich/energy-poor economies have a strong long-term incentive to invest in potential breakthrough technologies such as fusion, which could potentially change the entire energy landscape.

From a regional perspective, the following dynamics should be expected:

- **North America** looks very robust on all fronts. It combines large proven reserves in fossil fuels with favourable geographies across the renewable spectrum (wind, solar, geothermal),

as well as the availability of capital to invest into advancing nuclear as well as other future technologies.

- **China** looks similarly strong, even without as much fossil fuels of its own. Next to growing nuclear power capacities, China is well ahead in the build-out of renewable energy infrastructure and also dominates the global supply chain of the necessary technologies, not only in manufacturing (eg, solar panels, batteries), but also in mining and refining the critical minerals around it.
- **Europe** will need to use its capital richness to invest heavily in energy infrastructure: to secure the availability of the still-needed fossil fuels (eg, additional gas and oil storage) and grid capacity across Europe to support the build-out of renewables. The EU will want to keep neighbouring energy-rich countries close (Norway, Iceland) and may also potentially seek to tap solar power from North Africa. In principle, energy and commodity-rich Russia remains a natural trading partner for Europe, but geopolitics seem not to allow this for the foreseeable future.
- The **Middle East** will likely be pulled economically eastward by Asia's still-rising need for oil and gas. At the same time, however, the Gulf region could become an indirect exporter of energy (including from renewables) through potentially selling data centre services, including to Europe.
- **Africa, LatAm, and parts of Asia** will look to gain from the growing global energy needs, including from the focus on renewable and electrification. First, those countries with oil and gas (eg, Nigeria, Libya, Angola, Venezuela, Brazil, Ecuador, Malaysia) could benefit from supply diversification away from the Middle East and Russia. Second, those with critical mineral reserves (eg, Chile, DR Congo, Indonesia) will likely gain leverage over investments from capital-rich countries. The **rivalry between China, the US and Europe** for securing access to these critical mineral supply chains is likely to become a crucial geo-economic power play, which could lead to increased capital flows into the relevant countries.
- **Australia** stands out as being a capital-rich advanced economy blessed with an abundance of renewable, fossil, and nuclear fuel deposits and large and diverse reserves of critical minerals. This should make it an important geo-economic presence in the new global energy ecosystem.

The inflation dynamic

Pooja Sriram⁽ⁱⁱⁱ⁾ BCI, US | Jonathan Hill, CFA⁽ⁱ⁾ BCI, US

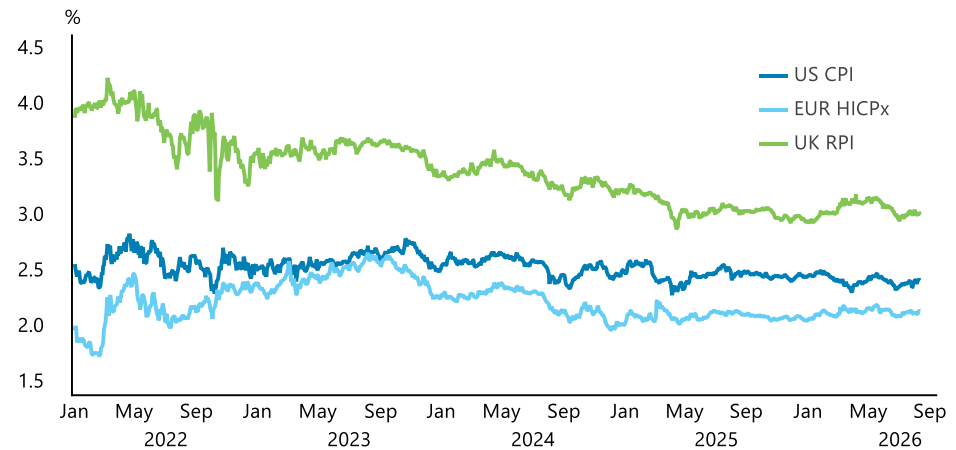
The mismatch between energy supply and demand creates upward pressure on three core factors for inflation risk: baseline expectations, price volatility, and asymmetric risks. Energy demand is rising faster than supply as geopolitical fragmentation increases the premium on energy security. This creates a tension in the market narrative.

**Key question: does the upside inflation risk from
the energy supply/demand imbalance
overwhelm other disinflationary forces?**

While AI is often viewed as near-term inflationary but medium-term disinflationary, we see growing upside inflation risks from energy that may be underappreciated.

One theme from this report is that the energy supply/demand imbalance should support commodity prices, implying inflationary pressure may originate from raw materials, rather than labor market-related wage increases. This suggests that an over-reliance on Phillips-curve-style logic could understate upside inflation risks.

FIGURE 17. 5y5y inflation swaps cross-market

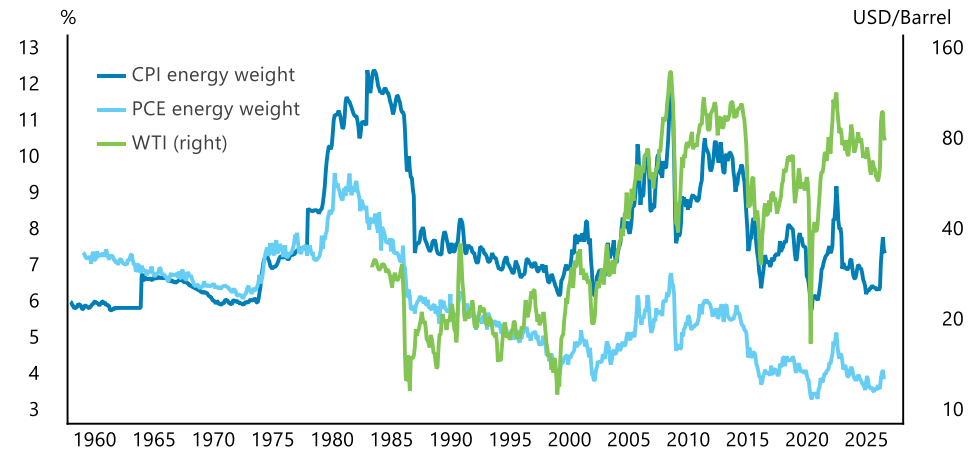


No conversion adjustment made to 5y5y RPI swaps.
Source: Barclays Research

Baseline expectations

Excess energy demand should translate into sustained upward pressure on energy commodities (see [Commodities in focus](#)). If the supply/demand imbalance leads to higher raw material prices, this should ultimately translate into higher commodity futures prices and higher baseline inflation forecasts. In addition, futures curves may be underpricing upside commodity price risk. Rising energy prices can also increase the weight of energy within CPI and PCE baskets, amplifying the inflation impulse.

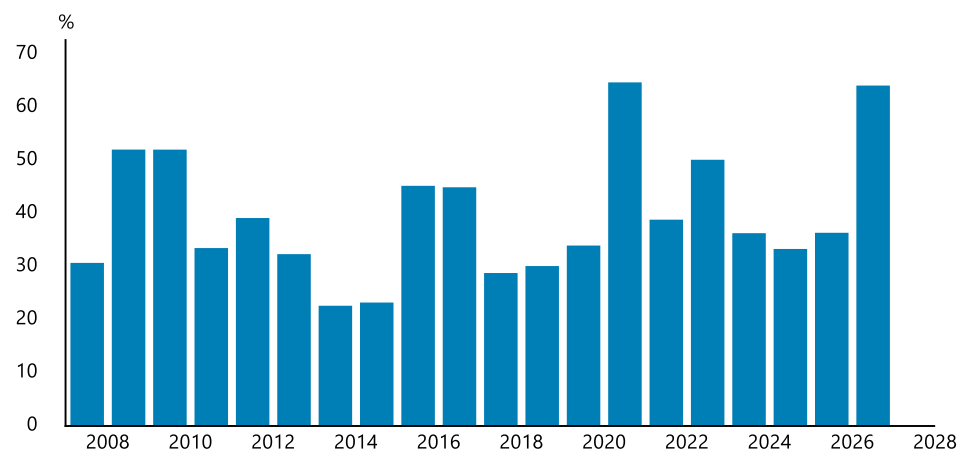
FIGURE 18. Energy weights in US inflation index baskets vs WTI futures



Source: BEA, BLS, CME, Macrobond, Barclays Research

Price volatility

Higher expected energy volatility should translate into increased inflation uncertainty and upward pressure on inflation risk premia. The CBOE's OVX index, a measure of oil volatility, has averaged near record highs in 2026, highlighting elevated energy market uncertainty.

FIGURE 19. Crude Oil ETF Volatility Index

Annual average.
Source: CBOE, Macrobond, Barclays Research

Asymmetric risks

Volatility alone does not create upside inflation risk unless price shocks are skewed to the upside. We believe that is the world investors face today. Geopolitical events such as the Russia-Ukraine war and the US-Iran conflict demonstrate how a more fragmented geopolitical environment can generate more frequent upside energy price shocks.

Put together, we see the above dynamics as helping to justify investors positioning for a rise in inflation risk premium.

Commodities in focus

Amarpreet Singh⁽ⁱⁱ⁾ BCI, US

Higher for longer

- **Growth in freedom molecules is slowing.** Access to molecules should not be taken for granted, as the growth in non-OPEC supply (freedom molecules) slows. Rystad expects non-OPEC supply to peak in 2031. We would not be surprised if that happens a bit sooner.
- **Demand remains resilient, not peaking:** Despite EV adoption and recurring peak-demand narratives, gasoline, jet fuel, industrial, and other oil demand segments continue to grow. We see no evidence of an imminent oil-demand peak and forecast **global oil demand growth of ~1.1 mb/d annually from 2027-30**, reaching roughly **110.6 mb/d by 2030**.
- **Refining highlights potential consequences of this setup.** Net refining capacity additions have been concentrated in Asia and Middle East (outside of the west) over the past couple of decades. As a result, the Russia-Ukraine and Middle East conflicts have pushed gross margins on European diesel to almost 90% right now.
- **Position for higher-for longer-oil.** Barring a step-change in technology, prices need to induce more upstream investments. We think the mid-cycle equilibrium price for Brent should be \$85/b in 2030, compared with the forwards curve at \$70/b.
- **Tighter cushions and fragmentation argue for higher volatility.** Production capacity has been declining globally and US SPR is trending at multi-decade lows, while the race for resources is making a comeback. This is consistent with higher oil vol. We recommend expressing this view using **long Brent call spreads** in the futures space.

For most of the period since 2018, barring the early days of the COVID pandemic and the Russia-Ukraine war when a potential ceasefire deal seemed imminent, and the first few months following the OPEC+ pivot last year, we have consistently been constructive on oil prices relative to the forward curve, primarily because we thought **non-OPEC supply growth was slowing faster than demand**, which left the market more exposed to geopolitical volatility.

As things stand today, with above-ground inventories trending around the lowest level in recent years, the SPR cushion significantly eroded, continued depletion in below-ground inventories implying the need for more upstream investments amid resilient demand, and significantly elevated geopolitical right tail risk, given the situation in the Middle East, we find ourselves sticking to that view.

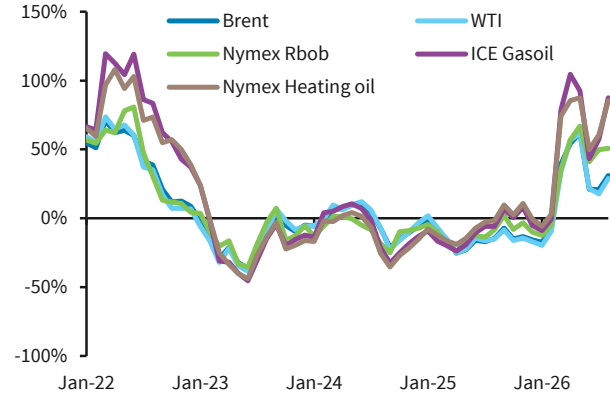
Some argue that the oil price response to the 2026 supply shock has been underwhelming and highlights structural weakness in underlying fundamentals. We would note two things. First, it is important to consider the effect of the supply shock on refined product prices as well, because SPR releases and refining capacity constraints have resulted in significant dislocations.

FIGURE 20. Forwards-implied 2026 Brent is up ~50% year-to-date



Note: Realized Brent price versus that implied by the forward curve at the end of the corresponding previous year. Last data point is 2026 and is based on forwards-implied average 2026 Brent price as of 2 September 2026.
Source: Bloomberg, Barclays Research

FIGURE 21. Refined product prices have jumped much more



Note: Y/y change in monthly average crude oil and refined product prices. Last data point is August 2026.

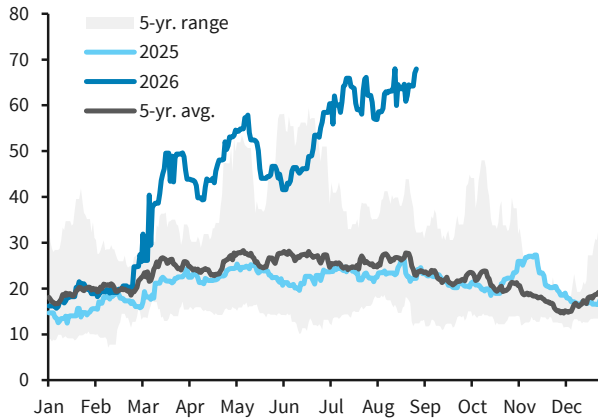
Source: Bloomberg, Barclays Research

Second, in 2020, global oil demand fell by slightly over 9 mb/d and Dated Brent averaged 33% below the forward-implied 2020 average at the end of 2019. If we go by the EIA's latest forecast, OPEC crude supply will decline by slightly over 5 mb/d this year and the forward-implied 2026 Brent is up 50% year-to-date.

We think the mid-cycle equilibrium price for Brent in 2030 should be \$85/b, compared with the forwards-implied average of \$70/b at the time of writing. The path, however, will be bumpy, as we expect volatility to remain elevated.

Oil market volatility has been relatively elevated over the past several years. In addition to the effect on fundamentals of externalities such as COVID-19 and wars in Europe and the Middle East, the anticipated effect of the ongoing energy transition on underlying fundamentals has also boosted uncertainty.

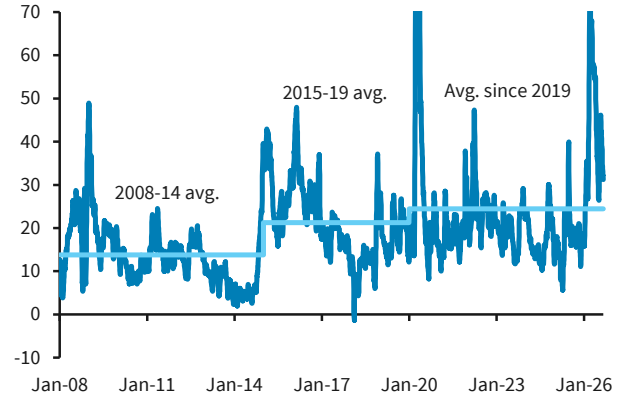
FIGURE 22. SPR, refining capacity and quality constraints have boosted refining margins



Note: USGC products minus WTI Cushing (3-2-1 crack spread), \$/b

Source: Bloomberg, Barclays Research

FIGURE 23. Geopolitical uncertainty has been exacerbating the long-term increase in excess implied volatility in oil markets



Note: Excess implied volatility in oil markets (OVX minus VIX), 7dma. The light blue line shows highlighted period averages.

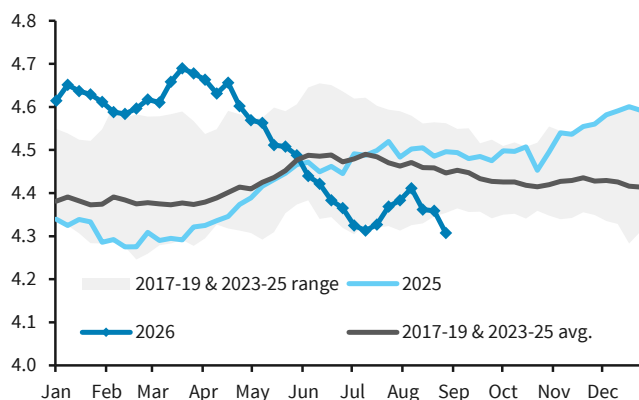
Source: Bloomberg, Barclays Research

For perspective, between 2008 and 2014, excess implied volatility in oil (OVX minus VIX) averaged 13.8. Between 2015 and 2019, it increased to 21.3, as fundamentals deteriorated with the rapid expansion in US tight oil output and an increase in spare capacity. US tight oil production growth has been slowing for years but excess implied volatility in oil has averaged 24.4 since 2019.

Let's unpack these factors.

- **Low above-ground inventories:** A potential deal to resolve the Strait of Hormuz standoff remains elusive, and based on our assessment of flows through the region, above-ground inventories will likely tighten further over the coming weeks. Our weekly global total oil indicator suggests that observed inventories are already trending at multi-year lows, and the situation looks worse when adjusted for the accumulation of reserves in China over the past several years. More importantly, ex-China inventories continued to tighten throughout 2023 and 2024 despite slower cyclical demand growth and the surprise increase in Iranian supplies despite sanctions. The OPEC+ pivot last year was the turning point for inventories.

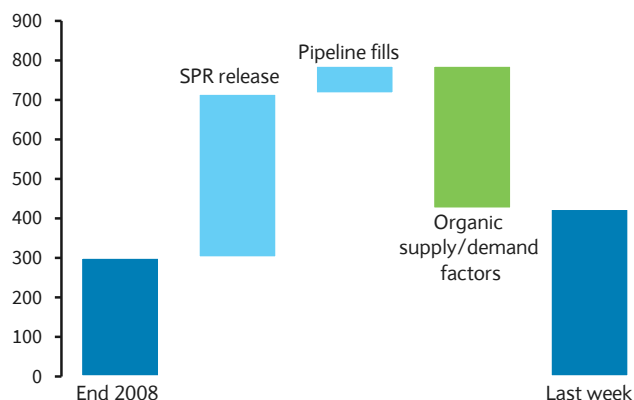
FIGURE 24. Above-ground inventories have tightened significantly since the beginning of the war



Note: Barclays' Weekly global total oil inventory indicator, bb

Source: EIA, Kpler, Barclays Research

FIGURE 25. SPR releases have cushioned the effect on commercial inventories (last week: 28 August 2026)

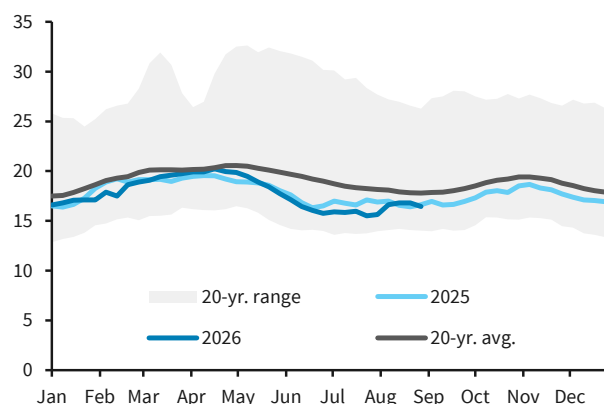


Note: Breakdown of the change in US commercial crude oil inventories since 2008, mb. Last week here is based on the EIA's data for the week ending 28 August 2026.

Source: EIA, Barclays Research

- Oil prices have been relatively well behaved because commercial crude oil inventories have been cushioned by SPR releases. If not for SPR releases and the increase in pipeline fills to facilitate production growth, commercial crude oil inventories in the US would be much lower. Adjusting for refining demand, commercial crude oil inventories in the US look relatively tight despite ongoing SPR releases.
- Refining demand is likely to remain strong. Core refined product inventories, which have not had a similar level of cushion from SPR releases, are much tighter than commercial crude oil inventories; along with constrained refining capacity because of years of underinvestment outside of China and the Middle East and growing attacks on refining infrastructure in Russia, limiting its exports, that is primarily behind the dislocation in crude oil and refined product prices of late.

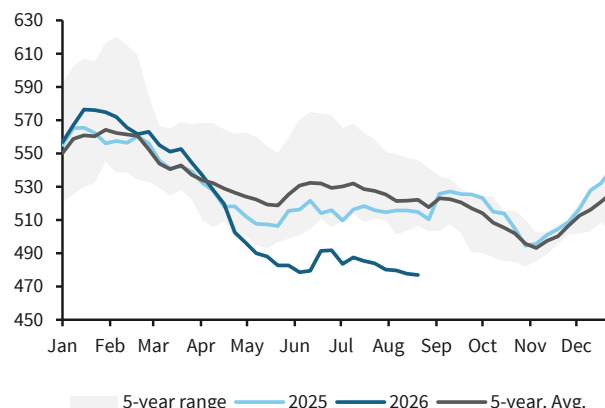
FIGURE 26. Yet commercial crude oil inventories in the US remain tight...



Note: Commercial crude oil inventories in the US excluding pipeline fills in days of refining demand

Source: EIA, Barclays Research

FIGURE 27. ...and global refined product inventories are much tighter

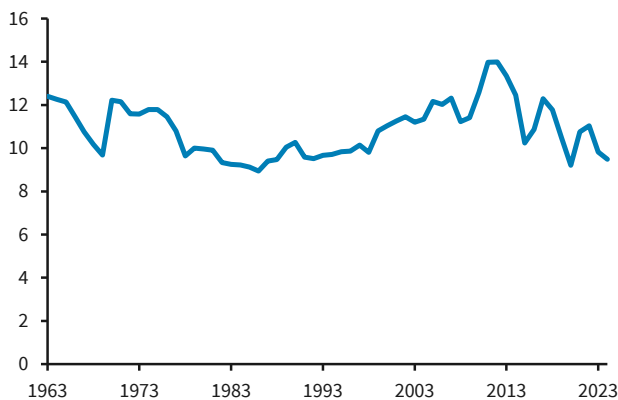


Note: Barclays' Weekly global core refined product inventories, mb

Source: EIA, ARA, IES, Barclays Research

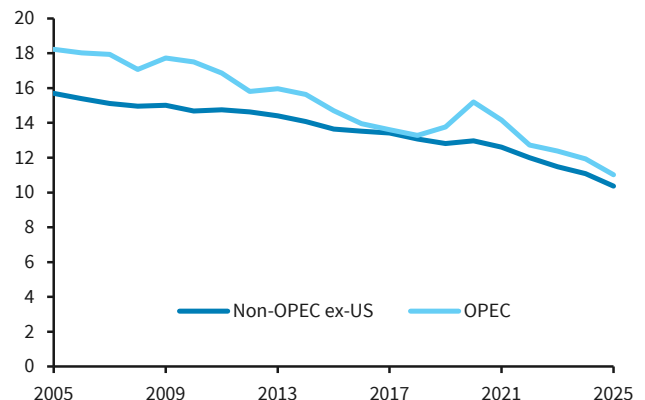
- Shrinking below-ground inventories: Above-ground inventories are a function of the balance between well-head supply and demand. Well-head supply of oil is a function of both economic and political drivers. OPEC supply is more politically driven in most scenarios whereas non-OPEC supply is more a function of economic drivers.
- Proved reserves, akin to below-ground inventory, relative to production, are generally a good leading indicator for well-head supply. They indicate the level of production capacity relative to current production and a higher number means greater potential for growth in well-head supply and lower base decline rate.
- If the number has been declining over time, that indicates sufficient investments are not being made to maintain productive capacity relative to production and therefore the outlook for production growth is deteriorating. This would also indicate that the development is getting more intensive leading to higher decline rates.
- Globally, the proved reserves-to-production ratio has been declining steadily over the past several years. In the US, which has accounted for ~75% of total non-OPEC supply growth since 2005, the ratio stood at 9.5 years in 2024, down from 11 years in 2022 and 14 years in 2012.
- Proved reserve estimates are influenced by realized prices and can be backward looking at times but WTI Midland averaged \$83/b between 2022 and 2024, significantly ahead of the marginal cost of supply for US tight oil production. Yet, despite continued efficiency improvements, the decline in production capacity in the US indicates a resource base coming of age.

FIGURE 28. Proved reserves have not kept pace with production in the US



Note: US proved crude oil reserves in years of production
Source: EIA, Barclays Research

FIGURE 29. Even outside of the US, productive capacity has been declining

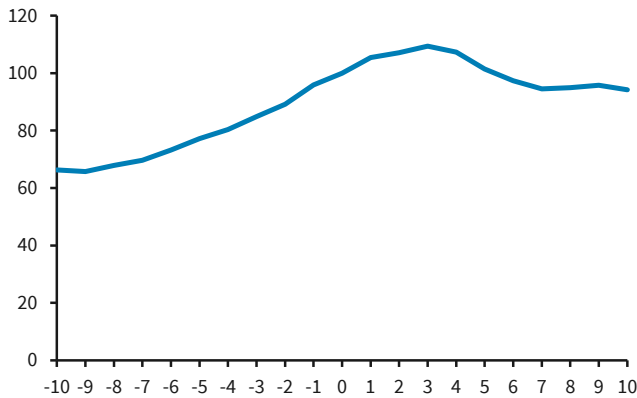


Note: Proved crude oil reserves outside of the US in years of production
Source: Rystad, Barclays Research

- A decline in the proved reserves-to-production ratio below the 10-year level has historically been a strong leading indicator for an inflection point in well-head production. For example, in Mexico, the R/P ratio fell below 10 years in 2002 and production peaked in 2004. In Nigeria, it fell below 10 years in 2011 and production peaked the same year. In Angola, it fell below 10 years in 2005 and production peaked in 2008. In Malaysia, it fell below 10 years in 2003 and production peaked in 2004.
- Even in the US, production formed a local peak in 1970, a year after the R/P ratio fell below 10 years. However, technological improvements in resource development helped reverse the

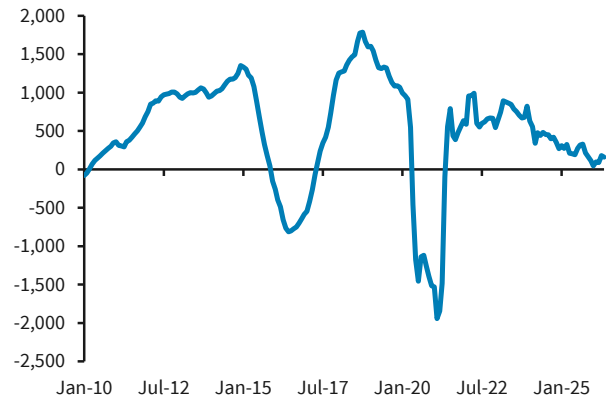
decline in production in the late 2000s, which is quite similar to what happened in the UK in the 1990s after production there formed a local peak in 1985, three years after R/P in the country fell below 10 years. Last, in Norway, it fell below 10 years in 1993 and production peaked in 2000.

FIGURE 30. R/P falling below 10 years has historically been a strong signal for an inflection point in production



Note: Average oil production profile for the US, UK, Norway, Mexico, Nigeria, Angola and Malaysia around the time when proved reserves to production ratio in the country first fell below 10 years (indicated by T=0 on the X-axis). The Y-axis shows production at T=0 indexed to 100.
Source: EIA, Rystad, Barclays Research

FIGURE 31. US crude oil production growth remains relatively weak

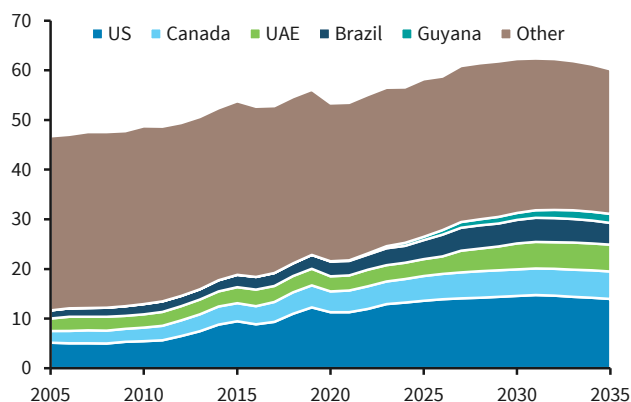


Note: Y/y change in L48 onshore crude oil production, kb/d, 3mma

Source: EIA, Barclays Research

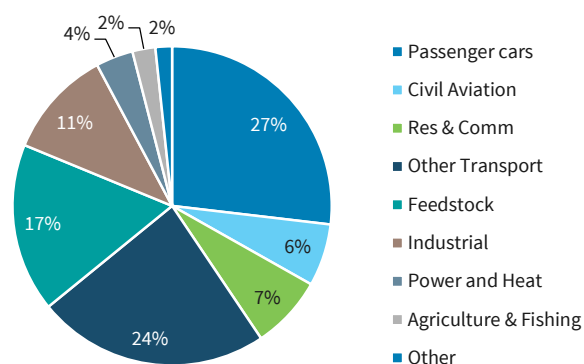
- Therefore, barring a step-change in development technology, it is safe to assume that US crude oil production will peak in the coming years. Rystad expects this in 2031, while our own estimates suggest it could happen a bit sooner. For perspective, L48 onshore crude oil production growth has been steadily slowing over the past few years and stood at just ~150 kb/d y/y in Q2 26.
- Unfortunately, investments in productive capacity outside of the US have also not kept pace with production. The aggregate R/P ratio for non-OPEC countries ex-US, which includes the newest member of the group - the UAE - stood barely above 10 years in 2025. Even in OPEC, where production has done nothing over the past few years, productive capacity has been steadily declining. In 2025, OPEC production stood at 26.6 mb/d, compared with 26.9 mb/d in 2019 but the aggregate R/P ratio for the group fell from 14 years to 11 years over the same period. With that backdrop, it should not be surprising that most OPEC producers failed to meet production targets last year.

FIGURE 32. Rystad expects non-OPEC oil production to peak in the coming years



Note: Non-OPEC crude oil and condensate production by country, mb/d
Source: Rystad, Barclays Research

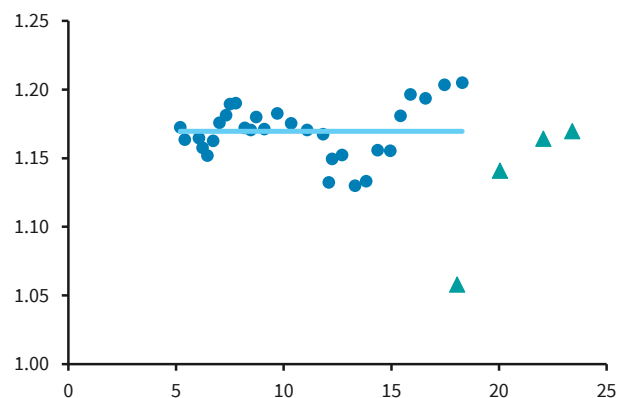
FIGURE 33. Two-thirds of global oil demand is largely price-inelastic...



Note: 2023 global oil demand breakdown by end-use
Source: IEA, Barclays Research

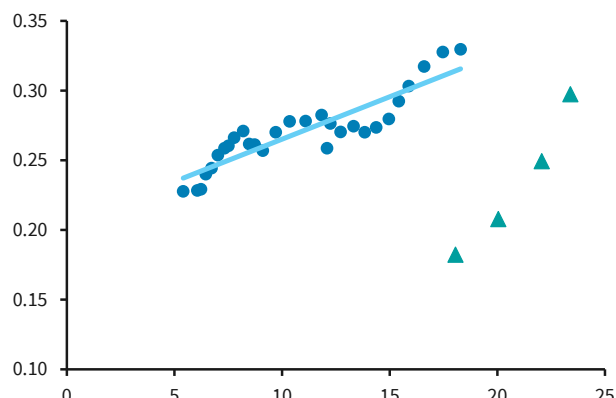
- Resilient demand: If demand were slowing in tandem with supply, prices would not need to induce more upstream investments, but demand has not been slowing much at all. First COVID-19, then monetary tightening and housing sector stress in China and the US have helped create an impression that peak oil demand is imminent but we do not see any evidence of that.
- We like to break down oil demand into four broad categories: passenger cars and other personal ground mobility - primarily gasoline; civil, commercial and defense aviation - primarily jet fuel; oil demand tied to industrial activities like petrochemical production, ground and sea freight, industrial processes and power generation; and residential and commercial demand, primarily for space/water heating and cooking, agricultural and fishing demand and other smaller end uses.

FIGURE 34. Global gasoline demand has been resilient despite growing EV sales and lasting behavioral changes due to the pandemic...



Note: Y-axis: Global per-capita gasoline demand, barrels per year; X-axis: Global per-capita PPP GDP, thousands of USD. Blue line shows the long-term pre-pandemic average. Green triangles show data since 2019.
Source: IEA, IMF, US Census, Barclays Research

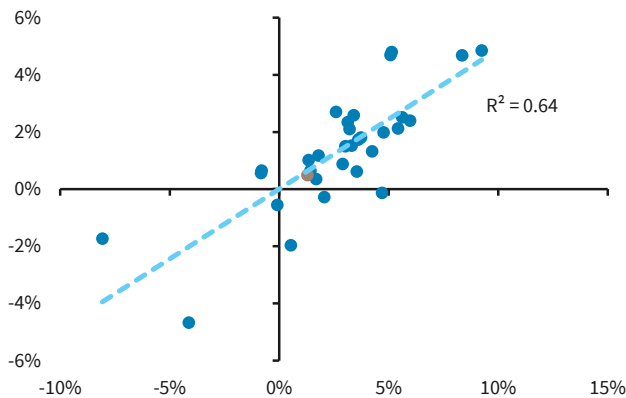
FIGURE 35. Jet-kerosene demand growth remains strong



Note: Y-axis: Global per capita jet-kerosene demand, barrels per year; X-axis: Global PPP GDP per capita, thousands of USD. Blue line shows the pre-pandemic trend and green triangles show data since 2019.
Source: IEA, IMF, US Census, Barclays Research

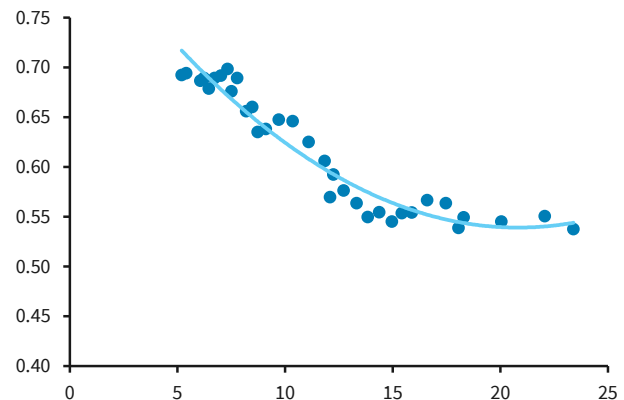
- Despite growing EV penetration, global gasoline demand is still rising with the population. This is because the offsetting effects of continued increases in the popularity of SUVs, slower efficiency improvements in conventional drive trains and growth in addressable markets (there is evidence of an [exponential increase](#) in car ownership when per-capita PPP GDP surpasses \$4,500) have helped to stabilize demand.
- On the other hand, jet fuel demand continues to grow strongly amid increasing income levels and populations, and demand linked to industrial activity is not showing any signs of accelerating efficiency improvements. Last, residential and commercial and other demand has also seen efficiency gains slow down a bit of late.

FIGURE 36. Demand for industrial end-uses has been tracking in line with the historical relationship with global industrial activity



Note: Y-axis: Y/y change in global oil demand tied to 'Other transport', 'Feedstock', 'Industrial' and 'Power and Heat' end-uses shown in Figure 33; X-axis: Y/y change in global industrial output. Brown dot shows 2023.
Source: IEA, World Bank, Barclays Research

FIGURE 37. Efficiency improvements in other end-uses have slowed of late



Note: Y-axis: Log of global oil demand (mb/d) tied to 'Res & Comm', 'Agriculture & Fishing' and 'Other' end-uses shown in Figure 33 X-axis: Global PPP GDP per capita, thousands of USD.
Source: IEA, IMF, US Census, Barclays Research

- Therefore, we do not see an oil demand peak on the horizon and barring a broader slowdown in the interim, expect it to grow at an average pace of 1.1 mb/d between 2027 and 2030 to average 110.6 mb/d by 2030.

FX: Data sovereignty, energy security and dollar exceptionalism

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One of the most dominant, resilient and persistent investment trends of the past decade (if not more) is the rise of US exceptionalism, which has coincided with a long period of a strong (potentially even somewhat overvalued) dollar.

So strong is this trend that it has helped the US dominate global capital flows. The decade before the 2008 crisis was marked by capital flows to international destinations (BRICs, EMs, etc.) in search of growth and capital deepening. Since about 2012, the flow of capital has reversed. By and large, capital has flown mainly into the US and specifically into corporate financing markets (credit and equities). The US equity market represents a dominant and rising share of global equity capitalization, single companies in the US are worth about as much as the capitalization of entire developed country indices, while sponsorship for similar sectors, themes and investment vehicles abroad has waned.

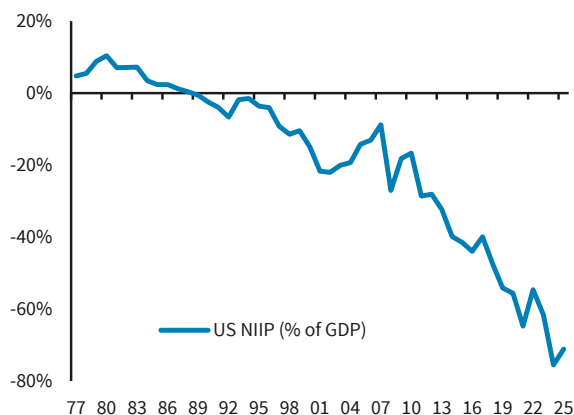
As things stand, the net share of US assets held by foreigners (ie, net of foreign assets held by Americans) is c.75% of US GDP, when 14 years ago net non-resident holdings amounted to a much smaller share ([Figure 38](#)). Naturally, this timespan coincides with a US dollar that has largely appreciated against major and emerging currencies. Dollar appreciation has transformed the FX market from one of diverse themes (such as commodity and central bank differentiation, carry, etc.) to one of (primarily) dollar sensitivities.

The mere scale of US asset ownership has given rise to contrarian narratives. Particularly around "Liberation Day", the concern about capital flight from the US drove a sharp dollar sell-off. But such narratives keep missing two of the most robust fundamental factors driving returns for capital higher and drawing the flow of capital from around the world to the US: energy security and data sovereignty. As we wrote in [The Rise of the Mighty Dollar](#), the past decade has seen the rise of the US as a major energy exporting destination and as a technology exporting hub, which has underpinned higher neutral rates and currency strength.

As we discuss in this note, the rise of AI in a world of geopolitical uncertainties and fragmentation makes these two pillars of USD exceptionalism not only more robust, but also more interlinked. The success of the AI experiment of such scale crucially hinges on energy abundance and security. Increasing demand on energy resources makes energy-supplying nations (eg, in the Middle East, Latin America but notably also the US, per [Figure 39](#)) crucial participants in the arena of international politics and boosts volumes and their terms of trade.

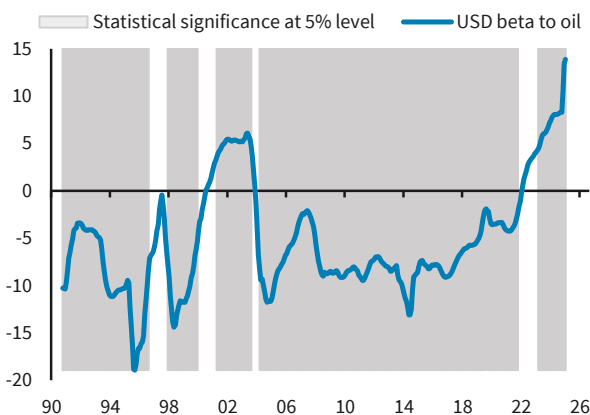
Inversely, countries and regions with digital and energy deficits (eg, the EU or Japan) [run crucial dependencies](#). Others (such as China) have embarked on a race for their own domestic version of technological and energy independence.

FIGURE 38. The US's Net International Investment Position is a symptom of US exceptionalism



Source: Haver Analytics, Barclays Research

FIGURE 39. The dollar is now positively correlated to oil



Note: The chart shows the semi-elasticity of the dollar to oil prices derived via a linear-log regression linking the two according to the methodology outlined in [The new dollar-oil relationship](#). The estimated beta translates to 0.5-1% dollar upside for a 10% increase in oil prices in recent years

Source: Haver Analytics, Bloomberg, Barclays Research

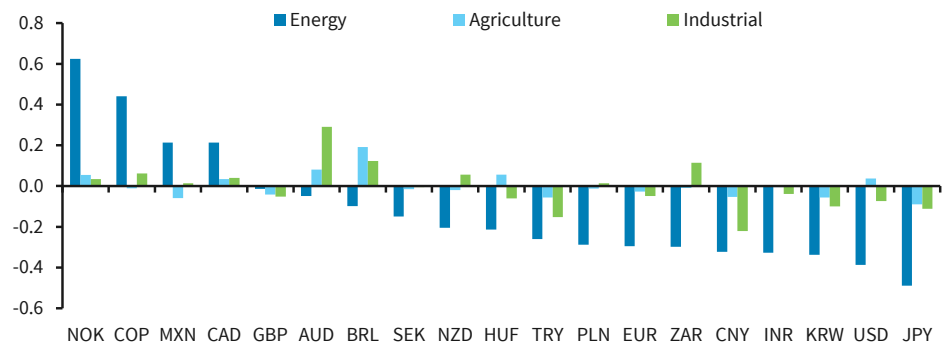
The same factors driving demand for energy are also driving demand in other commodities (including copper and other AI-sensitive ones), benefiting resource-rich nations (eg, Australia).

Once again, the US looks well positioned for its exceptionalism to extend. The base is by now different, which means that non-resident claims on US assets cannot grow at the same pace as they did over recent years. Yet it is also unlikely that strong outflows will emerge soon and drive a persistent dollar undervaluation trend, at least barring recessions in the US corporate sector.

There is also a clear wedge to terms of trade and overall competitiveness between the energy (and commodities more broadly) "haves" and "have-nots". Energy-exporting nations are expected to support higher local yields and more resilient currencies. On the other hand, rising digital deficits in industrial nations such as Europe and Japan could prove to be long-term factors for low growth and volatile terms of trade (Figure 40).

Last, on a geopolitical level, it is pretty clear that the world is splitting between those who are better linked to the US and those who are opting for the Chinese ecosystem of trade and information intelligence. This shift is redrawing the global energy map (eg, the Russia-China oil link), creating fresh areas of competition and risk. Rising trade tensions between the US and China and the Iran situation can be seen as expressions of these emerging areas of pressure.

FIGURE 40. Terms of trade betas to shifts in energy and other commodities



Note: The chart shows estimated long-term elasticities of countries' ToT to energy, agriculture and industrial metal indices. Estimates are derived from a regression of ToT percentage changes on their one-period lag, percentage changes in the three commodity price indices, and their squared terms to account for non-linearities
Source: Bloomberg, Barclays Research

A new era for energy expansion

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Jordan Isvy⁽ⁱⁱⁱ⁾ Barclays, UK

Transition: A secular capital-allocation cycle

We see the current moment as a critical juncture in the global energy transition. The first phase was defined by climate ambition, falling clean-technology costs and rapid deployment, supported by a geopolitical environment broadly conducive to global coordination. This was effective in bringing down green premia, establishing clean power as the predominant source of incremental generation (see [Taking stock of the IEA World Energy Outlook](#) and [EI statistical review of world energy - key takeaways](#)). We believe the transition is entering a new phase, increasingly underpinned by secular growth drivers beyond climate.

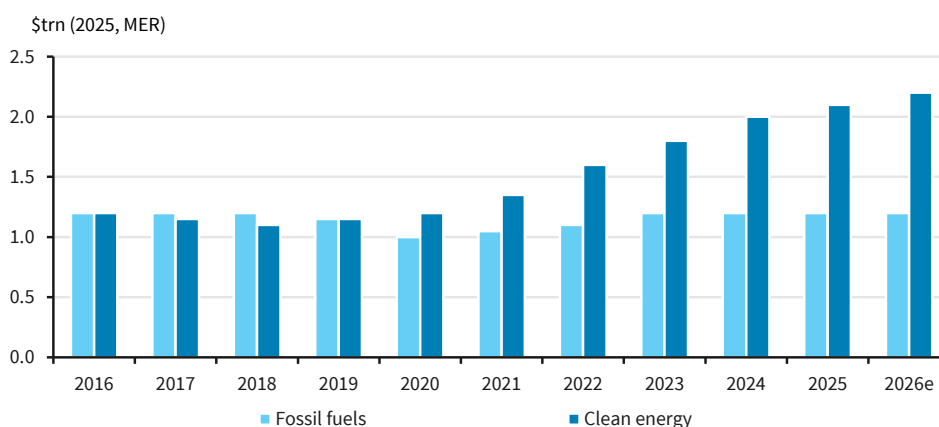
Energy security and competitiveness are emerging as transition drivers on par with sustainability. As we discussed in [Sustainable Investing 2026 Outlook: Security, Sustainability, Scalability](#), the transition is increasingly tied to energy sovereignty, AI infrastructure, defence and domestic industrial capacity. Countries are pursuing pathways shaped by their resource endowments, industrial priorities and energy-import vulnerabilities, while rising power demand is colliding with transmission constraints and geopolitical fragmentation is reshaping energy and technology flows. For the US, rising power demand and the pursuit of the "energy

dominance" agenda continue to support investment, despite a less favourable federal policy environment for parts of the clean-energy complex ([Taking stock of US power post-OBBA](#)). Across Asia and Europe, the economics of electro-tech and the push for energy sovereignty remain important transition catalysts. While the lexicon varies by region, the overall direction is the same with clean energy increasingly being deployed as strategic infrastructure. This broadens the transition's foundations beyond climate policy and, in our view, makes the underlying capital cycle more durable.

We view this as a structurally bullish signal for both clean and conventional energy.

Investment in the entire energy system is rising, with clean energy and electrification solutions increasingly gaining share in the global energy portfolio. Clean energy investment reached parity with hydrocarbons only in 2019, but is expected to be twice as large in 2026. The IEA expects total energy investment to reach \$3.4trn in 2026, up 5% year-on-year, with almost two-thirds directed towards clean energy technologies and infrastructure ([Figure 41](#)). The acceleration since 2022, despite a narrative shift away from climate, supports our view that the clean energy investment cycle is increasingly structural.

FIGURE 41. Global investment in energy, 2016-2026e



MER = Market Exchange Rate. 2026e = estimated. Data are approximated from the IEA.
Source: World Energy Investment 2026, CC BY 4.0, Barclays Research

A multi-decade investment gap, not a temporary capex lift

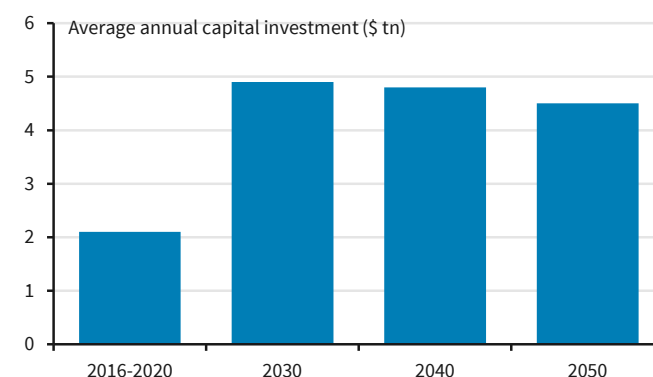
Rapidly rising energy transition capex is still insufficient to meet global energy needs, meaning hydrocarbons will likely form part of the global energy mix beyond 2050.

The energy transition is a multi-decade megatrend. It will require capital not only for greenfield assets, but also for transforming the existing high-emitting capital stock. In [The macroeconomics of the green transition](#), we and our Global Economics colleagues estimated that reaching the Paris Agreement would imply additional net investment of 2-8% of global GDP annually. This estimate comes with a wide range given the high degree of variance across different energy transition scenarios ([Figure 42](#)). However, the common thread is that the energy transition will require a sustained increase relative to the current c.\$2trn/year, highlighting the need for additional investments in the energy system. For example, the IEA's NZE scenario sees annual clean energy transition investment rising to c.\$5trn by 2030, before levelling off at c.\$4.5trn by 2050 ([Figure 43](#)).

FIGURE 42. Significant transition capex is required under a range of scenarios...

Av annual investment to 2030 (\$ tn)	Source	Scenario	Scope
3.5	NGFS	Net Zero 2050	Total transition investment in 1.5 °C scenario
4.5	IEA	Net Zero 2050	Broad clean-energy investment requirement across power, grids, EVs, efficiency, low-emissions fuels, CCUS and related infrastructure
>5	IRENA	1.5C pathway	Broad annual investment requirement to stay on a 1.5°C-aligned energy transition pathway.
5.6	BNEF	Net-zero pathway / Paris-aligned transition	Broad low-carbon energy-transition investment requirement to get on track for global net zero by 2050

Source: NGFS, IEA, IRENA, BNEF, Barclays Research

FIGURE 43. ... and will need to be sustained out to 2050IEA 2021 NZE energy-sector capital investment.
Source: IEA, Barclays Research

The composition of this investment matters as much as the total. The first phase of the transition was principally about adding clean generation and lowering technology costs. The next phase is likely to be more system-wide and capital-intensive: connecting clean generation, balancing variable output, electrifying final demand, transforming high-emitting industrial assets and protecting infrastructure against physical risks. This is consistent with our [2026 Global Energy Transition Barometer](#), which finds transition momentum improving across every region covered, with clean-energy supply the strongest and most broadly based tailwind. However, it also identifies rising system friction, with renewable capacity expanding while grid investment lags, clean-generation costs falling while integration costs rise, and electrification advancing without sufficient clean power or system flexibility. The constraint is shifting from whether clean technologies can scale to the ability of the wider energy system to absorb that scale reliably and economically.

In the new energy equation, reliability is attracting a premium

An increasingly electrified economy raises the premium for clean, firm baseload

power. Variable renewables will retain a leading role in new capacity additions, but nuclear, hydro, geothermal, long-duration storage and other forms of flexible low-carbon capacity should be viewed as complements, rather than competing transition narratives. We are particularly constructive on nuclear's strategic role, with its position at the intersection of decarbonisation, energy security and powering AI, informing our view that we are in the midst of a nuclear renaissance ([Investing in nuclear: From baseload to breakthroughs](#)). We remain selective on investability, seeing more attractive opportunities in the wider value chain ([Uranium: Make or break for the nuclear supercycle](#)), which is not as exposed to execution risk as downstream companies ([Nuclear progresses from conviction to construction](#)). In the longer term, we see nuclear fusion as a technology with the potential to revolutionise the energy system and solve the energy trilemma ([Fusion: Tomorrow's solution to today's problem](#)).

This selectivity illustrates a wider principle for clean infrastructure: strategic relevance does not automatically translate into attractive returns.

China's manufacturing scale and technology innovation have lowered the cost of solar, batteries and other electro-technologies, accelerating deployment but compressing margins across commoditised supply chains. Being constructive on clean-energy deployment is therefore not the same as being uniformly bullish on clean-tech manufacturers, with value potentially accruing more consistently to developers

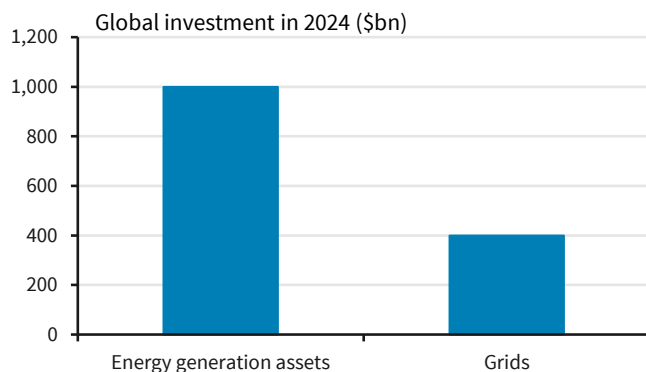
with route-to-grid and contracted demand or to scarcer enabling assets such as transmission, transformers, storage and grid software. Geopolitical fragmentation further complicates capital allocation, as countries are balancing low-cost Chinese equipment with supply-chain resilience and industrial sovereignty, while retaining access to US-linked hydrocarbons for flexibility ([Weekly Spotlight: Clean tech versus oil and gas exports](#)). The clean-investment gap will not be closed simply by identifying the technologies required in a transition pathway. Projects need investible revenue models, viable offtake, manageable construction risk, predictable regulation and a cost of capital consistent with their duration. Where these conditions are absent, even economically necessary infrastructure can remain unfunded.

Transmission is becoming as important as generation

The most acute shortfall is not necessarily in clean generation itself, but in the infrastructure required to make that generation usable. Cheap generation does not necessarily imply reliable power. Its value depends increasingly on route-to-grid, storage, demand flexibility, interconnection and the ability of the power system to manage changing patterns of supply and demand. The Iberian blackout illustrates the challenge. As we wrote in our [Network Security](#) series, the Spanish government's investigation identified insufficient voltage-control capacity as a principal contributor, rather than high renewable penetration or a simple lack of system inertia. Renewable generation was not inherently the problem, but its rapid growth had exposed weaknesses in a power system that had not been modernised sufficiently quickly. The investment is also needed for large renewable adopters such as [China](#) and [India](#), evidenced by their latest capex budgets.

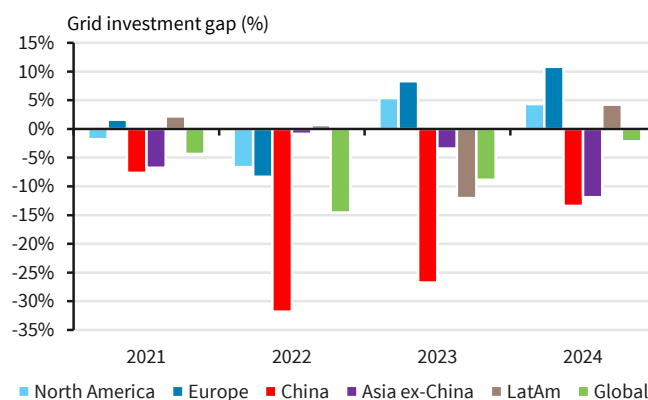
Bringing network investment closer to generation could define the next phase of the transition. Global investment into grids amounted to roughly \$400bn globally in 2024, compared with about \$1trn in generation assets, highlighting the potential scale of the investment opportunity as this ratio moves closer to parity ([Figure 44](#)). Our [Energy Transition Barometer](#) similarly finds renewable capacity growth outpacing grid investment in several regions, contributing to congestion, negative wholesale prices and greater volatility where network capacity and flexibility have failed to keep pace ([Figure 45](#)). We do not view this as a sign that the transition is failing, but rather that it is entering a more infrastructure-intensive phase that will require transmission, distribution, storage, transformers, cables, interconnection and digital grid management.

FIGURE 44. Global investment in grids significantly lags that into energy generation assets



Source: IEA (2025), World Energy Investment 2025, CC BY 4.0

FIGURE 45. Rapid renewables growth has outstripped grid investment growth, particularly in China and wider Asia



Grid investment gap is computed as the difference between grid investment and renewable capacity growth rates.

Source: BloombergNEF, Ember, Barclays Research

The financing challenge: Mobilising capital at scale through ESG bonds

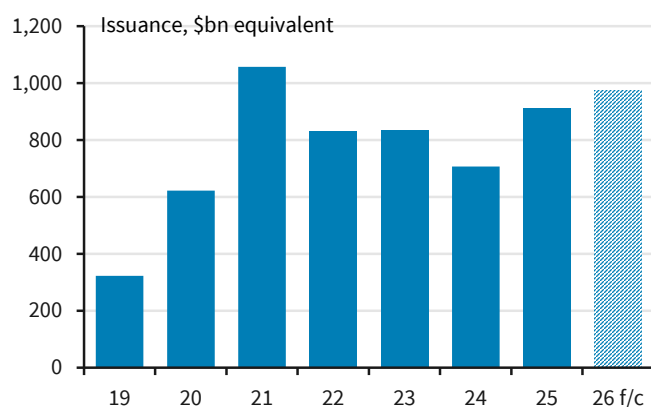
ESG bonds have helped scale low-carbon investment

Delivering the next phase of the energy system will require trillions of dollars of investment across generation, transmission, storage, industrial infrastructure and energy-security assets, presenting a significant financing challenge. Mobilising sufficient capital may prove just as important as identifying the technologies needed to deliver the transition.

The ESG bond market is one mechanism through which this capital can be deployed. Over the past decade, green, sustainability and sustainability-linked bonds have evolved into a multi-trillion-dollar market (Figure 46) and an important source of financing for corporates, sovereigns and SSAs in particular. To date, however, both issuance and use-of-proceeds have been concentrated in a relatively narrow set of sectors and activities. Banks, utilities and real estate have dominated corporate issuance (Figure 47 and Figure 48), while green bond financing has been directed primarily towards renewable energy, clean transportation and green buildings (Figure 49). As a result, labelled debt markets have played an important role in financing renewable energy, clean transportation and green buildings, but have been used less extensively in many of the sectors and activities facing the greatest transition financing needs.

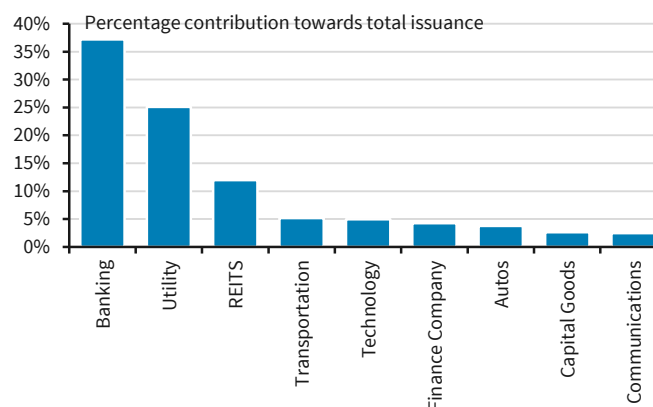
ESG-labelled bonds can also offer issuers benefits beyond financing specific projects. They can help diversify and broaden the investor base, provide a mechanism for demonstrating transition and sustainability commitments, and may offer issuance-related advantages through stronger order books and, in some cases, modest pricing benefits ([Secondary market greeniums fade, primary benefits remain](#)).

FIGURE 46. Total ESG bond issuance across all asset classes

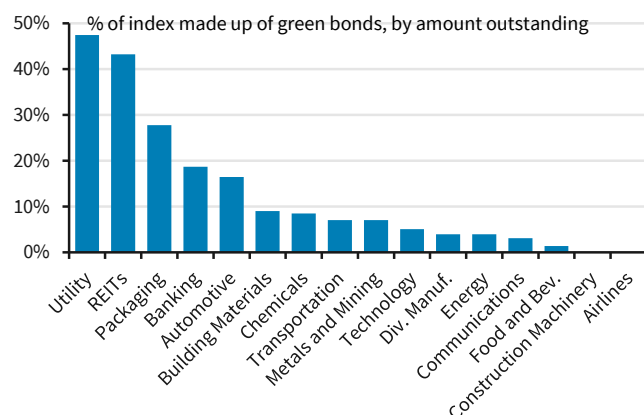


DM hard currency bonds only. Excludes transition bonds.
Source: Bloomberg, Dealogic, Barclays Research

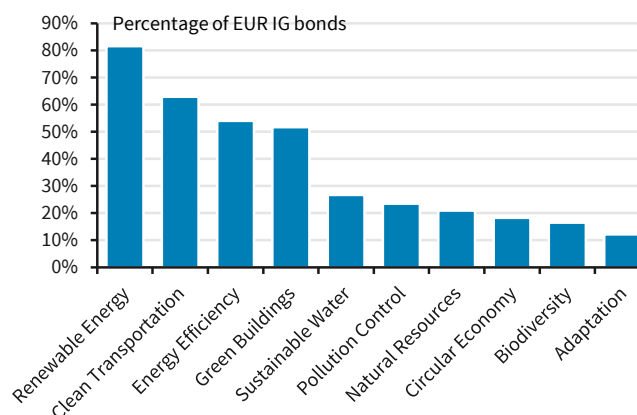
FIGURE 47. H1 26 issuance by sector



DM hard currency corporate bonds only. Excludes transition bonds.
Source: Bloomberg, Dealogic, Barclays Research

FIGURE 48. ESG bonds as a percentage of total, by €-IG index sector

Index as at end-2025.
Source: Bloomberg, Barclays Research

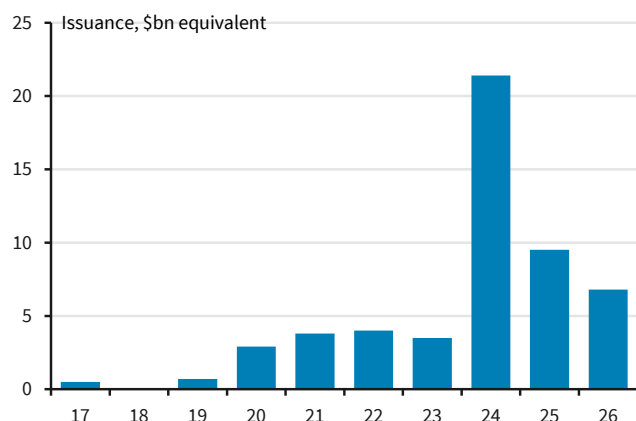
FIGURE 49. Percentage of €-IG green bonds including each project category as eligible

Index-eligible €-IG green bonds only.
Source: Bloomberg, Barclays Research

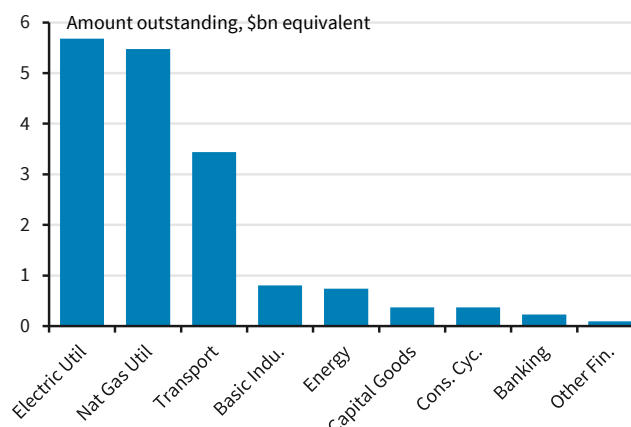
Transition finance could help close the gap

The challenge is that many of the sectors facing the greatest capital requirements in the next phase of the energy transition have made relatively limited use of ESG-labelled financing. Hard-to-abate industries, including parts of the energy, industrial and materials sectors, remain under-represented despite being central to the decarbonisation challenge. This matters because the new energy equation is about more than renewable deployment alone. It requires investment across a broader set of assets, including industrial transformation, energy-security infrastructure and technologies that reduce emissions from existing systems.

This is where transition finance could become increasingly important. Transition bonds have the potential to channel capital towards decarbonisation activities in sectors that cannot easily issue traditional green bonds because their activities are not fully aligned with a low-carbon end-state today. However, adoption remains relatively limited, with cumulative issuance of just over \$50bn since 2017 (Figure 50), a fraction of the more than \$3trn green bond market, and most issuance to date has been from utilities rather than industrial hard-to-abate sectors such as steel, cement, aviation, shipping, etc. (Figure 51). In our view, further market development will require greater policy and market clarity about what constitutes a credible transition investment, including guidance on issues such as carbon lock-in, the use of transitional technologies and appropriate sunset dates for emissions-intensive assets. Greater certainty could help issuers access the market with confidence while providing investors with clearer frameworks for assessing transition credibility ([Transition bonds: Assessing the path to maturity](#)).

FIGURE 50. Annual transition bond issuance

Includes all currencies. 2026 issuance as at 20 July.
Source: Bloomberg, Barclays Research

FIGURE 51. Corporate transition bonds outstanding, by sector

Includes all currencies. Excludes non-corporate issuers. As at 20 July 2026.
Source: Bloomberg, Barclays Research

Grid infrastructure could be the next frontier

Electricity networks may provide a useful example of where the ESG and transition bond markets can complement one another. Although grid infrastructure already features within many utility green bond frameworks, labelled financing has historically been more heavily concentrated on renewable generation than on the transmission and distribution networks needed to support it. Yet grid investment is increasingly becoming a critical constraint on electrification, renewable deployment and energy-system resilience.

The treatment of electricity networks highlights the complexity of sustainable finance classifications. In practice, not all green bond frameworks require network investments to be dedicated solely to renewable energy projects. Frameworks from electricity network operators including National Grid and Redeia allow broader transmission and distribution investments to qualify where networks meet criteria linked to facilitating the transition to a lower-carbon electricity system. This approach is broadly consistent with the EU Taxonomy's transmission and distribution of electricity activity (Activity 4.9), which recognises electricity networks as an enabling activity, provided certain system-level decarbonisation criteria are met. Renewable-specific grid connections and network upgrades designed to integrate higher shares of renewable generation are also commonly included as eligible expenditures.

Nevertheless, questions can arise in jurisdictions where fossil fuels continue to account for a significant share of electricity generation. Because transmission and distribution assets facilitate the delivery of all electricity on the system, some investors may view certain network investments as transition activities, rather than purely green ones, particularly where system decarbonisation remains at an earlier stage. We therefore see scope for both green and transition bond markets to play a larger role in financing the infrastructure required to support a more electrified, resilient and lower-carbon energy system.

Public Policy: Can the US build power fast enough?

Michael McLean^(v) BCI, US

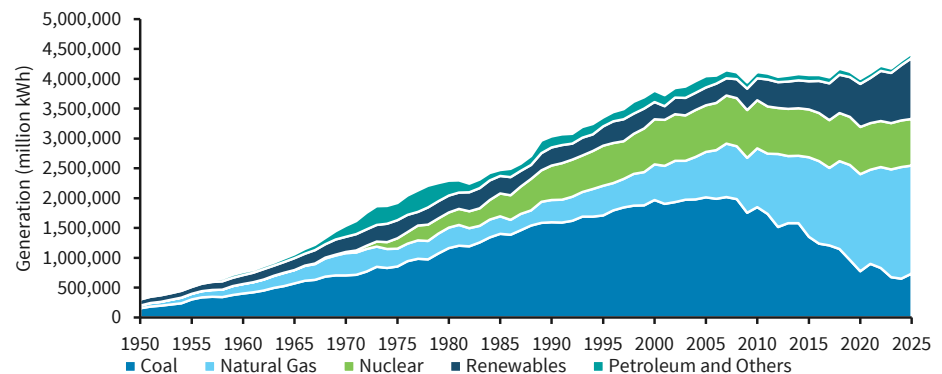
Meeting rising energy demand will require the rapid expansion of generation and transmission capacity. In our view, policy and permitting constraints could prevent the US from expanding energy infrastructure quickly enough and at sufficient scale to keep pace with demand.

A postwar energy boom

Much of today's US energy infrastructure was built during the 1950s, 1960s and early 1970s. Electricity generation more than doubled in the 1950s and nearly doubled again in the 1960s, while the US built roughly 8,000 miles of transmission lines annually between 1965 and 1975. Half of today's natural gas transmission network was installed during that period. Half of today's natural gas transmission network was installed during that period.

Critically, much of this build-out occurred before the modern federal permitting regime was established. We question whether the US could replicate anything close to that pace under today's regulatory framework.

FIGURE 52. US Electricity Generation by Major Source, 1950-2025



Source: Barclays Research, US Energy Information Administration (EIA)

Permitting has become increasingly complex

Permitting is often discussed as a single process, but in practice it involves multiple federal, state and local reviews that can add years to project timelines. For large infrastructure projects, we estimate developers may face more than 60 permits and reviews across numerous agencies.

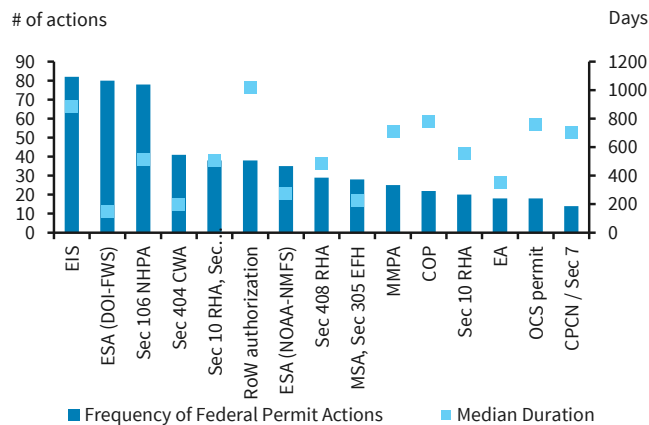
Our analysis suggests delays are common. Across 104 energy-related projects in the [federal](#) permitting system, nearly one-third of completed milestones have been late. State and local approvals can further extend timelines, while litigation risk remains significant. Almost 30% of large energy projects have been cancelled or paused.

Prior permitting reforms have under-delivered

Congress has repeatedly attempted to streamline permitting, but progress has been modest. The 2023 [Fiscal Responsibility Act](#) introduced a two-year deadline for environmental impact statements, yet only 39% have met that target.

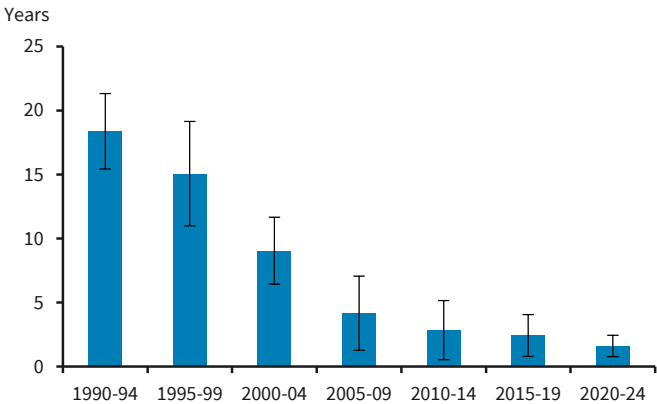
While several new reform proposals are under discussion, the track record of previous efforts suggests meaningful change may prove difficult to achieve quickly.

FIGURE 53. Top Federal Permits for Energy Projects and Median Duration of Permit Issuance



Source: Barclays Research, [Federal Permitting Dashboard](#)

FIGURE 54. Median Number of Years Between NOI and EIS



NOI is Notice of Intent. EIS is Environmental Impact Study.
Source: Barclays Research, [CEQ](#)

In 2024, bipartisan negotiations in the Senate failed to yield an agreement. Most recently, in July, the House Energy and Commerce Committee [released](#) the POWER ACT, a package of permitting reforms, but senators missed a soft deadline to release their own proposal of reforms before the August congressional recess.²

The jury is out on the effect of executive actions

The Trump administration has sought to accelerate permitting for energy and data centre projects through executive action, and some permitting metrics have improved. However, it remains too early to conclude that review times for major projects have materially shortened.³

More importantly, executive action has inherent limitations: it cannot fully address the federal, state and local constraints that slow infrastructure development. Without broader legislative reform, we believe permitting will remain a significant barrier to expanding US energy infrastructure at the speed and scale required.

² E&E News by Politico. [It's official: Permitting deal won't happen this week](#). August 5, 2026.

³ Executive Order: Unleashing American Energy, Executive Order: Declaring a National Energy Emergency, Executive Order: Unleashing Alaska's Extraordinary Resource Potential, Executive Order: Updating Permitting Technology for the 21st Century, Executive Order: Accelerating Federal Permitting of Data Center Infrastructure, Executive Order: Zero-Based Regulatory Budgeting to Unleash American Energy

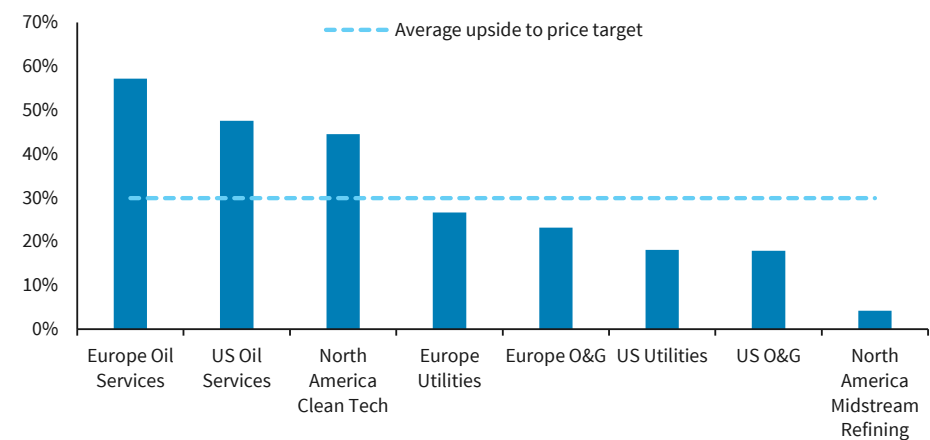
Part 2: Equity and Credit views

Summary of recommendations

Our preferred equities OW names across the global energy complex offer upgrades and/or re-rating potential

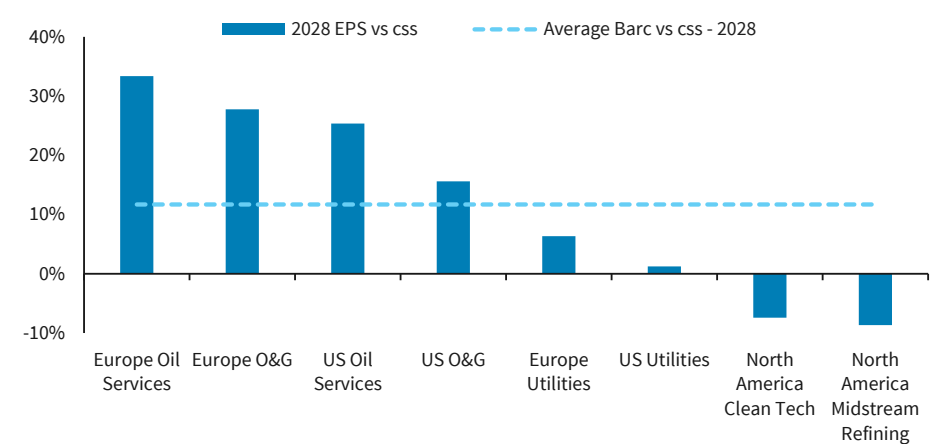
In Figures 55-56, we focus on our preferred OW equity ideas (typically three per sector). We take the average of the upside to our 12-month price targets and the average of the upside/downside vs. consensus EPS for 2028.

FIGURE 55. Average upside to Barclays Equities Research target prices, by sector



Source: Bloomberg, Barclays Research

FIGURE 56. Barclays 2028 EPS estimates vs. the consensus: average upside by sector



Source: Bloomberg, Barclays Research

Summary of sector views

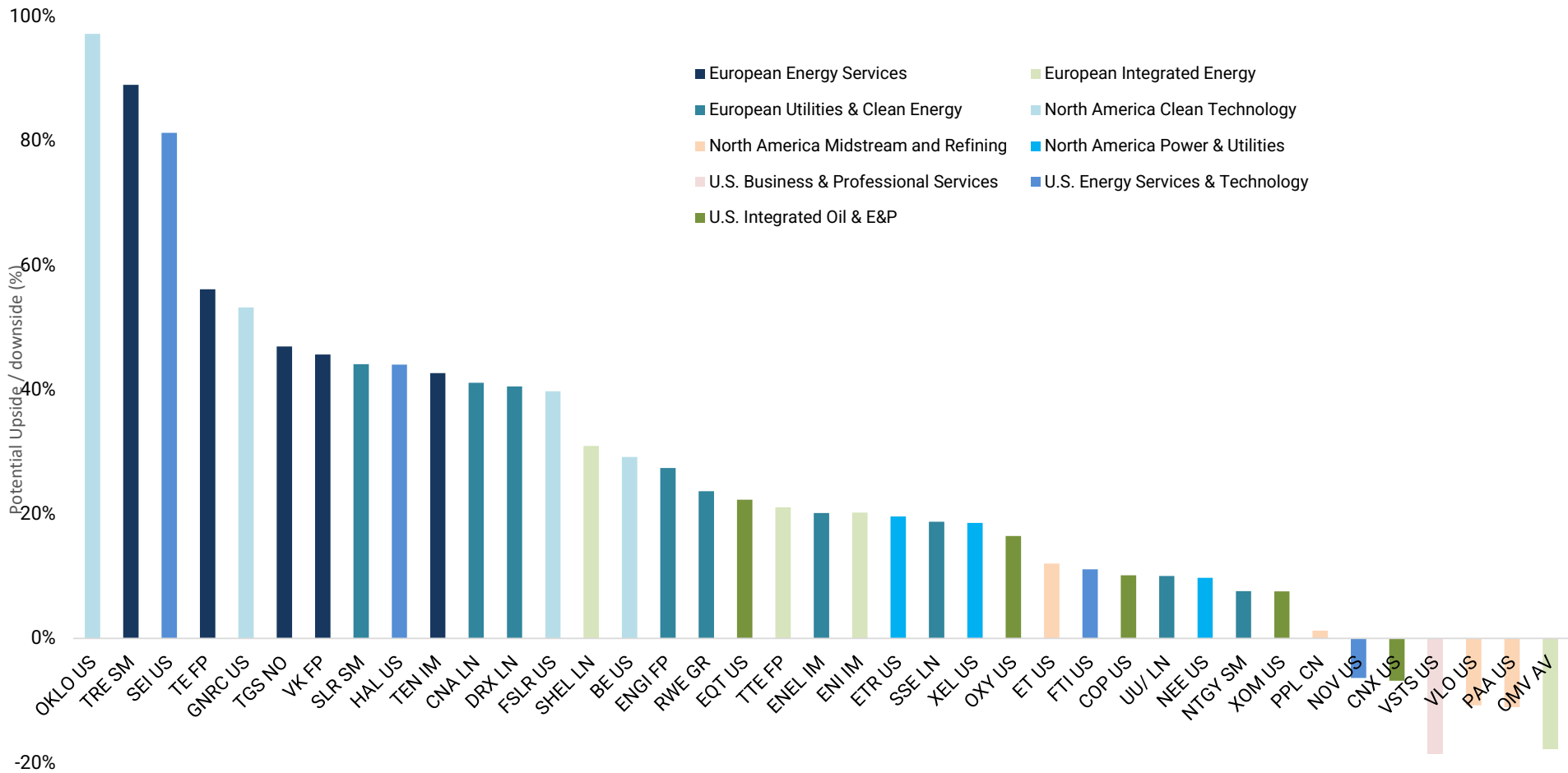
FIGURE 57.

Sector Name	Why Sector Matters	Energy Security & Return of Capital	AI, Power Demand & New Electricity Economy
US Oil & Gas	Key supplier of global energy & global LNG growth.	Security supports selective capex.	AI boosts gas and LNG demand, which supports cash flows.
N. America Midstream & Refining	Critical export infrastructure, core energy transport network.	More spend on LNG and export assets.	Gas infrastructure gains from AI demand.
Europe Integrated Energy	Key LNG and trading franchise, systemically important suppliers.	More upstream and LNG investment.	Gas remains critical transition fuel. Energy demand underpins cash generation.
LatAm Energy	Low-cost resource base.	Security lowers cost of capital.	LNG and power investment increasing.
US Energy Services	Critical service provider ecosystem. Enables resource and power growth.	Underinvestment supports pricing.	Geothermal and power create demand. New power markets support growth.
Europe Energy Services	Essential energy enabler. Supports LNG and offshore growth.	Security expands backlog visibility.	Grid and power investment supportive. Power demand improves visibility.
Utilities	Critical electricity infrastructure, central to electrification and AI.	Major grid and generation spend.	Direct beneficiary of AI demand. Higher demand improves cash flows.
Asia-Pacific Energy	Largest demand-growth region.	Focus on domestic security to improve resilience	Grid and storage investment rises.
Clean Tech & Power Supply Chain	Supplies power and grid buildout, supporting energy transition.	Capital flows to nuclear and storage.	Largest AI-related growth opportunity.

Source: Barclays Research

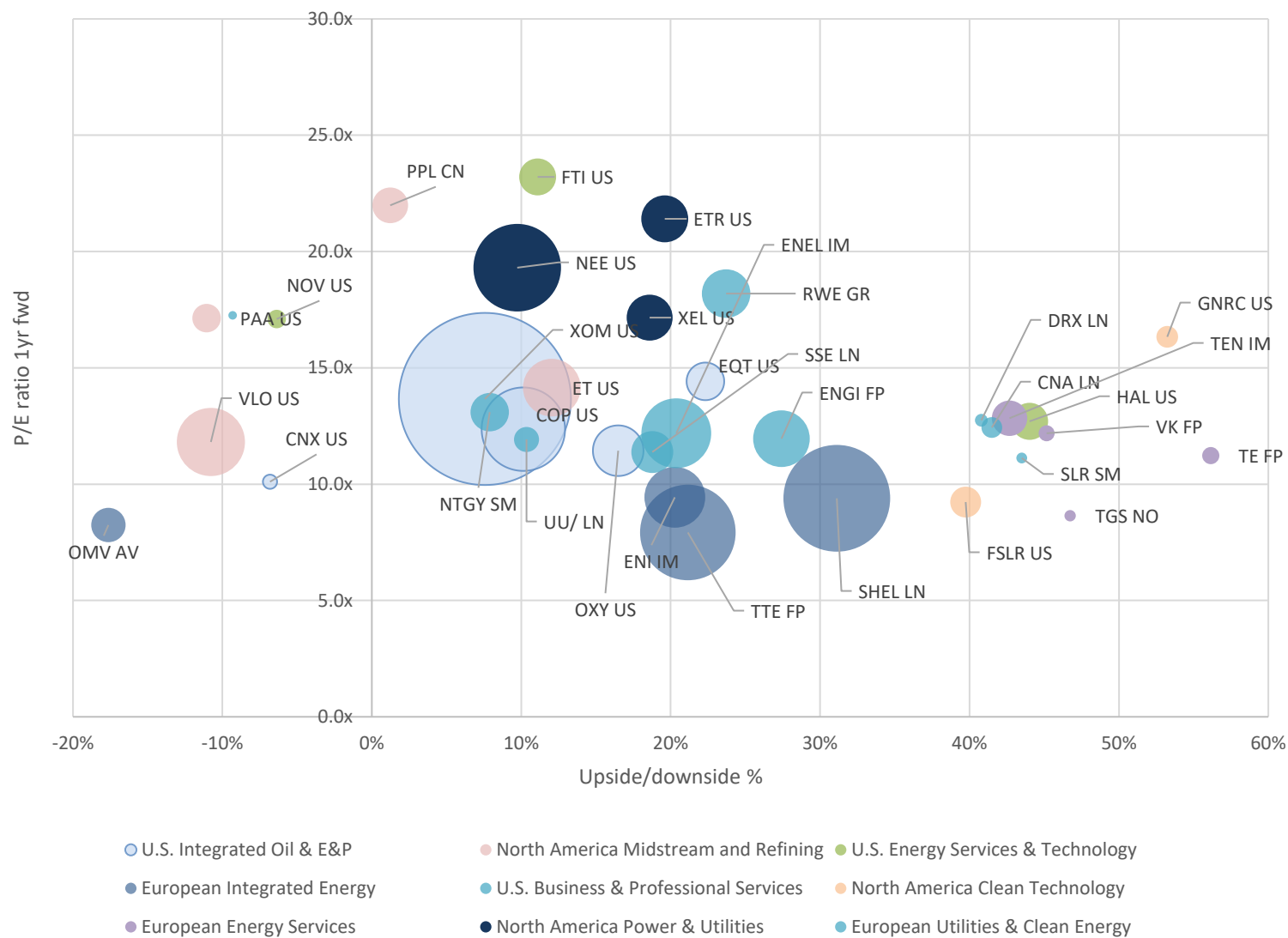
Equities

FIGURE 58. Equities: We highlight Global opportunities across the value chain



Source: Bloomberg, Barclays Research

FIGURE 59. Equities: Where upside meets value



Bubble size: Market Cap
Source: Bloomberg, Barclays Research

Equities - Preferred names

FIGURE 60. The Equity Energy Playbook

Ticker	Industry	Industry View	Analyst Rec.	Mkt Cap \$bn	PT	% Potential Upside	PE 1yr fwd	PE 2yr fwd	Net Debt / EBITDA 1yr fwd	Net Debt / EBITDA 2yr fwd	Rationale
XOM US	U.S. Integrated Oil & E&P	Positive	Overweight	675	177	8%	13.7x	14.1x	0.1x	0.0x	Deep resources, scale and execution strength support outsized FCF growth through the energy-addition cycle.
COP US	U.S. Integrated Oil & E&P	Positive	Overweight	161	150	10%	12.4x	12.2x	NM	NM	Low-cost inventory and capital discipline offer clean exposure to shale scarcity and FCF growth.
OXY US	U.S. Integrated Oil & E&P	Positive	Overweight	61	71	16%	11.4x	14.6x	0.2x	0.0x	Permian scarcity, EOR expertise and deleveraging create upside as quality shale tightens.
EQT US	U.S. Integrated Oil & E&P	Positive	Overweight	34	68	22%	14.4x	9.9x	0.7x	0.3x	Appalachian scale and demand-led contracts give EQT leverage to gas scarcity.
ET US	North America Midstream and Refining	Neutral	Overweight	75	24	12%	14.2x	13.6x	3.5x	3.2x	Underappreciated natural gas pipeline assets, Permian liquids and export links make ET a direct beneficiary of power demand and security.
PPL CN	North America Midstream and Refining	Neutral	Overweight	29	69	1%	22.0x	20.8x	2.9x	2.9x	NGL, LNG and Alberta power exposure give PPL multiple routes into security-led infrastructure demand.
VLO US	North America Midstream and Refining	Neutral	Overweight	104	323	-11%	11.8x	17.5x	0.1x	0.0x	Gulf Coast scale, feedstock flexibility and export reach position VLO to capture global margin disruption.
HAL US	U.S. Energy Services & Technology	Positive	Overweight	31	53	44%	12.7x	9.8x	0.9x	0.6x	North America recovery and VoltaGrid optionality make HAL a differentiated capex-cycle beneficiary.
FTI US	U.S. Energy Services & Technology	Positive	Overweight	31	87	11%	23.2x	NM	-0.8x	-1.0x	Subsea leadership, rising orders and margin upside position FTI for the next offshore infrastructure cycle.
SEI US	U.S. Energy Services & Technology	Positive	Overweight	4	90	81%	19.6x	9.6x	4.6x	3.0x	2GW+ contracted distributed power backlog gives Solaris direct exposure to hyperscaler speed-to-power demand.
SHEL LN	European Integrated Energy	Positive	Overweight	256	4500	31%	9.4x	10.0x	0.5x	0.5x	LNG scale, trading strength and FCF/share targets make Shell a clear energy-security beneficiary.
TTE FP	European Integrated Energy	Positive	Overweight	205	94	21%	7.9x	7.3x	0.5x	0.4x	Hydrocarbon growth, power buildout and long reserves give TTE balanced energy-addition exposure.
ENI IM	European Integrated Energy	Positive	Overweight	83	28.5	20%	9.4x	8.7x	0.6x	0.4x	Gas-led growth, LNG exposure and exploration strength align Eni with security-driven supply investment.
FSLR US	North America Clean Technology	Positive	Overweight	21	279	40%	9.2x	6.3x	-1.2x	-1.8x	US solar manufacturing scale makes FSLR a friend-shoring winner as power demand accelerates.
OKLO US	North America Clean Technology	Positive	Overweight	7	76	97%	NM	NM	NM	NM	Advanced nuclear optionality gives OKLO scarce exposure to reliable power for AI-led demand.
BE US	North America Clean Technology	Positive	Equal Weight	58	276	29%	49.5x	31.5x	-0.5x	-0.9x	Fuel cells offer a speed-to-power solution where grid constraints delay new industrial load.
TRE SM	European Energy Services	Positive	Overweight	2	49	86%	8.2x	6.1x	2.0x	2.1x	Backlog momentum, margin recovery and early AI-power wins support TRE into security-led project demand.
TE FP	European Energy Services	Positive	Overweight	7	48	53%	11.2x	8.1x	-4.5x	-3.8x	Record backlog and LNG awards give Technip visible earnings as global gas infrastructure investment rises.

Ticker	Industry	Industry View	Analyst Rec.	Mkt Cap \$bn	PT	% Potential Upside	PE 1yr fwd	PE 2yr fwd	Net Debt / EBITDA 1yr fwd	Net Debt / EBITDA 2yr fwd	Rationale
TGS NO	European Energy Services	Positive	Overweight	3	200	45%	8.6x	6.2x	0.2x	0.1x	Frontier exploration recovery and deleveraging potential make TGS a geared resource-security play.
TEN IM	European Energy Services	Positive	Overweight	28	34.5	41%	12.8x	11.7x	-1.2x	-1.1x	US pricing, deepwater activity and geothermal demand support Tenaris as energy spend broadens.
VK FP	European Energy Services	Positive	Overweight	6	27	44%	12.2x	7.9x	-0.2x	-0.3x	Middle East recovery, deepwater demand and geothermal growth support Vallourec across scarce pipe supply.
ETR US	North America Power & Utilities	Positive	Overweight	49	128	20%	21.4x	18.7x	6.3x	6.4x	Limited rate-case risk and open capacity for new load give ETR growth optionality beyond plan.
XEL US	North America Power & Utilities	Positive	Overweight	47	90	19%	17.2x	15.5x	5.7x	5.6x	Discounted valuation, balance-sheet quality and grid capex create re-rating potential as wildfire fears fade.
RWE GR	European Utilities & Clean Energy	Positive	Overweight	53	72	23%	18.2x	16.8x	2.4x	2.4x	Flexible generation, trading and data-centre exposure make RWE a leading European AI-power beneficiary.
SSE LN	European Utilities & Clean Energy	Positive	Overweight	39	2860	19%	11.4x	9.7x	2.8x	3.3x	Networks and renewables growth give SSE direct exposure to grid bottlenecks and electrification capex.
DRX LN	European Utilities & Clean Energy	Positive	Overweight	4	1120	40%	12.7x	10.5x	2.2x	1.9x	Dispatchable generation and biomass optionality position Drax for power scarcity and firm capacity demand.
ENGI FP	European Utilities & Clean Energy	Positive	Overweight	72	31	27%	12.0x	11.4x	NM	NM	Merchant, trading and flexible assets give Engie leverage to power scarcity and commodity volatility.
CNA LN	European Utilities & Clean Energy	Positive	Overweight	10	215	41%	12.4x	10.9x	-0.1x	0.1x	Centrica's flexible energy portfolio benefits from commodity volatility and demand for resilient supply.
SLR SM	European Utilities & Clean Energy	Positive	Overweight	3	24	42%	11.1x	9.0x	4.4x	3.9x	Merchant generation exposure gives Solaria leverage to higher power prices and electricity scarcity.
ENEL IM	European Utilities & Clean Energy	Positive	Overweight	109	11	19%	12.2x	11.5x	2.8x	2.9x	Grid scale and data-centre-linked demand support Enel as electrification capex accelerates.
UU/ LN	European Utilities & Clean Energy	Positive	Overweight	14	1535	9%	11.9x	11.3x	6.9x	6.9x	Defensive regulated growth and UK water re-rating potential make United Utilities a value idea.

Source: Bloomberg, Barclays Research

Credit preferred names & trade ideas

FIGURE 61. Credit Energy Play Book (at Issuer level)

Ticker	Issuer	IG/HY	Sector	Analyst Rec.	Credit rating	Net Debt / EBITDA (last reported)	Rationale
BPLN	BP	IG	Oil & Gas – Europe	OW	A-/A1	0.9x	Lower capex, debt reduction and stronger commodity backdrop support spread tightening.
TTEFP	TotalEnergies	IG	Oil & Gas – Europe	OW	A+/(P)Aa3	0.7x	Integrated resilience, diversified LNG, trading and refining earnings; low leverage and strong cash generation.

Ticker	Issuer	IG/HY	Sector	Analyst Rec.	Credit rating	Net Debt / EBITDA (last reported)	Rationale
VENLNG / VEGLPL / VENTGL	Venture Global HoldCo	HY	Energy – US	MW	BB/BB/B1	3.9x	One of the most active companies with respect to adding LNG capacity before 2030.
STOAU	Santos	IG	Energy – Asia Pacific	OW	Baa3 Pos/BBB-Stb/BBB Stb	1.9x	LNG-heavy producer; Asia energy security and project ramp-up support ratings upside.
LGENSO	LG Energy Solution	IG	Energy – Asia Pacific	OW	Baa2 Stb/BBB Neg/NR	NM	Positioned for ESS and data-centre power demand growth.
VST	Vistra	IG	Utilities & Power – US	OW	BBB-/Ba1	3.0x	Merchant power scarcity and load growth support leverage improvement.
EDF	EDF	IG	Utilities & Power – Europe	OW	BBB+/Baa1	2.2x	State support, nuclear exposure and strong financeability profile.
TENN	TenneT Holding	IG	Utilities & Power – Europe	OW	BBB+/Baa2	12.4x	State-backed funding framework and supportive liability-management dynamics.

Source: Bloomberg, Barclays Research

FIGURE 62. Trade Ideas

Ticker	Name	Sector	Trade idea	Rationale
RIG	Transocean	Energy – US	We view the 8.5% 2031s and longer-dated 2038s as the most attractive options.	Offshore cycle strengthening; merger creates scale, backlog and upgrade potential from rating agencies.
FLQIII / FLIQII	Freeport LNG	Energy – US	We recommend buying FLQIII 5.55s (2039 final, 2033 average life) and/or FLIQII 5.92s (2039 final, 2035 average life).	Attractive ~6% yields; contracted cash flows and lower refinancing risk.
PLNIJ	PLN Indonesia 2036s	Energy – Asia Pacific	Buy PLNIJ 36s	Electrification capex theme; improving state compensation cycle.
PETMK	Petronas 2050s	Energy – Asia Pacific	Buy PETMK '50s	Strong liquidity, low refinancing risk and energy security beneficiary with ~58% proven oil and gas reserves located in Malaysia.
PHILIP 5y CDS	Philippines 5Y CDS	Energy – Asia Pacific	PHILIP 5y CDS	Attractive hedge against oil-import vulnerability and supply shocks.
AYDEMT	Aydem Renewables	EEMEA	Buy AYDEMT 2030s	Renewable growth exposure with attractive ~10% carry.
AZURGY	Azule Energy	EEMEA	Buy AZURGY 2030s	Low leverage; benefits from new discoveries and stronger oil demand.
KOS-GTA	Kosmos GTA	EEMEA	Buy KOS (GTA) 11.25% 2031s	LNG optionality via Greater Tortue Ahmeyim and future gas sales upside.

Source: Bloomberg, Barclays Research

Equities - Where we are cautious

FIGURE 63.

Ticker	Industry	Industry View	Analyst Rec.	Mkt Cap \$bn	PT	% Potential Upside	PE 1yr fwd	PE 2yr fwd	Net Debt / EBITDA 1yr fwd	Net Debt / EBITDA 2yr fwd	Rationale
CNX US	U.S. Integrated Oil & E&P	Positive	Underweight	5	35	-7%	10.1x	7.9x	1.5x	1.4x	Less attractive relative growth and scarcity exposure.
PAA US	North America Midstream and Refining	Neutral	Underweight	18	23	-11%	17.1x	16.1x	2.6x	2.5x	Less leverage to preferred export growth themes.
OMV AV	European Integrated Energy	Positive	Underweight	26	57	-18%	8.2x	9.1x	0.5x	0.4x	Higher chemicals exposure; weaker strategic positioning.
NOV US	U.S. Energy Services & Technology	Positive	Underweight	8	20	-6%	17.1x	13.8x	NM	NM	Greater exposure to mature oilfield activity.

Ticker	Industry	Industry View	Analyst Rec.	Mkt Cap \$bn	PT	% Potential Upside	PE 1yr fwd	PE 2yr fwd	Net Debt / EBITDA 1yr fwd	Net Debt / EBITDA 2yr fwd	Rationale
NEE US	North America Power & Utilities	Positive	Equal Weight	172	91	10%	19.3x	17.7x	5.0x	4.8x	Valuation and growth concerns versus peers.
NTGY SM	European Utilities & Clean Energy	Positive	Equal Weight	33	32	8%	13.1x	13.2x	2.4x	2.2x	Less attractive positioning in power capex cycle.
GNRC US	North America Clean Technology	Positive	Equal Weight	11	278	53%	16.3x	13.7x	0.6x	0.1x	Less exposed to structural AI power beneficiaries.
VSTS US	U.S. Business & Professional Services	Neutral	Underweight	2	11	-9%	17.3x	17.0x	NM	NM	Merchant power exposure makes VST a direct beneficiary of AI-driven electricity scarcity.

Source: Bloomberg, Barclays Research

Credit - Where we are cautious

FIGURE 64.

Ticker	Issuer	IG/HY	Sector	Analyst Rec.	Credit rating	Net Debt / EBITDA (last reported) 1yr fwd	Rationale
NEE	NextEra Energy Inc	IG	Utilities & Power – US	UW	A-/Baa1	6.8x	Heavy funding requirements expected to weigh on spreads.
NTGY	Naturgy	IG	Utilities & Power – Europe	UW	BBB/Baa2	2.2x	Lower growth profile and limited exposure to electrification investment cycle.
KORGAS	Korea Gas Corp	IG	Energy – Asia Pacific	UW	AA/Aa2	NM	National oil companies and power utilities that cannot promptly pass on higher fuel costs are potentially vulnerable to bond spread volatility.
KORELE	Korea Electric Power Corp	IG	Energy – Asia Pacific	UW	Aa2	NM	Likely to incur higher capex and debt, as it is expected to lead Korea's power infrastructure build-out.

Source: Bloomberg, Barclays Research

Oil services - US Equity views

J. David Anderson, CFA BCI, US | Eddie Kim BCI, US

Why our sector matters

We view the current environment as the best setup for the Energy Services sector in 20 years. Over the past 12 months, we have witnessed twin “black swan” events that have dramatically disrupted global energy markets with ramifications that will last through the end of the decade. The War in Iran has triggered the largest oil supply shock in history, curtailing almost 20% of global oil supply from the blockage of the Strait of Hormuz and shutting in producing fields across the Middle East. This will have a lasting impact as energy security concerns shift oil and gas supply routes, new sources of supply are created, and inventories are rebuilt and expanded. Concurrently, the United States faces an unprecedented surge in power demand as hyperscalers build out data centres to support the global expansion of AI. However, in our view, the combination of industrial turbines' long lead times and underinvestment in grid infrastructure is likely to create a significant power gap by 2030, putting further strain on the energy complex.

Energy Services plays a critical role in the race for energy resources, providing the technology, equipment, and labor needed to extract energy, irrespective of the source.

Since the days of the first oil well drilled in the late 1800s, the services industry has been an integral part of the energy complex. Whether in exploration, evaluation, drilling, completing or producing, energy services plays a critical role in the lifecycle of the oil well. It provides the technology to interpret the subsurface, locate the hydrocarbons, and plan the wells for maximum recovery; the drilling tools to efficiently and effectively reach the target; the completion techniques to maximize flow; and equipment needed to control pressure and optimize production rates. Furthermore, the industry provides the assets needed to facilitate the activity – the offshore and onshore rigs, vessels, workover equipment, and testing facilities. In other words, oil cannot be produced without a fully capable, technically forward, and robust global Energy Services industry.

But it's more than just oil, Energy Services encompasses all energy sources. This is a sector that has proven to have a transferrable set of skills well beyond the oil markets. Natural gas markets are directly related as much of the equipment, technology, and infrastructure in oil is also needed for natural gas production. However, one major difference is the transportation of natural gas, which requires pipelines to move regionally in a gaseous form or must be liquefied in a complex process to be exported as LNG, supported by equipment from service companies. As new energy sources emerge, Energy Service companies will likewise have opportunities to support their development. Geothermal is perhaps the best example of the transferrable skill sets as it requires the same subsurface analysis, drilling technologies, and completion techniques as it does in oil and gas. More recently, many Energy Service companies are now providing distributed power for behind-the-meter solutions, transferring the experience of powering electric hydraulic fracking into supplying speed-to-power for data centres.

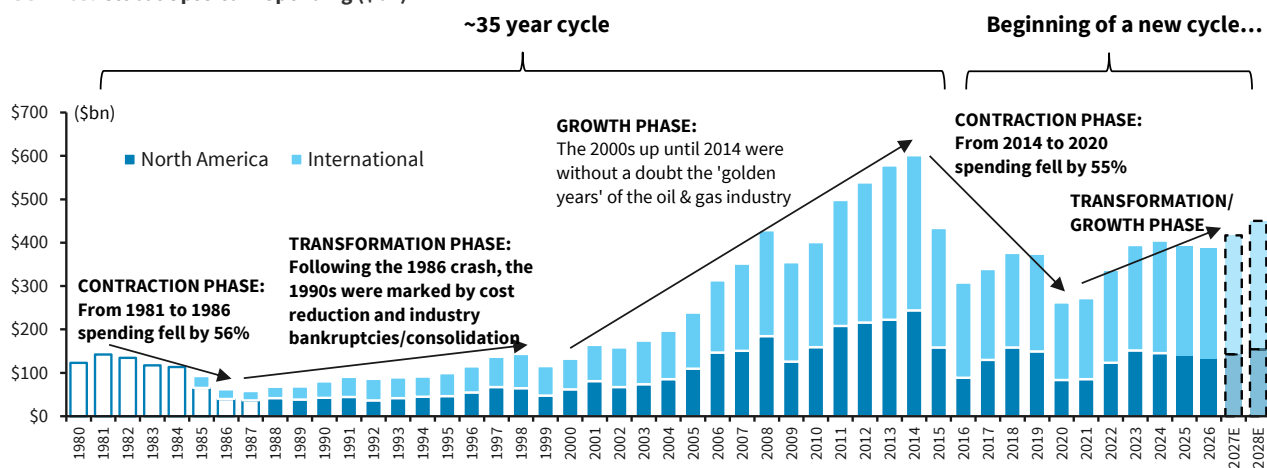
Energy Services touches every global upstream oil and gas market. The majority of revenue for Energy Services companies is directly tied to the upstream capital expenditures of oil companies. Globally, three primary markets account for the majority of upstream spending: North America, International Land, and Offshore. Each market is distinct in terms of the type of customer, the complexity of operations and the size of the projects. Shale activity is the primary driver of **North America** upstream spending, mostly from exploration & production companies (E&Ps) who tend to be more sensitive to commodity prices. **International land** is driven principally by national oil companies (NOCs), roughly weighted two-thirds Middle East, one-third Latin America. As NOC production and revenue directly support fiscal budgets, spending

decisions tend to be longer term. **Offshore** is the longest duration market and therefore the least sensitive to commodity prices. International oil companies (IOCs) account for the majority of deepwater activity that require long times to contract rigs, order equipment, and ultimately produce for more than a decade.

Energy security and the return of capital

Following a decade of underinvestment in oil & gas, we expect a rebound in the years ahead as energy security priorities/concerns come to the forefront. As shown in the figure below, globally the oil & gas industry has underinvested in Energy over the past decade as E&Ps have pushed the capital discipline narrative. More recently, over the past three years, the upstream spending cycle has been in a plateau, weighed down by global oversupply and upstream capital discipline in North America driven by investors demanding cash flow over production growth. Now with the oil supply shock of the Iran war likely reverberating for many years, oil prices structurally higher, and energy security prioritized by governments and countries alike, we expect substantially more capital investment in upstream over the next several years. As a result, we are forecasting global upstream spending to accelerate to 9% y/y in 2027 and 10% y/y in 2028.

FIGURE 65. Global Upstream Spending (\$bn)



Source: Barclays Research

Middle East – building out capacity, new supply routes to emerge as a result of the Iran conflict. Structurally higher oil prices will likely cause the international upstream spending cycle to inflect, though timing is unclear as it depends on a resolution to hostilities with Iran. Representing about ~20% of international spending, an estimated 9 mmbpd of Middle East production has been shut in. Combined with lower Iran oil exports, an estimated 2-3 mmbpd of oil has been displaced, effectively eliminating the market oversupply. Though OPEC is trying to reassure markets of its spare capacity, the reality is that only Saudi Arabia has the ability to bring back barrels over the next 12 months. Saudi was already in the process of adding ~40 land rigs by mid-2026 as it was bringing back voluntary production cuts, now we believe another 40-50 rigs will need to be added to further increase production. Furthermore, we expect Kuwait and UAE to also ramp up activity and accelerate capacity expansion plans, all in an effort to make up for displaced barrels.

Offshore – arguably the barrel with the most "security." After several years of delays, we had been anticipating the offshore market to inflect some time in 2027, but now we believe the timing will move up several quarters, possibly as early as the end of 2026. Over the past several years, there has been a steady decline in the total amount of offshore FIDs announced annually

as the timing of a number of larger projects has been pushed out. Concurrently, the deepwater rig count experienced a fair amount of “white space” over the past few years and the floating rig count fell from a recent peak of 132 rigs in 3Q24 to 116 rigs in 4Q25, though has since rebounded to 122 currently. Essentially, a backlog of deepwater development has been building over the past several years, projects are largely planned and most are economic well below \$50/bbl (if not below \$40). But because these are long cycle projects with first oil at least 2-3 years away, oil companies need confidence in long-dated oil prices and with the 2030-35 strip now at ~\$70-75/bbl, this should be the signal to sanction projects and ramp up offshore activity. With energy security now a priority, Offshore deepwater becomes the most secure/attractive barrel, the most viable and economic solution to diversifying global oil supply.

Speed to power solutions to fill a potential US gap. A year ago, the Distributed Power industry barely existed, comprised of medium sized turbines and reciprocating engines predominantly used to serve the needs of the oil and gas industry as it electrified the Permian Basin, including building out eFrac operations. Fast forward to today and Distributed Power is now critical to filling the significant power gap facing the United States over the next 3-5 years – due in part to constraints/limitations of the electric grid – and has become a speed-to-power solution for hyperscalers signing long-term contracts for off-grid solutions. We expect off-grid solutions to become increasingly attractive as rising power prices become a ballot issue and grid interconnection issues push out large-scale power additions past 2030.

Oil price volatility is negative for upstream spending plans. Commodity price volatility – and more specifically, oil price volatility – is, in our view, a long-term negative for upstream spending by E&Ps/operators. We believe oil prices in the \$75-95/bbl range is the ‘sweet spot’ for the Energy Services sector, below which activity typically declines, and above which it tends to increase in the near term but also incentivizes an equipment capacity build-out that ultimately creates higher peaks and lower troughs in activity levels.

Winners and losers in a fragmented energy world

Covid proved to the world the risk of global supply chains, which has since shifted countries to become more self reliant and less dependent on the global economy... the same thing is happening to oil markets. Similar to how the demand shock from Covid showed the risk of an interconnected global supply chain that forced countries to rethink sources and suppliers, we believe the supply shock of the Iran war will similarly force each country to reconsider its energy supply.

Energy fragmentation should result in higher upstream spending. As each country reassesses its energy supply, this will create a greater diversification of sources, which in turn will result in higher upstream spending to build out those resources. We believe offshore in particular should see a prolonged period of higher spending, not only because it is arguably the barrel with the most security (low geopolitical risk, no transportation constraints) but is also the lowest cost oil production with breakevens well below \$50/bbl in deepwater. We expect energy-independent countries like the US and those in the Middle East will add capacity to reinforce these advantages, fuel their own economies, and increase exports.

We expect the US will continue to be a long-term energy winner. Over the next 3-5 years, we believe the energy-independent US will continue to be a winner and see stable-to-modestly-increasing oil production (with potential upside from Venezuela flows, with that country’s heavy oil now filling up US refineries) – though we note this could be partially offset by shale resources now increasingly being constrained by Tier 1 acreage/inventory which continues to be exhausted – and growing natural gas production driven by higher demand for US LNG.

Asia demand will become less dependent on the Middle East, likely shifting supply toward offshore. Asian countries that are heavily reliant on the Middle East for oil will likely source

from elsewhere, and could even look to develop their own offshore resources. OPEC and the Middle East could be forced to react/adjust and pour investment into establishing new transportation routes that bypass the Strait of Hormuz.

AI, power demand, and the new electricity economy

AI and data centres are a structural, not cyclical, source of new electricity demand, and the US is facing risk of inadequate supply. After two decades of essentially flat capacity growth, the grid now faces simultaneous load from AI data centres, reshoring, and electrification just as coal retirements accelerate. We estimate a supply/demand gap on the order of 50 GW by 2028, as gas turbine deliveries, renewables, and storage fall well short of what's needed, based on current projections. The bottleneck to the AI trade is no longer chips, it's electrons: interconnection queues stretch into multi-year delays, firm capacity is scarce, and large-scale grid additions get pushed past 2030. For utilities and power-linked sectors this means sustained load growth, grid expansion, and heavy spending on gas generation, transmission, and firm-capacity solutions. The key question for every sector is whether AI turns power scarcity into an earnings stream or into a funding, affordability, and execution risk.

Projections of power scarcity are redrawing the Energy Services map. An industry long defined by drilling and pressure pumping has pivoted hard toward power generation, leveraging its heritage of providing on-site electricity to remote oil and gas operations. A growing number of service companies now run power businesses, ordering gigawatts of turbines and reciprocating engines to serve data centres, while capital has rotated out of the hyperscalers themselves and into the hard assets that power them. This also reshapes the energy and credit picture: more gas demand, more midstream and power-linked infrastructure investment, and growing competition for capital between energy and digital infrastructure. The discriminator for investors is contract quality, distinguishing operators with long-term hyperscaler agreements and operating megawatts from those still marketing order books.

Distributed power has graduated from bridge to baseline. In roughly a year, medium-sized turbines and reciprocating engines have gone from oilfield niche to a critical speed-to-power solution, validated by hyperscalers signing contracts of ten years and longer for off-grid supply. With large industrial turbines back-ordered into 2030, on-site generation delivers years faster than a grid connection, and the regulatory backdrop is an accelerant rather than a headwind: as electricity bills become a ballot issue, state and federal policymakers are likely to increase efforts to ensure that large new loads (especially AI/data centres) pay for their own power infrastructure and do not impose costs on other ratepayers.

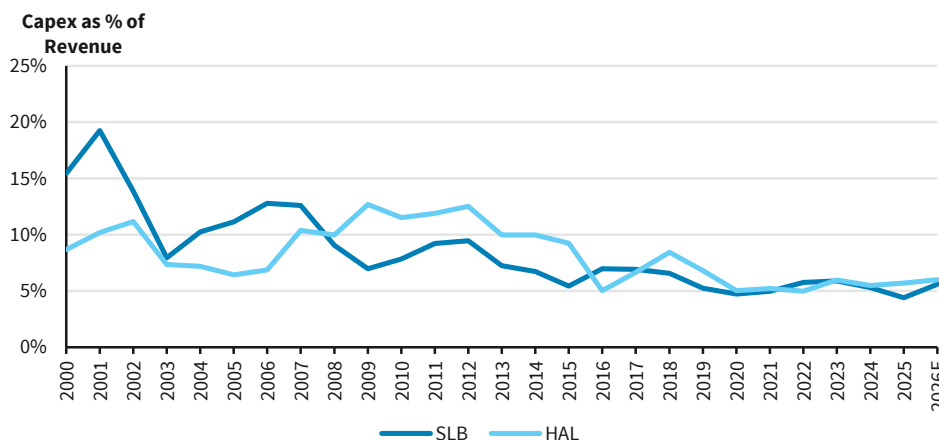
Demand-side debates have largely disappeared; the binding constraint is supply, with OEM manufacturing capacity the industry's biggest bottleneck.

Geothermal is emerging as the clean firm-power answer to the medium-term gap. While gas-fired distributed power fills the near-term void, geothermal offers dispatchable, zero-fuel baseload with round-the-clock capacity factors that solar and wind can't match, and it retains rare bipartisan tax support. The step change is enhanced geothermal systems (EGS), which apply the horizontal drilling and hydraulic fracturing playbook that unlocked shale to engineer reservoirs almost anywhere with hot rock, removing the geographic constraint that kept the technology niche. **Fervo** is the standard-bearer: leaning on the existing oilfield services supply chain rather than a constrained OEM order book, it has binding PPAs and a multi-gigawatt hyperscaler framework agreement in hand, with a cost trajectory from roughly \$7,000/kW today toward \$3,000/kW at scale, competitive with new-build gas and well below nuclear. It's not the only game in town, with incumbents like **Ormat** and closed-loop designer Eavor Technologies (private, not covered) also in the mix, but with nuclear likely a 2030s story at best, EGS is arguably the best-positioned clean firm power source of the next decade.

The race for capital: financing and investment ideas

We don't see a need for new capital to support services for traditional oil & gas, especially as service companies have significantly reduced capex over the years. Following the upstream spending trough in 2020/2021 immediately after COVID after which many companies restructured their balance sheets, the Energy Services sector is lean, balance sheets are strong and free cash flow is increasing. Additionally, capital discipline has brought down capex as a % of revenue significantly (see Figure below) as companies prioritized returns over overbuilding service capacity and equipment – a hallmark of the prior upcycle from 2010 to 2014 (which continued through 2019). As a result, we don't see a need for new capital to service traditional oil & gas services/equipment.

FIGURE 66. SLB/HAL capex as % of revenue



Source: Company Reports, Barclays Research

Where capital is most needed will be to support geothermal and distributed power. Where new capital is needed in the coming years will be to support geothermal and distributed power, two markets that traditional Energy Services companies have increasingly moved into. For geothermal, companies like Fervo Energy will need funding to support projects front-loaded with heavy construction capital (followed by decades of low cost O&M and generation). For distributed power companies such as Solaris, Voltagrid (private, not covered), Liberty Energy Services, ProPetro, and Atlas Energy (among others), financing will be needed to purchase power-generation equipment (engines or turbines) as well as 'balance of plant' equipment to provide a full turnkey solution for data centre developers.

Investment recommendations (US Energy Services & Technology industry view: Positive)

- Halliburton (HAL) - Overweight.** HAL remains the most levered big four name to short-cycle North American land and we think the setup into 2H26 is more constructive than consensus is giving credit for. Layered on top is HAL's 20% ownership stake in VoltaGrid and the joint distributed power partnership. We see HAL as the name in our coverage where the cyclical trough is being priced in on the core business while the power optionality is increasingly tangible but still under-monetized in the multiple. We are Overweight rated on HAL with a price target of \$53/share based on 9.0x 2027E EBITDA plus \$5/sh attributed to HAL's power business using current net debt.
- TechnipFMC (FTI) - Overweight.** Despite a 4x increase in the stock over the past three years, FTI still trades at a mid-cycle multiple from the prior deepwater cycle – in a far better subsea industry and in a much better market position than back then. Orders are trending toward

\$11bn in 2027E (from ~\$10bn this year), Subsea margins toward 25%, with SURF 2.0 as the next leg, and a quiet (but growing) Surface business. We are Overweight rated on FTI with a price target of \$87/share based on 12.0x 2028E EBITDA using current net debt. Our 12.0x multiple is based on our belief that FTI will continue to re-rate as the cycle gains momentum.

- **Solaris (SEI) - Overweight.** After recently announcing a 600MW contract (its third long-term contract with a hyperscaler), SEI now has 2GW+ signed under long-term contracts and indicated that opportunities going forward will be "more streamlined" with a framework/processes in place to sign up a 4th/5th customer at a faster pace. As distributed power demand continues to increase through the end of the decade, we view Solaris as being best positioned to take advantage. We are Overweight rated on SEI with a price target of \$90/share based on a sum-of-the-parts analysis, applying 11.0x to 2028E Power Solutions EBITDA and 5.0x to 2026E Logistics Solutions EBITDA, using 4Q29 net debt to reflect the capex burden of the pro forma fleet.
- **NOV Inc. (NOV) - Underweight.** It's hard for us to make the relative case for NOV against the rest of our coverage. NOV has executed well on the self help side – >85% FCF conversion for two consecutive years, \$876mn of FCF in 2025, and a strong balance sheet that gives CEO Jose Bayardo room to play offense – but the underlying earnings power remains capped by the late cycle nature of the business... and while we think the cycle will ultimately turn, it's still early. The 2026 EBITDA guide of in-line to slightly lower y/y with a back-half-loaded shape and book-to-bill near 1.0x, underscores that the structural reset is largely priced in while the cyclical lift remains a 2027 story at the earliest, in our view. Against WFRD, SLB, and now HAL – each of which offer either superior international leverage, a clearer NAM inflection, or a power optionality – we struggle to find the case to own NOV here. We are Underweight rated on NOV with a price target of \$20/share based on 7.5x 2027E EBITDA, with a negative adjustment of \$57mn reflecting our estimate of EBITDA from drillbit patent leaching related royalties, and using current net debt.

Oil services - Europe Equity views

Mick Pickup Barclays, UK

European Energy Services have seen a renaissance since 2025, outperforming the wider sector by over 10% since the April 2025 lows (EMEAOGP vs EMEAOG). The drivers have been three-fold – increased demand expectations comprising AI and the evolving energy landscape; increased expectations of conventional resources; and more importantly, the flattening of US shale expectations. Simply put, as Europe seeks a greater diversification of its energy supply by both geography and source, European Energy Services will become a key enabler, serving as a principal conduit through which conventional, unconventional and renewables are delivered to market. The set-up, in our view, is akin to the mid-2000s. A race for resources falls into the sweet spot for the sector, and while demand estimates can be debated, the gap that needs to be filled by activity from the oil services sector has grown. More importantly, the asset base of the service sector has shrunk over the last decade as conventional activity receded from focus. Consequently, the evolving energy equation is now clear to us. The Services sector is increasingly expected to do more with less, driving greater visibility and alignment with clients and, in turn, redefining the operator- contractor relationship.

Inverse Goldilocks scenario: – The three bulls driving the European sector

1. Demand estimates are higher

As discussed earlier in the note, we see energy demand growing across numerous vectors. These include more hydrocarbons than were once envisioned to be a part of the energy mix, as

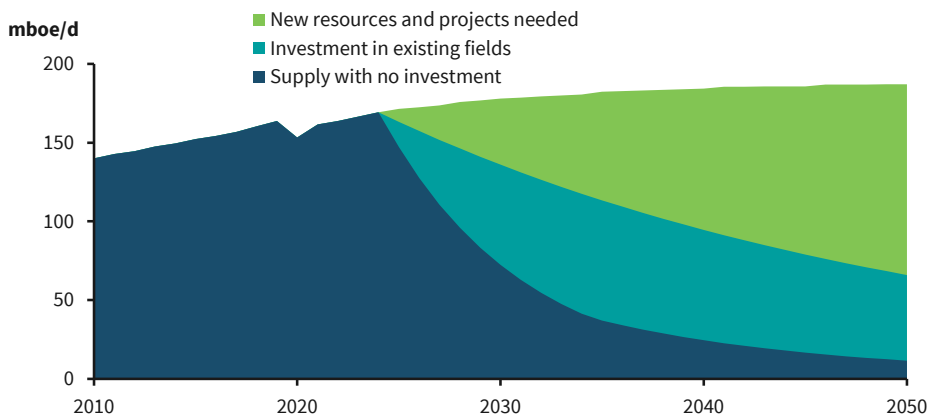
the physical limitations of transition hit home. We have long argued that transition is a multi-faceted beast, with the simple premise being "electrify all that can be electrified and generate the rest as cleanly as possible". This is not new, the hydrocarbon industry has consistently improved over time, removing lead from fuels, then sulphur, and over recent years the focus has been the carbon embedded in the molecules themselves. In that context, our message has mirrored that from the major national producers – there is an “All of the Above” requirement for energy with the overriding message of “And not Or” ([See On the Road in the Middle East – licence to drill 8 Nov 22](#)). Transition, in our view, has always been a journey not an end point, as we put it – a verb, not a noun or a dial, not a switch (see [Middle East trip - what we learned on the ground 22 Nov 21](#)) (see [A dial not a switch 30 Oct 23](#)). AI and Energy Security have arguably just crystallised a physical reality. Energy remains, by definition, the ability to do work and GDP is arguably the sum of the output of all work. The Energy Service sector makes the infrastructure that produces energy and facilitates GDP growth.

2. Decline rate estimates have grown

Conventional reservoirs produce up to a plateau level and then go into decline as they age and pressures begin to drop. These vary by country, with mature basins such as the UK running in excess of 10% and other areas, such as Canada, typically offering low decline long life assets. This, in our view, was the big change in the mid-2000s. Decline rates became an issue and we think they were largely ignored since 2014, as excess OPEC spare capacity, shale’s boom and efficiency upgrades moderated their influence. However, this all changed in 2025. The impact was first highlighted by Halliburton in its 2Q call, when it stated that the “decline curve is still working.” Then in September 2025, the International Energy Agency (IEA) published its first major update on decline rates in a decade in its [The Implications of Oil and Gas field decline rates](#) publication. It noted that it believed that the decline rate for conventional oil had moved up to 5.6% per annum, with gas hitting 6.8% for field post peak. This is a significant move up from the typical 3-4% figure quoted in the 2000s. It noted that 90% of all capex spent since 2019 has been to maintain existing production, and that if all capex were to stop overnight, production would fall by 8% per year on average. The analysis suggested that the fall would total 5.5mn b/d of oil and 270 bcm annually, equivalent to the combined oil production of Norway and Brazil and the whole of Africa’s gas production. This is up from 3.9 mn b/d and 180bcm a year in 2010. In essence, the oil sector is running faster to stand still, already making the Service sector more important, in our view, ahead of the AI demand increase (see [Implications of rising decline rates 28 Sep 25](#)).

Mind the gap: The combination of higher demand estimates and increasing decline rates has one clear consequence – more new volumes are required to fill the gap. This will require more exploration, which we have started to see, and more work on the existing base load. Exxon highlighted the dilemma in its Energy Outlook 2025, where it estimated oil production to naturally decline at a rate of about 15% p.a, even higher than the IEA estimate. With no new investment, it estimated global oil production to drop from 100mb/d currently to 30mb/d by 2030, a 14mb/d decrease p.a. to 2030, while the IEA has an estimated decline of 5.5mb/d p.a. on average to 2035 (although this is on a slightly different basis).

FIGURE 67. ExxonMobil global supply and demand forecasts, in mboe/d

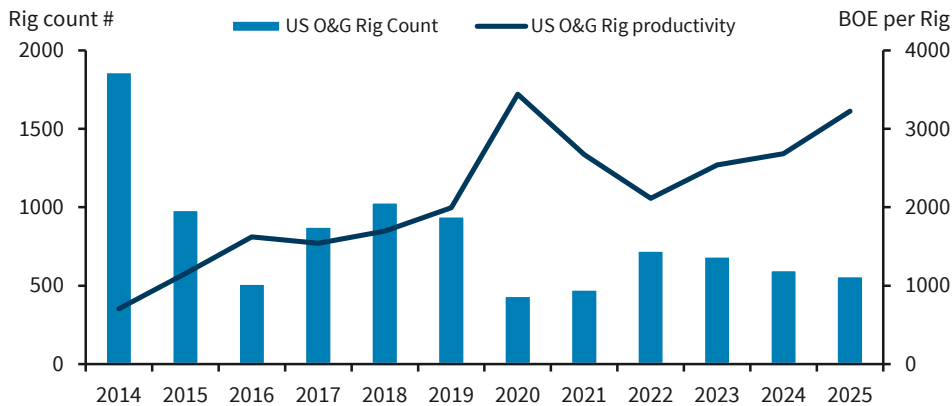


Source: Exxon Global Outlook 2025, Barclays Research

3. Back to the old stomping grounds

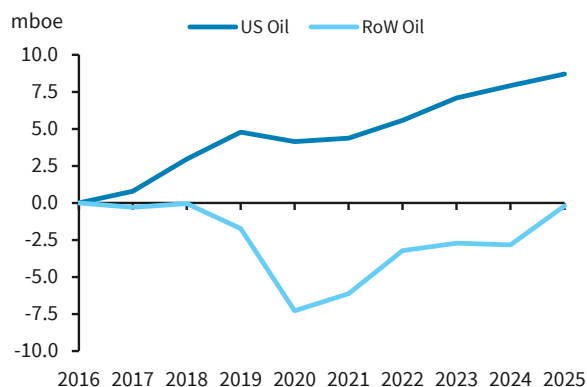
2016-25 saw a major shift in the hydrocarbon landscape. Prior to that, the predominant source of incremental molecules was emerging frontier countries, often in deep waters. In 2000, conventional oil accounted for ~97% of total oil output globally, however, by 2024, with the advent of US shale, it had dropped to 77%. Shale also made up 30% of all gas production. It was a decade where incremental demands were largely met by US shale. Indeed, over 2016-25, US shale accounted for 100% of all incremental barrels and 54% of all incremental gas supply. In this environment, the European Energy Services sector largely became irrelevant to global investors, as conventional energy, its key end market, was widely perceived to be in decline. This changed in 2025. The US rig count fell by 7.5% over the year with productivity only now approaching 2020 highs. This is compounded with questions over the quality of the remaining shale resources versus those which have already been produced.

FIGURE 68. Rig count vs Well productivity



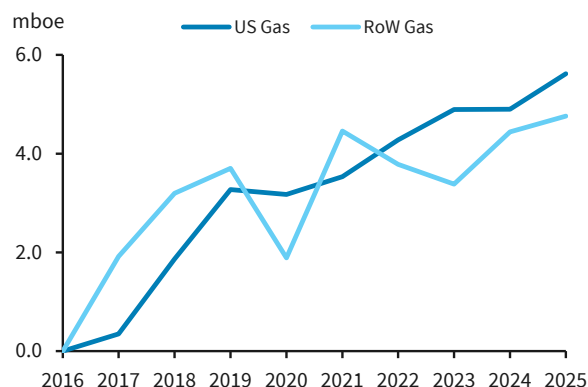
Source: EIA, Baker Hughes, Barclays research

FIGURE 69. Incremental Oil production since 2016



Source: EI Stats Review, Barclays research

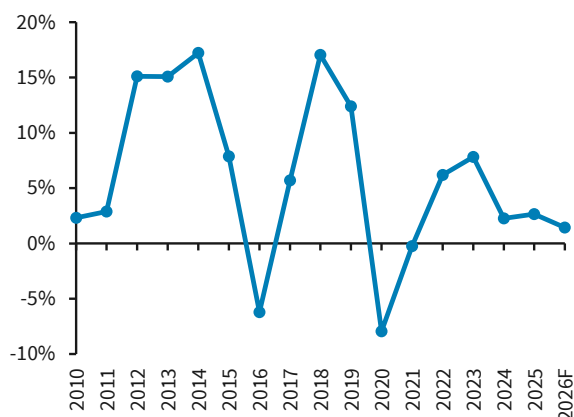
FIGURE 70. Incremental Gas production since 2016



Source: EI Stats Review, Barclays research

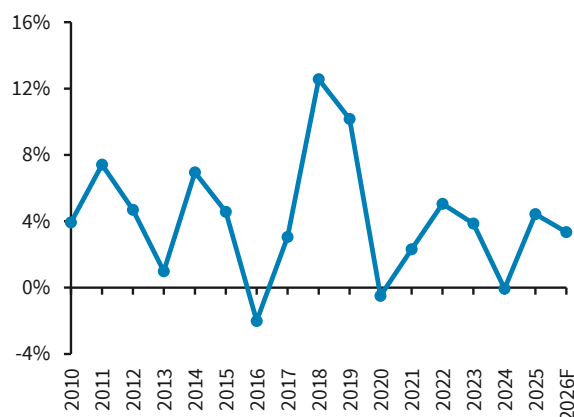
The net result is that, whilst the US is still growing, its rate of increase has slowed dramatically. It appears as if the era when the US was the foundation of the global supply bridge is over. As a result, we believe the industry has started to move back to the regions it had relied upon for the period running up to 2014. Shale was a major issue, for European Energy Services, but its impact is arguably now over as it has become part of the baseload, rather than the growth engine.

FIGURE 71. US Oil production growth



Source: Bloomberg, Barclays research

FIGURE 72. US Gas production growth



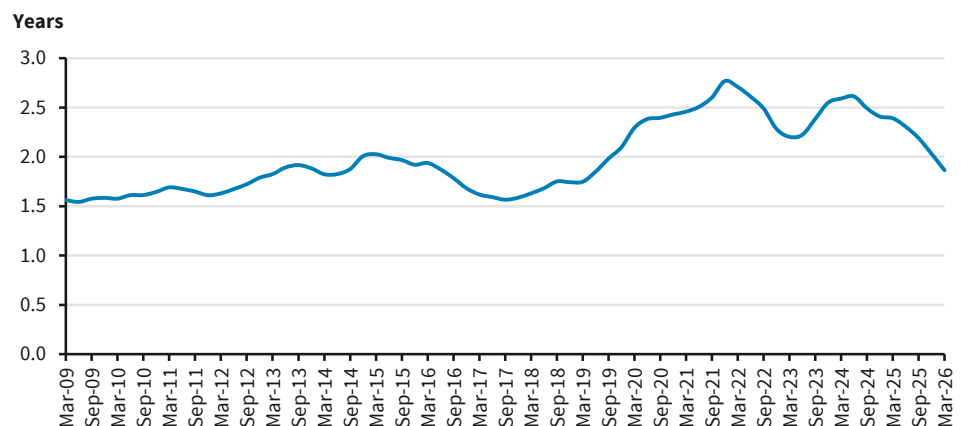
Source: Bloomberg, Barclays research

AI and Energy security: The new demands of AI and energy security, in our view, intensify a shift in energy that was already ongoing. Yes, there is an argument that the US may need more domestic production as it takes an early role in AI, but the need for AI is global. Every country, in our view, will want to have its own AI infrastructure as part of its national security. This comes with global energy needs and will, in our opinion, mean more natural gas in the form of LNG. We also add that while the focus on energy security has gravitated to oil and gas tankers exiting the Strait of Hormuz, it should not be forgotten that petrochemicals and fertilisers compete for the energy stored in the molecules of hydrocarbons, making up 12% of oil demand and ca. 20% of gas demand. These are also critical components of modern life and agriculture and have seen major disruption and will likely lead to countries debating whether to add their own production capacity. These, in our view, should lead to further demand for the Energy Service names.

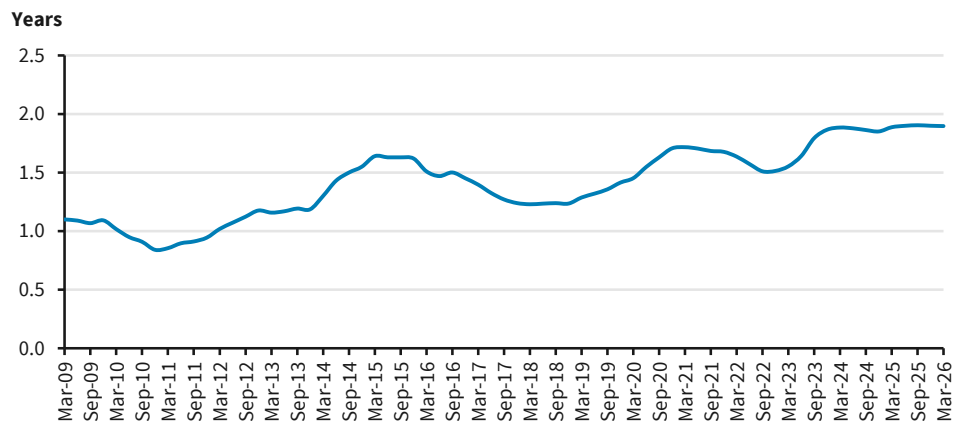
Back to the future: An argument that can be made is that AI and energy security accelerate a transition in energy that was already underway, taking it back to where it came from. For a significant period of time, the energy market was based on building the biggest, most efficient processing unit on a coastline, then taking supply from the global market. As a result, countries had refineries and petrochemical plants, and took feedstock from around the world. This arguably changed in the 2000s when the Middle East launched a series of major world scale processing facilities, dis-intermediating the refineries in regions like Europe, but the model was very much a hub-and-spoke approach to energy, with the Middle East firmly at the hub. Transition changed this, with the sources of energy becoming much more distributed. The UK, for example, in 2000 had ca 70 thermal power plants, which has fallen to ca 50. In the same period, the number of wind turbines (individual generators) has grown to over 12,000 from ca 1,000. As such, the number of generating sites has exploded, with distance to market key and the system becoming distributed. For other transition fuels, such as biofuels, this is even more the case. Distance to feedstock and market are key determinants of economic viability. Transition systems are very much, in our view, a distributed nodal system. Arguably this is now once again the case of hydrocarbons, with countries looking to secure their own domestic production and stock piles. This means an even greater push for geographic diversity than what was already driving the sector.

The industry is already busy – it's now about longevity: The increasing optimistic outlook, that we see for the sector, comes alongside a Services industry that has also embraced the capital discipline mantra over recent years. New assets are few and far between. We have seen the first signs of M&A, with TGS and PGS combining and Subsea 7 and Saipem proposing a combination — both looking for efficiencies. This paints a picture of a prolonged cycle. The boom-and-bust of the past has gone, in our view, with little appetite to fund speculative equipment in the industry. Instead, a longer plateau is therefore likely in our view. Our analogy is that it is demand that ultimately kills the cycle, but new capacity that makes it ill beforehand. This time the industry is healthy, albeit ageing, and supportive conditions towards the end of the decade promise longevity of business. This comes with backlogs already at or near record highs.

FIGURE 73. Onshore backlog, in years



Source: Company data, Barclays Research

FIGURE 74. Offshore backlog, in years

Source: Company data, Barclays Research

This optimistic picture that we see, in part augmented by AI power demand and Energy Security, flies in the face of current multiples, despite the sector performance. Our PE-based framework is that Energy Services stocks typically trade at c.10x earnings at the point in the forecast horizon where investors begin to perceive uncertainty, with that point moving further out as backlog duration, bidding momentum and confidence in the cycle improve. As the market gains conviction that the upcycle is durable, investors are prepared to capitalise earnings further into the future at a c.10x multiple. Across our coverage, the stocks are currently trading on average at c.15x/12x/10x 2026F/2027F/2028F PE, respectively. We are assuming the sector rolls forward at the similar multiple next year and thus are using 15x 2027F PE as a base case for valuation for the sector. This reflects the market is assigning benefit to improved earnings longevity, but there remains further scope for continued cycle visibility with backlog conversions, pricing discipline and Middle East/offshore activity and a new energy supply paradigm taking shape.

Our preferred investment themes in the current environment — further reinforced by AI-led power demand and energy-security priorities — remain exploration, offshore and the Middle East, with expansion post conflict a likely outcome.

Within our coverage, we see **Tecnicas Reunidas** as well positioned for backlog momentum from Middle East resolution, continued margin progression and opportunity around US power, with it recently securing its first North American AI power driven contract; **Technip Energies** as supported by recent LNG awards and a record backlog that underpins 2027 visibility; **TGS** as a beneficiary of a frontier exploration rebound, with scope for further deleveraging and potentially enhanced shareholder returns; and OCTG names **Tenaris** and **Vallourec** as increasingly well placed into 2027, helped by improving US pricing, deepwater activity and a resurgent Middle East cycle. They will also benefit from the repair and developing geothermal market which require their products.

Oil and Gas - US & Europe Equity views

Theresa Chen, CFA BCI, US | Lydia Rainforth, CFA Barclays, UK | Betty Jiang BCI, US

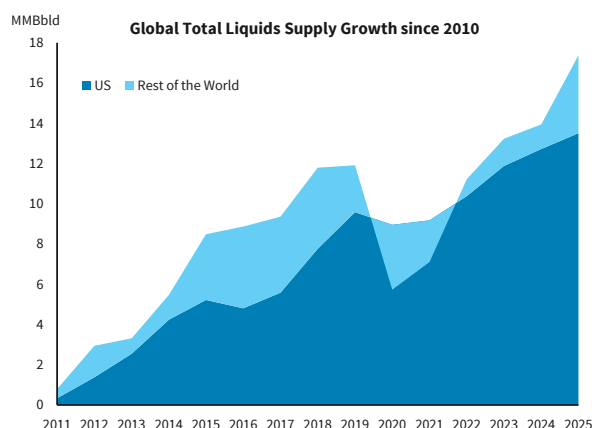
US Integrateds and E&Ps

Why our sector matters

From growth engine to strategic asset

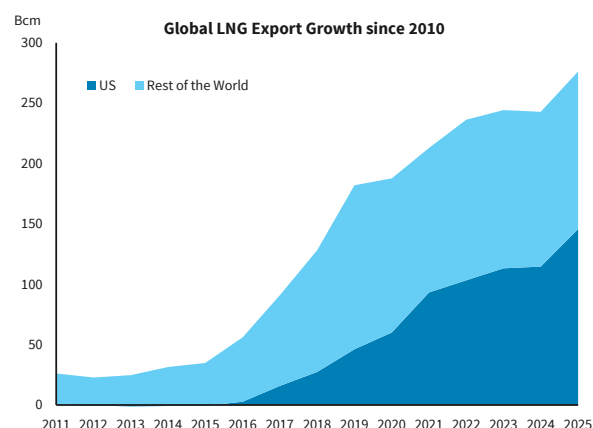
The most important development in global energy over the past 15 years was not the rise of renewables or the emergence of LNG as a global commodity. It was the shale revolution. The U.S. went from one of the world's largest energy importers to the world's largest producer of oil and natural gas, accounting for the vast majority of global oil, natural gas, and NGL supply growth since 2010. That growth fundamentally reshaped global energy markets, lowered energy costs, reduced OPEC's influence, enabled the development of a global LNG market, and provided the feedstock foundation for a renaissance in U.S. petrochemicals and manufacturing.

FIGURE 75. U.S. accounted for nearly 80% of global oil and NGL supply growth...



Source: Energy Institute

FIGURE 76. ...and over 50% of the LNG growth since 2010 and nearly 100% of the increase since 2020



Source: Energy Institute

Investors, however, experienced the opposite outcome. An industry fueled by abundant resources and cheap post-GFC capital pursued growth at all costs, destroying value even as it transformed the global energy system. As a result, markets became conditioned to view U.S. Upstream as an industry capable of delivering seemingly unlimited growth whenever prices rose.

The problem is that this assumption increasingly no longer holds. The same shale resource base that delivered an unprecedented growth cycle is maturing. Inventory quality is becoming more heterogeneous, resource replacement has re-emerged as a strategic concern, and the sector's ability to rapidly deliver multi-million-barrel-per-day production growth is materially lower than it was a decade ago. In many respects, the largest change facing global energy markets is not what the U.S. will do next, but what it is unlikely to repeat: another decade of unconstrained supply growth that consistently bailed out the rest of the world.

The implications are profound. A mature U.S. shale sector does not imply declining production, but it does imply greater scarcity value for future growth, higher strategic importance for secure molecules, and increasing competition for access to long-duration resources. We believe investors continue to underappreciate this shift.

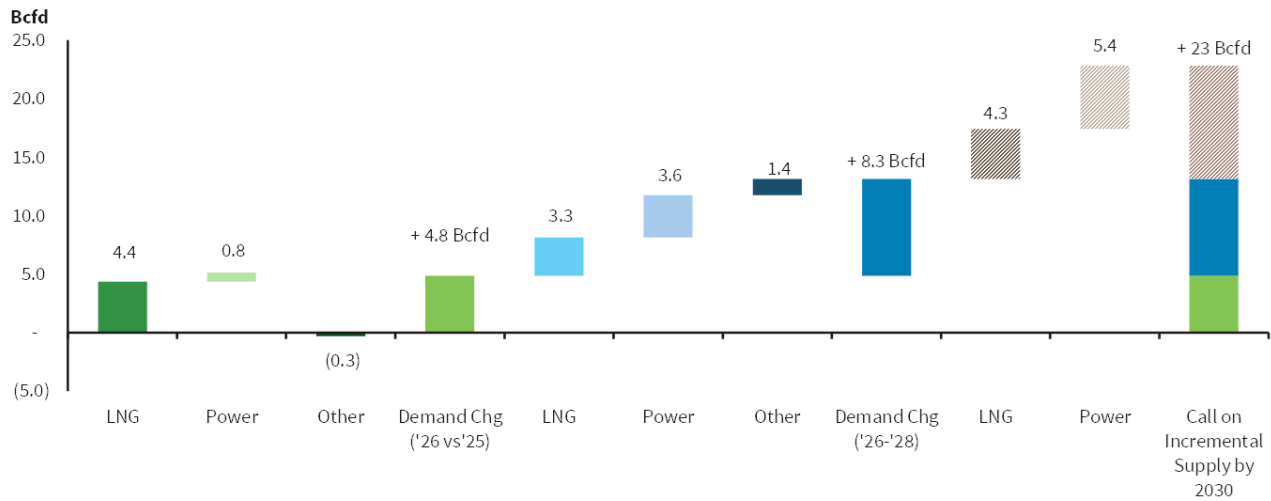
Demand for "secure" molecules is compounding the investment case for US Energy

The growing call on U.S. resources to balance global markets is reframing the investment case for U.S. Upstream. The same abundance that once depressed returns has become a strategic asset in a world increasingly focused on energy security, supply chain resiliency, and access to dependable sources of energy.

The clearest example is LNG. U.S. LNG buildout represents a structural, contracted call on U.S. natural gas that fundamentally changes the value proposition for the producers behind it. For more than a decade, U.S. gas was a market where abundance capped returns. The emergence of LNG increasingly inverts that relationship, as secure molecule abundance becomes a characteristic the market is willing to pay for. Henry Hub is becoming structurally more connected to premium global pricing benchmarks such as TTF, NBP, and JKM, while international buyers and global IOCs race to secure feedgas exposure through upstream positions in regions such as the Haynesville and Permian. What was once viewed as oversupply is increasingly being recognized as strategic supply.

The same dynamic extends beyond LNG. Debate continues regarding the ultimate role of natural gas in supporting AI and data centre growth, but the notion that gas will remain essential due to its ability to provide reliable 24/7 power generation has largely become consensus. With a substantial portion of global hyperscale and data-centre investment concentrated in the U.S., the AI theme further reinforces the call on U.S. natural gas and the strategic value of the resource base supporting it.

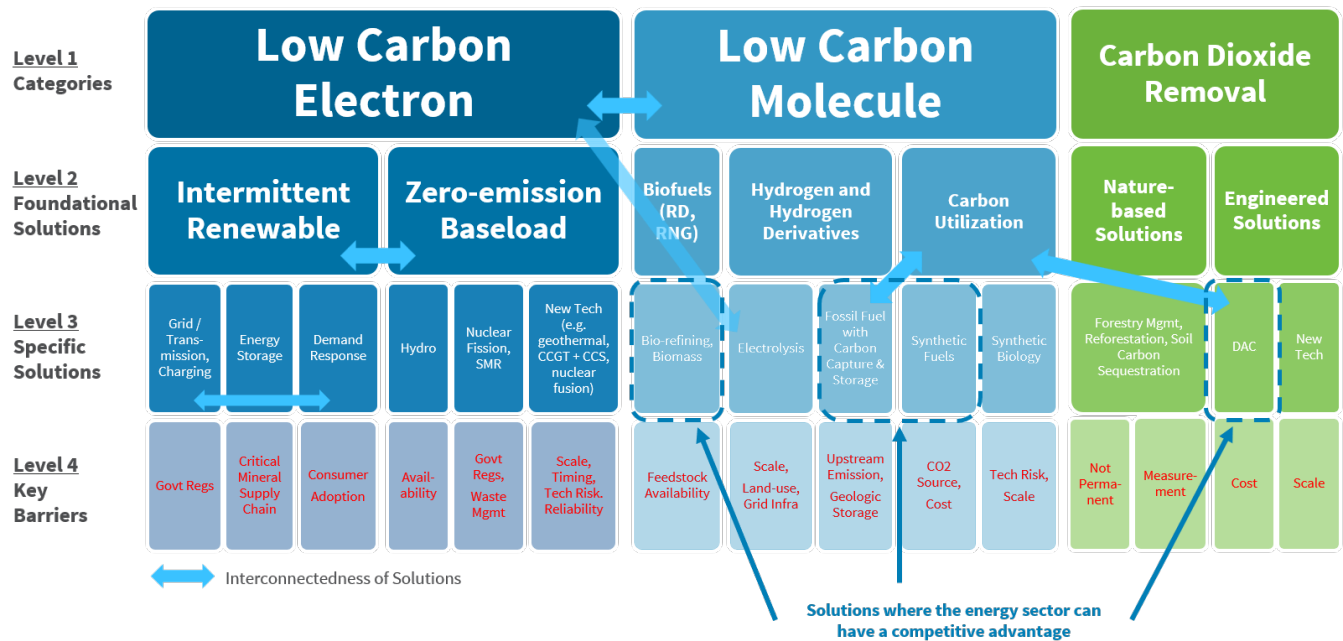
FIGURE 77. U.S. gas market is seeing structural growth from dual drivers of LNG export and AI/Power



Source: Barclays Research

Critically, the U.S. role extends beyond simply producing hydrocarbons. The combination of resource depth, infrastructure, technical expertise, capital availability, and project execution capability has created the world's most important energy ecosystem. The same capabilities that unlocked the shale revolution are now being deployed across LNG, petrochemicals, power infrastructure, carbon capture, lower-carbon fuels, and broader energy-transition initiatives. Regardless of where one stands on the transition debate, the world continues to require more energy, and the U.S. remains uniquely positioned to provide it.

FIGURE 78. Energy sector is an enabler of solutions no matter the direction for the global energy system



Source: Barclays Research

U.S. upstream is one of the only major supply bases not constrained by a chokepoint

In a world where OPEC+ spare capacity was effectively challenged overnight during ongoing Strait of Hormuz disruptions, the strategic importance of U.S. energy became undeniable. Despite years of criticism directed toward the hydrocarbon sector, U.S. energy production has repeatedly demonstrated its ability to stabilize global markets during periods of heightened uncertainty.

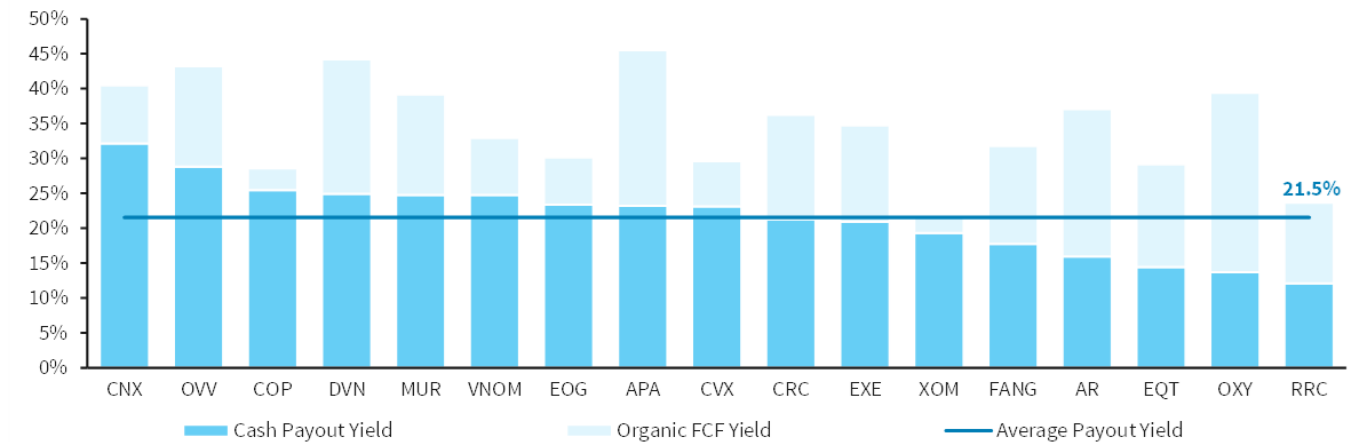
The U.S. remains one of the few large-scale supply bases capable of responding to changing global demand without relying on a single geopolitical chokepoint. Whether through ~13.9 MMBbld of crude production or ~26.9 Bcfd of natural gas exports, U.S. energy production has become a cornerstone of global energy security. While concerns surrounding shale maturity are increasingly valid, the strategic value of remaining resource depth only increases in a world where alternative sources of growth are becoming harder to find.

Robust financial health further strengthens the investment case

The investment case is further strengthened by the healthiest balance sheets in the sector's history. Following years of capital discipline, most U.S. Integrations and E&Ps now operate with sub-1x leverage, substantial free cash flow generation, and shareholder return frameworks that often distribute 50-100% of FCF through dividends and buybacks.

Yet despite stronger business models, increasing calls on U.S. molecules, and growing competition for resource access, sector valuations remain relatively compressed. In our view, the market continues to underestimate both the scarcity value of future U.S. production growth and the sector's strategic importance in an increasingly security-focused energy landscape.

The irony is that shale may be more valuable as it matures than it was during its peak growth phase. Investors are being offered exposure to a sector with improving scarcity, stronger capital discipline, increasing strategic relevance, and some of the strongest balance sheets in the market. That combination makes the investment case for U.S. Upstream more compelling than it has been in years.

FIGURE 79. US Majors and E&Ps can return ~22% of their current market cap over 2026-2028 at strip prices

Source: Barclays Research

Capital discipline is being reinforced by volatility

Shale capital discipline remains intact, but growing volatility and the potential for higher mid-cycle prices are creating increasingly difficult trade-offs. The energy sector faces an unusually wide range of potential outcomes tied to geopolitics, energy security, trade flows, decarbonization policy, and technological change. While energy security concerns could ultimately support higher commodity prices, the pathway remains uncertain. In that environment, management teams are understandably reluctant to make large, long-duration capital commitments. The result is an industry that remains biased toward flexibility, balance-sheet strength, and shareholder returns rather than broad-based production growth.

This stands in stark contrast to prior cycles, where higher commodity prices often translated almost mechanically into accelerating capital spending. Today, even against a backdrop of depleted global inventories, ongoing geopolitical conflict, and a growing call on U.S. molecules, the industry remains cautious. While some operators may argue they can simultaneously grow production and sustain attractive shareholder returns in a higher-price environment, we believe resource duration concerns will ultimately dominate decision making. Indeed, only a handful of companies with sufficient resource depth appear to have the "license to grow," and even then growth is generally expected to remain in the low-to-mid single-digit range. For most, preserving high-quality inventory and maintaining long-term optionality increasingly appear more valuable than maximizing near-term production growth.

Upstream companies are also learning to embrace the increasing volatility. One of the more notable developments in recent years has been the growing integration of U.S. Upstream companies across the value chain. This trend is particularly pronounced in natural gas, where producers are seeking greater exposure to the downstream margins created by LNG exports, transportation, storage, power demand growth, and marketing optimization. Rather than remaining pure commodity producers, companies are increasingly extending their reach into midstream infrastructure, LNG offtake arrangements, trading and marketing platforms, and end-market relationships in an effort to capture value across a broader portion of the molecule's journey.

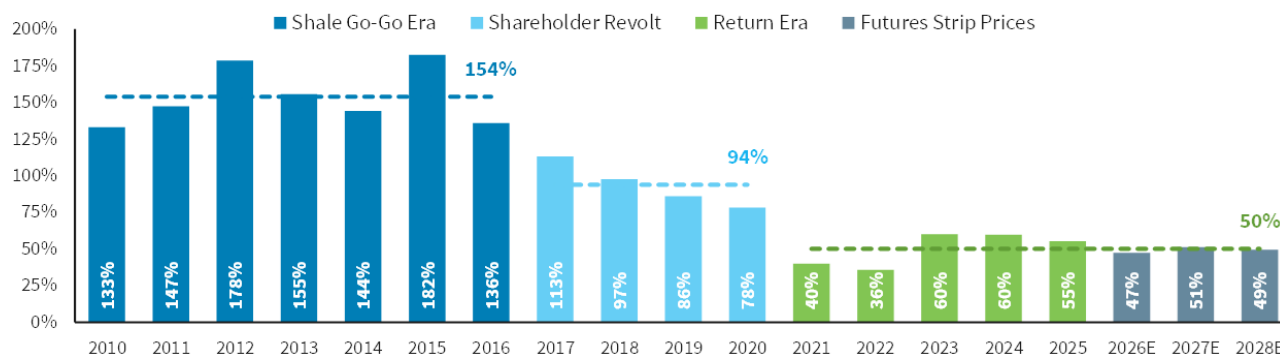
In our view, this represents the next evolution of the U.S. Upstream business model. If the first phase of shale was production growth and the second phase was capital discipline, the emerging third phase may be commercial optimization. As commodity markets become more volatile and end-market demand becomes more strategic, companies with the ability to secure premium market access, monetize volatility, and extract value across the value chain are likely to enjoy an increasing competitive advantage.

Energy security & capital allocation shifts

From return of capital to securing molecules

The return of history and concurrent geopolitical upheavals has thrust energy security and resource access back into the limelight. Since the COVID-19 pandemic, the U.S. Integrated Oil & E&P landscape's capital allocation priorities shifted to prioritize capital returns over growth, with corporates racing to strengthen balance sheets and adopt robust, formulaic shareholder return frameworks to the tune of 50-100% of FCF. Return of capital over return on capital had become the name of the game. Investors swiftly punished companies perceived as swaying from capital discipline and rewarded those that channelled marginal dollars into capital returns. While capital discipline priorities remain intact, the emerging energy security dilemma is redefining respective corporate strategies.

FIGURE 80. Majors & E&Ps have structurally changed their capital allocation priorities



Source: Barclays Research

Growing call on US molecules and companies is introducing some diversification in upstream spending.

The market is once more beginning to reward molecule optionality and resource access for Integrated Majors such as **CVX** and **XOM**, while inventory depth becomes an even greater focal point across the U.S. shale landscape. Resource access concerns associated with geopolitical upheavals are compounded by the maturation of the U.S. shale industry and a decade+ of global underinvestment in long-cycle, capital intensive Upstream oil & gas projects as companies shifted capital from hydrocarbon investments to shareholder returns, de-leveraging and transition-focused initiatives.

Resource-rich host governments are increasingly seeking U.S. partners

The clearest evidence that energy security has moved back to the top of the agenda is geographic. Resource-rich countries are increasingly turning to U.S. operators for capital, technical expertise, and project execution, creating opportunities that were largely unavailable a decade ago. Improved fiscal frameworks and geopolitical alignment are encouraging operators to revisit regions that had previously fallen out of favor.

- **Iraq** is perhaps the most notable example. **COP's** acquisition of a 42% stake in BP's Kirkuk entity provides exposure to ~3 Bboe of gross recoverable resources through redevelopment of Kirkuk, Bai, Hassan, Jambur, and Khabbaz. **CVX** has simultaneously expanded its footprint through agreements covering Nasiriyah, Balad, multiple exploration blocks, and a recent exclusive negotiation period for West Qurna-2, while also participating in efforts to revive a cross-border export pipeline to Syria. The breadth of activity suggests Iraq is actively courting U.S. operators as strategic partners for both resource development and export expansion. **XOM** has similarly re-engaged with the country, signing a Heads of Agreement to develop the Majnoon field and improve recovery rates at other Iraqi assets just a few years after exiting West Qurna.

- The same dynamic is playing out elsewhere. **EOG's UAE and Bahrain positions** reflect host governments' willingness to import U.S. shale expertise to unconventional resource development. Management has become increasingly constructive on the UAE opportunity, describing its 900,000-acre unconventional concession as a potential long-term growth platform analogous to the Eagle Ford.
- Meanwhile, **Venezuela** is once again attracting industry attention following regulatory changes. **CVX** has the greatest leverage to the reopening, with production already increasing materially under its self-funded development program, while both **XOM** and **COP** continue to evaluate longer-term opportunities. The resource potential remains substantial, although contract durability, fiscal terms, and political stability will ultimately determine whether the opportunity can attract long-term capital.

More broadly, the willingness of operators to revisit countries such as Iraq, Venezuela, and even Libya underscores how energy security concerns are reshaping the relationship between IOCs and host governments. Countries are increasingly competing for U.S. capital and expertise, while operators are broadening their opportunity sets beyond mature North American resource plays.

Frontier exploration is returning

Alongside improving access to international resources, the industry is also showing a renewed willingness to spend on exploration. After a decade in which exploration budgets were systematically reduced in favor of short-cycle shale development, management teams increasingly view exploration as a necessary component of long-term inventory replacement.

Investors are beginning to focus less on near-term production growth and more on the quality of companies' 2030+ resource pipeline. As a result, exploration spending is rising from historically depressed levels, and management teams are becoming more willing to publicly defend exploration capital. **CVX's** commentary around strengthening its exploration capabilities following the HES acquisition is one indication of this shift. The renewed activity across **Namibia's Orange Basin**, the **Eastern Mediterranean, offshore West Africa**, and other frontier basins highlights a broader industry realization that future resource growth cannot rely exclusively on existing shale inventories.

Among U.S. E&Ps, **MUR's Vietnam and Côte d'Ivoire programs** represent some of the high-impact exploration stories. Investors are increasingly focused on the Hai Su Vang appraisal program in Vietnam, alongside follow-up activity after recent success offshore Côte d'Ivoire. Similarly, **APA's Suriname portfolio** remains a key source of longer-term upside, with investors focused on extending the opportunity set beyond GranMorgu and the timing of future exploration wells.

Resource renewal is becoming the domestic response to shale maturation

While international opportunities and frontier exploration are receiving greater attention, resource renewal remains the dominant theme across U.S. shale. As the sector grapples with maturing inventories and a growing call on U.S. molecules, operators are increasingly focused on extending inventory life through both organic and inorganic means.

The approaches vary by company. **FANG** continues to focus on adding prospective development zones, including opportunities such as the Barnett within the Midland Basin. **EOG** has pursued resource expansion through both acquisition and international diversification, including Encino Resources and opportunities across Bahrain and the UAE. **APA** has discussed the potential to materially expand its economic inventory base, while **OXY** continues to pursue resource-extension opportunities through international ventures, enhanced recovery programs, and unconventional EOR initiatives.

Resource renewal is also occurring through new domestic plays and basin extensions. **Alaska** has emerged as a growing area of focus for APA following its acreage acquisition around existing infrastructure, where the company has already made discoveries and plans additional drilling activity. The **Powder River Basin**, secondary benches, enhanced recovery technologies, and unconventional EOR are all increasingly viewed as ways to expand recoverable inventory without relying solely on acquisitions.

Winners and losers in a fragmented energy world

Well-capitalized companies with differentiated resource duration and technical expertise are poised to outperform. In an increasingly resource-scarce world, well-capitalized companies with differentiated resource duration and project execution capabilities are more likely to see the market recognize their respective value propositions. Among Integrates, we favor **XOM** for its robust 2030+ resource access and major project portfolio that is poised to deliver outsized FCF well into the 2030s, its unmatched technical and project execution capacity and novel emerging business segments. The Guyana success story serves as a prominent example of XOM's ability to deliver.

Among oil-weighted E&Ps, we favor **COP** for its combination of low-cost resource depth, capital discipline and one of the clearest free cash flow growth profiles in the sector. As major projects come online and spending rolls off, COP's free cash flow breakeven is expected to fall from the mid-\$40s/Bbl WTI today to the low-\$30s/Bbl by the end of the decade, among the lowest in the sector. At the same time, we believe a combination of LNG start-ups, Willow and ongoing cost optimization should drive a meaningful step-up in free cash flow, even in a flat commodity price environment. Recent international moves such as Iraq provide exposure to large-scale resource opportunities without materially increasing corporate capital.

We also prefer **OXY**, which combines one of the most attractive Permian positions in the sector with differentiated exposure to enhanced oil recovery. We believe the market remains under-appreciative of OXY's improving capital efficiency, strengthening balance sheet and growing international optionality. The company's long-standing presence in Oman and the UAE positions it well to benefit as Middle Eastern producers increasingly seek technically capable partners to expand resource development and improve recovery rates.

Within gas-weighted names, we favor **EQT** given its scale, integrated Appalachian footprint, long-duration inventory, margin expansion, and growing portfolio of demand-driven commercial opportunities. As the Northeast demand thesis continues to develop, we believe EQT is increasingly separating itself from peers through its ability to secure differentiated supply agreements and leverage its integrated upstream, midstream and marketing platform. This should allow EQT to grow FCF per share through modest and sustainable top-line production growth and buyback.

We are more cautious on **CNX**. While management has consistently demonstrated strong operational execution and capital discipline, the company has the shortest resource runway among the larger gas-weighted E&Ps in our coverage. As development increasingly migrates toward the Utica, investor focus is likely to shift toward the basin's higher-cost and higher-risk profile. While CNX's technology initiatives and ancillary businesses provide support, we believe differentiated resource depth and inventory quality will become increasingly important differentiators as investors look further into the next decade.

North America Midstream and Refining

Why our sector matters

As global energy demand continues to scale, midstream operators provide the infrastructure necessary to unlock production growth upstream. By investing in gathering,

processing, transportation, and storage assets, midstream enables producers to access premium markets, improve realizations, and bring incremental supply online that might otherwise remain constrained.

We think developments in the Middle East and broader geopolitical turbulence have structurally augmented the value of US energy infrastructure assets, and especially export infrastructure, which remain free from the threat of potential bombardment and/or geographical restriction. Whether or not the Iran conflict and resulting disruptions to the Strait of Hormuz normalize over the near term, we expect the geopolitical risk premium to endure, highlighting the enhanced reliability of US energy supply.

If crude prices eventually settle higher, we think a potential uptick in US onshore production could further strengthen the growth outlook across our midstream coverage, driving incremental demand for infrastructure assets across energy value chains. From Permian NGL infrastructure, to residue egress, to long-haul crude transportation, we would expect most (if not all) of our midstream operators to benefit from recontracting tailwinds on existing assets as well as opportunities for additional expansions.

Energy security and the return of capital

Midstream companies continue to lean into organic growth. Against a constructive backdrop for North American energy infrastructure, midstream operators by and large are prioritizing growth capex spend, complementing returning cash to shareholders via dividend growth and share repurchases to a lesser extent. This is true across commodity value chains.

Gas infrastructure incumbents, for example, continue to commercialize and construct transmission expansions at rapid pace in response to structural demand growth for gas-to-power and LNG feedgas. Meanwhile, NGL-focused operators continue to make large-scale investments in Permian NGL infrastructure to accommodate liquids-rich domestic production growth within the basin.

Accordingly, we expect 2026-28 to represent peak years for growth capex for many of our midstream names. Whether capital spend inflects lower beyond this time-frame will largely depend on the producer backdrop. If production ultimately steps higher amid the emergent call on US energy products, we think the incremental bid for energy infrastructure could result in elevated growth capex spend through the end of the decade (and potentially beyond).

While **Cheniere** continues to deploy capital toward growth projects underway, as well as opportunities advancing toward FID (SPL Expansion Phase 1), the company remains one of the only names within our midstream coverage able to execute on significant share repurchases, due to its highly contracted business model and robust cash flow generation.

Returning cash to shareholders remains paramount for refiners. We anticipate refiners in our coverage will continue to return cash to shareholders in line with their respective capital return commitments. While we expect little change to these capital allocation frameworks, we believe outsized margins as a result of volatility will likely continue to lead to excess free cash, at least in the near term. Within our coverage, VLO has been the most judicious and consistent in returning capital, and we note the company has no major growth capex plans at this point.

Of note on the capital allocation front, within our coverage **PSX** and **MPC** (via **MPLX**) stand out as refiners investing significant capital into midstream growth. We anticipate **PSX** will continue building out its Permian NGL value chain, having FID'd a new Permian processing plant and fractionator during 2Q. **PSX** has said publicly it plans to add a Permian processing plant approximately every year. For its part, we expect **MPLX** will continue investing in its core areas

of operation in the Permian and Northeast as it continues to execute on a sizable project backlog.

Volatility, energy security, and the effect on our sector

Within midstream, we think the macro backdrop remains most important for our NGL-leveraged operators. Despite limited direct commodity price exposure across midstream in general, recent and ongoing upside volatility in crude prices have underscored the significant liquids torque among our NGL-leveraged names (**EPD, ET, KNTK, PSX, TRGP**), and especially those with positioning on second-tier and/or relatively fringy acreage where producers are more price sensitive (**KNTK**).

As elevated crude prices support expectations for an eventual uptick in US onshore production, these names stand out as meaningful beneficiaries of incremental volume growth, which will bolster demand for assets up and down the NGL value chain (and beyond). Whether production ultimately steps higher will be critical to offsetting a number of known headwinds set to impact the company over the next few years.

Separately, we continue to view **SUN** as well positioned amid ongoing commodity and geopolitical volatility. The partnership's Fuel Distribution segment benefits from its ability to optimize supply and product flows across a broad network, while Burnaby provides incremental exposure to attractive West Coast refining economics.

For refiners, limited refined product supply as a result of the ongoing volatility has bolstered margins in the space. We think the current set-up will be relatively durable, given significant Middle Eastern refining capacity currently offline, Russia banning diesel exports in response to ongoing Ukrainian strikes on refining infrastructure, as well as currently limited global inventories. This constructive macro outlook is reflected in the stock prices among the group, which have continued to climb higher.

The key risk to margins at this point would be a durable peace, re-opening of the Strait, and a faster-than-anticipated return to normalcy on the supply front. More near term, we continue to monitor the pace at which China as well as other North Asian countries resume incremental refined product exports, which could begin to loosen an otherwise tight global S/D balance. Though our covered refiners are still observing resilient demand, we continue to watch for early signs of demand softness, owing to elevated prices at the pump, where we find little hope for relief, given the current macro set up. Read more [here](#).

Winners and losers in a fragmented energy World

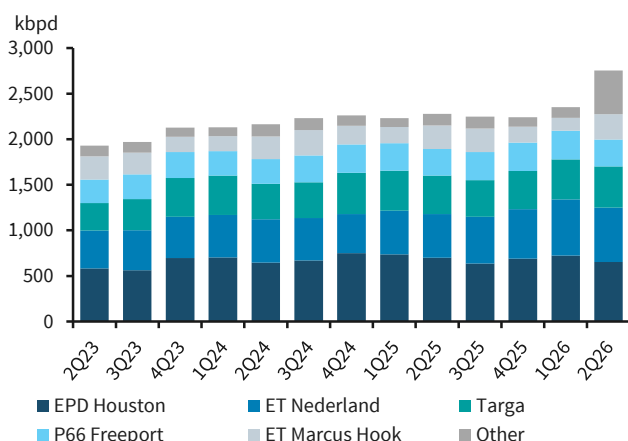
Growing demand for US energy export infrastructure. Supply disruptions catalyzed by the Iran war have augmented the value of US Gulf Coast energy export infrastructure in general, which remains unencumbered by the conflict at large. We expect global demand for US NGLs, LNG, and refined product exports to remain structurally higher going forward, giving rise to contracting tailwinds and incremental expansion opportunities for our midstream operators with exposure to these assets.

Recent demand growth for US NGL exports among Asian and European buyers will likely remain sticky. As international markets have turned to the USGC to offset lost NGL supply from the Middle East, our covered NGL exporters (**EPD, ET, PSX, TRGP**) have noted that Asian and European buyers are seeking to extend existing contracts and shift from spot purchases to term agreements where possible. **ET**, for example, recently extended the vast majority of its ethane export agreements out of Nederland into 2041, adding 10 years to current contracts. The partnership has also highlighted the potential to extend LPG export contracts well into the 2030s.

Meanwhile, at **EPD's** Morgan's Point export facility, multiple customers that previously purchased ethylene on a spot basis have migrated to term contracts, driven by the economic competitiveness of US ethylene as a secondary feedstock for European petchem facilities. To this point, it is worth **reiterating** that elevated naphtha prices coupled with disruptions to Middle East feedstocks are forcing international petchem operators to lean more heavily into US ethane and ethylene exports to feed their crackers. Given its exposure to petchem, we view **EPD** as a notable beneficiary of this shift.

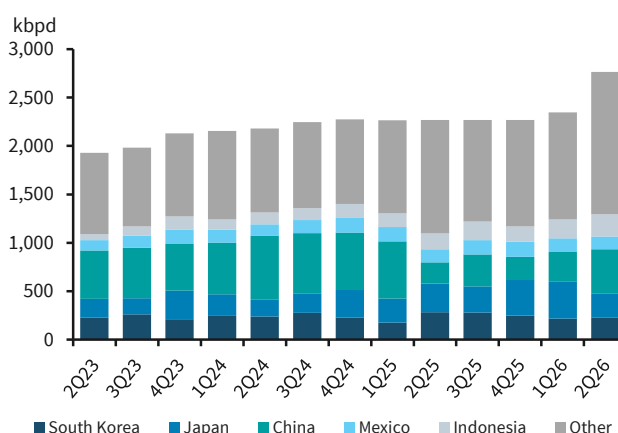
To accommodate higher NGL export demand, midstream operators have provided visibility toward incremental export capacity expansions, including **MPLX's** JV Texas City LPG export dock (400 kbpd; ISD 2028). **TRGP** has also discussed the potential for further expansions to its own LPG export capacity, which will reach 19-20 mmbbls/month by 3Q27 (vs. 14-15 mmbbls/month currently).

FIGURE 81. US LPG exports by origin installation



Source: Barclays Research, Kpler

FIGURE 82. US LPG exports by destination country



Source: Barclays Research, Kpler

Amid an uncertain outlook for Qatari LNG supply, we view North American LNG as the most reliable long-term value option. The US and Qatar are the two largest suppliers of LNG, accounting for ~25% and ~19% of global LNG exports in 2025 (~437 mtpa total per WoodMac), respectively. Through a large-scale, multi-phased expansion at North Field, Qatar was previously on track to expand production capacity to ~142 mtpa by decade end, up from ~78 mtpa in 2025. However, in the wake of **Iranian strikes** on Ras Laffan in March, 2 of Qatar's 14 liquefaction trains will remain offline for 3-5 years (~17% of total capacity), and the aforementioned expansions are likely to face material delays.

The initial phase of Qatar's North Field expansion was previously slated to achieve First LNG later this year (North Field East; +32 mtpa), followed by subsequent phases in 2027 (North Field South; +16 mtpa) and 2028 (North Field West; +16 mtpa). Per commentary from Qatari Energy Minister Saad al-Kaabi in March, North Field East could be delayed by over a year, pushing back the timeline for North Field South and West as well.

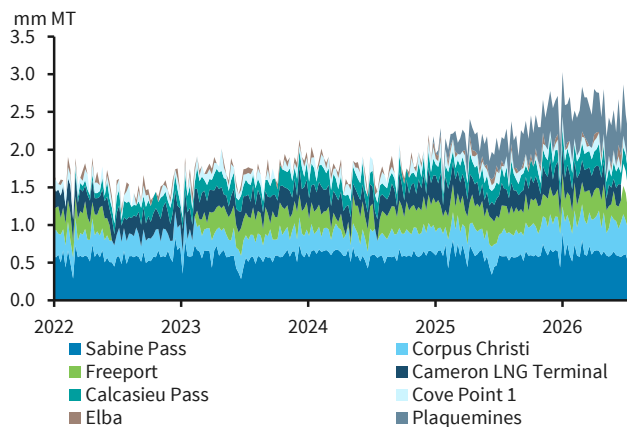
We think these realities pave the way for a protracted LNG supply shock that will likely outlast the conflict and any disruptions to the Strait. While US liquefaction operators have limited (if any) ability to meaningfully increase production capacity in the near term, we are already observing an uptick in commercial support for future expansions as customers seek more reliable US supply on a long-term basis.

Since the conflict began, additional US liquefaction projects have passed FID, including Venture Global's CP2 project and Caturus' Commonwealth LNG. Within our coverage, [we think Cheniere](#) is also gearing up to FID SPL Train 7 soon (~6 mtpa), which will mark the first phase of its larger-scale planned expansion at Sabine Pass (+~20 mtpa total). **Cheniere** is developing a multi-phase expansion project at its Corpus Christi facility as well (+~24 mtpa). Commercial tailwinds aside, however, we do not think any of these projects can be meaningfully accelerated given the lengthy construction and commissioning process associated with liquefaction.

In the absence of a near-term relief valve to ease current S/D tightness, LNG consumption patterns are increasingly volatile across Asia, and especially among price-sensitive countries beholden to the spot market. As global gas prices continue to trend well above the ~\$10/mmbtu threshold at which EM demand tends to rise, Asian countries are leaning further into alternative energy sources such as coal and liquid fuels and limiting industrial use where possible. At the same time, European gas storage inventories are trending at precariously low levels ahead of winter 2026-27, setting the stage for increased competition between Europe and Asia for available cargoes.

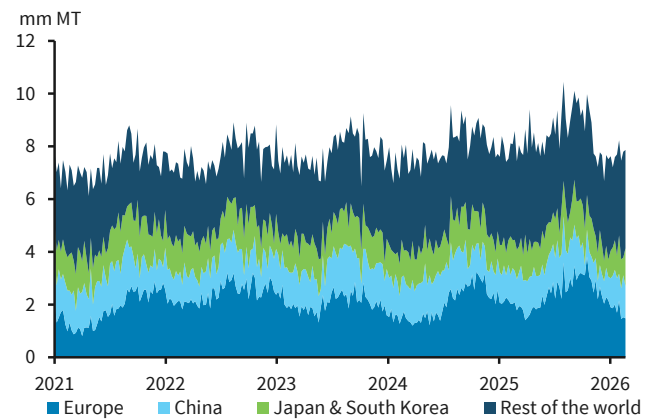
We wait to see how these dynamics impact longer-term LNG adoption rates across Asia, which has generally been perceived as the engine of global demand growth once gas prices are low enough to catalyze more elastic demand.

FIGURE 83. US LNG exports



Source: Barclays Research, Kpler

FIGURE 84. LNG imports



Source: Barclays Research, Kpler

On the refined products front, US refined products have emerged as a key source of global supply since the outset of the conflict between the US and Iran. With significant Middle Eastern and Russian refining capacity down, coupled with limited access to Middle Eastern feedstocks among Asian and European refiners, all our covered refiners have observed robust export demand. For example, as we have written about [previously](#), **MPC** has seen new markets open up, reporting it exported ULSD from LAR to Australia. It also observed particularly strong demand from Latam and Europe. With the Strait again mostly closed and infrastructure abroad still under threat, we expect US refined product exports will continue to be a crucial source of supply for global markets as US refiners remain both unencumbered by the risk of bombardment and able to access feedstocks.

AI, power demand and the new electricity economy

Structural drivers of gas-to-power demand growth, including the rising data centre load, are underwriting significant expansion opportunities for our gas infrastructure

incumbents. As the proliferation of AI and data centres drives material growth in gas-to-power demand, we expect the incremental bid for natural gas infrastructure and power generation capacity to continue to benefit our midstream operators with exposure to these assets (**DTM, ET, ENB, KGS, KMI, TRP**).

Given hyperscalers' sustained emphasis on speed-to-market, we see the most near-term upside for midstream players willing to provide gas to data centres via direct supply agreements and associated pipeline laterals (eg, **ET**), as well as those offering quick-cycle, behind-the-meter (BTM) power generation capabilities. Our covered gas infrastructure operators are also pursuing more traditional transmission expansions in support of longer-lead utility grid expansions. Irrespective of the approach, we expect stock price performance among these names to be increasingly dictated by commercialization progress and ultimate project FIDs.

With respect to the BTM route in particular, we wait to see if additional entrants begin to populate this space and take on exposure to this higher-risk business in support of the data centre buildout. Whether returns for existing participants are affected as a result also remains to be seen. Given that **KGS** was able to make its own foray into BTM by raising secondary equity to buy turbines on spec (with no negative effect on valuation), we think it is conceivable that other companies could be equipped to follow a similar path. We think there is a degree of uncertainty regarding the true height of barriers to entry for this business as a result. As BTM projects begin to enter service, we expect to gain more clarity on what it means to actually operate and maintain these assets. Whether operational execution separates the winners from losers in this space remains to be seen.

The race for capital: financing and investment ideas

Where is capital most needed and who can provide it?

Midstream companies remain well capitalized to self-fund organic growth. Although prior expansionary cycles were often dependent on equity issuance and/or incremental leverage, midstream companies today are generally positioned to internally fund organic growth thanks to improved balance sheet capacity and significant cash flow visibility. Even if growth opportunities accelerate from here, we expect our midstream companies to remain judicious capital stewards and only sanction projects that adhere to their stringent return frameworks.

Still, external funding sources remain an option. **ENB's** C\$2.7bn financing JV with KKR in collaboration with Apollo is the latest example of a midstream operator tapping into the growing availability of insurance-backed capital to fund midstream growth opportunities at an attractive cost of capital. Against a strong fundamental backdrop for energy infrastructure, across commodity value chains, we will wait to see if other midstream names within our coverage pursue a similar approach to support incremental growth capex.

Investment recommendations

- **Energy Transfer, LP (ET), Overweight:** We think the market has yet to appreciate ET's unique exposure to demand-pull natural gas pipeline assets as well as its Permian-focused liquids footprint, which provide both defensive and offensive attributes. Across its diversified and well-connected asset base, ET continues to benefit from tailwinds on multiple fronts, including macro themes such as incremental power demand, liquids-rich domestic production growth spearheaded by the Permian, and the rising need for energy export infrastructure. Even the often less appreciated components of ET, compression (USAC) and fuel distribution (SUN), also continue to generate healthy cash flow for the partnership.
- **Pembina Pipeline Corp. (PPL.TO), Overweight:** PPL's diversified, vertically-integrated footprint positions the company well to benefit from several LT growth drivers. In particular,

PPL's NGL assets are levered to growing demand for condensate as diluent for incremental WCSB production. With several projects currently underway and/or under consideration for WCSB egress, PPL is poised to benefit without taking construction risk on building major, cross-border pipeline infrastructure (read more [here](#)). Diversification into LNG via PPL's 49.9% owned Cedar LNG facility (expected ISD 2028) is another key growth avenue amid the current macro volatility. As construction on the project progresses, the outlook has only been strengthened by major disruptions to Qatar's LNG capacity and the resulting premium place on North American LNG infrastructure. Also of note, PPL recently FID'd Greenlight Phase I, as it expands into power generation to serve major data centre development in Alberta.

- **Valero Energy (VLO), Overweight:** VLO's advantaged Gulf Coast footprint positions it uniquely well to capitalize on strong USGC refined product export demand, supported by disruptions to refining infrastructure in the Middle East and Russia and constrained feedstock access across Europe and Asia. Its location also provides flexible access to the most cost-advantaged feedstocks, a benefit recently reinforced by increased Venezuelan crude runs across its system (read more [here](#)). Beyond its geographic location, VLO's long track record of best-in-class operational execution gives us confidence in its ability to capture the current margin upswing while remaining resilient through cycles, supported by a strong balance sheet and disciplined approach to capital allocation.
- **Plains All American Pipeline (PAA), Underweight:** Although PAA's Permian crude footprint has been a direct beneficiary of elevated crude prices since the onset of the Middle East conflict, we think the partnership's growth outlook remains largely beholden to the macro environment and whether crude prices remain structurally higher to incentivize incremental production. Moreover, even in a more bullish production scenario, we see better risk/reward in Permian NGL names such as KNTK and TRGP, which exhibit significant liquids torque.

Energy - Europe

The global energy system is entering a new phase. While decarbonisation remains an important long-term objective, recent years have highlighted the importance of ensuring that energy remains secure, affordable and reliable. At the same time, energy demand continues to grow, driven by emerging markets, rising living standards, electrification and increasingly digital infrastructure such as AI and data centres.

In our view, the challenge facing policymakers and industry is therefore not simply replacing one energy source with another. Rather, it is ensuring that sufficient energy is available across multiple sources to meet growing demand while maintaining security of supply and supporting economic growth.

This creates an increasingly important role for integrated energy companies. The sector provides oil, gas, LNG, refining, trading and increasingly power solutions. These capabilities position it at the centre of the themes that we believe will shape the next decade: energy security, energy addition, AI-driven electricity demand and the need for large-scale capital deployment.

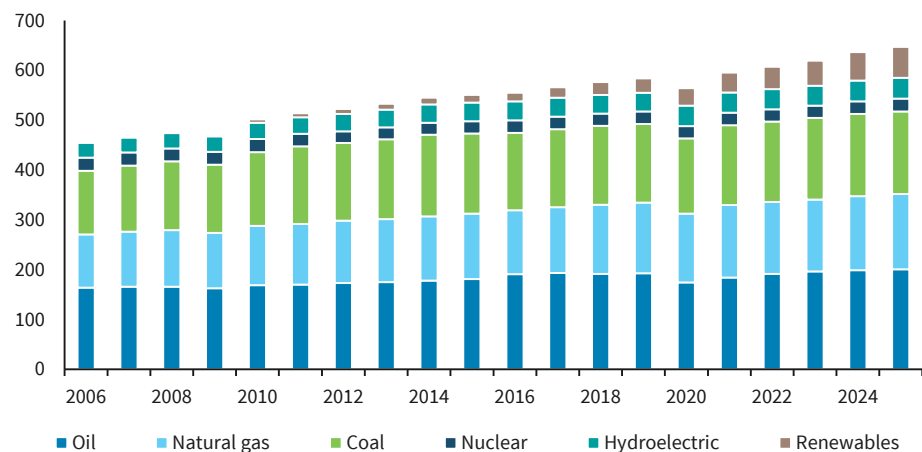
Why our sector matters

We expect global energy demand to continue rising through the next decade despite improvements in efficiency and increasing deployment of renewable technologies because of GDP growth, populations getting wealthier, and improving living standards. Some parts of the world still do not have a reliable energy source.

Historically, discussions around energy have often focused on substitution. Increasingly, however, the challenge is one of addition. As shown in [Figure 85](#), over the past decade renewables have grown but have not replaced other energy sources; instead, they have been

added on top. As new energy sources enter the system, existing energy sources remain essential to support reliability, affordability and security of supply and to meet growing energy consumption.

FIGURE 85. World energy consumption by fuel, in exajoules



2024 and 2025 are based on the old accounting method of the EI Statistical Review of World Energy
Source: EI Statistical Review of World Energy, Barclays Research

a) Oil still matters and needs capital

Oil remains a cornerstone of the global economy. It continues to underpin transportation, aviation, petrochemicals, shipping and a large share of industrial activity, with few large-scale alternatives currently available.

Importantly, the industry's challenge is not only meeting future energy demand, but also replacing production from existing fields. The global resource base is declining quicker than before, with decline rates estimated at around 8% per year due to the rise of unconventional (IEA). Nearly 90% of annual upstream O&G investment since 2019 has been dedicated to offsetting production declines rather than to meet demand growth. If all capital investment in existing sources of oil and gas production were to cease immediately, global oil production would fall by 8% p.a. on average over the next decade, or around 5.5 mb/d each year. See more details in our note [Implications of rising decline rates](#).

As a result, the energy system is engaged in a constant race for resources. Continued investment is required not only in existing assets, but also in exploration, appraisal and new project development to discover and bring forward the next generation of production.

b) Electrification is driving renewables and gas needs

One of the most important trends shaping the energy system is the continued electrification of society. Growth in electric vehicles, cooling demand, industrial electrification and digital infrastructure is increasing electricity consumption across both developed and emerging markets.

More recently, AI and data centres have emerged as an additional source of structural power demand. The scale and reliability requirements of these facilities are driving investment across generation, transmission and power infrastructure, while reinforcing the need for reliable and flexible energy supplies.

As a result, electricity demand is becoming an increasingly important driver of overall energy consumption. While renewables are expected to capture a significant share of this growth, ensuring reliable power supply will require investment across the broader energy system,

including generation, networks, storage and flexible capacity. Natural gas will play a critical role alongside renewables, providing the flexibility and reliability needed to support a more electrified economy

c) Natural gas and LNG are increasingly strategic

As electricity demand rises, we expect gas to remain an important complement to renewables, helping maintain reliability and security of supply. We see LNG demand being supported by three structural themes: energy security, decarbonisation and AI-related power demand.

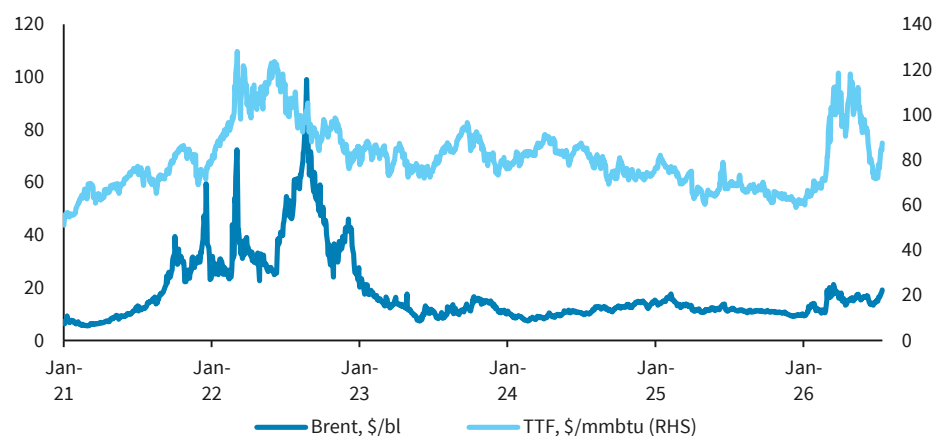
For many regions, particularly across Asia, gas is seen as a transition fuel. Governments are looking to reduce reliance on coal, while supporting rising electricity demand from urbanisation, electrification and, increasingly, AI and data-centre growth. This was one of the key conclusions from our recent trip to Southeast Asia (see our note [New Horizons: Southeast Asia – the race for resources](#)). Coal still accounts for a significant part of the energy mix in Asia (India 59%, China 57%, Indonesia 45%...) and even modest coal-to-gas switching can create substantial incremental gas demand. As an example of scale, moving 1% of electricity demand in China from coal to gas, would add c30mtpa to LNG, on our estimates (see our note [LNG – supply creates its own demand](#)).

As a result, we expect LNG demand to grow significantly - Shell expects global LNG demand to grow by +65% by 2050, to nearly 700mt, led by Emerging Asia and supported by energy security needs, diversification across end-use sectors and the continued role of Europe, China and Japan in market balancing. See our note [Shell LNG Outlook 2026](#).

Energy security and the return of capital

One of the clearest developments in recent years has been the growing importance of energy security in investment decisions. The Russia-Ukraine conflict in 2022 and the Middle East crisis in 2026 both triggered sharp increases in oil and gas prices ([Figure 86](#)), disrupted global energy flows and exposed vulnerabilities across the energy system. The Russia-Ukraine conflict accelerated Europe's shift away from Russian pipeline gas and drove a significant increase in LNG demand. More recently, the Middle East crisis removed roughly 10% of global oil and gas supply from the market, highlighting the importance of supply security and diversification.

FIGURE 86. Brent and European gas prices



Source: Bloomberg, Barclays Research

Governments, consumers and corporations increasingly place value on reliability and resilience alongside sustainability objectives. This is supporting investment across a wide range of energy infrastructure, including upstream oil and gas, LNG, storage, refining and power systems.

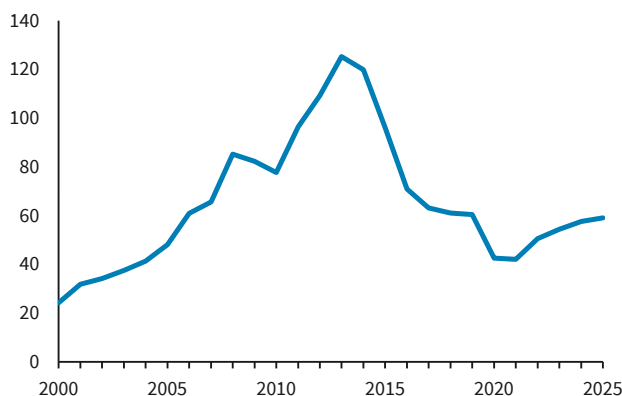
The shift is particularly visible in regions that remain dependent on imported energy. Our recent trip in Southeast Asia found that energy security has become one of the dominant themes shaping policy across the region, supporting domestic resource development, LNG infrastructure and strategic storage investment. Oil still accounts for c.40% of the region's energy mix, yet Southeast Asia consumes around four times more crude oil than it produces domestically. With approximately 70% of crude imports sourced from the Middle East, reducing import dependence and increasing diversification have become key policy priorities across the region. The region also holds significant undeveloped oil and gas resources, and one of the clearest messages from our discussions was the desire to accelerate the development of domestic supply. Governments are increasingly looking to make investment frameworks more competitive, including simplifying fiscal regimes and PSC terms to attract capital and reduce reliance on imports. See details in our note [New Horizons: Southeast Asia – the race for resources](#).

Back to upstream

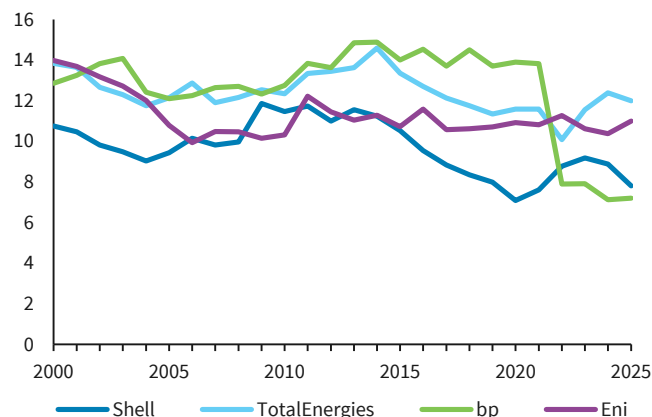
Following a period when capital allocation increasingly focused on renewables and low-carbon opportunities, we are seeing a renewed emphasis on upstream oil and gas. Several European energy companies have increased their focus on hydrocarbons, with a slow down in low carbon spendings:

- bp - As part of its 2025 strategy reset, bp increased annual upstream investment to around \$10.5bn for 2026–27, up from c.\$9bn previously, while reducing planned low-carbon capex from \$3–5bn per year to around \$0.8bn.
- Equinor - At its latest Capital Markets Day, Equinor reduced expected renewables investment from \$2–3bn per year to around \$1bn per year after 2026, while increasing upstream capex by \$1bn.
- Shell - It has increasingly prioritised its Integrated Gas business, raising capex by around \$1bn to \$6bn per year for 2025–28, while scaling back spending on lower-return low-carbon businesses and focusing capital on areas where it sees stronger shareholder returns.

Access to long-life, high-quality resources is becoming increasingly valuable. A few companies have been underinvesting in the upstream division and have seen their resources life shrink - this is the case for Shell and bp notably ([Figure 88](#)). As a result, we are now back into a race of resources with exploration returning to the fore. Recent examples include the Namibian offshore discoveries (TotalEnergies, Shell and Galp), bp's Bumerangue discovery in Brazil and Eni major gas discoveries in Indonesia.

FIGURE 87. European upstream capex, \$bn

Source: Companies reports, Barclays Research

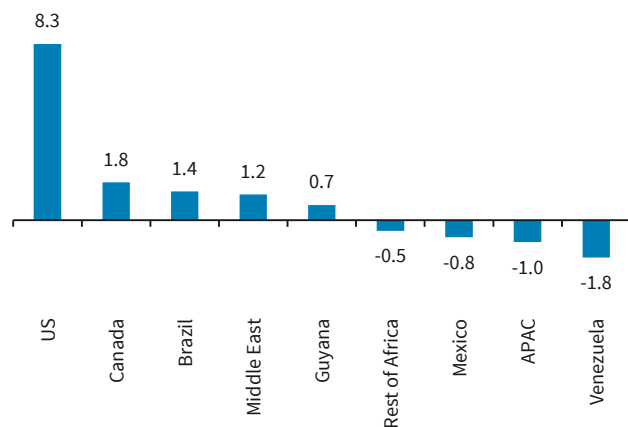
FIGURE 88. Reserves life, in years

Source: Companies reports, Barclays Research

Part of this shift is also driven by the industry's natural decline challenge. The global resource base is declining quicker than before, with decline rates estimated at around 8% per year. Nearly 90% of annual upstream O&G investment since 2019 has been dedicated to offsetting production declines rather than to meet demand growth. This creates a constant race for resources, supporting investment in exploration, appraisal and new developments.

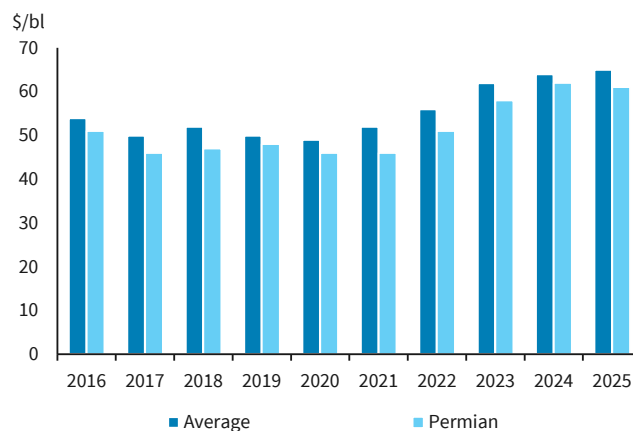
Winners and losers in a fragmented energy world

The geographic centre of growth within the energy system is also evolving. The US has been the dominant source of oil and gas supply growth over the past decade, driven by the shale revolution. However, as the sector matures, production growth is slowing and costs have increased, suggesting a greater need for investment in new sources of supply elsewhere. This is increasing the importance of OPEC producers, the Middle East and new resource developments across regions such as Latin America, Africa and Southeast Asia. See details in our notes [What if... a multi-year bull run is on its way?](#) ; [A focus on non-OPEC supply.](#)

FIGURE 89. Non-OPEC growth has all been US

2015-2025

Source: 2026 Statistical Review of World Energy, Barclays Research

FIGURE 90. WTI price to profitably drill a new well in the US, in \$/bl

Source: Dallas Fed survey, Barclays Research

As a result, we have witnessed a regaining of interest in international IOCs, whose global upstream and LNG portfolios, trading businesses and diversified supply chains position them to benefit from international growth.

AI, power demand and the new electricity economy

AI represents one of the most significant new sources of electricity demand emerging across the global economy. The growth of data centres is increasing requirements for:

- electricity generation;
- transmission infrastructure;
- firm capacity;
- backup power;
- gas supply.

As a result, AI has the potential to strengthen demand for natural gas, LNG and flexible power generation while increasing investment requirements across electricity systems.

Increasingly, AI is becoming a matter of national competitiveness and energy security, with countries seeking to develop their own data-centre and AI capabilities. The recent example of the export ban of Fable 5 and Mythos 5 of Anthropic by the US highlight how important AI sovereignty is becoming. The sovereignty of data is also becoming critical with countries now building data centres. This is creating new centres of energy demand, from Singapore, Malaysia and Indonesia in Southeast Asia to Abu Dhabi in the Middle East. The scale of power consumption is significant, with projects such as Stargate UAE 5GW that will consume 3-5% of the domestic ADNOC Gas production.

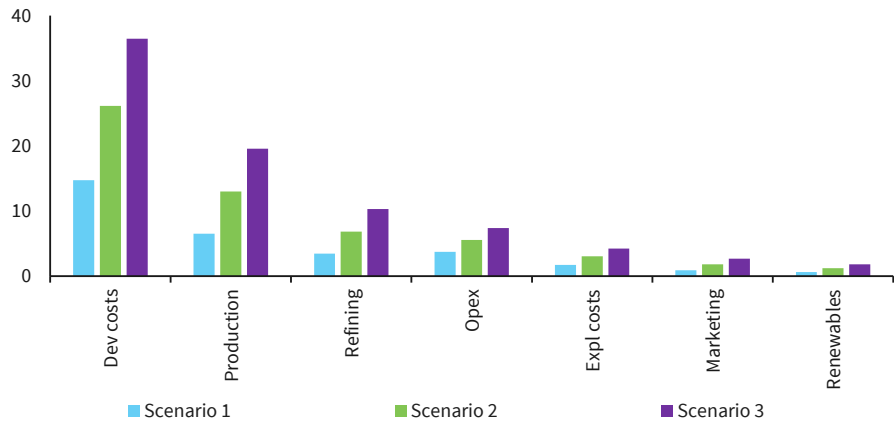
Another important consideration is that data centres are increasingly seeking access to low-carbon but highly reliable power. In practice, this often means combining renewable generation with flexible gas-fired capacity. Reliability is becoming a key differentiator, and operators appear willing to pay a premium for firm, "fit-for-purpose" power solutions. TotalEnergies has indicated that it was able to sell its solution at a price 10% higher than PPAs ([source](#)).

We believe this creates a meaningful long-term opportunity for natural gas, LNG and flexible generation, as reliable power increasingly becomes a prerequisite for digital growth, thus benefitting Integrated Energy companies.

AI is also a source of productivity gains for the sector

Our recent work suggests AI could create a productivity opportunity of up to \$80bn across the global energy industry by 2030, including approximately \$60bn associated with upstream activities. Potential benefits include lower development costs, improved operational performance, reduced downtime, higher production efficiency and refinery optimisation. See our note [New Horizons – Agentic AI, the \\$80bn game-changer for energy](#).

In our view, large integrated companies are well positioned to capture these benefits given their scale, technical expertise and access to operating data.

FIGURE 91. Upstream sector productivity opportunities, \$bn

Group includes Exxon, Chevron, Shell, TotalEnergies, bp, Equinor, Eni, Repsol, OMV, Galp, Aker bp, Var Energi and Neste (for refining only). Scenario 1-3 depends on the level of AI adoption, n3 being the highest.
Source: Barclays Research estimates

As a result, AI may support both sides of the value equation from increasing demand for energy; and improving the efficiency of energy production and delivery.

The race for capital

We see significant investment requirements across:

- upstream oil and gas;
- LNG liquefaction;
- LNG import infrastructure;
- gas-to-power projects;
- electricity networks;
- storage infrastructure;
- AI-related power systems;
- energy services capacity.

In our view, companies that combine strong balance sheets, asset scale and capital-allocation flexibility will be best placed to fund this next phase of energy development.

Investment recommendations

- **Shell (OW)** We see Shell as one of the clearest beneficiaries of energy security and LNG growth. As the world's largest LNG trader and with LNG expected to represent c.60% of group production by 2030, Shell offers unique exposure to growing global gas demand, while management targets 10% annual FCF/share growth through the decade. We are Overweight with a target price of GBP45/sh.
- **TotalEnergies (OW)** TotalEnergies offers exposure to both energy addition and electrification. The company targets >3% annual hydrocarbon production growth and ~1% electricity growth through 2030, largely driven by organic projects, supported by an industry-leading 12-year reserve life. It has built one of the largest integrated power platforms among the majors, targeting 100TWh of profitable net power generation by 2030 and a 12% ROACE

by 2027. Management is targeting ~10% annual FCF/share growth through 2030. We are Overweight with a TP of EUR94/sh.

- **Eni (OW)** Eni combines one of the longest reserve lives among European peers (c.11 years) with a differentiated exploration-led growth model and increasing LNG exposure. Its ability to discover, develop and monetise resources efficiently positions it to benefit from growing energy security concerns and the need for new sources of oil and gas supply. We are Overweight with a target price of EUR28.5/sh.
- **OMV (UW)** OMV remains more exposed than peers to the structurally challenged petrochemical sector. Historically, chemicals have accounted for roughly one-third of group operating earnings, primarily through Borealis. The industry faces persistent pressure from Chinese capacity additions and Middle Eastern feedstock-advantaged producers, resulting in global polyolefin overcapacity and weaker margins. We are Underweight with a target price of EUR53/sh.

Oil and Gas - Europe & LatAm Credit views

Parikshit Budhe⁽ⁱ⁾ Barclays, UK | Franck Bataille⁽ⁱ⁾ BBI, Paris | Badr El Moutawakil⁽ⁱ⁾ BCI, US

European Energy: Resilience through the cycle

Recent volatility across oil and gas markets has reinforced the strategic importance of secure hydrocarbon supply. As outlined in [Energy markets in flux: credit implications](#) and [Case study: TotalEnergies and the Middle East escalation](#), we believe the current cycle will be defined less by commodity direction alone and more by resilience of business models, balance-sheet flexibility, and capital allocation discipline.

Europe's energy-security challenge remains structural. Bloomberg Intelligence estimates that fossil fuel import reliance could fall to c.40% by 2035 from c.60% today only under an accelerated investment scenario, while oil import dependence would remain around 78%. In our view, this reinforces the continued role of diversified hydrocarbon supply and LNG within the European energy system. Energy security is also increasingly influencing capital allocation, with investment gravitating towards upstream, LNG and integrated value chains where return visibility and strategic relevance remain strongest.

Against this backdrop, we see three themes shaping European energy credit performance: commodity volatility is back but balance sheets are stronger; integrated models continue to provide superior earnings resilience; and capital discipline is increasingly supporting credit quality through the cycle.

Commodity volatility is back, but balance sheets are stronger

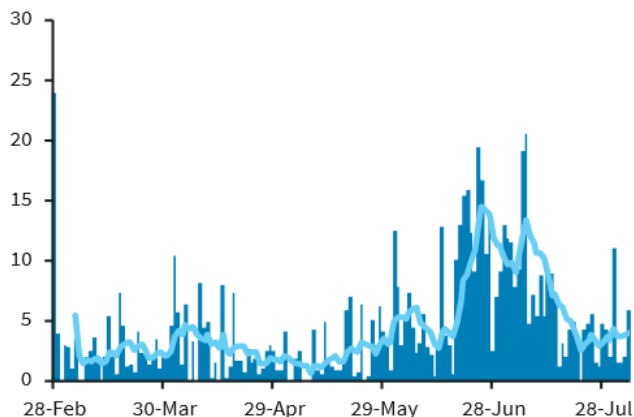
Commodity markets have repriced materially around renewed supply-risk concerns. [Barclays Commodities Research](#) forecasts Brent at \$96/b in 2026 and \$85/b in 2027, above forward-implied averages of \$88/b and \$77/b, respectively. In a prolonged Strait of Hormuz disruption scenario, Barclays estimates \$2-10/b upside risk to its 2026 forecast, with spot prices potentially testing \$150/b under a more severe outcome.

Physical markets also remain tighter than headline balances imply. Barclays Commodities notes that oil flows through the Strait of Hormuz averaged 46% of pre-war levels during the latest escalation phase, while inventories excluding China remain tight and refining margins resilient. This highlights how geopolitical events continue to have disproportionate effects on commodity markets despite a structurally stronger sector balance-sheet backdrop. Europe's

domestic oil and gas production continues to decline, with Bloomberg Intelligence forecasting that regional gas production could fall to 92bcm by 2035 while imports rise to 235bcm.

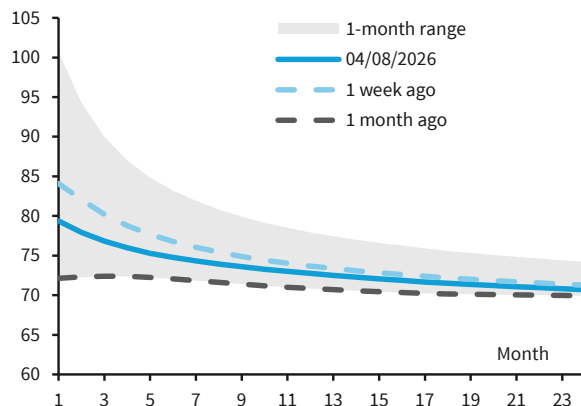
However, in our view, stronger prices are now meeting stronger balance sheets (Figure 94). Several years of deleveraging, elevated cash generation and portfolio optimisation have left European majors better positioned to absorb commodity shocks, making volatility more likely to drive earnings dispersion than broad credit deterioration.

FIGURE 92. Oil flow through the Strait of Hormuz remains depressed



Note: Net daily crude oil and refined product (including LPG) exports through the Strait of Hormuz, mb/d. Line shows 7dma.
Source: Kpler, Barclays Research

FIGURE 93. ICE Brent futures curve over time, \$/b



Source: Bloomberg, Barclays Research

Portfolio quality is becoming the key differentiator

Integrated business models remain structurally advantaged in a more volatile commodity environment. Exposure across upstream, LNG, trading, refining, marketing and power activities provides multiple cash-flow levers and reduces dependence on a single commodity, geography or asset base. The value of integration increasingly sits in cash flow diversification rather than production scale, with LNG, trading, refining and downstream earnings helping offset commodity, regional and operational volatility.

The natural hedge is most visible during supply disruptions. In [Case study: TotalEnergies and the Middle East escalation](#), we highlighted that Qatar, Iraq and UAE offshore assets represented c.15% of production but only c.10% of upstream cash flow. Management also indicated that an \$8/b increase in Brent would offset the expected 2026 cash-flow contribution from those assets under a \$60/b oil price assumption. This illustrates how stronger commodity prices can partially offset operational disruption through a price-versus-volume hedge embedded within integrated portfolios.

The sector implication is broader than TotalEnergies. Higher commodity prices, LNG optimisation, trading gains and refining margins can partially offset lost production volumes or regional disruptions, creating a price-versus-volume hedge that is largely unavailable to more concentrated business models.

LNG remains a key differentiator. As domestic production declines and import dependency increases, scale, trading capability and supply flexibility become increasingly valuable across the gas value chain. Industry forecasts suggest that European and UK gas production could continue to decline materially over the coming decade, reinforcing the strategic role of LNG within Europe's energy system.

Portfolio quality is also increasingly about returns rather than scale. European majors continue to prioritise investment across oil, gas, LNG and integrated value chains where return visibility remains strongest while maintaining greater discipline across lower-return opportunities.

Capital discipline is supporting credit resilience

The most important structural change across the sector may be financial rather than operational. Unlike in previous upcycles, stronger commodity prices are not translating into a broad reset higher in capex. Bloomberg Intelligence expects European energy majors' 2026 capital spending to remain broadly disciplined, with investment weighted towards upstream, integrated gas and downstream activities.

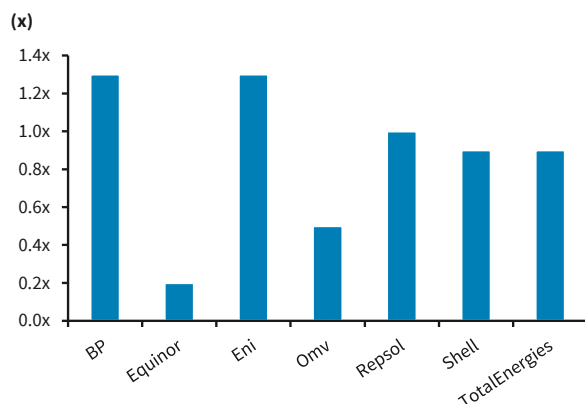
Importantly, stronger commodity prices are not translating into a broad sector capex reset. Instead, capital is increasingly being directed towards LNG, upstream resilience and integrated value-chain opportunities, while balance-sheet strength and shareholder flexibility remain core financial objectives.

This discipline is visible across the sector. TotalEnergies has maintained annual investment guidance around \$15bn despite stronger cash-flow expectations, while BP has tightened 2026 capex guidance to \$13-13.5bn as it prioritises balance sheet repair.

Shareholder returns have also become more flexible. BP has suspended buybacks and redirected surplus cash flow towards debt reduction, while TotalEnergies continues distributions alongside gearing of 13.1% and a stated objective of moving closer to 10% gearing.

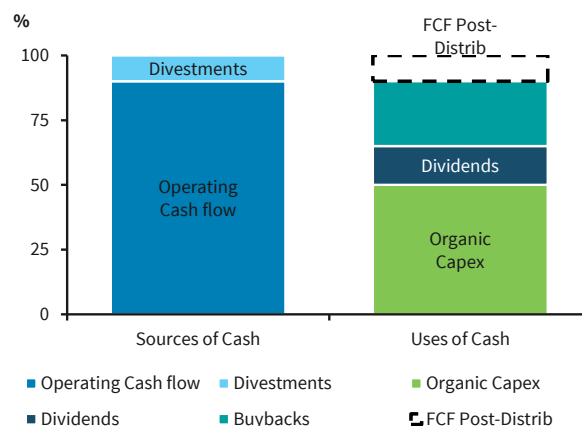
The sector-wide message is that buybacks are increasingly acting as the variable component of capital allocation, while dividends remain the core shareholder commitment. Combined with disciplined capex and stronger balance sheets, this should help preserve rating headroom and limit credit deterioration through future commodity cycles.

FIGURE 94. Net debt-to-EBITDA peer comparison (x)



Source: Bloomberg, Barclays Research

FIGURE 95. European oil majors: Capex allocation (%)



Note: Chart data based on Bloomberg estimates.
Source: Bloomberg, Barclays Research

Investment recommendations

As detailed in [Revisiting single-name views: mid-year 2026 update](#), we remain constructive on European energy. Recent results have reinforced our positive sector view, with integrated majors continuing to deliver resilient earnings, strong cash generation and disciplined capital allocation despite heightened geopolitical volatility. Against the backdrop of the themes discussed above, our preferred positions within the sector remain TotalEnergies and BP.

- **Overweight BP (BPLN):** BP remains a preferred recovery and deleveraging story within European Energy. Since announcing its strategic reset, management has prioritised balance-sheet repair through lower capex, suspended buybacks, planned disposals and debt reduction. We view recent governance noise as secondary to the improving credit trajectory, with stronger commodity prices and disciplined capital allocation supporting further leverage improvement through 2026-27. In our view, BP offers one of the strongest combinations of improving fundamentals, rating momentum and spread-tightening potential within the sector.
- **Overweight TotalEnergies (TTEFP):** Within our coverage, TotalEnergies remains one of the clearest expressions of the resilience themes discussed above. The group combines a highly diversified earnings base across upstream, LNG, trading and refining with low leverage, strong cash generation and disciplined capital allocation. Recent Middle East disruption has reinforced the value of this integrated model, with stronger commodity prices, LNG and trading performance helping offset operational effects. In our view, TotalEnergies remains well positioned to benefit from commodity volatility while maintaining balance-sheet flexibility and shareholder returns.

Latin America corporate energy: Diverging fortunes in the race for capital

Energy addition: Why LatAm matters. Latin America is one of the few regions globally where both oil and gas production can still grow materially through 2030–35, making it a direct beneficiary of the "energy addition" thesis. Argentine Vaca Muerta producers — Vista Energy, YPF, Tecpetrol, Pluspetrol — operate with lifting costs of c.\$4–5/boe and can sustain profitable operations even at \$60/bbl crude, well below the breakeven of most conventional global peers. The contrast with Brazil and Colombia's junior and mid-cap producers is instructive: both groups are far more sensitive to crude price downside given higher-cost conventional operations. Brazil's independents (Prio, Brava Energia/Ecopetrol, Vast Infraestrutura) hit record output levels in 2025 but remain burdened by high leverage after aggressive M&A — Brava was at 2.3x LTM in Q3 25 on reported numbers, rising to 3.5x excluding oil inventories. Colombian juniors such as Gran Tierra Energy or GeoPark may carry lower absolute leverage, but their conventional cost structures and heavy-crude exposure make them acutely vulnerable to oil price declines — particularly from increased global supply — with companies such as Gran Tierra explicitly relying on oil prices to manage financial pressures in 2026. The bottom line: in a world where oil prices could face structural pressure from the return of Venezuelan supply over 5–10 years and OPEC+ unwinds, Argentine unconventional producers are the only LatAm E&P credits with a clear margin of safety at sub-\$60 Brent, while Brazilian and Colombian juniors need sustained \$70+ prices to remain comfortable. At the weakest end, Pemex's production is "holding steady" at 1.6m bpd only because the company adds c.280,000 bpd of condensates to the figure, masking continued underlying declines.

Energy security and the return of capital. The Iran war and Strait of Hormuz disruptions have accelerated capital flows towards LatAm hydrocarbon assets, particularly in Argentina, where the policy environment is now actively encouraging investment. Argentine corporates issued \$4.5bn across nine deals in Q2 26 alone — up 45% q/q — at an average YTM of around 9%, versus 7.2% for LatAm corporates overall. Crucially, Pluspetrol, Pampa Energía, MSU Green and Vista Energy all placed bonds at 10–12 year tenors — an exceptionally long duration for Argentine issuers and a clear signal of market confidence in the country's energy trajectory over the coming years. Pluspetrol priced a \$450m 10-year at 7.85% yield, with proceeds directed to Vaca Muerta development under RIGI. YPF's CEO has stated that the company aims for \$30bn in annual energy exports by 2031, with multiple RIGI filings planned for LNG and midstream infrastructure. And the pipeline keeps building: RIGI has opened c.\$95bn in potential investment opportunities, though only a dozen projects have been approved so far, with infrastructure bottlenecks in transport, power and logistics remaining binding constraints.

Meanwhile, Petroquímica Comodoro Rivadavia is preparing a debut international bond (up to \$550m), and Provincia de Neuquén and Genneia are expected to follow.

Regional divergence. The credit differentiation within LatAm energy is as wide as anywhere in EM. On the constructive side, Argentine producers benefit from low-cost shale economics, RIGI-driven capital formation, and the fact that Vaca Muerta's light tight oil faces no direct competition from a potential Venezuelan supply recovery (Venezuelan extra-heavy crude competes with Canadian and Brazilian grades). On the negative side, Mexico is structurally impaired: Pemex carries \$20bn in supplier debt and accounts payable (as of 1Q26), and the government has injected close to \$55bn since 2025 to revive production and throughput. Moody's downgraded the Mexican sovereign to Baa3 in May 2026, citing rigid spending, a narrow tax base, and persistent Pemex bailouts — placing Mexico on the lowest rung of investment grade with two agencies (Moody's and Fitch), suggesting that the current support policy to support Pemex directly from the budget could change. Colombia faces a structural gas crisis: Ecopetrol's production in Colombia is in decline while distressed producer Canacol (the country's largest independent gas supplier) is attempting to terminate gas contracts via Canadian court proceedings, leaving the country increasingly reliant on imported gas at rising global prices.

Gas, power and the emerging electricity story. Natural gas is becoming LatAm's critical transition fuel. Brazil is building gas-fired generation to reduce hydro dependency, Argentina is positioning for LNG exports (YPF's RIGI filings target LNG infrastructure specifically), and Colombia's gas deficit is forcing emergency procurement. The Iran war has created LNG export opportunities for the region — Argentina's RIGI could expand beyond Vaca Muerta upstream into LNG, and Colombia's deep-water gas potential remains untapped because of self-inflicted policy wounds. In the power sector, Panama's Generadora de Gatún — accounting for about 20% of the country's firm capacity — successfully placed a \$1.05bn 18-year senior secured bond at 6.875% to refinance its 640 MW gas plant, illustrating investor appetite for LatAm baseload infrastructure. Cox Mexico's \$2bn dual-tranche issuance for Iberdrola Mexico's 2.6 GW platform further evidences capital flowing toward firm power assets. While Latin America is not yet a primary hyperscaler data-centre destination at US/Asian scale, growing electricity demand from urbanisation and early-stage digital buildout is creating incremental gas and grid investment needs across the region.

The race for capital: Credit positioning. The key credit differentiation between issuers/countries is straightforward: Argentine unconventional producers (Vista, Pluspetrol, Pampa, YPF) offer a rare combination of sub-\$5/boe production costs, reform-driven capital access, and 10+ year tenor — effectively trading based on expectations of structural energy demand growth at single-B-level yields (~9%). Brazilian independents (Prio, Brava, Vast) offer operating leverage to oil prices but carry stretched balance sheets that make them vulnerable in a lower-for-longer scenario — Prio appears best positioned given spot-rate sales exposure, while Brava's high leverage amplifies downside risk. Pemex (B1) remains a sovereign-linked story where credit analysis is secondary to fiscal policy — the spread to the sovereign has narrowed on state support, but the company has dropped its flagship 20-field production strategy after billions spent and fast declines, and its new CFO-turned-CEO has signaled that the Finance Ministry is prioritising creditor payments over operational recovery. The service providers in LatAm could also offer an attractive opportunity: FPSOs have shown a very strong uptime track record with healthy DSCR (including for the weakest operator such as MV24s), drillers are benefiting from the focus on deep water growth from Petrobras and higher day rates to extend their maturity (such as Constellation Energy), and subsea service such as Oceanica have taken advantage of the better market momentum to refinance, with clients expecting the service providers in Brazil to participate in Petrobras's growth in deepwater, as the company is expected to have more than 20 new FPSOs by 2030.

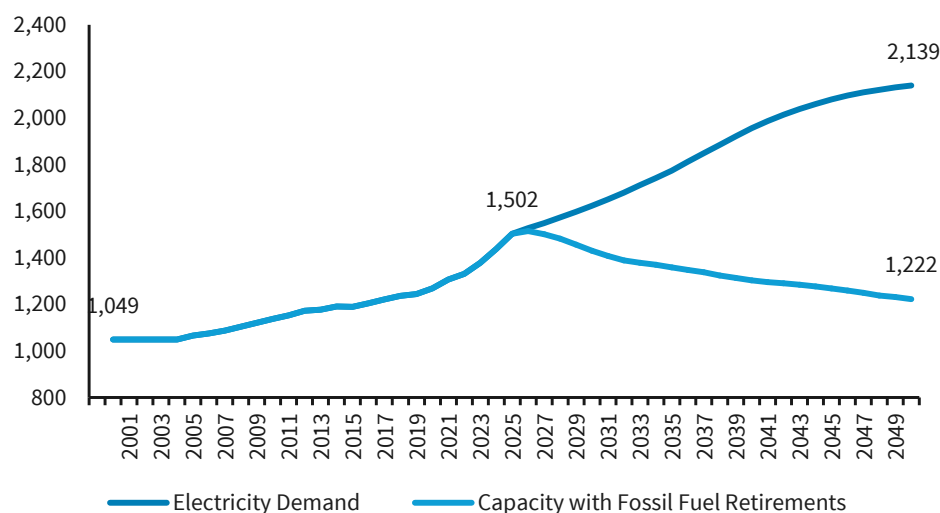
Clean Tech - North America Equity views

Christine Cho, CFA BCI, US

Why our sector matters

Renewable energy has had a growing share of a largely stagnant power market across North America in recent decades. Electricity needs had decoupled from economic growth since the late 20th century, in large part due to increases in efficiencies in power delivery and use. However, the rise of artificial intelligence, hyperscalers, and continued grid electrification have upended the market and created a sudden and stark need for new power generation and delivery. And while natural gas and other more traditional power sources are now expected to play a larger role in meeting future electricity demand than previously anticipated, this is not expected to be at the expense of alternative energy sources. Rather, the rapidly expanding pool of demand is providing new avenues for growth for our coverage, with many having only recently found mainstream investor appreciation for their specific niches.

FIGURE 96. US electricity gap (GW)



Source: BloombergNEF

Perhaps the most significant separator for the clean technology space is speed to power. At a time when US grid interconnection queues can stretch up to five years, and gas turbine waitlists have extended past six years, many of the solutions offered by companies in this sector are able to provide power on relatively accelerated timelines (such as fuel cells manufactured by BE, generators provided by EROC, or solar panels put together by FSLR). Many of the products and services that are offered by the companies in our coverage also work alongside, or in conjunction with, a grid interconnection. Some (BE, EROC) offer solutions that can be used by customers as "bridge power", providing primary electric capacity while full grid interconnection is on hold. When grid connection is available, the customers can either sell the products, or keep them as backup power sources. This eases the constraints on the grid while also providing users with much needed current power.

Lastly, the numerous generative technologies in the space provide a growing list of customers with more variety than ever to meet their needs. Hyperscalers can have a very different set of requirements than traditional commercial and industrial users, and there can be a wide range of priorities even between data centres themselves. Solar, generators, fuel cells, batteries, and

potentially nuclear are all very differentiated technologies all being considered by data centres on different timelines.

Energy security and the return of capital

One of the defining characteristics of the current power regime has been the shift of attention to behind-the-meter (BTM) investments and applications. BTM refers to generation that happens "behind" the grid, meaning it bypasses grid interconnection to provide power directly to one or multiple sources. As the wait for interconnection has ballooned to several years (depending on location), and hyperscalers in particular are time sensitive with regards to power access, there has been a lot of recent discussion about the prospects for BTM power solutions. In some cases, these could be permanent, "islanded" power plants, never connected to the grid. In others, they could be temporary measures before formal grid interconnection could be secured. Either way, BTM optionality has opened up a new source of build and investment.

How energy security is changing capital allocation

As hyperscaler demand strains utilities, the US power grid has become, and will continue to be, increasingly less stable. Hyperscalers' AI training data centre locations in particular are very sensitive to power outages or fluctuations, and can even cause problems for the grid themselves (AI compute's inherent volatility can create issues for the more stable grid load). To combat these issues, data centres need grid stabilizers and backup solutions. This has created a mostly new and large market opportunity for both battery energy storage (FLNC), as well as backup generators (GNRC, EROC). Additionally, as grid interconnection wait times have soared, customers have turned to solutions that can bring power generation without the need for grid connection, such as fuel cells (BE). As we will discuss in further detail below, a large amount of capital expenditure is being spent on building out capacity.

Also stemming from the emerging gap between power demand and power supply is a renewed interest in nuclear power. Long a stagnant industry in North America (and a declining one in Europe), nuclear's ability to bring clean, stable, baseload power is unique in the marketplace. Capital expenditures and cost overruns associated with building GW-scale plants has long been a deterrent for new investment. Some small modular reactor (SMR) companies, such as OKLO and SMR, are looking to change that by deploying more modular and repeatable designs that should drive down costs over time (although these remain more of a story for the 2030s than today).

How energy security is affecting volatility

The volatility in both energy security and energy prices is a net positive for the clean technology sector in the US. For commercial and industrial users (including hyperscalers), a lack of trust in the grid's ability to maintain energy creates the need for redundancy, which means additional purchases of our coverage's products. This comes in the form of extra storage and backup units, or perhaps onsite generation to supplement the grid, even if interconnection is secured. Notably, more volatile (and generally higher) energy prices are generally a positive for names in our residential solar space (RUN, SEDG, ENPH), as elevated utility rates and energy bills mean the value proposition of a home solar system increases (and the payback period shortens for consumers). Critically, the increase in utility rates is happening in conjunction with the coming loss of residential solar tax credits, cut short by the passage of the One Big Beautiful Bill Act (OBBA). Falling home battery prices and elevated and volatile utility rates will continue to make residential solar an attractive solution.

Winners and losers in a fragmented energy world

The US has been applying both the "carrot" (tax credits for domestically manufactured clean energy products) and the "stick" (tariffs or outright bans of foreign made competition) in the clean tech sector for some time now, but it has ratcheted up the level in recent years,

particularly under the Trump administration and since the passage of the OBBBA. Historically, purchasers of clean energy products (such as solar, wind, and storage) had been eligible to receive a 30% investment tax credit (ITC) of the purchase price, as well as some "adders", such as an additional 10% for domestically manufactured products (domestic content). However, the OBBBA tied the foreign entity of control (FEOC) requirements to the ITC. Essentially, any clean energy product made substantially of products from China or China-related entities will not be eligible to receive the ITC or domestic content adder. This creates a 30%+ moat from just a tax credit perspective. On top of that, many tariffs have been levied against Chinese and other foreign clean energy imports. In addition to the ~10% rate currently applied to all Chinese imports through Section 301, there are numerous more specific taxes that apply to the clean tech space. We have laid out some of the more notable ones in [Figure 97](#).

FIGURE 97. A selection of clean tech tariffs on Chinese imports

Relevant Clean Tech Tariffs on China		
Product Category	Tariff Program	Rate
Solar cells & modules	Section 201 + Section 301	Can exceed 50%+ depending on product/origin structure
Polysilicon	Section 232	Final decision pending
Battery components	Section 301	Up to 25-50% depending on component
Steel products	Section 232	50%
Steel derivative products	Section 232	25%
Aluminum products	Section 232	50%
Copper products	Section 232	50%

Source: Barclays Research, ITC

As a result of these policies, Chinese products, which can be significantly cheaper than their American counterparts, have been effectively shut out of the US market. Inversely, because of their higher manufacturing costs, American products have made little headway in Chinese markets, as the two countries' clean tech markets have been essentially cut off. Europe has remained more of a battleground between Western and Chinese made offerings. Two problems related to the clean energy space have arisen in Europe: a general equipment scarcity, as well as issues related to curtailment (an excess of renewable energy on the grid means too much power at peak times, so power plants are "curtailed" during the day). These problems have made Europe more welcoming of Chinese imports, but energy security concerns have meant that it still predominantly relies on European products and American imports. Generally speaking, most companies in our coverage sell and manufacture most of their products in the US, with some ancillary sales in Europe and other developed, international markets. It is rare for any sales to occur in China.

Of note here, from our discussions with companies, American hyperscalers are sensitive to using any equipment, systems, or services that can be traced back to China, due to company and national security concerns. As the US is home to many of the largest hyperscalers and data centre buildouts, this has given US companies in our coverage a significant leg up in this space as of now.

AI, power demand and the new electricity economy

Data centres have created a new and large market for many of our companies to sell into, as these customers want power, backup, and energy security in a timely manner. Compared to the traditional utility-scale developers or C&I customers that have been the clients in this space in years past, data centre developers are generally less cost-sensitive and more time-sensitive, willing to absorb a higher cost if it means they can fire up their sites a couple years sooner. Data

centre developers also often have to abide by company emissions mandates and are very visible in their own communities, meaning they often require town approval before substantial production can begin. Generally, this favors the clean energy space as opposed to more traditional gas powered solutions, as emission levels are smaller and power generation is often quieter. The need for speedier power solutions is also the driving force behind the growing level of investment in behind-the-meter solutions, whereby companies forego grid interconnection (for the time being) in order to get needed power generation faster.

It is crucial for some data centre operations that they have a constant stream of power (many data centres express a desire for "5 9s" availability, meaning products that can provide 99.999% guaranteed uptime). Neither the grid, nor any individual power source, can promise these uptimes, so this has created a large market for backup power, specifically for battery energy storage solutions (BESS), such as those offered by FLNC, or backup generators, like those assembled by GNRC and EROC.

The electrical work in and around data centres also provides some of our coverage with an opportunity to take some meaningful share, either through new or existing products. As data centre racks become increasingly energy dense, the wiring needed to provide them with adequate power has become more complex, leading to some new solutions around how to streamline the process, such as SHLS' Air Link. Higher voltage levels will also require advanced solid state transformers (SSTs), a market with no current constituents that a couple of our companies (SEDG, ENPH) are in the design stages of creating (the traditional transformer market has historically been dominated by larger, multi-industrial companies).

Lastly, the urgent need for new power solutions has led to a rapid deployment of capital into new, emerging, or dormant technologies, such as fuel cells and nuclear power. Fuel cells, long seen as too costly compared to reciprocating engines or gas turbine alternatives, have seen a pickup in interest due to quick scaling and deployment ability, as well as relatively low carbon and noise emissions. SMR interest has also picked up on the back of the promise of clean, baseload power.

The race for capital: financing and investment ideas

a) Where is capital most needed and who can provide it?

The energy infrastructure space had been in a state of disinvestment for the past few decades, as electricity requirements stalled despite continued economic and population growth. The hyperscaler power shock, combined with increasingly insular energy policies, has re-accelerated investment in clean tech. As data centres want power now, significant investment is needed, and has been made, in scaling capacity for equipment providers. In this space, this has occurred mostly through public or private fundraisings (the nuclear market in particular has been an active one for IPOs as of late). There has not been much consolidation or M&A activity in the space yet - a handful of companies (notably NXT and ARRY) have made strategic investments into acquiring smaller, ancillary businesses to help grow wallet share - but most clean tech markets activity has remained primary and secondary offerings.

Plenty of capital has been raised in the nuclear space in particular, with SMR and OKLO both relatively recent IPOs. Unit economics is of the utmost importance for nuclear, as there is still much to be learned about the cost structure and Nth of a kind costs of modular reactors. Nuclear has also seen an influx of government grants, as the Trump administration has positioned itself as a champion of nuclear energy. We lay out some of the government programs for nuclear programs in [Figure 98](#) below.

FIGURE 98. Governmental nuclear assistance programs

Program	Benefit / Amount	Description
HALEU Availability Program	Cost Sharing	The program is meant to spur investment in commercial HALEU enrichment supply by providing capital needs and purchase commitments.
Enrichment Task Orders	\$3.4bn	This program is designed to help enrichment facilities and centrifuge manufacturers accelerate deployment timelines by providing funds. Originally two separate programs (\$2.7bn for LEU and \$700mm for HALEU), the funds are now intermingled, and three \$900mm enrichment contracts have been awarded. The remainder will help other aspects of the fuel supply chain.
Fuel Line Pilot Program	Regulatory Fast Track	The Fuel Line Pilot Program supports the Reactor Pilot Program and establishes a domestic nuclear fuel supply chain for testing new reactors. The program leverages the DOE authorization process to build and operate nuclear fuel fabrication lines to serve for research, development, and demonstration purposes and to provide a fast-tracked approach to commercial licensing.
Surplus Plutonium Utilization Program	20 tons of Pu	DOE is establishing a program to make excess Plutonium available for advanced nuclear technologies. Applications were due in late November, with allocation decisions expected soon.
DUECE Program	\$1.5bn	Started in 2016, this program seeks to develop advanced centrifuge technology to produce adequate enriched uranium for defense needs. BWXT was awarded these funds in September 2025, through an indefinite delivery, indefinite quantity (IDIQ) contract.
Reactor Pilot Program	Regulatory Fast Track	The Reactor Pilot Program establishes a new DOE pathway for advanced reactor demonstration to fast-track commercial licensing. The program aims to leverage DOE's authority to expedite the research and development of advanced nuclear reactor technologies with the larger goal of reaching criticality for at least three advanced nuclear reactor concepts located outside of the national laboratories by July 4, 2026.

Source: Barclays Research, DoE

Investment recommendations

Although the recent developments in the market have generally been good for our sector across the board, we give a few standout investment recommendations below, from our perspective, as well as one that may not see as much upside as the others.

- Oklo, Inc (OKLO)** - OKLO (OW) is a small modular reactor (SMR) company that has been at the cutting edge of developing the next generation sodium-cooled fast reactors. These 75MW modules each will be able to provide customers, both traditional utilities as well as hyperscalers, with reliable baseload power at near zero levels of carbon emissions. These modules are powered by high assay, low enriched uranium (HALEU), a fuel that is further enriched than fuel used by traditional reactors. The company has not yet reached commercialization, and is still awaiting full NRC approval before it is able to do so. Therefore, this is admittedly a technology that we will not know the outcome of until the early 2030s at the earliest. However, if it is successful and economical in scaling nuclear power, it will be difficult for other technologies to compete, in our view.
- Bloom Energy (BE)** - BE (EW) is a distributed power company that develops and manufactures solid oxide fuel cell systems capable of providing reliable, always-on electricity for data centres, utilities, and commercial customers. Unlike intermittent renewable technologies, BE's fuel cells produce dispatchable power with high availability and lower emissions than traditional combustion-based generation, making them increasingly attractive as grid constraints and data centre power demand accelerate. While the technology is commercially proven and already deployed at scale, its long-term opportunity depends on management's ability to drive down costs (a 30% tax credit goes away post-2030), expand manufacturing capacity, and how durable demand is if and when turbine and grid interconnection queues are eased. In the meantime, BE's fuel cell platform is acting as an

important bridge between rising electricity demand and an increasingly constrained utility grid.

- **First Solar (FSLR)** - FSLR (OW) is the leading US-based solar manufacturer and one of the largest utility-scale solar module suppliers in the world, with annual manufacturing capacity measured in the tens of gigawatts and a backlog that supports deployment across a substantial portion of the US utility-scale solar market. The company utilizes differentiated thin-film technology rather than traditional crystalline silicon, providing reduced exposure to Chinese polysilicon and wafer supply chains while benefiting from a largely domestic manufacturing footprint. As trade tensions between the US and China continue to reshape global solar supply chains, FSLR stands to benefit from tariffs, import restrictions, and domestic manufacturing incentives that increase the relative competitiveness of US-produced modules. While utility-scale solar remains dependent on transmission expansion and complementary storage deployment, FSLR is uniquely positioned to capitalize on growing electricity demand and the push for domestic energy infrastructure, making it one of the most strategically important solar manufacturers in the Western market.
- **Generac (GNRC)** - GNRC (EW) is best known as the leading provider of standby and mobile diesel generators, technologies that have historically played a critical role in backup power applications across residential, commercial, and industrial markets. However, as power reliability becomes increasingly important for hyperscale data centres and AI workloads, the market is beginning to favor solutions capable of providing continuous or near-continuous power behind-the-meter, especially as grid connections become harder to come by and we find that diesel back up generation is mostly paired with grid connections. As behind-the-meter applications grow, there tends to be redundancy baked in with fuel cells, gas-fired microgrids, batteries, and other distributed energy solutions, rendering diesel back-up generation as less necessary. As a result, GNRC may be less directly exposed to the emerging multi-gigawatt data centre power buildout than investors anticipate. In our view, the key question is whether GNRC can expand beyond its legacy generator franchise and establish a differentiated position in the next generation of distributed power infrastructure.

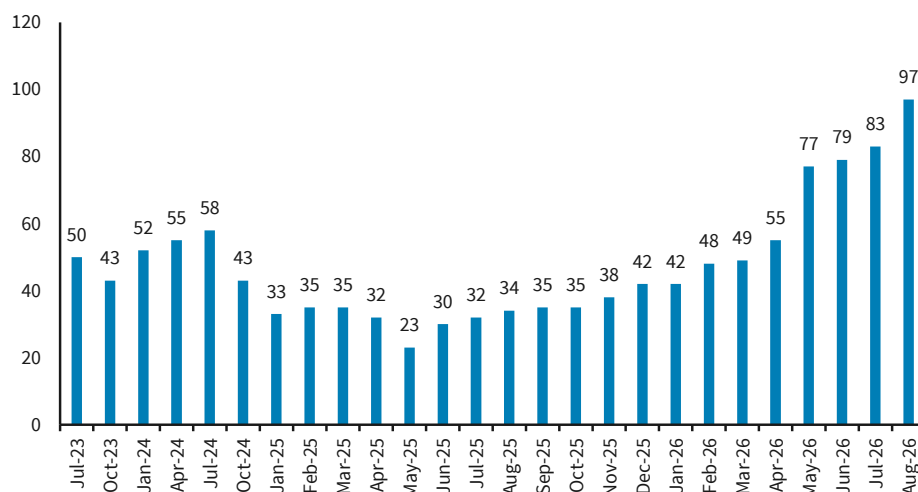
Energy - US Credit views

Paul Chambers⁽ⁱ⁾ BCI, US | Harry Mateer⁽ⁱ⁾ BCI, US

Why our sector matters

To support growing demand, the North American energy credit ecosystem is driving the expansion of traditional upstream sources (mainly by unlocking resources in the Permian Basin), facilitating an ongoing consolidation wave in US shale, supporting export infrastructure (mainly through LNG), and building infrastructure to supply an expected ramp in gas-fired power generation. We see potential emerging opportunities in Canada, as several oil pipelines are under consideration to both increase output and enable either the expansion of existing or the development of new export routes.

One area where there are some (early-stage) signs of growth supporting the belief that oil will stay higher for longer is the deepwater market, where lead times for first oil can span from three to five years (or more) and require investments upwards of \$1bn depending on the scale. With the success of large offshore fields in the Gulf of Mexico, Ghana, Nigeria, Brazil, and Angola, the number of countries seeking to profit from their own natural resources is on the rise as well. Recent offshore success in Guyana, Namibia, Colombia, and the Ivory Coast has created an increase in demand for deepwater drilling rigs, with both Transocean and Seadrill stating on 2Q26 earnings calls that deepwater drilling utilization could approach 95-100% in 2027.

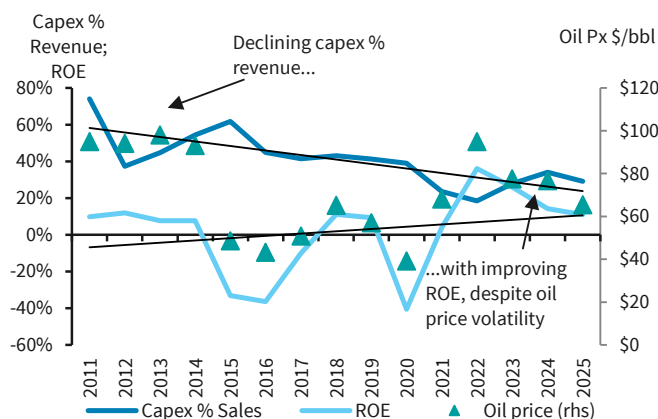
FIGURE 99. Deepwater open tender activity for floating rigs - Rig years

Source: Petrodata

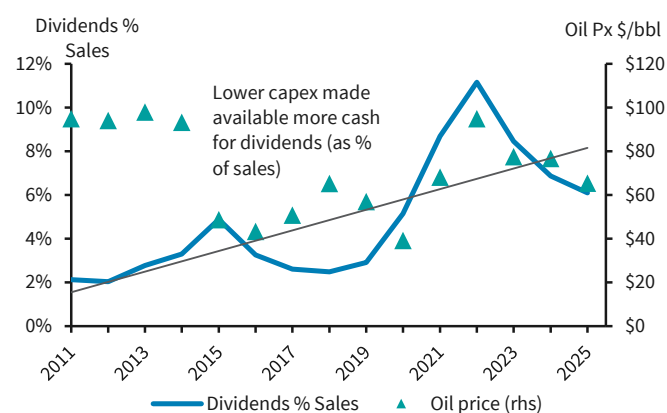
For credit investors, we view the RIG/VAL merger as a unique opportunity to own bonds in the largest offshore drilling company in the world, with pro forma EBITDA exceeding \$2bn in 2027 and a combined backlog of around \$10bn. As RIG seeks to simplify its capital structure, moving from secured to unsecured, and with growth in EBITDA and scale from its pending merger with VAL, we expect substantial upgrades in ratings from the credit agencies. **We view the 8.5% 2031s and longer-dated 2038s as the most attractive options. Our base case remains that RIG will refinance the VAL second-lien 2030s after the deal closes, still estimated before year-end.**

Energy security and the return of capital

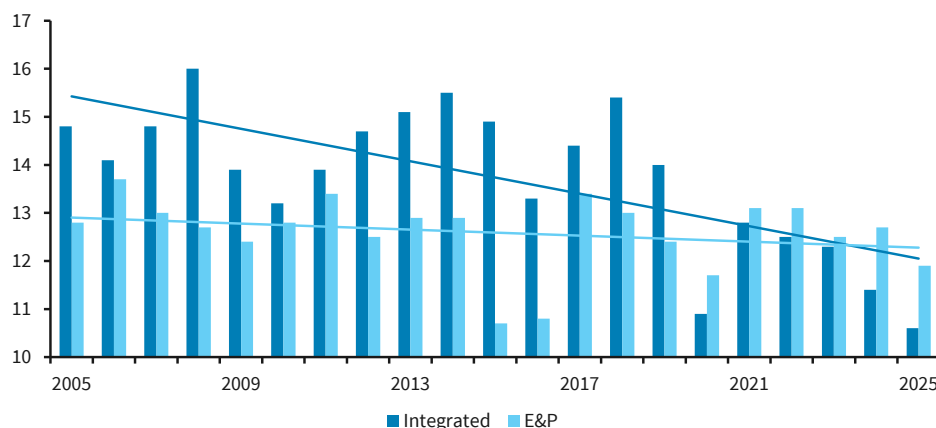
Within the upstream sector, the heightened (global) call on US hydrocarbons has not yet translated into a material increase in spending, which we think reflects the dynamic shown in [Figure 100](#) and [Figure 101](#): namely, after suffering through poor returns in the early years of the shale boom, producers have migrated to a return-of-capital, moderated approach to spending that emphasizes free cash flow generation rather than output growth. Changing this will require a more sustained price signal; in general, **our conversations with producers suggest that an \$80/bbl strip price would be required to adjust spending plans materially.**

FIGURE 100. US Upstream capex percent of sales falling while ROE improves, despite commodity volatility

Note: S&P 1500 E&P companies on a per share basis.
Source: Bloomberg, Barclays Research

FIGURE 101. US upstream dividends percent of sales increasing as capex declines

Note: S&P 1500 E&P companies on a per share basis.
Source: Bloomberg, Barclays Research

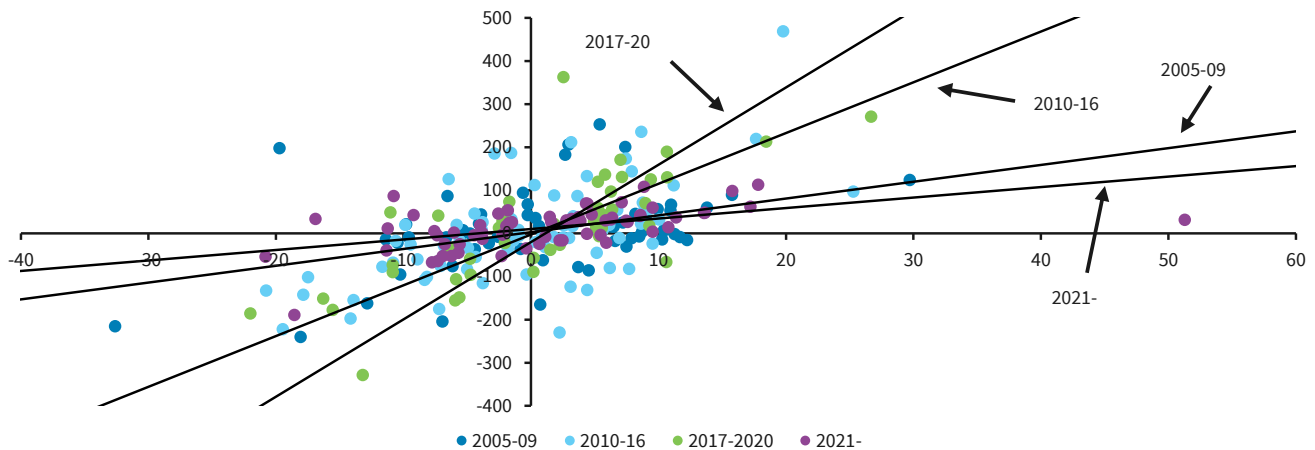
FIGURE 102. Although North American upstream reserve life has ebbed and flowed with commodity prices, in general it has been declining (proved reserves/production, years)

Notes: Average of total proved reserves divided by 12-month production for North American producers over time.
Source: Company reports, Barclays Research

Where we do see a shift in capital spending for global energy security is US LNG, stemming from an unprecedented build-out of liquefaction capacity on the US Gulf Coast, a dynamic we discussed in [The Credit Landscape in US LNG](#) (April 13, 2026). As an example, **NextDecade's Rio Grande LNG project recently raised \$3.5bn in the US bond market, and we see potential for significantly more debt issuance in the next few years to support the construction of additional US LNG trains.**

To date, challenges to global energy security have occurred outside of North America, meaning that producers in the US and Canada have benefited from the price reaction, without a corresponding deterioration in volumes. Positively, a number of issuers have taken advantage of periodic spikes in commodity prices to accelerate balance sheet improvement, reducing financial leverage and ultimately reducing financial statement volatility through the cycle, as evidenced by [Figure 103](#), which shows **reduced returnsensitivity for a given change in the price of oil**. The flip side is that we think **energy companies are much better positioned to return capital counter-cyclically** when commodity prices inevitably experience a downturn, along with investing capital to heed the call for additional volumes.

FIGURE 103. The sensitivity of investment grade energy sector returns to changes in oil prices has declined as balance sheets have strengthened and capital spending has eased



Notes: Percent month-over-month change in oil price on x-axis against monthly excess returns (bp) on y-axis.
Source: Bloomberg Indices, Barclays Research

With international gas price volatility increasing amid challenges to energy security, **we see increased opportunities for companies that are able to market US gas to international buyers**. This is occurring in both the midstream (eg, Cheniere, Venture Global) and upstream (eg, EQT, EXE) sectors as North American producers increasingly seek ways to attach global gas prices to their domestic production. Over time, this could improve gas producer economics, but is **likely also to require increased investment in storage** to navigate mismatches between localized demand and opportunities in the US, production levels, and international exports. In addition, we see a risk that a glut of global LNG capacity later this decade could put downward pressure on international gas prices, potentially increasing volatility further.

Energy fragmentation

We think increased volatility tends to benefit commodity marketers and trading firms, although with greater potential profitability comes the risk of operating losses. We also think that **working capital needs could structurally increase in a more fragmented environment**, creating an opportunity for capital providers to step in and help bridge short-term liquidity needs. As above, we think that storage capacity could prove increasingly important in order to help manage through market cycles and act as a buffer during times of constraint. We have also seen gas producers invest in marketing entities, with a goal of maximizing price realizations and navigating what remains a highly volatile (and seasonal) commodity.

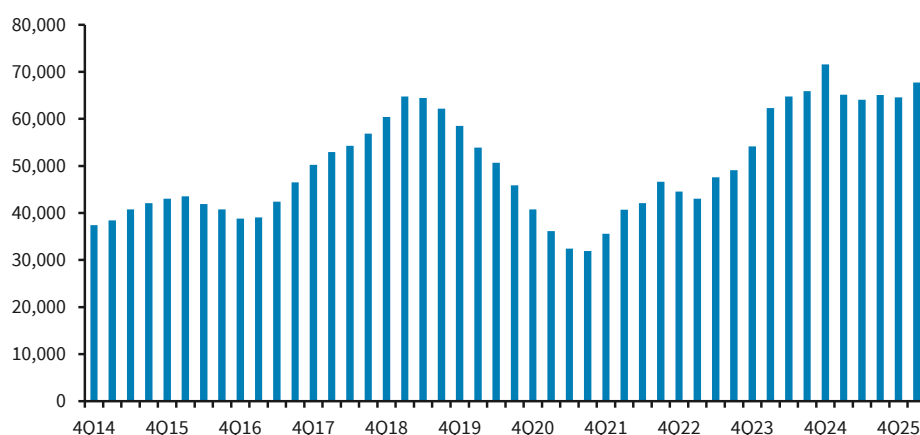
With the exception of China's independent refiners, which operate at limited run-rates, **the world is tight on refining capacity**, and downstream markets are interconnected, which can make profitability spikes very acute when infrastructure is damaged somewhere in the world. Thus, we think that refiners with low leverage are best positioned to capture periods of very high margins. And unlike the LNG market, where we think global capacity expansions are outpacing demand, the refining capacity expansion pipeline is more modest, and with multi-year construction timelines and a mixed execution track record. Given this, we see potential for new refining capacity, most likely through brownfield expansions, to be sanctioned in the next couple of years.

AI, power demand, and the new electricity economy

Ramping utility and data centre capex is driving investment in parts of the midstream sector, spanning both traditional (eg, gas transportation pipelines) and less traditional (eg, behind-the-meter power generation) parts of the value chain. With respect to the former, capex backlogs

are building for natural gas pipeline companies (eg, KMI, BWP, ET) and we expect financing to include some debt, as balance sheets are generally healthy.

FIGURE 104. Investment grade midstream capex has been moving higher since the 2018-20 deleveraging cycle, and we expect further increases (LTM capex, \$mn)



Source: Company reports, Barclays Research

To date, Williams has been the most active midstream credit in terms of behind-the-meter generation, recently securing JV financing. Elsewhere in the sector, we see potential for "equity-like" JV structures (often with embedded repurchase rights for the project sponsor) and hybrid capital to maintain investment grade ratings while obtaining some equity credit (without diluting common equity investors).

The race for capital

Within North American investment grade energy, **we see capital needs to finance new gas liquefaction capacity through 2030, for the build-out of pipelines to supply gas-fired generation (with growth largely underpinned by AI and data centre needs), and for working capital to support a more volatile commodity price backdrop amid energy scarcity in certain parts of the value chain.** In light of this, we think there are opportunities to deploy both short- and long-duration capital into energy debt markets.

For US LNG, [Figure 105](#), shows that greenfield project costs have moved to over \$1,000/mtpa of capacity. In general, project financing can range from 60% to 80% of a project, implying more than \$100bn of debt capital to be raised for the projects listed. Even though many of these have already secured initial finance commitments, we expect term-outs in the bond market over time, providing another opportunity for investors to commit capital.

FIGURE 105. More than \$160bn of US LNG projects are under construction and/or have a final investment decision; costs for greenfield projects have risen to over \$1,000/mt

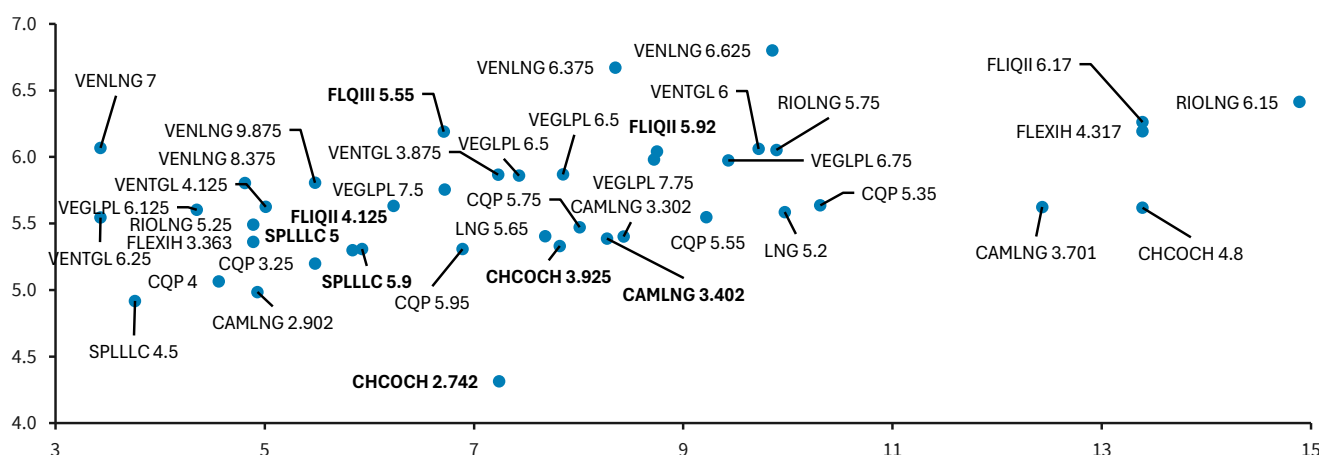
US LNG Projects	Budgeted Cost (\$bn)	Nameplate Capacity (mtpa)	Cost/mtpa (\$mn)
Corpus Stage 3	7.5	10.4	719
Golden Pass	11.6	18.1	641
Louisiana LNG	15.9	16.5	964
Port Arthur LNG Phase 1	13.0	13.0	1,000
Port Arthur LNG Phase 2	14.0	13.0	1,077
Rio Grande Phase 1	12.0	18.0	667
Rio Grande Train 4	6.7	6.0	1,117

US LNG Projects	Budgeted Cost (\$bn)	Nameplate Capacity (mtpa)	Cost/mtpa (\$mn)
Rio Grande Train 5	6.7	6.0	1,117
Venture Global CP2	29.0	20.0	1,450
Venture Global Plaquemines	24.0	20.0	1,200
Commonwealth LNG	13.0	9.5	1,368
Delfin Floating LNG	5.0	4.4	1,136
Sabine Pass Stage V Phase 1	4.7	6.0	782
Average	163.1	160.9	1,018

Source: Company reports, Barclays Research

Given our view that the global LNG market could overbuild capacity by 2030, existing projects may offer some of the more attractive investment opportunities, as they were financed in less competitive environments and may be structured more conservatively, with incumbency benefits in terms of both operating track records and customer relationships. **In particular, we like amortizing structures**, which often trade cheap because of the investment grade market's preference for bullet maturities and index-eligible debt, but are better suited for opco-level debt, which often includes contract cliffs. By owning amortizers, investors can reduce refinancing risk, particularly as contract expiration dates approach.

FIGURE 106. US LNG bond relative value: Some of the amortizing notes trade cheap (years to maturity/average life on x-axis; yield on y-axis, %)



Notes: Amortizing structures are in bold.
Source: Bloomberg, Barclays Research

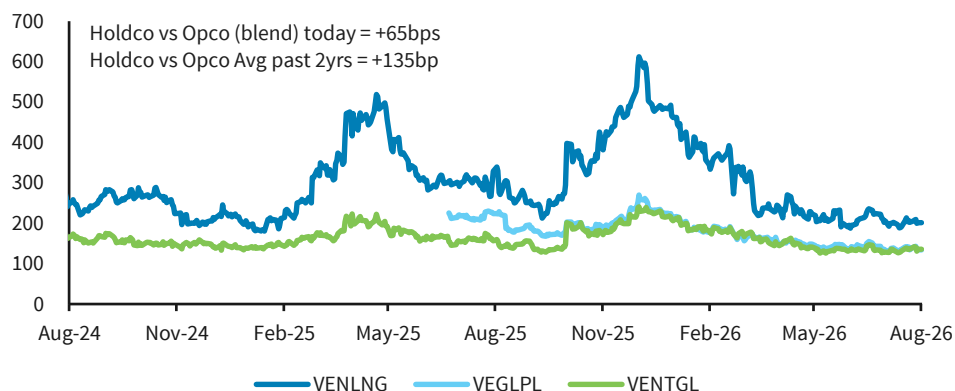
Specifically, we like parts of the Freeport LNG structure, which was beset by operating issues several years ago, but has been running consistently since then, has a tolling contract structure (ie, no gas procurement risk), and offers yields that are often comparable to or higher than those of bullet facilities. **We recommend buying FLQIII 5.55s (2039 final, 2033 average life) and/or FLIQII 5.92s (2039 final, 2035 average life), which offer roughly 6% yields for relatively short-duration, opco credit.**

One of the most active companies with respect to adding LNG capacity before 2030 is Venture Global (VENLNG/VEGLPL/VENTGL), which has capacity ramping from its current 40mtpa to around 85mtpa in 2030. With Phase I of the Plaquemines facility expected to go under contract during early 4Q26 (or late 3Q26) and Phase II in 3Q27, VG has more limited spot exposure (91% of cargoes hedged/under contract in 2H26 and 75% for 2027) until the start-up of currently under construction Calcasieu Pass 2 in early 2028 (or end of 2027). As global capacity adds outpace estimated demand, spot margins are likely to be much lower than current levels.

The same will likely be true in 2029 and 2030, when spot margins based on current spreads are even lower and in some cases not materially better than current contracted margins.

While this is not an immediate issue and these factors could revert, the VENLNG holdco bonds have traded to a level that make the relative value to the two opco boxes a bit more challenging given the longer-term headwinds. We recently downgraded the VENLNG holdco bonds to Market Weight as a result. As can be seen from Figure 107, the holdco/opco spreads are trading tighter than we have seen over the past two years, reflecting the market opportunity through 2027, though perhaps not fully discounting the lower margins at the end of this decade.

FIGURE 107. VENLNG holdco versus VEGLPL/VENTGL opco relative bond spreads



Source: Bloomberg

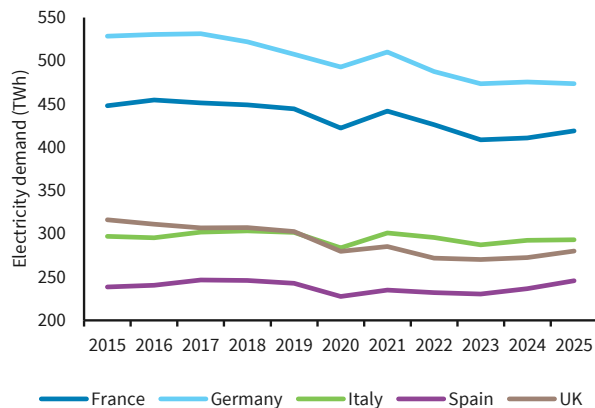
Utilities and Power - US & Europe Equity views

Dominic Nash, PhD Barclays, UK | Nicholas Campanella BCI, US | Theresa Chen, CFA BCI, US

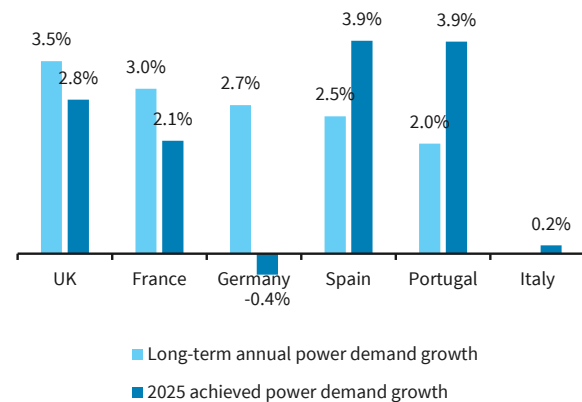
Europe

Why our sector matters

European utilities provide the electrons and power networks needed to supply an electrified economy. After more than a decade of flat if not negative power demand growth, we have now reached a fundamental inflection point. 2025 was the first year we have seen power demand growth across Europe, and the first half of 2026 continued this trend.

FIGURE 108. After a decade of flat power demand growth, Europe is picking up...

Source: EuroStat, UK Government DUKES data

FIGURE 109. ... and utilities expect this trend to continue

Source: Barclays Research estimates, Utility company forecasts

Electrifications efforts have fundamentally changed Europe's power market, moving it from a centralised system relying on a number of large scale, dispatchable power generation assets to a decentralised system balancing countless smaller renewables with less power utilisation and correlated, intermittent power generation. Renewables are not a reliable supply source at all times of the day and have to be accompanied by dispatchable assets or batteries by design. In a world with growing power demand, this is becoming increasingly evident.

We expect this growing power demand to affect all utility subsectors: It is driving up the value of gas turbines, legacy gas-fired power plants and existing grid connections while accelerating investment in both power transmission and distribution networks and keeping renewable ambitions high.

Energy security and the return of capital

How energy security is changing capital allocation

We believe European utility capital allocation has undergone a notable shift away from a singular focus on decarbonisation and towards a broader energy-security agenda. Within Europe's energy trilemma, security of supply and affordability have risen sharply in importance following recent supply disruptions and grid availability shortcomings.

As a result, capital has been directed towards regulated networks, flexible generation, batteries, inertia-providing assets and life extensions for existing nuclear capacity, rather than purely maximising renewable deployment. We see this as a structural reallocation of value towards assets that support system reliability and resilience, with integrated utilities emerging as the primary beneficiaries given their exposure to networks, dispatchable generation and balancing services. Tight reserve margins across several European markets, long lead times for new thermal capacity, and concerns about system stability have been reinforcing investment in assets that guarantee reliability of supply.

Effects on volatility

The combination of tighter reserve margins, growing dependence on weather-sensitive renewable generation, rising power demand from electrification and data centres, and heightened geopolitical risk means that commodity and power markets are becoming structurally more sensitive to supply disruptions. Energy security is seeking to mitigate this. In our view, Europe is moving from an era of surplus capacity towards one where security of supply carries an increasing premium.

Winners and losers in a fragmented energy world

Importantly, we do not view this volatility as uniformly negative for European utilities.

While volatility can create challenges for energy consumers, many utilities own assets and infrastructure that become more valuable when markets tighten. We see increased volatility creating a clear divergence between companies: utilities with merchant power exposure, trading operations, storage assets and flexible generation fleets are well positioned to benefit, especially as European utilities have virtually no direct exposure to disrupted Qatari LNG volumes.

US exposure provides an additional buffer against volatility. We also see a strategic benefit for European utilities with meaningful US operations, where even stronger power demand growth is being driven by electrification and data centres. European utilities with US network and generation exposure benefit from geographic diversification while participating in a market where security-of-supply concerns are already translating into higher investment, capacity additions and earnings growth. In our view, US exposure allows utilities to participate in a more advanced version of the same security-driven investment cycle emerging in Europe, reducing reliance on any single regulatory or commodity market.

AI, power demand and the new electricity economy

We believe AI and data centres are the most important new sources of structural electricity demand for Europe, creating a step-change in utility investment requirements and long-term growth. Our latest [data centre tracker update](#) identifies a pipeline of over 89GW of announced datacentre capacity across Europe, with even partial delivery capable of materially tightening European power balances, at a time when reserve margins are already under pressure.

In our view, this increases the value of dispatchable and flexible generation, particularly existing gas-fired assets, as data-centre developers increasingly seek firm power supply because new gas turbines are hard to obtain before 2030, as the US has already contracted over 100GW. Higher and more persistent baseload demand should drive greater utilisation of thermal fleets and support a normalisation of spark spreads, with front-of-the-meter data-centre connections lifting baseload power prices meaningfully over time.

At a company-specific level, we see two primary utility opportunities to benefit from data centres:

1. **The sale of legacy sites with established grid connections and development rights:** This is the "simplest" way to capitalise, as it doesn't require any new developments. RWE provided a compelling example last year (see our take [here](#)), having sold a 27ha site at its former Didcot A power station to CloudHQ for €225mn (c.£7.3m/ha) while also indicating that land, site infrastructure and grid connections can be worth c.€1m/MW. In our view, this demonstrates that legacy thermal sites can create significant hidden value through redevelopment and site monetisation.
2. **We view co-location as an even larger long-term opportunity.** Datacentre developers are increasingly seeking direct access to reliable power. Existing power stations often benefit from spare grid capacity, transmission connections, land banks and cooling infrastructure, allowing utilities to host data centres alongside their flexible generation assets through a direct private wire connection. In our view, this is beneficial for both DC developers and the utility, as the former can access electricity at a much cheaper price (between wholesale and retail level), while power plants can operate at higher utilisation, capturing more value.

The race for capital: financing and investment ideas

Where capital is needed and who can provide it

We believe the greatest capital requirements in European utilities are still concentrated in regulated power networks and baseload power assets, but note that the sector has already gone through a substantial funding cycle. Over the past three years, European utilities have successfully raised over €40bn of equity through 13 capital raises, with all but one undertaken to fund growth investment rather than balance sheet repair. We think the market's generally positive reception to these transactions demonstrates that investors increasingly view utility capital expenditure as value accretive rather than dilutive. We believe the sector has transformed from what was traditionally perceived as a low-growth value sector capable of delivering c.7-8% TSR into a growth sector with expected TSRs of roughly 11-12%.

FIGURE 110. European utilities raised more than €40bn in equity over the past three years

Utility	Date announced	Type of equity raise	Amount raised	Amount raised in €m	Initial share price reaction*
Severn Trent	9/29/2023	Equity placing + retail offer	£1bn	1,163	4.4%
National Grid	5/23/2024	Rights issue	£7bn	8,140	-10.9%
Pennon	1/28/2025	Rights issue	£490m	570	1.2%
Elia	3/7/2025	€850m private placement + €1.35bn rights issue	€2.2bn	2,200	17.1%
Italgas	4/10/2025	Rights issue	€1bn	1,000	1.3%
Iberdrola	7/23/2025	Private placing	€5bn	5,000	-4.7%
Orsted	8/11/2025	Rights issue	DKK60bn	7,800	-29.7%
SSE	11/12/2025	Equity placing + retail offer	£2bn	2,326	16.8%
Engie	2/27/2026	Private placing	€3bn	3,000	-2.0%
PPC	4/23/2026	Public offer in Greece + private placing abroad	€4.25bn	4,250	-2.8%
Solaria	4/29/2026	Equity placing	€300m	300	-3.7%
United Utilities	4/30/2026	Equity placing + retail offer	£800m	930	11.1%
RWE	6/22/2026	Private placing	€4bn	4,000	0.1%
Total raised				40,678	

* First market close post-announcement versus last market close pre-announcement.

Source: Company data, Bloomberg

Investment recommendations, separated by theme

- Preferred merchant-exposed power generators that are set to profit from increased power prices: RWE, SSE, Drax, Engie, Centrica and Solaria (all OW).
- Trading-oriented utilities that are set to profit from increased commodity price volatility: Centrica, Engie, RWE (all OW) and Naturgy (EW).
- Preferred AI and data-centre-exposed names (our 'Magnificent Eight'): Drax, Enel, Engie, RWE, SSE and Solaria (all OW) plus Fortum and Naturgy (both EW).
- Cheap utilities with re-rating potential: United Utilities (OW), based on the UK water theme.

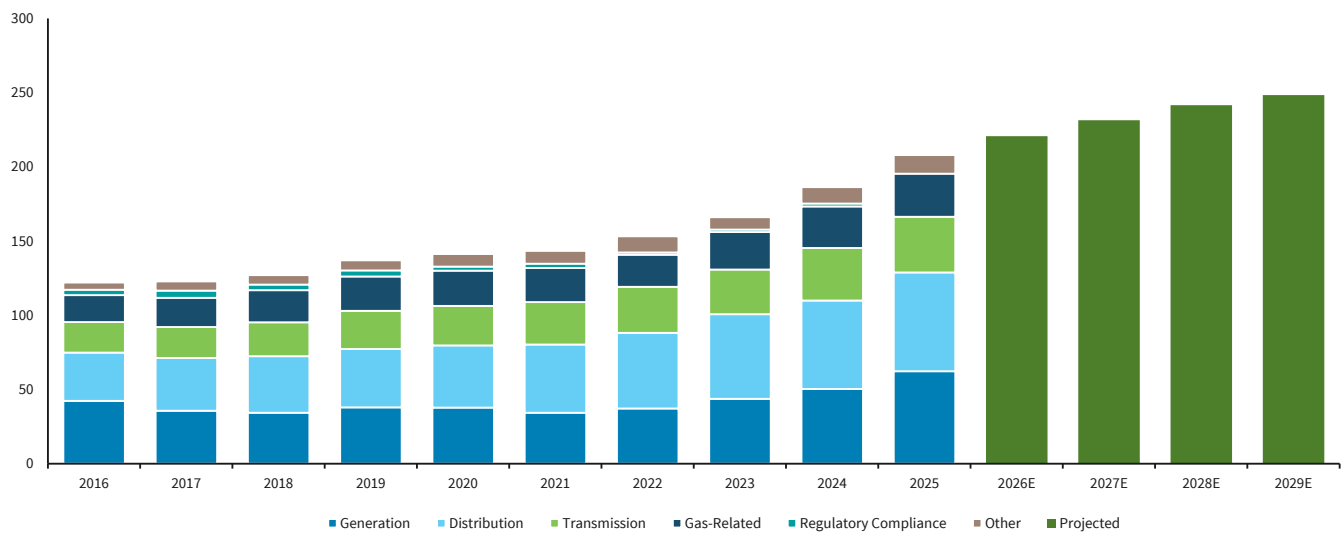
US

The US utility and power sector is undergoing the largest demand shift since the invention of air conditioning. This is driving several dynamics.

Acceleration in rate base growth: US utility capex increased steadily from 2016 to 2025 because of several structural drivers: grid hardening against extreme weather, replacement of

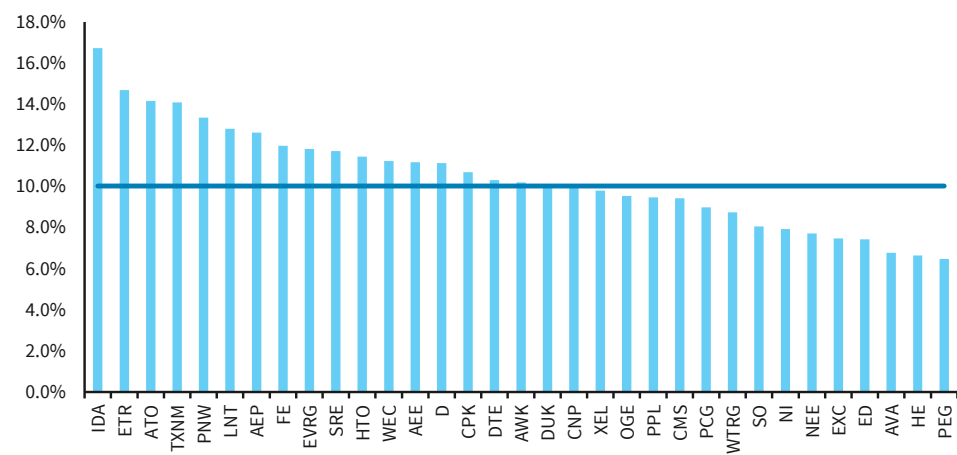
aging infrastructure, renewable buildout and interconnection spending, and smart grid and AMI roll-outs. Lately, there has been a fresh wave of significant investment tied to data centres. Given robust capex plans, the average company's five-year rate base growth forecast has accelerated from a CAGR of less than 7% before 2025 to 10% now.

FIGURE 111. EEI industry capex 2015A-2029E (\$bn)



Source: EEI

FIGURE 112. Rate base CAGR for 2026E-2030E



Source: Barclays Research, company reports

Affordability: Higher prices have broadly raised affordability risks for utilities nationally, increasing regulatory uncertainty and political scrutiny across key jurisdictions. Recent developments in Indiana, Texas and Pennsylvania are particularly notable.

- The Indiana Utility Regulatory Commission (IURC) has an ongoing affordability review that could increase regulatory pressure on utilities through formal investigations into authorized returns and rate design. The most significant proceeding is the ROE investigation, which will examine whether the lower risk associated with performance-based ratemaking and multi-year rate plans warrants lower allowed ROEs. A second investigation will review whether utilities are effectively double recovering costs through trackers such as TDSIC while also operating under multi-year rate plans, with the potential outcome of rolling trackers into base rates to simplify customer bills. The IURC also recommended eliminating the 50bp RTO

adder. In addition, regulators are considering expanded oversight of utility mergers and acquisitions at the holding company level and have been asked to provide guidance on how pending TDSIC filings fit within the evolving regulatory framework.

- Texas recently delayed its Batch Zero process. Batch Zero is ERCOT's new system-wide framework, approved by the Public Utility Commission of Texas (PUCT) in June 2026, for studying and allocating transmission capacity to large load interconnection requests of 75MW or greater, replacing the prior project-by-project review process. The delay of this process is not only a reliability pause but also an affordability safeguard. On August 3, Governor Abbott issued an audit directive, following his earlier June 10 letter instructing the PUCT and ERCOT to structure large load interconnections so that data centre growth lowers, rather than raises, residential bills. The core concern is forecast risk: if ERCOT commits transmission capacity to speculative data centre projects that never materialize, the resulting stranded infrastructure costs could flow through to residential and commercial ratepayers. PUCT Chairman Thomas Gleeson has framed the agency's approach explicitly around ensuring that data centres must fund their own transmission infrastructure while contributing to bill reduction. Separately, there has been opposition in Texas to 765-kV Permian transmission lines. In addition to landowner concerns, critics argue that the high cost of the projects (to flow through to ratepayers) is unnecessary, stating that they believe lower-voltage alternatives or incremental upgrades could meet reliability needs at a lower cost.
- Pennsylvania has also had significant affordability developments. Governor Shapiro has spoken out about reducing utility returns. In April 2026, Exelon's PECO withdrew its rate case request after Shapiro publicly demanded the pull-back on affordability grounds. The concerns have also reached the legislature, as embodied in HB224, which would cap the ROE as the 10-year US Treasury bond yield plus two percentage points. While the bill is not expected to pass the Republican-controlled Senate, its introduction and passage in the House exemplify the affordability concerns in PA.

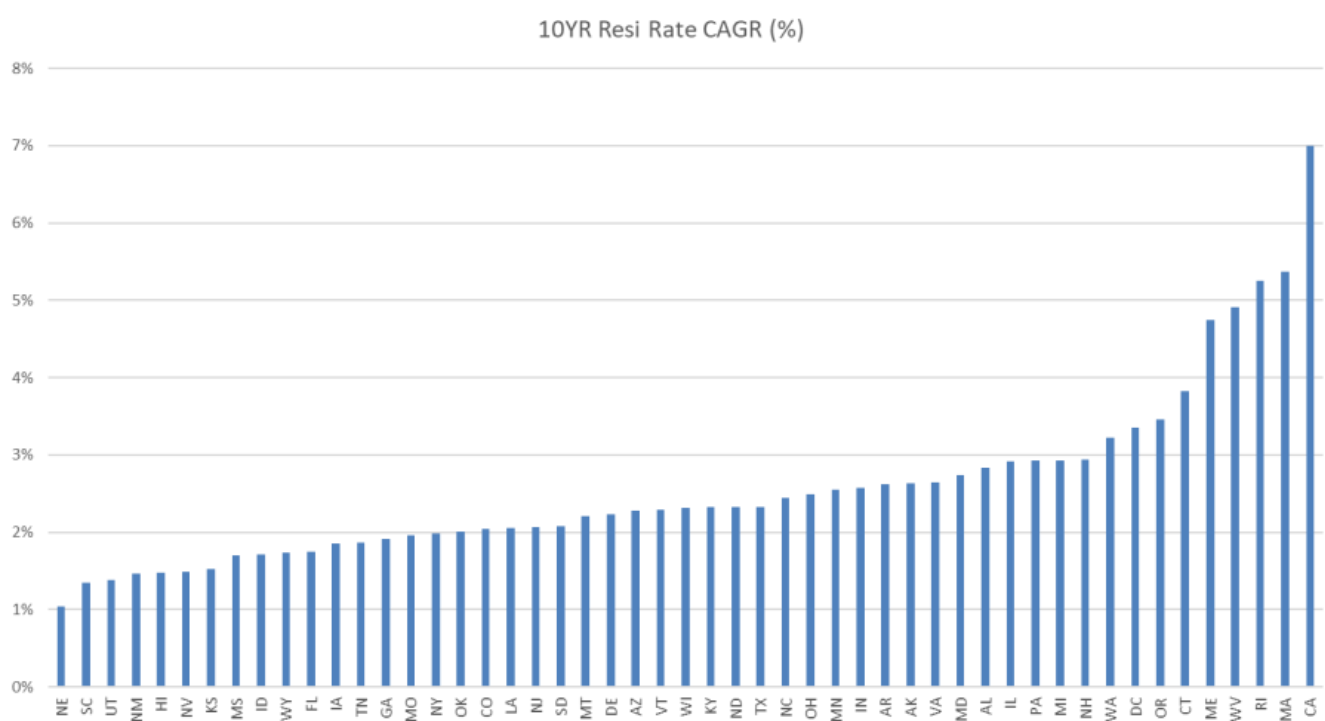
Below, we show our rate affordability matrix by company, as well as residential rate increases (10-year CAGR) by state.

FIGURE 113. Rate affordability matrix

Ticker	State	3Yr Average Residential Rate (cent/kWh)	Total Residential Sales (TWh)	Residential Customer Count	Average Customer Usage (kWh/month)	Average Customer Bill (\$/month)	Median Household Income (\$/year)	Wallet Percentage (%)
ETR	LA/AR/TX, etc	12.4	36.3	2,596,043	1,166	145	64,693	2.7%
SO	GA/AL/MS	15.4	49.3	3,919,624	1,047	161	74,400	2.6%
NEE	FL	14.1	70.9	5,287,101	1,118	157	75,630	2.5%
OGE	OK/AR	11.7	9.9	768,341	1,071	126	65,276	2.3%
PNW	AZ	15.2	15.6	1,256,120	1,034	157	84,700	2.2%
HE	HI	44.1	2.3	472,536	405	179	98,240	2.2%
AEP	OH/OK/VA, etc	15.0	36.6	3,866,826	790	119	75,070	1.9%
D	VA/SC/NC	14.1	39.0	3,188,589	1,019	143	91,828	1.9%
PCG	CA	34.9	27.2	5,047,462	449	157	100,600	1.9%
NI	IN	18.0	3.4	428,844	662	119	76,710	1.9%
DTE	MI	19.4	15.1	2,067,758	610	118	79,460	1.8%
CMS	MI	18.7	12.5	1,657,843	628	117	79,460	1.8%
PPL	PA/KY/VA	14.7	18.6	2,154,127	719	105	72,041	1.8%
DUK	NC/FL/IN, etc	14.1	65.3	7,409,922	735	104	72,973	1.7%
EVRG	KS/MO	13.3	15.7	1,468,313	892	118	84,016	1.7%
POR	OR	15.7	7.7	809,573	796	125	89,700	1.7%
WEC	WI	17.7	10.8	1,493,175	603	107	82,560	1.6%
IDA	ID/OR	11.4	6.0	539,849	921	105	81,902	1.5%
AGR	NY/ME/CT	19.8	13.9	2,051,002	565	112	89,291	1.5%
NWE	MT/SD	13.4	3.4	368,037	760	102	81,576	1.5%
ED	NY/NJ	30.5	14.0	3,403,241	342	104	87,676	1.4%
AEE	MO/IL	13.2	18.3	2,156,513	708	93	80,082	1.4%
OTTR	ND/MN/SD	11.0	1.2	105,692	926	102	89,144	1.4%
AVA	WA/ID/AK	10.9	4.2	361,606	965	105	92,113	1.4%
BKH	SD/CO/WY	15.9	1.5	192,186	638	102	90,690	1.3%
EIX	CA	29.8	19.5	4,599,981	354	105	100,600	1.3%
XEL	MN/CO/TX	14.9	25.7	3,321,140	645	96	94,118	1.2%
AES	IN/OH	13.8	6.5	945,460	573	79	77,563	1.2%
ALE	MN/WI	14.5	1.1	138,968	641	93	91,566	1.2%
MDU	ND/MT/WY	11.4	1.1	119,261	766	87	85,790	1.2%
EXC	IL/PA/MD, etc	16.2	57.3	8,343,744	572	93	93,221	1.2%
PEG	NJ	18.9	13.3	2,056,231	541	102	103,500	1.2%
SRE	CA	42.3	3.9	1,499,139	219	93	100,600	1.1%
ES	CT/NH/MA	28.3	11.4	2,910,129	326	92	104,168	1.1%
FE	OH/NJ/PA, etc	14.3	36.9	6,000,000	512	73	85,918	1.0%
TXNM	TX/NM	15.1	3.4	834,000	339	51	64,140	1.0%

Source: Company data, Bureau of Labor Statistics, Standard & Poor's, Barclays Research

FIGURE 114. Residential rate increases by state



Source: EIA Data – Residential Bill CAGR 2024 from 2014, Barclays Research

Generation mix shift: Data centre growth has reshaped the utility sector's generation strategy. The need for large amounts of reliable, around-the-clock generation has strengthened the case for gas and is accelerating interest in nuclear. Gas gives utilities a relatively fast and flexible source of firm generation, while nuclear provides high capacity factor, carbon-free power that can support hyperscaler loads. At the same time, renewables remain an important part of utilities' plans, but their intermittency makes them less suited to serving data centres on a standalone basis, though they become more attractive when paired with storage.

Load growth: The outlook for utility load growth has changed materially as data centres have shifted the industry from a long period of flat demand to sustained growth. We increasingly now expect 4-5% long-term load growth across the sector, although the magnitude will vary by utility and geography. Importantly, utilities are now planning significantly higher generation, transmission and distribution investments in this higher-demand environment, creating longer-duration capex opportunities. The key debate is no longer whether load will grow, but how much of the announced data centre pipeline will ultimately materialize and when.

Higher financing needs: Higher load growth translates into the higher capital requirements and, therefore, greater financing needs. This creates higher equity needs to maintain balance sheet health and credit ratings.

Top three current long-term Overweights

- **PPL Corp (PPL):** We like PPL because it offers increasingly visible above-average EPS growth (near the top end of 6-8%) with upside potential and has a robust economic development pipeline and a strong balance sheet. The company recently identified a meaningful \$10-12bn of incremental capex opportunities through 2032, related to generation investments with the Invitum JV (between PPL and Blackstone Infrastructure) and regulated Kentucky. Management expects one or more commercial agreements with the JV by year-end and noted that negotiations are progressing independently of PJM's RBP process. The data centre pipeline for the utilities in Pennsylvania and Kentucky continues to advance. Pennsylvania politics has weighed on the shares, but with the recent rate case settlement approved, we think the company will be able to avoid a rate case for a while using the DSIC mechanism and cost management. Prior to this rate case, PPL was able to stay out for ten years. The Invitum JV is where the big updates are likely to come, and we think it represents an attractive opportunity. Meanwhile the stock trades at a c.5% discount to peers based on 2028 P/E.
- **Entergy Corp (ETR):** While the stock trades at a 10-15% premium 2028 P/E multiple, we see upside potential in the multiple given limited rate case risk and room for additional growth beyond what is in the industry-leading EPS growth plan, such as 7GW of capacity open for future customer growth from 2029 to 2031.
- **Xcel Energy (XEL):** We believe XEL's current 5% discount 2028 P/E multiple exaggerates the potential wildfire liability that could come from Smokehouse fire, while investors can own a premium utility, management team, and balance sheet at a discount. Following the Marshall settlement, investors can focus on XEL's premium attributes: balance sheet, RAB and diversification. The Q2 call with XEL was positive, and management highlighted an increase of additional investment pipeline from +\$10bn to +\$15bn based on MN IRP/RFP settlement approval, SPS recommended portfolio, MISO Tranche 2.1, and SPP ITP portfolio.

Utilities and Power - US, Europe & LatAm Credit views

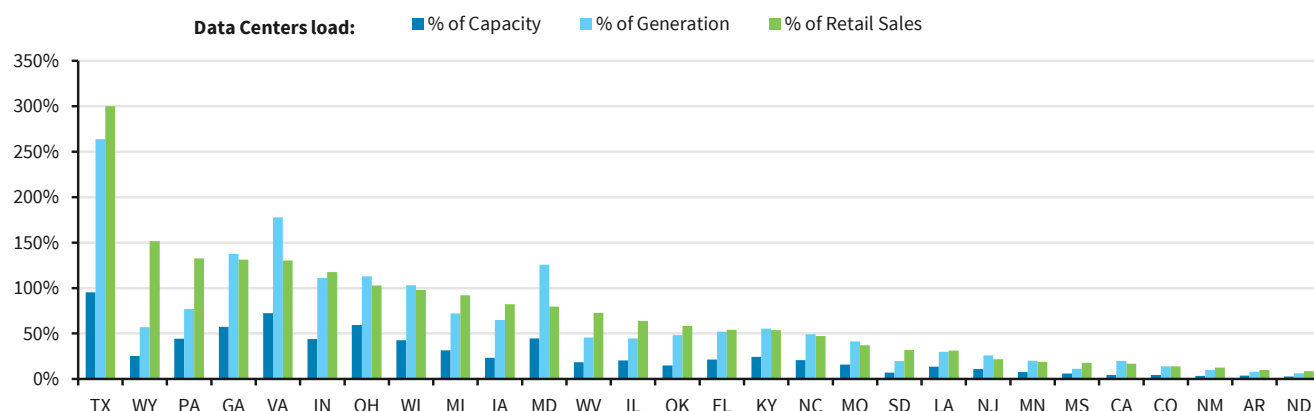
Srinjoy Banerjee, CFA⁽ⁱ⁾ BCI, US | Franck Bataille⁽ⁱ⁾ BBI, Paris | Parikshit Budhe⁽ⁱ⁾ Barclays, UK | Stefan Fenistyk⁽ⁱ⁾ BCI, US

US credit

The US regulated utility sector is seeing an acceleration in investment needs, driven by three primary pillars:

- 1) **Generational Shift in Load Growth:** There has been a structural acceleration in load growth, largely driven by data centre demand, where utilities have communicated multi-year high-probability pipelines of 400GW ([Leap in Data Centre Pipeline Drives Market Reform](#)).
- 2) **Energy Transition/Resilience Spend:** This depends on state-level priorities and regional extreme weather exposure (storms, wildfires), with the federal focus de-emphasized under the second Trump administration (see [Taking stock of US power post-OBBA](#)),
- 3) **Electrification:** While de-emphasized from a federal policy standpoint, the electrification of vehicle fleets and buildings remains a key driver of investment at the state and city level.

FIGURE 115. Utility Communicated Data Centre Pipelines as a % of Total Capacity/Generation/Retail Sales by State



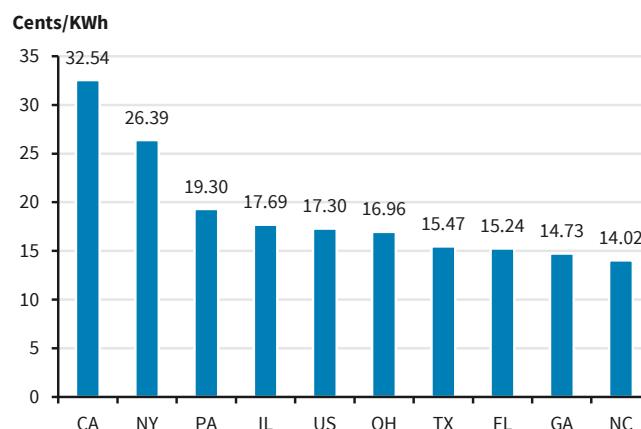
Source: Company reports

Affordability concerns

With stakeholders mindful of affordability, US state and federal policy is evolving, with input from regulated utilities, to ensure that data centres pay for the investment needs. In practice, this means utilities are looking for state regulators to approve special large load tariffs – see [Data Centres Drive Over 250GW Load Growth Opportunity](#). Components of these tariffs include minimum capacity thresholds for large loads (typically 50-100MW), minimum contract terms (often 15+ years, including a ramp-up period), minimum contract-capacity billing (eg, 80%+), and termination rules and collateral-posting requirements.

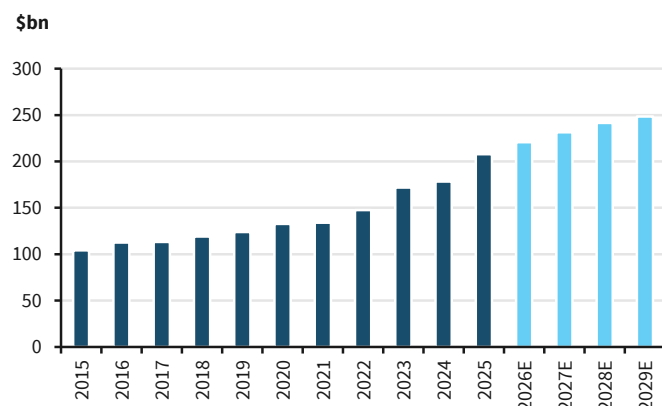
FIGURE 116. US: Residential Utility Bills as a % Median Income

Source: EIA

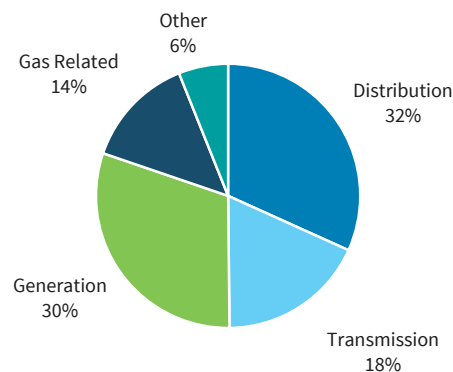
FIGURE 117. Meaningful Range of Utility RatesTop 10 states by population
Source: EIA

Regulated utility investment breakdown

We view the step-up in investment as manageable, with transmission and distribution (grid-level) capex modest at this stage (~\$150-300mn/GW). In certain cases, virtually no initial investment may be required given existing spare grid capacity). Generation spend is greater (\$2-3bn per GW), though with some key markets (ERCOT in Texas, PJM in Mid-Atlantic) de-regulated, not all the investment need is for the utility sector. We expect there is a ~\$50bn per annum opportunity set from data centre development, compared to ~\$250bn of annual sector capex.

FIGURE 118. Expected Increases in Aggregate Utility Capex

Source: EEI

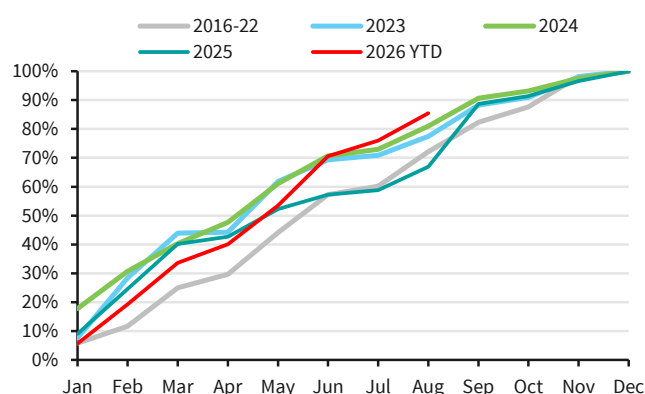
FIGURE 119. Breakdown Split of US Utility Capex Spend in 2025

Source: EEI

Diversification of funding needs

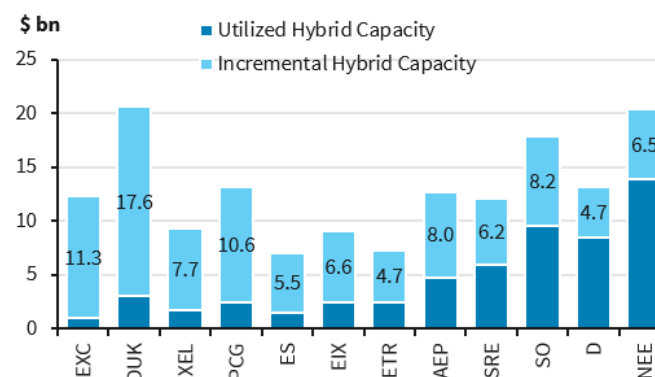
Combined with equity funding, along with growing cash flows and diversification of debt funding, inclusive of Department of Energy loads, alternative currencies and products (eg, convertibles), we think USD debt issuance needs will grow at a measured pace y/y (~\$10-20bn/year) – see [En Route to a \\$1 Trillion IG Utility Index](#). Accordingly, we estimate \$140bn of US Utility funding needs for 2026 (with ~70% complete YTD). Utilities have also been utilizing hybrid capital, which benefits from 50% equity treatment.

FIGURE 120. Cadence of 2026 Utility Issuance versus Prior Years



Source: Barclays Research

FIGURE 121. Hybrid Issuance Capacity for Large Cap Utilities



Source: Company Reports, Barclays Research

Merchant power structurally improved

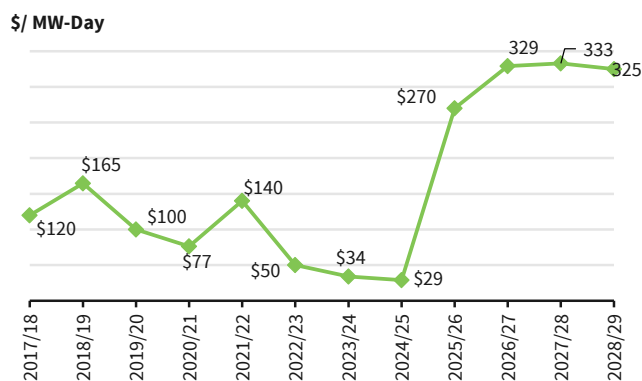
We view the merchant power market as structurally well positioned over the next 3-4 years given a supply -demand imbalance driven by higher data centre power demand and lack of incentives (and supply chain constraints) to build out more merchant generation capacity. The [2028-29 PJM capacity auction result](#), the third consecutive auction to clear the cap, with a 6.8GW shortfall versus PJM's targeted 20% reserve margin reliability target, is clear evidence of that.

The substantial pipeline of interconnection requests is driving an uptick in market reform efforts in both ERCOT and PJM. At ERCOT, the batch process is used to group large projects (75MW or more of peak demand) into a single study so it can assess electricity demand additions and transmission upgrades. ERCOT is assessing 438GW of large load requests (89% from data centres), compared to peak load of ~91GW. At PJM, there are a mix of near-term ("connect and manage" and the reliability backstop auction) and long-term (market structure, facilitating long-term bilateral contracts) changes being proposed. There is also push back at the state level, for example Texas Governor Abbott's data centre audit⁴ and Pennsylvania Governor Shapiro's focus on greater guardrails for data centre development,⁵ though both appear to stop short of a moratorium, and both states face gubernatorial elections in November. We expect these efforts to accelerate refinement of the exceptional load pipelines into more realistic load growth prospects.

⁴ <https://gov.texas.gov/news/post/governor-abbott-directs-comprehensive-data-center-audit>

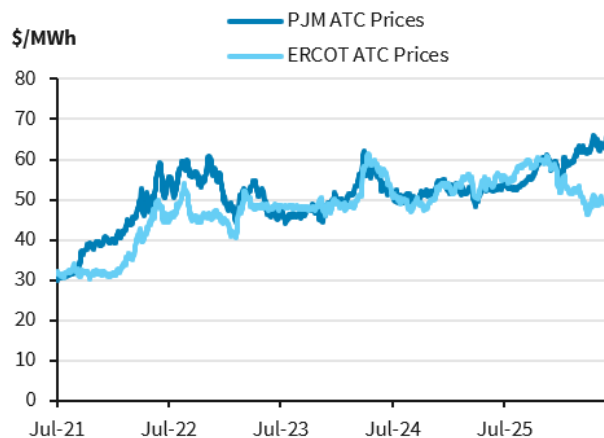
⁵ https://www.pa.gov/content/dam/copapwp-pagov/en/governor/documents/eo2026_05_protecting%20pennsylvania%20consumers%20from%20data%20center%20impacts_final_executed.pdf

FIGURE 122. PJM Capacity Prices



Source: PJM

FIGURE 123. PJM and ERCOT Power Prices

Prices for Calendar Year 2027
Source: Bloomberg

Investment recommendations

- Overweight Vistra (VST):** We think VST is on a solid upgrade path to mid-BBB by year-end 2027, as it continues to benefit from power demand growth tailwinds and sluggish supply. Management is charting a pathway to low-2x leverage by year-end 2027, well below the 3x upgrade threshold set by S&P and Fitch. Importantly, unlike regulated utilities that need to invest into the load growth narrative, VST can benefit from higher margins on its incumbent fleet due to greater scarcity of power resources.
- Underweight NextEra Energy (NEE):** We view NextEra as committed to its credit ratings, and we expect rating agencies to [maintain their credit rating](#) following the Dominion acquisition. That being said, NEE, pro forma for Dominion, will become one of the largest corporate tickers in the HG Index (we calculate second largest corporate ticker after ORCL, albeit broken into five different entities). With NEE intending to issue all future holdco debt for the combined entity (including Dominion Holdco refi), we expect persistent funding needs to weigh on spreads such that it trades wide relative to its rating level.

European HG Utilities: Financial innovation required to fund super powered Europe

Franck Bataille BBI, Paris | Parikshit Budhe Barclays, UK

As outlined in our [EU HG Utilities & Energy 2026 Outlook: Your Country Needs You!](#), European utilities are entering a new investment super cycle, driven by electrification, energy-security requirements and the need to reinforce critical power infrastructure. Beyond renewable deployment, the binding constraint is shifting towards additional baseload needs, networks, system resilience and financeability. In our view, the next phase of the European energy transition will be defined less by technology choice and more by the ability to fund, permit and deliver strategic infrastructure while preserving investment-grade balance sheets.

From renewables to networks, the strategic shift is accelerating, but baseload investment remains needed

The first phase of the European transition was largely focused on building out renewable capacity. The next phase is increasingly about making the system investable, connected and reliable. Grid connection queues, transmission bottlenecks and rising system-balancing requirements have made networks the critical constraint, with utilities reallocating capital towards regulated transmission and distribution assets. This is not simply a decarbonisation

story. Electrification, AI-related power demand and security-of-supply objectives all require a materially larger, more resilient and more flexible grid.

Networks, however, are only part of the solution. Rising power demand, growing system-balancing requirements and declining dispatchable capacity are reinforcing the need for baseload generation alongside grid investment. While Europe has added significant renewable capacity over the past decade, current market-price signals remain insufficient to support large-scale investment in new generation without regulatory support or risk-sharing mechanisms. As a result, network expansion, storage, flexibility assets and derisked baseload investments are increasingly moving to the forefront of utility capex plans.

From a credit perspective, the shift is constructive but capital-intensive. Regulated networks and regulated/derisked generation offer stronger earnings visibility, lower commodity exposure and better cash-flow predictability than merchant generation, but they also require sustained upfront investment. While Europe knows what needs to be built, the key question is whether supportive regulation, robust risk-sharing schemes and efficient market design can be implemented so that the sector can finance and deliver this investment cycle in a well balanced manner.

FIGURE 124. Achieved power demand growth (2015-2025)

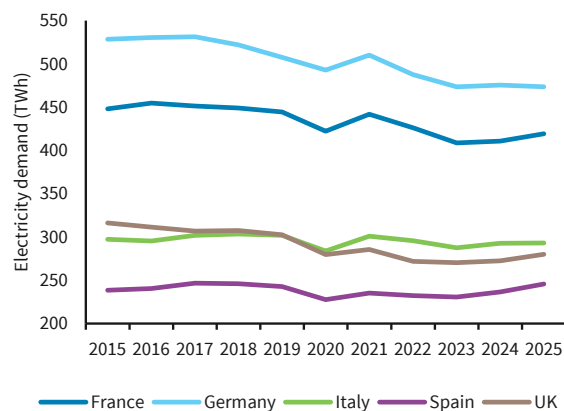
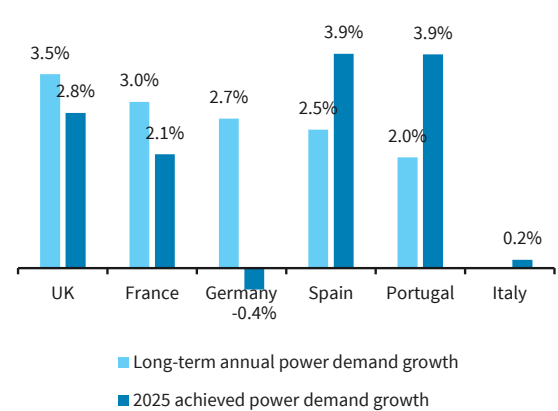


FIGURE 125. Forecasted long-term annual power demand growth



Source: Barclays Equity Research

Source: Barclays Equity Research

FIGURE 126. Network capex allocation: national power transmission plans

Country	Document	Document publication	Investment	Investment window	Implied annual investment
Germany	2023 Federal Network Development Plan	12-Jun-23	€350bn	2024-45	€16.7bn pa
	2025 Federal Network Development Plan (2nd draft)	16-Mar-26	€365-392bn	2025-45	€18.5-19.5bn pa
UK	2025 RIIO-T3 Final Determination	4-Dec-25	€76bn	Mar 2026 - Mar 2031	€15.2bn pa
France	2019 Strategic Development Plan for the French transmission grid (SDDR)	17-Sep-19	€33bn	2020-35	€2.2bn pa
	2025 Strategic Development Plan for the French transmission grid (SDDR)	14-Feb-25	€100bn	2025-40	€6.7bn pa
Spain	2025 Draft Network Development Plan	12-Sep-25	€13.59bn	2025-30	€2.7bn pa
Italy	2025 Grid Development Plan	14-Mar-25	€23bn+	2025-34	€2.6bn pa
Belgium	Federal Network Development Plan	5-May-23	€9bn+	2024-34	€0.9bn pa

Source: Barclays Research

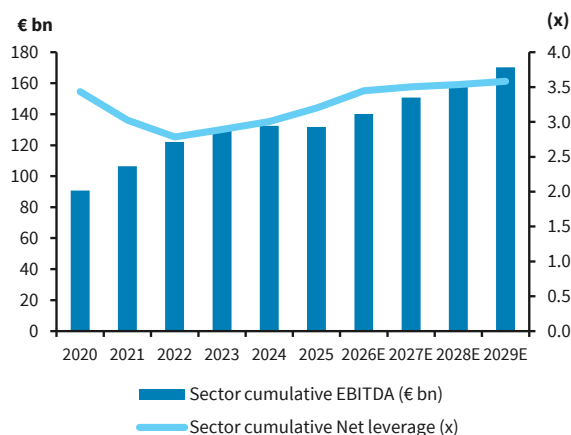
Financeability becomes the key credit differentiator

European utilities are entering a more capital-intensive phase. Network expansion and additional baseload needs are driving a sustained increase in sector capex, while leverage is already trending higher across much of the sector. As a result, investment needs are rising at a time when rating headroom remains finite.

This is making financeability an increasingly important consideration. **The key challenge is no longer identifying investment opportunities, but ensuring utilities can fund them while maintaining access to capital markets and preserving investment-grade ratings.** In our view, future investment decisions will increasingly be assessed through the lens of balance-sheet capacity, regulatory visibility and funding flexibility.

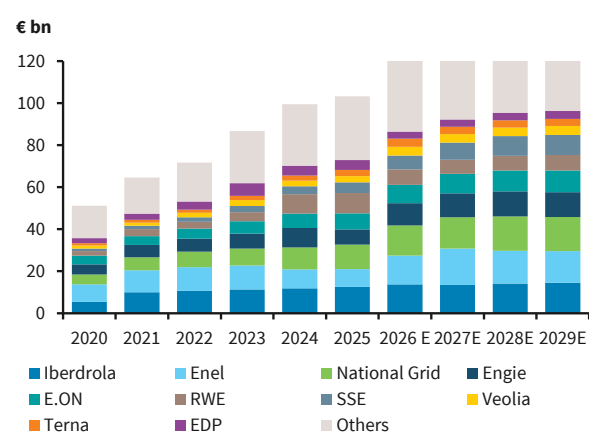
The response is already visible across the sector. Utilities have increasingly turned to equity raises, hybrid capital, asset recycling and disciplined capital allocation to strengthen balance sheets ahead of the investment cycle. Recent actions by National Grid, Iberdrola, SSE, EnBW, Elia and others illustrate a growing focus on preserving financial flexibility rather than allowing leverage to absorb the full burden of investment.

FIGURE 127. Stoxx 600 Power & Gas Utilities: Bloomberg consensus EBITDA (€ bn) and leverage (x) forecast



Note: data is based on Bloomberg consensus – includes companies in the Stoxx 600 Power & Gas Utilities index
Source: Bloomberg Intelligence, Barclays Research

FIGURE 128. Stoxx 600 Power & Gas Utilities: Bloomberg consensus capex forecast (€ bn)



Note: data is based on Bloomberg consensus – includes companies in the Stoxx 600 Power & Gas Utilities index
Source: Bloomberg Intelligence, Barclays Research

From project support to public/private ownership solutions – the New Utility financing playbook

The scale of Europe's utility investment cycle is expanding the financing toolkit. Traditional debt funding remains important, but increasingly sits alongside equity raises, hybrid capital, project-level support mechanisms and direct state intervention. The common objective is preserving financeability while enabling private investment in strategic infrastructure.

- **Sizewell C illustrates project-level risk sharing⁶.** The UK framework combines a regulated asset base (RAB) model with government liquidity support, contingent financing arrangements and other protections designed to improve project financeability and reduce funding risk.
- **CEZ and EDF illustrate alternative ownership and funding solutions.** In 2025, the Czech state acquired an 80% stake in the Elektrárna Dukovany II nuclear project and approved state

⁶ See "Credit view: New nuclear requires state support and regulatory derisking", from the broader note

financing for construction, effectively removing project funding obligations and contingent liabilities from CEZ. [More recently](#), the government has backed plans to separate generation from distribution and customer-facing activities, potentially creating a structure that would allow greater state control over strategic generation assets while retaining access to private capital through regulated and retail businesses.

Meanwhile, EDF represents the most direct form of intervention. France completed the full nationalisation of EDF in 2023 by acquiring the remaining minority stake, giving the state full ownership of one of Europe's most strategically important nuclear operators. The French government has also committed to fund around 50% of future new-build nuclear investment, further strengthening EDF's financeability as it enters a major baseload investment cycle.

Together, EDF and CEZ demonstrate how governments can move along a spectrum of intervention, from project-level ownership and funding support to broader corporate ownership solutions.

- **TenneT is the clearest grid example.** The Dutch model goes beyond traditional project financing, with the Kingdom of the Netherlands providing an institutional state-guarantee to TenneT Netherlands (TENNNL) and supporting funding capacity at the operating-company level. The transaction ultimately resulted in guarantee-eligible bonds being upgraded to AAA/Aaa, illustrating how state support can be used to protect market access and facilitate large-scale grid investment.

While the mechanisms differ, the direction of travel is similar. TenneT protects corporate funding capacity, Sizewell C derisks project delivery, CEZ combines project ownership support with a potential restructuring of strategic assets, while EDF demonstrates full state ownership. Together, these examples suggest that financeability is increasingly becoming a policy objective in its own right, with governments adopting a broader range of tools to support strategic energy infrastructure.

FIGURE 129. New utility financing options

Funding model	Example	Credit relevance
Corporate-level state guarantee	TenneT Netherlands	Protects market access and funding capacity
Project ownership + state funding	Dukovany II / CEZ	Transfers project funding risk away from the utility
Ownership restructuring	CEZ proposed separation	Balances strategic control with access to private capital
Full nationalisation	EDF	Direct state control of strategic infrastructure
RAB model + government support	Sizewell C	Improves project financeability and reduces funding risk

Source: Barclays Research

Investment recommendations

- **Overweight EDF (EDF):** Within our coverage, [EDF remains a preferred core holding](#)⁷. The transition from recovery to execution is well underway, but we continue to see EDF as one of the strongest examples of the broader financeability theme discussed above. Full state ownership, a formal leverage framework, improving regulatory visibility and risk-sharing mechanisms around future nuclear investment provide a supportive backdrop as the group enters a capex-intensive phase. In our view, EDF is particularly well positioned to navigate the infrastructure cycle while maintaining investment-grade metrics.
- **Overweight TenneT Holding (TENN):** As detailed in [European Utilities - TenneT: 3D glasses needed](#), TenneT is one of the clearest examples of how utilities are adapting ownership,

⁷ Refer to the 'European Utilities & Energy' section of the broader publication, [European Hybrid Capital - Corporate Hybrids: Summer Heat](#)

funding and support frameworks to preserve investment-grade ratings through a period of elevated infrastructure investment.

Following the restructuring, the Dutch business benefits from a sovereign-backed funding platform and state guarantee, while TenneT Germany has secured committed equity support and a standalone investment-grade financing framework. Against this backdrop, we remain OW on TenneT Holding, where the hybrids represent the only debt remaining at the HoldCo level, with no new HoldCo issuance expected through 2030 and supportive liability-management dynamics. While we are MW on both TENNLL and TENNGR (OpCos), we continue to see value in selected unguaranteed TENNLL bonds.

- **Underweight Naturgy (NTGYSM):** As detailed in [Revisiting single-name views: mid-year 2026 update](#), we are UW on Naturgy. The balance sheet remains solid and event risk has eased following the reduction in private equity ownership. However, a lower-growth, gas-heavy business mix, limited EBITDA expansion and a shareholder-focused financial policy leave the credit less well positioned for the next phase of Europe's infrastructure cycle. With spreads now broadly aligned with BBB fair value following the repricing of restructuring risk, we see limited scope for further compression versus peers with stronger network and electrification exposure.

LatAm credit

Unlike other regions, Latin America's energy-security challenge is less about securing fuel imports and more about maintaining reliability. Renewables account for roughly 60% of the region's power mix, among the highest shares globally, leaving most countries relatively well positioned from a resource perspective. Instead, energy security is shifting towards the ability to move electricity from resource-rich regions to demand centres and maintain system reliability during climate-related disruptions. Mexico remains a key exception given its reliance on US natural-gas imports.

We view the next decade of investment as directed less toward new generation capacity and more toward transmission expansion, storage deployment, and grid resilience. Across several countries, the principal constraint is the ability to deliver electricity rather than generate it. Transmission bottlenecks, renewable curtailment, and climate volatility are becoming the key drivers of investment needs.

The challenges vary by market. Mexico remains exposed to natural-gas imports from the US and faces growing transmission constraints in industrial regions. Chile has abundant renewable resources but requires additional storage and grid infrastructure to reduce curtailment and support reliability. Colombia's hydro-dominant system is increasingly exposed to drought risk, while Brazil's rapid renewable buildout is creating transmission and congestion challenges that require significant network investment.

Chile

Chile is simultaneously one of the world's most renewable-heavy markets, though curtailment remains a key issue. Chile's power system is well advanced in the transition from fossil fuels generation to renewables: renewables account for close to 70% of generation and the country has committed to retire all coal by 2040. However, this decarbonization is colliding with grid geography. Chile's main investment need is increasingly storage and transmission rather than additional renewable generation. Solar and wind are concentrated in resource-rich zones away from central demand, and new solar projects face curtailment during times of peak supply. BESS is therefore becoming the nearer-term monetization tool for the utilities. The country has approved and under-construction BESS projects totaling about 2.6GW, though the system operator's long-term diagnostic points to at least 6.9GW of additional BESS need by 2044.

Hydro variability and gas sourcing are also important to the country's energy security. Prolonged droughts threaten Chilean hydropower output and resilient grids are a core pillar of its 2050 transition roadmap. Chile also has 5.3GW of installed gas-fired capacity and is a net importer of natural gas. We think that Argentina's Vaca Muerta could change Chile's supply calculus. Argentina's resource base and planned transport look to be more sufficient for Chile's incremental flexible-power and industrial-gas needs.

Within our coverage, Enel Chile has already exited the coal business, and has long-term LNG contracts with Shell and access to Argentina gas imports to provide backup generation during renewables shortfalls. The company is heavily focused on battery project with additional storage under development. Engie Energia Chile retired coal capacity in 2025 and is moving its generation towards renewables, storage, and gas. A previous mismatch between generation and contracted supply led to a high reliance on spot-market energy purchases, but the company is increasingly self-sufficient in generation by letting existing contracts lapse and building generation. Colbun is careful to only retire thermal assets when the grid can absorb more renewables, and is focusing on storage projects to address intermittency issues with renewables.

Mexico

Mexico is highly dependent on US natural gas, with over 60% of the generation mix from natural gas and the country heavily reliant on gas imports from the US. CFE itself is heavily exposed to natural gas, though it has expanded its hedging program following the 2021 Texas freeze. Other key themes are nearshoring and state-led energy policy. The state is not retreating from a centralized energy policy under the current Sheinbaum administration. Mexico's 2025-30 electricity plan foresees around MXN624.6bn of investment, including MXN424.6bn in generation, MXN124.5bn in transmission and MXN72.5bn in distribution, with mixed-investment generation maintaining 54% CFE participation within the generation market, while CFE maintains its monopoly on transmission and distribution. These efforts will keep CFE dominant within the Mexican electricity market. The private sector now has greater clarity on the mixed-schemes for the generation segment, which are critical to expanding generation capacity in the country without encumbering CFE's balance sheet. CFE will contribute items in kind such as permits, land, or other potentially non-cash items to maintain its majority stake in projects. CFE has already received >200 proposals from private developers for projects.

The IPP universe is positioned differently than CFE. FIEMEX (the former Iberdrola portfolio acquired by Mexico Infrastructure Partners) effectively functions as an extension of the state's generation fleet, providing CFE with access to a large portfolio of combined-cycle assets located near industrial demand centres. Valia remains one of the most CFE-linked private generators, with approximately 99% of contracted capacity tied to CFE and meaningful recontracting exposure. Saavi is more diversified, with exposure to other customers via the wholesale and private counterparties markets, gas compression, and California power exports, while still benefiting from long-duration contracted generation assets with CFE such as Tierra Mojada. New issuer Cox is pursuing opportunities under the new mixed-investment structure and appears focused on growing generation. Other infrastructure providers such as Esentia remain critical to the system because they own the natural gas transportation backbone connecting the Permian Basin to Mexico's major industrial and power-generation centres. Roughly 80% of Esentia's revenues are ultimately linked to CFE, underscoring CFE's dominance within the electricity market.

Mexico offers high demand visibility but lower regulatory optionality. Estimates of electricity reserve margins during times of peak national demand have periodically fallen below 3%, which indicate very low slack in the Mexican electricity system. Northern and Bajío industrial growth is the demand engine, but the constraint is transmission and deliverability with these

areas having electricity demand growth but transmission expansion lagging. One major hurdle for the country's nearshoring efforts include getting power to where it is needed rather than aggregate generation alone, especially for northern border and central industrial states.

Lastly, the country also has faced challenges with hydro generation during the dry season and during droughts, though El Niño does not have a strong causality on the electricity market compared to Colombia.

Colombia

Colombia faces one of the largest energy-security concerns in the region: a hydro-dominant system that is facing climate volatility and declining domestic gas flexibility. Hydro accounts for about 75% of Colombia's generation, though the recent El Niño event in early 2024 drove reservoir levels to 28.6% by April of that year, only marginally above the 27% rationing-risk threshold. During that time period, hydropower output fell by more than 25% y/y, gas-fired generation and LNG imports rose significantly.

The new Hidroituango dam by EPM reduces but does not eliminate the security problem, since the country is exposed to severe drought events. The project is 2.4GW of generation capacity. At full operation, the project is expected to supply 13% of Colombia's energy depending on hydrology and demand growth. The project is currently operating at half capacity with four of eight units online and the second stage due by 2027. Gas is the marginal reliability fuel source for the country, but imported LNG economics are painful in dry years. Domestic gas production is declining, increasing the reliance on imports. Venezuelan gas is a potential long-term option, but a pipeline between the two countries is currently stalled.

Brazil

Brazil benefits from the broadest scale in the region plus regulated infrastructure depth. Brazil's electricity mix is almost 90% renewable, which is largely hydropower though with increasing contributions from other sources. Wind and solar together provide around 25% of generation. However, the country is facing increasing solar curtailment and transmission congestion relating to the rollout of these assets. Droughts have similarly impacted hydrological generation, which has impacted dispatch and power prices.

The country has focused on a broader transmission buildout to accommodate renewables growth and reduce congestion between resources and demand centres. The regulator, ANEEL, has a clear framework including inflation-linked, availability-style transmission cash flows, which has attracted investment into the sector. Axia Energia is the largest power company in LatAm. After its privatization from a state-owned company, the company has been simplifying its corporate structure, reducing contingent liabilities, improving operating efficiency, and accelerating investment in transmission. The company has ~44GW of installed generation capacity and nearly 75,000km of transmission lines, Axia is pursuing regulated transmission revenues and grid modernization to support long-term earnings growth integrating increasing volumes of renewable generation. The company is also migrating its hydro generation from legacy quota contracts into the free power market.

Asia Pacific Credit views

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Why our sector matters

Asia Pacific is the largest consumer of energy globally, accounting for 44% of global demand in 2024 and growing, according to the IEA⁸. The agency estimates that Asia's share will reach 46-47% by 2050, keeping the continent as the key driver of global energy demand for the foreseeable future.

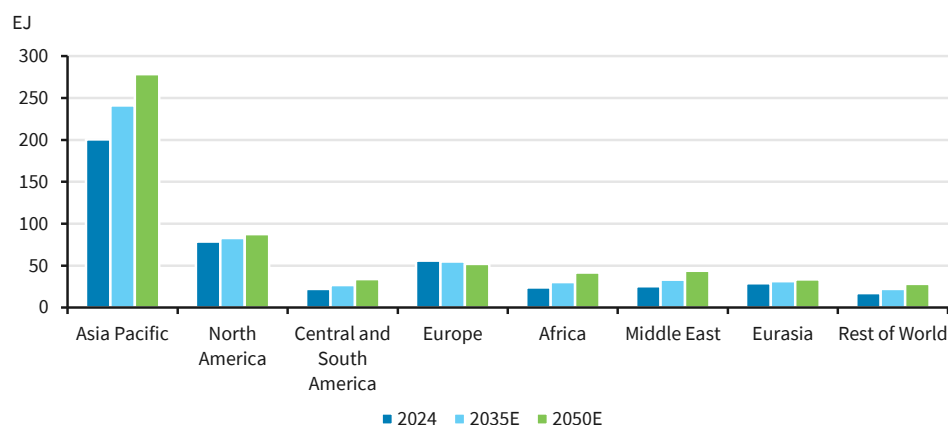
Within Asia, China will likely remain the largest consumer of energy in Asia in 2050. However, demand growth (8-16%) is expected to lag the broader region's 26-39% estimates. Instead, the IEA expects India and Southeast Asia to be the growth engines in Asia: India's demand is expected to grow 80-106% by 2050, with Southeast Asia's expected to grow 63-80%.

The IEA expects China's energy consumption growth to decelerate due to a combination of slower GDP growth expectations, a shift in growth from energy-intensive heavy industries such as steel and cement towards services and high value-added manufacturing, improved energy efficiencies, and a shrinking population. Under the IEA's Stated Policies Scenario, China's energy consumption is expected to peak in 2035 and plateau thereafter.

In contrast, energy demand growth in India, which is expected to be the largest globally, will be propelled by rising economic activity. This reflects India's being in an earlier stage of industrialisation than China, and the IEA expects its GDP to grow 6.1% annually to 2035. Population growth, infrastructure build-out, urbanisation, improved access to electricity and growing affluence will also contribute to energy demand growth.

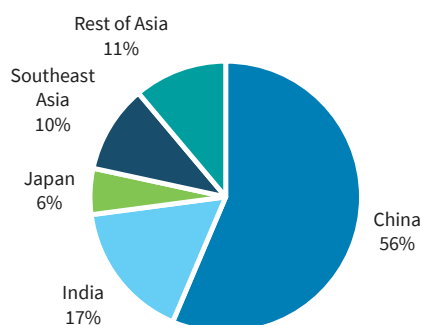
Similarly in Southeast Asia, energy demand is expected to be driven by economic growth, rising affluence and population growth. This is supported by its position as an alternative manufacturing hub to China, still-rising urbanisation rates, and progress towards universal electricity access.

FIGURE 130. Asia Pacific expected to remain the largest consumer of energy

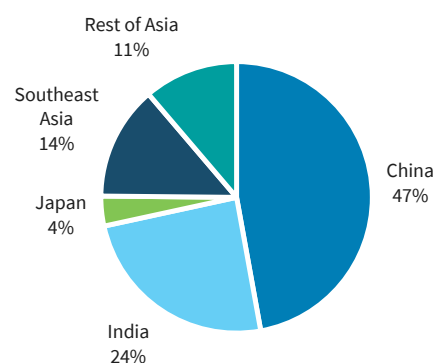


Note: 2035 and 2050 estimates are based on current policies.
Source: IEA, Barclays Research

⁸ World Energy Outlook 2025, International Energy Agency

FIGURE 131. China remains the largest energy consumer in Asia...

Asia's energy demand in 2024.
Source: IEA, Barclays Research

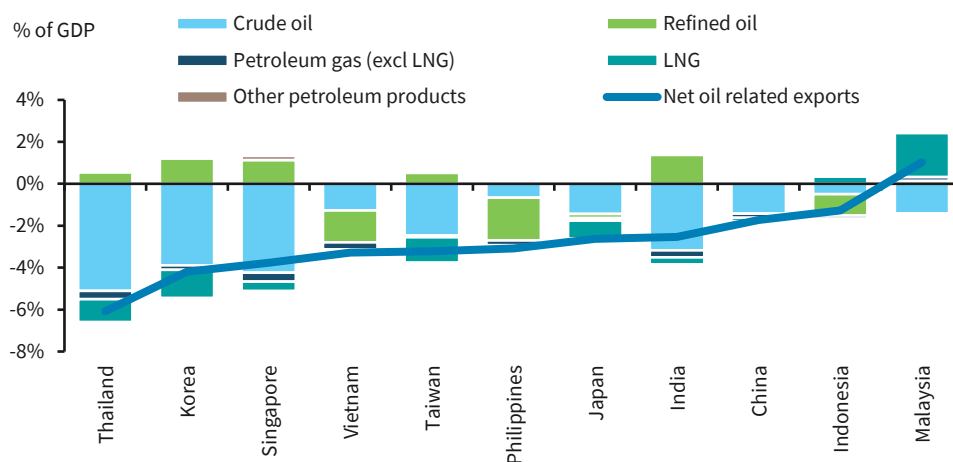
FIGURE 132. ...but India and Southeast Asia will be the growth engines

Asia's estimated energy demand in 2050.
Source: IEA, Barclays Research

Energy security and the return of capital

How is energy security changing capital allocation in each sector?

Key Asian countries are net importers of energy; in particular, Japan, the Philippines, Singapore, Korea, and Taiwan depend entirely on imports of oil and gas (Figure 133), with significant portions supplied from the Middle East.

FIGURE 133. Asia is broadly a net importer of energy

Note: We used HS codes 2709 for crude oil, 2710 for refined oil, 2711 excluding 271111 for petroleum gas, 271111 for LNG, and 2712 and 2713 as other petroleum products. India, Taiwan, Vietnam, and Philippines data are as of 2024.
Source: US ITC, Barclays Research

Asian countries have been able to overcome the near-term effects of oil and gas supply disruptions in the Middle East by sourcing alternative, higher-priced cargoes from other regions (the US, Africa and Asian energy exporters) and measures to reduce energy consumption. Asian countries have also substituted towards coal by raising the utilisation rates of existing coal-fired plants and restarting previously idled nuclear ones.

Longer term, we expect Asian countries to diversify their energy sources by:

- Reducing their dependence on energy products from the Middle East by diversifying purchases to other countries. The US, Australia and Africa will likely be beneficiaries of this

shift. Singapore is also likely to benefit as a key transshipment and refining hub for fuel products;

- Extending the useful lives of existing coal-fired plants and/or building new ones; and
- Increasing investments in renewable energy and nuclear power plants.

Australia a potential source of supply

The pivot away from Middle East energy supplies will encourage exploration and development activities, although realistically, there are few large opportunities available in the region.

Australia stands out as a potential source of new supply. It is already a key exporter of LNG to Asian countries, particularly in North Asia. Australia's largest upstream oil and gas operators, Woodside Energy (Market Weight) and Santos (Overweight), have projects that have come online recently or will do so over the next two years that will provide incremental supplies to the region in the near term (Barossa, Pikka, Scarborough, and Trion). Woodside is also developing the Louisiana LNG Project (LLNG) in the US and targeting first LNG in 2029. The majority of LLNG's future output has not been committed to-date.

Over a longer period, Woodside and Santos have developmental projects where the Final Investment Decisions (FID) could be brought forward: Browse, NWS Project Extension and Sunrise for Woodside; Beetaloo, Narrabri, Bedout Basin and Papua LNG for Santos. Environmental challenges and evolving regulations in response to energy security, including a potential scheme in Australia to reserve 20% of gas production for domestic use, will need to be considered as well.

If/when these projects are approved, the companies will likely require a combination of debt and equity funding, with detrimental effects on shareholder returns and credit quality.

Southeast Asia another potential source of oil and gas supply

Exploration and development activities in Indonesia, Malaysia and Vietnam are also potential sources of greenfield supply⁹. However, these projects are likely to be focused on strengthening domestic energy supply, with smaller proportions allocated towards exports.

On a smaller scale, Asian corporates have also acquired minority stakes in existing energy companies or producing assets either to benefit from the heightened focus on energy security or to secure energy supplies for their own use. These acquisitions have tended to be funded internally.

Clearly, Asian countries and corporates will allocate more capital to secure energy resources.

Coal power extension to benefit local producers and exporters of seaborne coal

In response to recent energy supply disruptions, South Korea lifted restrictions over coal power plants' utilisation rates and delayed the planned retirement of three coal-fired plants, while Taiwan sought to restart a retired coal-fired plant¹⁰. At the same time, China, India and Indonesia continue to construct new coal-fired plants.

The latter have substantial domestic coal supplies to support their power plants, although China and India also rely on seaborne coal imports and are among the largest buyers globally.

⁹ Asia Pacific Upstream 2026: Offshore Projects Ramp Up as Frontier Drilling Returns, 2 March 2026, *Offshore Engineer*

¹⁰ Asia's Gas Shock Playbook: Coal, Curtailment, and Competing for LNG, 9 April 2026, *The Center on Global Energy Policy at Columbia University*

All three countries have also continued to develop new coal mines, with China and India accounting for over 90% of new coal mine developments in Asia, according to EnergyTrackerAsia¹¹.

We expect these to be funded domestically through a combination of internal funds, state support and long-term lending led by policy banks, alongside major commercial banks.

Chinese authorities generally support such developments through policy facilitation, rather than direct fiscal investment, including expedited approvals, infrastructure provisions, and coordination with financial institutions.

Within Asia Pacific, Australia and Indonesia are the key exporters of thermal coal. However, there have been fewer greenfield thermal coal projects in Australia and Indonesia to date. Offshore funding is often difficult due to concerns over carbon emissions, although onshore funding channels remain open in Indonesia.

Renewables and nuclear as longer-term solutions

Asian countries have ambitious goals to grow their non-fossil fuel energy sources in order to address energy security concerns, meet growth in energy demand, and reduce carbon emissions.

China seeks to increase the share of non-fossil fuels to 25% by 2030 and more than 30% by 2035.

In its 15th Five-Year Plan for Renewable Energy Development (2026-30), released on 23 July, China targets about 3.5TW of installed renewable power capacity and 6,000TWh of annual renewable electricity generation by 2030, with combined wind and solar capacity exceeding 2.8TW, up from about 1.84TW in 2025, accounting for more than half of the country's installed power generation capacity. The plan also targets renewable energy consumption of approximately 1.8bn tonnes of standard coal equivalent (tce) by 2030, up from about 1.18bn tce in 2025, while placing greater emphasis on grid flexibility, storage and system reliability to improve renewable energy utilisation and support peak demand¹².

China's ability to execute its renewables roll-out at speed is underpinned by the country's dominance in the renewables value chain: it controls over 80% of installed manufacturing capacity for solar PVs and 70% for wind turbine blades. We expect the necessary investment to be funded primarily through internal funds, preferential green loans, and capital market instruments, including green bonds and renewable energy REITs. Government support is likely to be concentrated on policy facilitation, power grid and related infrastructure development, and initiatives to improve renewable energy integration and system reliability.

India's strong demand for energy means it will likely need more of every source.

Its renewable energy sector continues to play a central role in the country's clean-energy build-out and remains relatively defensive, supported by strong policy backing and long-term contracted cash flows. India is targeting 500GW of non-fossil fuel power generation capacity by 2030 and 50% of the generation mix to be renewable or non-fossil-based capacity. Installed renewable energy capacity totalled 274.7GW at end-March 2026. This included 150.3GW of solar and 56.1GW of wind generation capacity. Renewable capacity additions in FY26 totalled 54.6GW, up 24.8% y/y, and included 44.6GW of solar (+42.2% y/y) and 6.1GW of wind (+12.2% y/y). Capacity towards rooftop solar also increased in FY26, with 8.7GW added under PM Surya Ghar

¹¹ Asia's Proposed Coal Projects May Reverse Recent Declines, Significantly Increasing Methane Emissions, 18 August 2025, EnergyTrackerAsia

¹² China Aims to Boost Renewable Energy Consumption 53% by 2030, 23 July 2026, Bloomberg

Muft Bijli Yojana in FY26; this was a 51% increase y/y¹³. The scheme is aimed at accelerating residential solar adoption through government-backed subsidies.

Although capacity additions continue at pace, and the focus is shifting towards grid readiness, storage deployment and the roll-out of firm and dispatchable renewable energy (FDRE).

India's support for nuclear energy is also increasing through the Nuclear Energy Mission, which includes INR200bn for the development of small modular reactors (SMRs) and targets at least five indigenously developed SMRs by 2033. In addition, India has outlined a long-term nuclear expansion roadmap, targeting 22.4GW of nuclear capacity by FY2031-32 and 100GW by 2047, compared with c.8.8GW currently. The government is also seeking to attract greater private-sector participation and investment in nuclear power through ongoing policy reforms¹⁴. Private firms are expected to play a critical role by providing capital, developing captive small-reactor projects, expanding domestic manufacturing capacity and partnering with public sector enterprises in new reactor ventures. Key risks are not technological but execution-related, including land acquisition, project delays, fuel security, supply-chain development, financing costs and the challenge of deploying more than 90GW of new nuclear capacity over two decades, a pace far exceeding India's historical experience.

India's Union Budget 2026-2027¹⁵ includes a 30% y/y increase in allocation towards renewable energy, to INR329bn, with solar alone accounting for INR305bn, up 32% y/y. A further INR300bn was allocated towards power (+39% y/y) and INR241bn towards nuclear energy (unchanged y/y). The allocation underpins the government's focus on clean-energy deployment, rooftop solar and grid integration, although we expect execution bottlenecks to be the key driver of sector outcomes in the near term.

South Korea's 11th Basic Plan for Electricity Supply and Demand (2024-2038) aims to grow its electricity supply 71% over the next 15 years. The plan envisages new LNG power plants replacing retiring coal-fired plants. The plan also targets raising renewables' share of its power mix to 35%, from 25% currently, and nuclear to 35%, from 17%. Most of these capacity additions will be undertaken by Korea Electric Power Corporation (KEPCO, Underweight), which is majority-owned by the government. KEPCO is a regular issuer of USD bonds; we expect the company will need to step up its capital market activities to fund this capex.

Southeast Asian countries including Indonesia, the Philippines, Singapore and Vietnam are at various stages of considering nuclear power in their energy mix as a source of stable, low-emission baseload capacity. Most of these projects are led by the governments or their SOEs with foreign partners, and we expect the respective states to provide most of the funding.

How is energy security affecting volatility in each sector?

According to our economists, Thailand, Korea, Singapore, Taiwan, Vietnam, and Philippines are more exposed to the energy price shocks. The Philippines, Japan, Vietnam, Singapore, and Korea stand out in terms of dependence on the Strait of Hormuz for crude oil imports, while India and Singapore are most exposed for supply of LNG.

In the months following the Middle East conflict, Asian countries and corporates were able to source alternative fuel supplies at higher prices, particularly from the US, Africa and Asian energy exporters such as Malaysia. Clearly, the near-term focus has been on access to supply, rather than the cost of supply.

¹³ Year wise Achievements, *Ministry of New and Renewable Energy*

¹⁴ A New Chapter in India's Nuclear Journey, 7 April 2026, *Government of India*

¹⁵ Expenditure Budget 2026-2027, February 2026, *Government of India*

Longer term, we expect the reliability and stability of energy supply to become more important to Asian countries and corporates. Cost will remain an important but not the overriding consideration, in our view.

We also expect increased pressure to apportion energy supplies for domestic use. This is potentially detrimental to the region's energy producers if domestic prices are capped below export prices:

- Australia said it plans to reserve 20% of its LNG output for the domestic market from 1 July 2027¹⁶;
- Indonesia has asked its oil and gas companies to prioritise sales in the domestic market¹⁷;
- Indonesia is reviewing its existing Domestic Market Obligations (DMO) to boost domestic coal supply after recent shortages resulted in power outages¹⁸. As a positive, the review will also consider raising the existing cap on the domestic coal price to USD80/t, from USD70/t.

Winners and losers in a fragmented energy world

Among the USD sovereign bond issuers, we highlighted the potential effects of oil market disruption in [Sri Lanka](#), [Pakistan](#) and [Indonesia](#) due to higher oil import bills, the potential headwinds to economic activity and in the case of [Indonesia](#), its ability to sustain fuel subsidies if oil prices remain elevated for a prolonged period. Conversely, [Malaysia is a net exporter of energy in the region](#) and will benefit from an increased focus on energy security.

For Asia Pacific corporates, oil and gas companies are obvious beneficiaries of heightened energy security, although potential protectionist measures could limit the positive effect on earnings. Within the region, Australia's oil and gas companies stand out with their LNG-heavy production mix, the ability to increase export volumes materially with new projects coming online, and their proximity to Asia's energy importers.

Renewable power companies should benefit as well, especially if heightened focus on energy security translates into higher discharge tariffs for renewable power. China and India lead in both existing renewables capacities and growth plans. According to the IEA, China's renewables generation could almost triple in the next decade and quadruple by 2050. India's renewables generation is expected to quadruple in the next 10 years and increase tenfold by 2050. Southeast Asia's renewables generation is expected to double over the coming decade and quadruple by 2050, but would still pale in comparison to China and India.

Investments in renewable capacity should increase demand for energy storage systems (ESS) produced by Chinese and Korean battery makers, as well as battery materials producers (nickel, lithium, anodes and cathodes). In the EM Asia USD bond universe, key battery makers are Contemporary Amperex Technology Co Limited (CATL) in China and LG Energy Solution (LGES, Overweight) in Korea. Battery materials producers include Nickel Industries (Overweight NICAU '30s), LG Chem (Market Weight) and POSCO Holdings (Market Weight), although battery materials contribute only a small portion of LG Chem's and POSCO Holdings' total earnings.

On the flip side, increased electrification could translate into lower fuel demand, with a negative effect on oil refiners. Asian refiners and petrochemical companies that are highly dependent on imports are also vulnerable to supply disruption. In particular, [Korean refiners](#) and petrochemical companies import most of their feedstock from the Middle East.

¹⁶ Australia requires LNG exporters reserve 20% of gas for east coast market, 7 May 2026, *Reuters*

¹⁷ Indonesia asks LNG, crude oil producers to prioritize domestic market, 4 June 2026, *Gas Processing & LNG*

¹⁸ DEN Urges DMO Review as Blackouts Hit World's Biggest Coal Exporter, 25 June 2026, *Jakarta Globe*

National oil companies and power utilities that cannot promptly pass on higher fuel costs are potentially vulnerable to bond spread volatility. These include [KEPCO and Korea Gas](#) (Underweight) in Korea and [Pertamina](#) in Indonesia. KEPCO is also likely to incur higher capex and debt, as it is expected to lead Korea's power infrastructure build-out. However, as state-owned enterprises (SOEs), their ratings are equalised to their respective sovereign ratings and do not reflect their stand-alone credit profiles.

Coal-related corporates are potential losers under the IEA's estimate of lower future demand. These include miners in Indonesia (Indika Energy, Overweight INDYIJ '29s) and Australia (Whitehaven Coal). It would also include Australia's coal terminal and port operators (Newcastle Coal Infrastructure and Port of Newcastle).

AI, power demand and the new electricity economy

Beyond demographics, urbanisation and industrialisation, AI efforts in Asia will add to the region's energy demand, presenting upside risks to existing demand estimates:

- **China's** 15th Five Year Plan (2026-30), which builds on an earlier AI Plus initiative, prioritises developing AI capabilities and integration and targets a 70% AI penetration rate in key sectors by 2027 and 90% by 2030. State support includes a CNY1trn (USD148bn) fund to aid industries such as robotics, AI and innovative technologies¹⁹. China has also reportedly allocated CNY2trn (USD295bn) to build a network of interconnected data centres over the next five years²⁰.
- **India** hopes to attract more than USD200bn in AI infrastructure investments, with Reliance Industries (Market Weight) announcing a USD110bn commitment over seven years while the Adani Group pledged USD100bn²¹.
- **South Korea** announced [KRW2,061trn \(USD1.35trn\) of investment plans](#) across semiconductor, physical AI and data centres. The plans will double its semiconductor capacity in five years, while 8.4GW of power supply will come online by 2029 and a further 10GW by 2035 to support data centres.
- **Southeast Asia's** AI activities are more disparate. Singapore, which is land and energy constrained, had previously imposed a moratorium on new data centre construction but has since selectively approved new developments. Instead, it is emerging as a key regional headquarters for tech companies, a centre for AI innovation, and a financing hub for AI investments²². Malaysia is experiencing a boom in data centre construction, with capacity expected to double to 2.53GW by 2030²³, and data centres are expected to account for a third of Malaysia's electricity demand by 2035²⁴. Indonesia's large domestic market, with the world's fourth-largest population, provides an attractive market for AI-driven e-commerce. However, power reliability is a key challenge.

Besides adding new generation capacity, Asian countries are also pursuing the following measures to manage the expected increase in power demand:

¹⁹ Evaluating China's AI ambitions under the 15th Five Year Plan, 23 June 2026, *Bloomsbury Intelligence & Security Institute*

²⁰ Xi Promises AI for All in Debut at China's Top Tech Summit, 17 July 2026, *Bloomberg*

²¹ India To Invest \$200 Billion In AI Infrastructure: AI Impact Summit, 20 February 2026, *Forbes*

²² AI in Southeast Asia: An era of opportunity, February 2026, *McKinsey & Company*

²³ Beyond Scale. Kuala Lumpur as the Centre of Malaysia's Digital Infrastructure, 22 January 2026, *NextDC*

²⁴ Data centres to guzzle one-third of Malaysia's electricity by 2035, govt taking measures to ensure supply, 1 July 2026, *The Edge Malaysia*

- **China's** State Grid Corp plans to spend CNY4trn (USD574bn) on grid upgrades between 2026 and 2030, with a focus on strengthening its west-to-east transmission network with UHV lines²⁵. China Southern Power Grid has also budgeted CNY180bn for 2026 alone, with total investment over the next five years expected to reach about CNY1tn. Together, the two major grid operators are projected to invest close to CNY5tn during the 15th Five-Year Plan period²⁶.
- **India** plans to spend INR9trn to expand its transmission network (220kV and above). Its National Electricity Plan envisages adding 191,000ckm of transmission lines to its existing ~500,000ckm and 1,270GVA of transformation capacity²⁷ by FY2031-32. However, execution remains a key challenge. India's domestic rating agency, ICRA, reported that of the projects commissioned by March 2026, only ~12% were completed on schedule, with delays related primarily to land acquisition, right-of-way issues and regulatory approvals²⁸. Timely delivery of transmission infrastructure will therefore remain critical to avoiding evacuation bottlenecks and supporting the continued growth of non-fossil-fuel generation.
- **South Korea** is upgrading its transmission grid and deploying battery storage systems under its 11th Basic Plan for Electricity Supply and Demand, with KEPCO expected to spend KRW73trn (USD54bn) on grid upgrades through 2038²⁹.
- **Indonesia**, through its state utility Perusahaan Listrik Negara (PLN), intends to invest USD188bn between 2025 and 2034 on new capacity (USD131bn), new transmission (USD24bn) and distribution lines (USD11bn)³⁰.
- **Southeast Asian** countries are developing the ASEAN Power Grid (APG) to connect electricity transmission networks across countries in the region. The APG was launched in 1999; however, completion of the Laos-Thailand-Malaysia-Singapore Power Integration Project (LTMS PIP) in 2024 has enabled Laos to sell renewable (hydro) power to Malaysia and subsequently to Singapore by interconnecting through Thailand and Malaysia³¹. The World Bank estimates that full regional integration by 2045 will require USD800bn of funding³².

The race for capital: financing and investment ideas

Where is capital most needed and who can provide it?

Significant funding is clearly required across Asia Pacific. However, we expect sources to differ by country and type of borrower. We think South Korean, Australian and Indonesian corporates are most likely to access the offshore capital markets, while Chinese and Indian utilities are likely to rely on onshore funding:

- **China:** We expect financing from a combination of government funds, loans from supranationals (eg, World Bank), onshore lenders and offshore capital markets, including green bonds and renewable energy REITs. China has a deep onshore capital market and its corporates have good access to offshore public and private markets.

²⁵ China's power grid investments to surge to record \$574 billion in 2026-2030, 15 January 2026, *Reuters*

²⁶ China to invest 5 trillion yuan in power grid over next 5 years, 2 February 2026, *China Daily*

²⁷ National Electricity Plan (Transmission) launched by Cabinet Minister for Power and Housing & Urban Affairs, 14 October 2024, *India's Ministry of Finance*

²⁸ Transmission sector set to witness multi-year upcycle in capex, supported by Government's plan to evacuate power from over 900-GW non-fossil fuel capacity by 2035-36, 8 July 2026, *ICRA*

²⁹ South Korea to pour 73 trillion won into power infrastructure, 28 May 2025, *The Chosun Daily*

³⁰ Indonesia: Electricity grid in transition, April 2026, *Global Transmission Report*

³¹ Laos-Thailand-Malaysia-Singapore Power Integration Project (LTMS PIP): Challenges and Lessons Learned, October 2023, *Tenaga Nasional*

³² ASEAN Power Grid Financing (APGF) Initiative, 20 October 2025, *World Bank Group*

- **India's** funding for power infrastructure is likely to be a mix of public and private capital. We expect central and state transmission utilities to continue to undertake projects, while private participation is largely supported through tariff-based competitive bidding and long-term contractual frameworks. Power Finance Corporation (PFC) and REC Limited (REC) are likely to remain key providers of financing across India's power and transmission sectors. For nuclear power, in addition to public funding, we expect foreign capital to play a role through export-credit agencies, vendor financing and minority FDI participation.
- **South Korea** has a deep onshore capital market, although its corporates are also active issuers in the USD bond market. Cost of funding is a key consideration for Korean borrowers, and they often select the funding channels based on the prevailing cost.
- **Australia's** oil and gas companies have historically relied on the offshore USD bond market for funding, and we expect that to be the case if/when they need to fund capex.
- **Indonesia's** public utilities borrow in both onshore and offshore markets. However, we believe the former is not sufficiently deep to support the required funding. Hence, we expect the majority of fundraising to occur in the offshore markets, as well as loans from supranationals. Indonesia's private corporates, including coal miners, have historically relied on a combination of onshore bank loans and offshore bond issuances for funding.
- Most other **Southeast Asian** utilities rely on onshore funding, with occasional issuance in the offshore USD bond market.

Investment recommendations

Overweight Santos (STOAU, Baa3 Pos/BBB- Stb/BBB Stb)

Santos is Australia's second-largest oil and gas company. We believe its gas-heavy production mix (82% of 2025 revenues) and Australia's location in Asia Pacific mean that Santos is well positioned to benefit from Asia's efforts to diversify its energy imports from the Middle East and to meet the region's increasing demand for LNG as a key source of energy. Santos's output is expected to grow 25-30% between 2024 and 2027, after recently starting production at the new Barossa LNG and Pikka Phase 1 projects. The company also has a stake in the Papua LNG project (to be operated by TotalEnergies), which is targeting FID by year-end, while development options include the Beetaloo and Bedout Basins. Moody's has indicated that successful ramp-ups of the Barossa and Pikka Phase 1 projects could result in the company's ratings being upgraded. We believe S&P could similarly reassess its BBB- rating if these projects reduce the company's exposure in Papua New Guinea to below 40%, from 60% in 2025. See [Santos: Tale of two halves; remain OW](#) for details.

Overweight LG Energy Solution (LGENSO, Baa2 Stb/BBB Neg/NR)

LG Energy Solution is one of the largest non-Chinese rechargeable battery producers globally, with ~310GWh of production capacity. The company produces batteries for EVs, energy storage systems (ESS) and mobility/IT devices. In response to growth in demand for ESS, coupled with slower US EV sales, in 2025 the company converted some of its EV battery production capacity into ESS lines. Since then, it has secured more than KRW3trn (>USD2.1bn) of ESS contracts and plans to establish more than 50GWh of ESS production capacity in North America by end-2026. ESS currently accounts for a high 20% of its revenues, compared with single-digit percentage points a year ago. The company believes it has the largest local battery production capacity in North America and is well positioned to meet customers' requirement to source from non-prohibited foreign entities (PFEs). We believe LG Energy Solution will be a key beneficiary of growing demand for power solutions, especially from AI data centres. See [LG Energy Solution: Looking towards 2H with hope; remain OW](#) for details.

Buy PLN1J '36s

PLN is Indonesia's largest vertically integrated utility and is 100% owned by the government. The company plays a crucial role in electrification, where in accordance with the 2025-2034 Electricity Supply Business Plan (RUPTL) that targets to add 69.5GW in new capacity with a total investment of ~USD180bn, PLN will develop 20.4GW (29%) directly or via subsidiaries. The remaining is allocated to private IPPs. At the same time, PLN provides subsidized electricity domestically, for which it receives subsidy and compensation payments from the government. Positively, the cadence of compensation payments from the Government to PLN has improved significantly year to date, to a monthly frequency with payout of ~70% of claims, from a prior quarterly frequency. This has smoothed pressure on PLN's working capital cycle.

The PLN1J '36s have widened 29bp since issuance, more than the INDON '36s, which are 15bp wider over the same period. Given the PLN1J '36s relative underperformance, we believe current valuations at an OAS of 150bp and cash price of 94pts provide an attractive entry point and offer exposure into Indonesia's electrification and energy security theme. The improved compensation cycle will support compression of the bond towards the INDON '36s, in our view.

Buy PETMK '50s

We believe Petronas is well positioned to benefit from longer term efforts from countries to diversify energy imports from the Middle East, given that the upstream and gas segment are the main contributors to Petronas' profit after tax, at ~50% each, and due to the geographic distribution of Petronas' proven oil and gas reserves with ~58% located in Malaysia as of start 2026.

The company's liquidity position is robust with cash balance of MYR204bn at end FY25 and limited upcoming bond maturities (USD500mn bond maturity in October 2026 and USD600mn in 2028). Positively, scheduled dividend payments to the Malaysian Government of MYR20bn in 2026 are also lower than the MYR32bn in FY25 and FY24, though we [acknowledge the potential of higher than slated dividend payments](#). The PETMK '50s are quoted at yield of 6%, cash price of 82pts, and OAS of 67bp.

Buy PHILIP 5y CDS

Within Asia, the Philippines stands out for its structural vulnerability to disruptions to the reliability and stability of energy supplies, given that the country is reliant on imports for most of its crude oil (96%), gasoline (66%), and diesel (71%) needs. Moreover, the country's CPI basket is one of the most sensitive amongst Asia peers to increases in crude oil prices, with our prior estimates indicating that CPI is expected to increase by 0.3pp per USD10/bbl increase. Reflecting these vulnerabilities, the PHILIP 5Y CDS has been amongst the most sensitive within HG Asia sovereigns to increases in oil prices this year, with the CDS soaring to highs of ~98bp in March 2026 from pre US-Iran war levels of roughly ~60bp. At current levels of ~67bp, we think the PHILIP 5y CDS is an attractive hedge.

EEMEA Corporates - Credit views

[Stella Cridge](#)⁽ⁱ⁾ Barclays, UK | [Luc Roberts](#)⁽ⁱ⁾ Barclays, UK | [Evgeniia Bystrova](#)⁽ⁱ⁾ Barclays, UK

Importers and exporters set their strategies

Across Emerging EMEA, economies can be generally categorised into exporters of upstream oil and gas products, and those that largely import to meet their domestic needs. Large economies in the region that have high import dependency (and which have a large number of corporate bond issuers) include Turkey, while exporters include multiple Sub Saharan African countries (eg, Angola, Nigeria) and those in the Middle East (UAE, Qatar, Saudi Arabia). Given the heightened global risks around fuel security in recent years, as well as pressures on the external

positions of those countries with high import dependency, a number of countries have put in place strategic plans to address such risks. These developments have implications for corporates in these countries. We assess those with the most market relevance below.

Israel: Investing further in the energy transition

Israel has found significant quantities of gas in the last two decades and has since brought onstream three large gas fields in Israeli coastal waters: Tamar, Leviathan and Karish (see [Leviathan Bond: Concentration risks overshadow strong track record](#), 18 October 2023). These discoveries, and subsequent commercialisation, have led to a move away from coal and gas imports for both power and industry. During 2025, Israel Electric (IEC) – the state-owned electricity company (active in generation, transmission and distribution) – for instance, generated 75% of power from gas and 25% from coal versus 51% and 43% in 2020, and plans the full phase out of coal in the future. The commercialisation and transition in the energy and power sectors was partly funded through issuance in the USD bond market, including from IEC, Energean (operator of Karish) and Leviathan Bond (share of the project owned by parent NewMed Energy). Having boosted domestic fuel security, and even exported excess gas within the region (to Jordan and Egypt – exports are set to grow in the coming years), Israel is now taking the next step in its energy transition by investing in grid capacity to permit further roll out and usage of renewable power in the country. IEC plans to invest \$2.5bn per year from 2026-2030 in transmission and distribution to enable the integration of renewables onto the grid as well as energy storage facilities, electric vehicles and AI. Separately, we expect issuers such as Leviathan/NewMed and Energean to remain active in the bond market to fund future investments in upstream reserves and resources (see [Israel Corporates: Moving forward post H1 disruptions: snapshot pack](#), 2 July 2026).

GCC: Two-pronged approach to investment in hydrocarbons, new technologies

GCC countries have an outsized role in global energy supply, accounting for a quarter of global crude oil exports and a third of proven reserves (and c.20% for natural gas). Their long-term strategies hence balance the importance of the sector for their economies and state budgets, with opportunities to participate in the energy transition both through natural gas as a transition fuel and investments in renewables. Saudi Aramco recently brought online the first phase of its \$100bn Jafurah shale gas project, which is set to boost gas availability for the domestic market, while QatarEnergy had been aiming to almost double LNG capacity by 2030, prior to the onset of the Iran war. The UAE's recent exit from OPEC+ is expected to lead to oil capacity expansion³³, while ADNOC Gas has announced it expects to spend c.\$28bn between 2026 and 2030 to deliver on growth plans. Although the Iran war has disrupted key export routes and delayed major projects, we expect resilience to increase in the coming years as new re-routing projects are brought online (although we see LNG exports as more vulnerable, see [QatarEnergy's LNG facilities extensively damaged: assessing credit buffers](#), 19 March 2026).

Collectively, these projects point to a large financing pipeline for state-owned hydrocarbon producers in the GCC. Healthy cash flows and large cash balances should provide significant funding capacity. However, the simultaneous execution of expansion plans and spend to bolster export routes could sustain external financing needs and drive ongoing issuance in bond markets.

The region has also been an increasing source and destination for investment in renewable projects, and more recently data centres. For instance, Masdar, Abu Dhabi's dedicated green energy investment vehicle, has been aggressively deploying capital globally to build a 100GW renewables portfolio by 2030 (gross basis). Several large-scale renewable projects have been built in the region, with further capacity under construction, while regional demand growth is

³³ Bloomberg, 'UAE Oil Head Says OPEC Exit Gives Ability to Speed Up Investment', 5 April 2026.

being driven by AI, data centres, and advanced manufacturing. This includes a 5.2GW solar and battery storage project in Abu Dhabi which is set to deliver 1GW in continuous power (this is notably a key requirement for energy supply to data centres). Saudi Arabia is also undertaking significant investments and is aiming to generate 50% of its electricity from renewables under its Vision 2030 strategy, with the remaining portion set to come from natural gas, displacing existing oil/liquid fuels capacity entirely. Growing investment in green projects has been accompanied by green labelled financing across GCC Corporates and Financials (see [GCC Credit: UAE hosts COP28: Green bond issuance in the spotlight](#), 30 November 2023).

Sub Saharan Africa: Tackling domestic power challenges as well as export opportunities

Multiple Sub Saharan African (SSA) countries are large exporters of upstream oil and gas products, including Nigeria (1.8mbpd) and Angola (1.1mbpd). There are also some smaller (eg, Ghana, c100kbpd) and emergent exporters such as Senegal (c70kbpd), which saw BP and Kosmos Energy commence LNG exports from the Greater Tortue Aymeyim (GTA) field in 2025. Amid the war in Iran, SSA exporters saw strong demand for their upstream output as a result of the closure of the Strait of Hormuz and we expect ongoing support as global importers seek to ensure reliable supplies of products. At the same time, some of these countries have relatively underdeveloped domestic power infrastructure and a significant need to improve grid availability (eg, Nigeria, which currently has high reliance on diesel generators) and to reduce import costs (eg, Senegal, currently reliant on coal powered electricity plants). This need creates opportunities for EEMEA corporates to benefit from demand growth, including Seplat Energy in Nigeria, which is set to see gas output grow in the coming years (see [Seplat Energy: A Nigeria growth story: Initiate at Market Weight on tight valuations](#), 8 May 2026) and entities such as Kosmos Energy in Senegal, which could see higher future gas revenues if GTA can secure gas sales by pipeline to onshore Senegal (see [Kosmos Energy: GTA: An asset with optionality](#), 19 January 2026).

Turkey: Renewable investments supporting energy independence goals

Turkey remains heavily dependent on imported hydrocarbons, including for natural gas, oil and coal, which has historically weighed on the country's current account balance. The geopolitical developments in the EEMEA region in 2022-2026 have also put an extra emphasis on the importance of Türkiye's energy security and independence ([Türkiye Corporates: Assessing gas flow risks](#), 26 March 2026). Consequently, expanding domestic renewable energy generation has become a key element of the country's energy strategy. Supported by market liberalisation, feed-in tariffs and dedicated renewable energy support mechanisms, Turkey has significantly increased renewable capacity over the past decade, with renewables now representing around 60% of installed capacity. Imported thermal sources now represent c30% of installed generation capacity. The supportive regulatory framework has also encouraged a growing number of Turkish corporates to invest in renewable energy assets, both as a source of cost savings and as part of broader sustainability and decarbonisation objectives ([EM Corporate Days 2026: Adapting to new challenges](#), 25 June 2026). For example, Limak Cement has an intensive capex programme targeting decarbonisation, including through renewable investments, which it financed via the bond issuance.

Focus on sustainability and emissions remains among EEMEA corporates

Outside of those companies that are directly exposed to the energy and power sector, our 7th annual Emerging Markets Corporate Days: A Sustainable Future ([EM Corporate Days 2026: Adapting to new challenges](#), 25 June 2026) held in June 2026 showed that EEMEA corporates continue to execute plans across sectors to reduce emissions and improve the cost and reliability of power. For instance, in the telecoms sector in Africa, firms continue to seek to replace diesel generator usage with more grid access and use of renewable power and battery

technology. In Ukraine, firms continue to invest in greater self-generation amid significant attacks on power assets. In countries such as Kazakhstan, where the sovereign has set national level targets, key quasi-sovereign entities are executing their transition to renewable power from fossil fuels. Lastly, even in countries where the transition is challenging, such as South Africa given high domestic reliance on coal (the bulk of the grid is coal-powered), there has been significant investments in renewables (solar, wind) both from national level IPP investment programmes and among multiple private sector corporates that have been replacing power purchases from state-owned Eskom in order to reduce cost and improve reliability. (see [Sustainable Investing Research: South Africa's big re-wire](#), 14 January 2026).

Trade ideas

Buy AYDEMT 2030s (Evgeniia Bystrova): The Turkiye Corporates credit complex provides several opportunities to gain exposure to the growing renewables sector, for example via Aydem Renewables (MW), which is one of the largest pure renewable generators in the country. While we view valuations of AYDEMT 30s as broadly fair due to the balance of risks, the bond provides an attractive carry opportunity at c10% yield in the context of EM USD Corp-Quasi index and in a wider global context ([Aydem Renewables: Limited deleveraging prospects balanced by high carry](#), 17 August 2026).

Buy AZURGY 2030s (Stella Cridge): Azule Energy (Overweight) is an Angola-focused upstream oil and gas company (shareholding 50/50 BP and Eni) which produces over 200kbpd per day of output. Angola is one of Africa's largest suppliers of oil to the global market (c1.1mbpd), and Azule is likely to benefit from recent new discoveries and investment plans that can boost its reserve base and support future output levels (see [Azule Energy: Q2 26 results: Step down in net debt q/q; stay Overweight](#), 28 August 2026)

. Azule's three bonds have been underperforming the Angola sovereign in spread terms, but offer compelling value on account of its low leverage (0.8x at H1 26 or 1.1x, adjusted) and FX protections (revenues received offshore), which insulate against sovereign risks. We recommend adding the AZURGY 2030s, which are most attractive on the curve relative to ANGOL.

Buy KOS (GTA) 11.25% 2031s (Stella Cridge): Kosmos Energy (Market Weight) is an upstream oil and gas company with operations across Ghana, North America (Gulf) and Mauritania/Senegal. Given prolonged disruptions to output in Qatar, we expect that there will be stronger long-term demand for supply of LNG projects outside of the Gulf region including the Greater Tortue Ahmeyim (GTA) project of KOS (c27% stake, alongside partners BP and the state oil companies of Senegal and Mauritania). GTA's capacity is currently around 3.0mtpa (gross), and margins in the business are currently tight for KOS given the fixed cost base. However, there are prospects to improve throughput through the sale of gas onshore to both Senegal and Mauritania, which are taking steps to prepare infrastructure onshore to receive gas (see [Kosmos Energy: Q2 26 results: Positive momentum towards refinancing](#), 4 August 2026). KOS as a company has relatively high leverage, which is a credit risk, but the KOS-GTA bonds have greater recourse to the operations in Mauritania/Senegal and hence benefit from selected covenant protections (see [Kosmos Energy: GTA: An asset with optionality](#), 19 January 2026).

Part 3: Equity and Credit Strategy views

US Equity Strategy

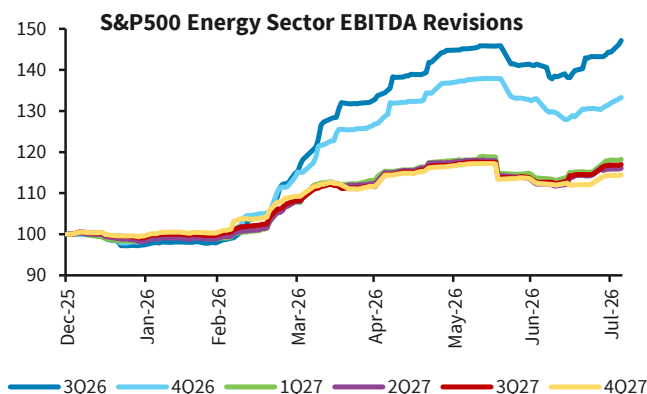
Venu Krishna, CFA BCI, US

Energy has been the best-performing sector in the S&P 500 YTD, reflecting investors' growing appreciation for a world characterized by geopolitical fragmentation, supply chain reconfiguration and increasing concern about resource scarcity. We upgraded the sector to Neutral in March, in deference to a potentially sticky geopolitical risk premium, and relative returns have been broadly flat since then, as oil price retracement from April highs was offset by exceptionally strong 2Q26 results. Energy sector EPS jumped triple digits y/y and O&G names are contributing more to FY26 index earnings revisions than Banks, Tech Hardware and Software combined—try imagining that nine months ago.

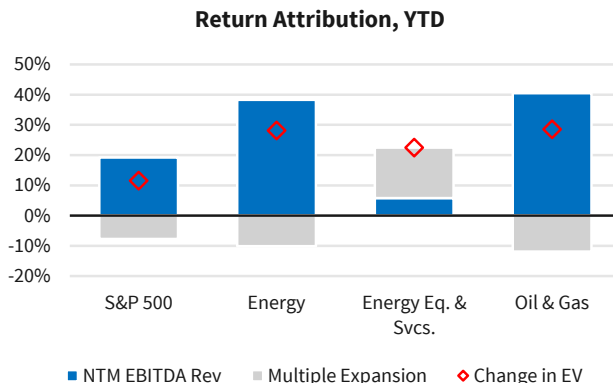
We share the view of our co-authors that the global energy system is likely to become more capital intensive, strategically important and politically consequential over the coming decade than many investors currently assume. The combination of rising power demand, electrification, AI-driven infrastructure build-out and increasing emphasis on energy security argues for a substantial acceleration in energy-related capex, which should create compelling opportunities across individual companies exposed to energy production, transportation, storage and grid modernization.

That being said, our sector views are more nuanced. The persistent relative outperformance of the Energy sector has generally coincided with sustained periods of high and rising oil prices. While our commodity team's forward strip view into 2030 is above consensus, that forecast also comes with the assumption that oil price volatility will remain high, given the state of geopolitical impairments. Supply chain normalization, increased spare capacity and eventual inventory rebuilding could meaningfully compress the risk premium for Energy equities. Moreover, ~65% of the Energy sector is Integrated/E&P names. Capital discipline in this space remains markedly stronger than in prior cycles, with healthier balance sheets and more attractive shareholder distributions. Still, investors are ultimately underwriting mature cash flow businesses whose earnings power remains heavily commodity leveraged, in a market where flows and generalist portfolio allocations have been shown to favor higher-growth compounders.

Ultimately, we agree that the broad energy ecosystem is in the midst of a growth phase and that accelerating electrification, AI and data centre expansion, and optimization for resilience instead of cost are likely to exacerbate energy bottlenecks and necessitate substantial investment throughout the power chain. But in terms of translating to equity returns, individual stocks and sub-industries within Energy likely offer more attractive opportunities to position for outperformance compared with a broad allocation to the Energy sector as a whole. At the sector level, we remain more constructive on Utilities, Industrials and Tech, where leverage to power transmission and generation, grid modernization, infrastructure build-out, electrification, power management and battery technologies offer a more widespread opportunity set.

FIGURE 134. Energy earnings estimate revisions are strongest for 2H26, but 2027 is heading higher as well

Source: Bloomberg, Barclays Research

FIGURE 135. With the exception of OFS, Energy sector returns have been driven by earnings revisions more than multiple expansion

Source: Bloomberg, Barclays Research

European Equity Strategy

Emmanuel Cau, CFA Barclays, UK

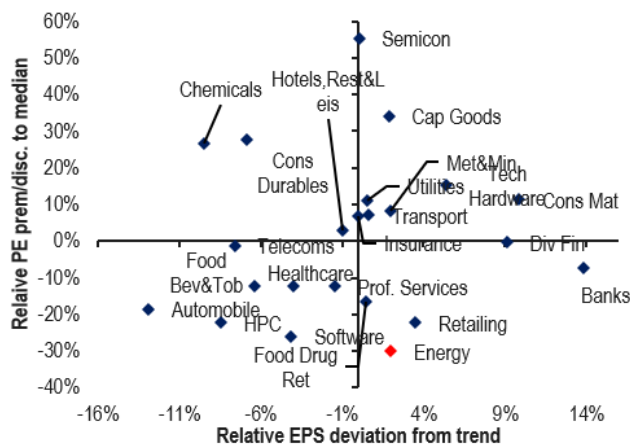
Energy's second coming. The Energy sector has come back into focus after spending much of the past decade out of favour. The narrative on peak oil and the rise of renewables led many investors to conclude that fossil fuels were facing a secular decline. As a result, the sector endured years of underinvestment, while investor exposure to the space steadily diminished. In some cases, ESG mandates further constrained capital allocation. However, we think the backdrop is changing. As our macro colleagues have argued, the world is moving into an era of energy addition, rather than substitution. Yes, renewables continue to gain share, but the mix of rising energy demand and supply risk means investment is required across generation, transmission, grids, storage and flexible power sources. The challenge is no longer simply replacing one source of energy with another, but expanding the entire energy system to support a more digital, electrified and resilient economy.

For Europe, this is increasingly an issue of both economic growth and security. The Russia-Ukraine war exposed the vulnerabilities created by years of underinvestment and dependence on external suppliers, while more recent tensions in the Middle East have reinforced the value of a secure and reliable energy supply chain. Energy security has therefore shifted from a policy objective to an economic imperative and is increasingly becoming a core pillar of strategic autonomy. At the same time, the investment case extends well beyond geopolitics. AI-related power demand, electrification, industrial reshoring and decarbonisation are creating multiple structural drivers of energy investment. Consistent with this backdrop, our economists expect a multi-year investment cycle to remain firmly in place, forecasting average annual investment growth of more than 6% in the US and 4.5-5% in Europe and the UK (see [Supersize me: The coming investment cycle](#), 26 Mar). While AI spending has already driven a significant acceleration in US corporate capex, Europe remains much earlier in the capex cycle. Meanwhile, improving credit demand for fixed investment backed up by supportive fiscal policy suggests the European investment cycle still has scope to broaden.

From an equity market perspective, the broader investment theme is well understood and many beneficiaries have already re-rated (see [Strategic autonomy – not cheap, but more to go](#) 22 Apr). Energy, Utilities, Capital Goods, Construction Materials and Banks have generally outperformed, as investors increasingly price in higher investment intensity and stronger

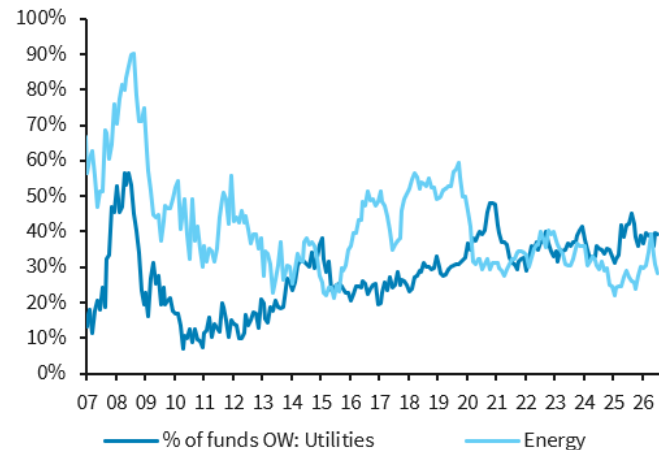
earnings growth. However, the earnings backdrop continues to justify much of this performance, with all the aforementioned sectors still delivering EPS growth above their long-term trends, as they keep benefiting from improving order books. While parts of the strategic autonomy trade now screen as expensive, albeit justifiably, the traditional Energy complex (and Banks) is unique in that it continues to trade below historical valuation averages. In our view, this offers a more attractive risk-reward profile within the broader investment theme. Near term, we prefer Oil Services and Utilities (OW) and remain Market Weight on Integrated Energy, reflecting its higher sensitivity to oil prices and the binary nature of US-Iran news flow. However, we see growing medium-term re-rating potential, given supportive structural tailwinds and still-muted investor positioning.

FIGURE 136. Energy continues to grow earnings above trend, while still trading cheaper than long-term history



Source: IBES, LSEG Data & Analytics, Barclays Research

FIGURE 137. Energy ownership remains depressed in portfolios, while Utilities is also below post-COVID highs



Source: EPFR, LSEG Data & Analytics, Barclays Research

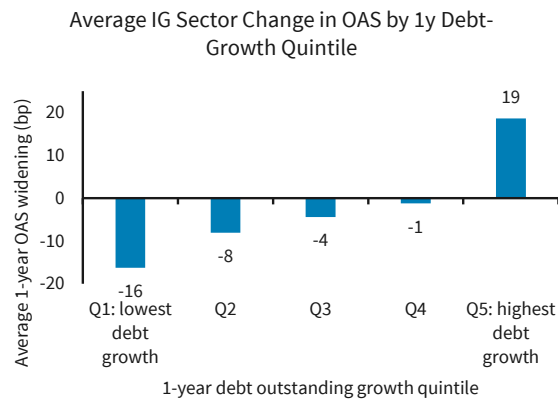
US Credit Strategy

Dominique Toublan⁽ⁱ⁾ BCI, US

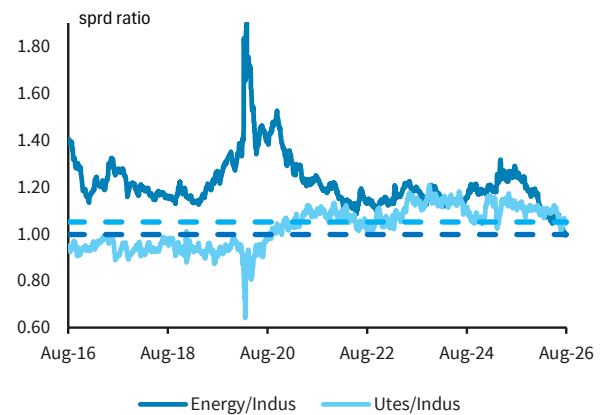
Large debt-funded capex puts pressure on spreads. History suggests markets do not wait for leverage to deteriorate before repricing sectors entering major investment cycles. TMT issuance surged more than 300% from 1999 to 2001, and Energy issuance increased about 50% from 2013 to 2015. In both cases, spreads drifted roughly 40bp wider versus the index as investors adjusted to a new issuance regime (see [here](#)).

More broadly, we have found that debt growth matters most at the extremes. As shown in [Figure 138](#), historically, sectors in the highest quintile of debt growth widened nearly 20bp over the following year, while every other quintile tightened. Put differently, rapid debt growth usually carries a spread tax (see [here](#) for more details).

Current valuations leave little room for forgiveness, especially for the Energy sector. IG Energy currently trades at its richest level versus IG Industrials in more than a decade, with the spread ratio sitting just below 1.0x ([Figure 139](#)). That richness was earned through years of balance sheet repair, capital discipline, limited issuance and higher energy prices recently. The challenge is that a significant buildout of LNG, gas infrastructure, and power related investment points to a rising cadence of debt issuance ahead. Utilities face an even steeper financing curve, which argues for continued relative underperformance there. Strong fundamentals can support tight spreads, yet history suggests sectors embarking on large capital cycles tend to underperform the rest of the index over time.

FIGURE 138. Extreme debt growth has historically led to sector widening

Source: Bloomberg, Barclays Research

FIGURE 139. The IG Energy sector trades at its tightest level vs. IG Industrials in more than a decade

Source: Bloomberg, Barclays Research

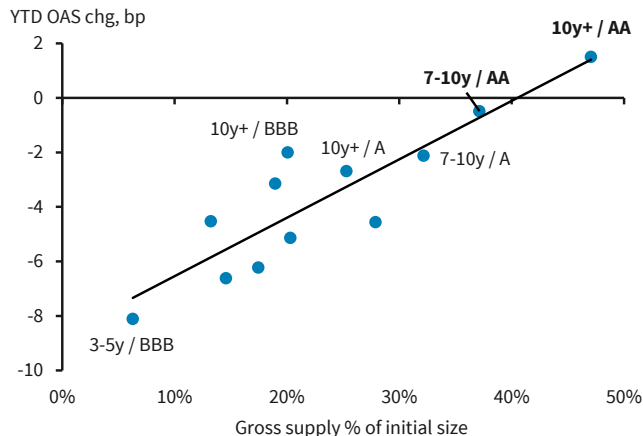
European Credit Strategy

Soren Willemann⁽ⁱ⁾ Barclays, UK

When equity investors hear "capex cycle", credit investors hear "supply headwinds". As illustrated clearly for the US credit market in x, sectors with the largest growth rate in supply tend to underperform. But we do not have to look into the history books to see how this affects European credit: as we discuss in [Supply Pressure Points](#), in 2026 so far there has been a clear imprint in YTD spread performance, with rating/maturity segments with the largest amounts of relative supply underperforming ([Figure 140](#)). We see little reason why a "supply super cycle" in Utility/Energy sectors should be any different in terms of acting as a localized, technical, headwind. Supply in these sectors will also likely come in the same segment, as is currently seen for hyperscalers: relatively high rated and longerdated. As such, these two pockets of prolific issuers should compete against each other for demand. This reinforces our [preference](#) for shorter-dated, lower-rated credit within IG.

"Things could be different for these sectors", and we are Overweight the Energy and Utility sector, for a few reasons. As discussed by our fundamental analysts, fundamentals are generally strong for these sectors and capex build-outs will be disciplined. The key difference from the hyperscaler supply cycle is that Energy/Utility sectors are - and will be - supported by the government, in terms of either government liquidity support or ownership stakes: "nationalisation" is a scary word for an equity investor, but absent haircuts to bonds, there is usually upside for a credit investor. Another aspect that will likely support these sectors, leaning against a technical headwind, is that they will likely maintain their non-cyclical attributes, with fundamentals less affected by any growth downside, unlike, say, hyperscalers. Finally, unlike the setup on the \$IG side, €IG valuations relative to the broader market are either mid-range for Energy or fairly wide for Utilities ([Figure 141](#)).

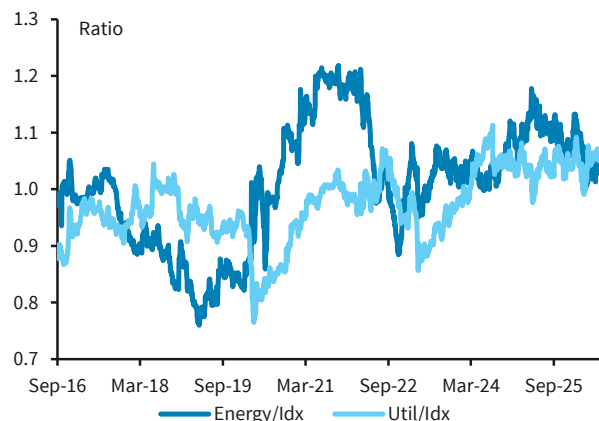
FIGURE 140. Rating/maturity segments with the highest supply rates have had the worst performance YTD in €IG



YTD supply by segment in pct of initial market size vs. YTD OAS changes. Senior non-financial bonds only.

Source: Dealogic, Barclays Research

FIGURE 141. €IG Energy is mid-range vs. the broader market on a ratio basis, while Utilities are at the wider end, 80th percentile on a 10-year horizon



Source: Bloomberg, Barclays Research

Case study - Physical AI and energy

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AI gets physical, powered by energy

The investment debate about robotics has so far largely focused on chips and AI models. Yet as AI moves from software into the physical world, energy is becoming increasingly important. Every humanoid robot, autonomous vehicle, warehouse robot and drone ultimately converts electricity into movement. As a result, the rise of physical AI could create a new source of electricity demand that extends well beyond data centres and into factories, warehouses and logistics networks. Put differently, physical AI is effectively creating a new category of energy consumer: machines capable of performing physical labour.

Robots are a new category of energy consumer

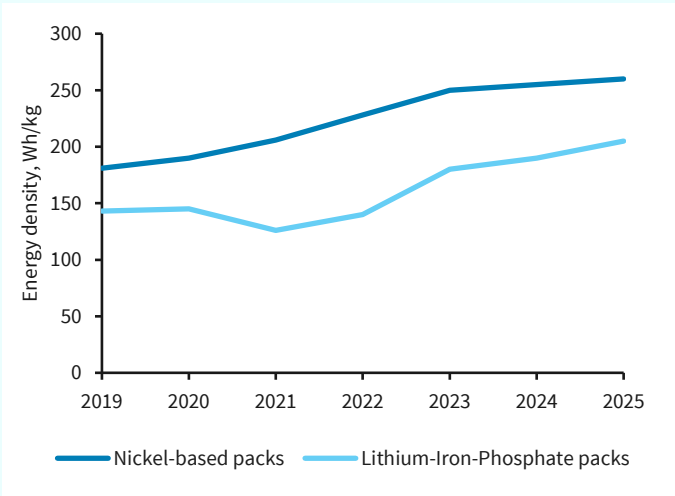
Physical AI is still a small market today, but our research suggests it could evolve into a **\$1trn ecosystem by 2035**, spanning humanoids, autonomous vehicles, industrial automation and drones. As deployment accelerates, investors may need to think about robotics as not only a technology theme, but also an energy theme. Unlike traditional software, physical AI creates a **two-layer energy demand**: cloud compute to train increasingly complex models and local power to run robots performing tasks in the physical world.

Training physical AI models is particularly energy intensive because they must learn not only reasoning, but also physics, motion, balance, navigation and manipulation, much of it through large-scale simulations and digital twins. Advanced AI model training currently consumes energy at a rate of 100-154MW, with projections exceeding 4GW by 2030. While data on physical AI models remain limited, we expect their energy requirements to be even greater, given the added complexity of learning and executing tasks in the physical world.

The second layer of energy demand comes from **physical execution**. Once deployed, robots continuously consume electricity through motors, sensors, processors and communication systems. Unlike large language models, robots must operate in the physical world, overcoming gravity, inertia and friction. Put simply, movement is energy intensive, making batteries, typically lithium-ion, a critical enabling technology.

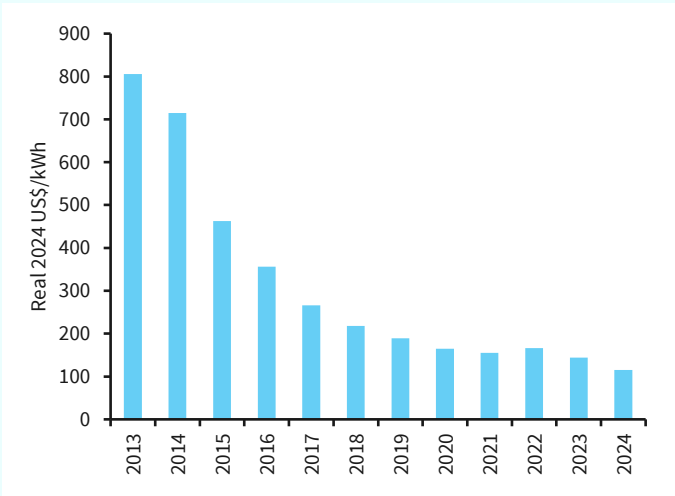
Batteries are improving, but remain a constraint. Battery density has increased by about 50% since 2019 (Figure 142), while battery prices have fallen almost eightfold since 2013 (Figure 143). These advances have been critical in making mobile robotics economically viable. Yet batteries remain one of the sector's biggest constraints.

FIGURE 142. Energy capacity per kg of battery pack has steadily improved



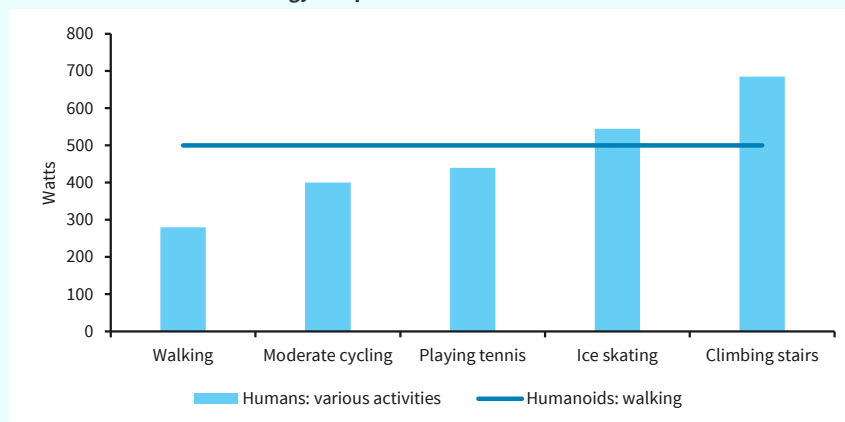
Source: Bloomberg BNEF (Bloomberg New Energy Finance), Barclays Research

FIGURE 143. Lithium-ion battery prices have dropped significantly over the past decade



Source: Bloomberg BNEF (Bloomberg New Energy Finance), Barclays Research

Current humanoids typically operate for only 4-5 hours before requiring recharging or battery replacement. More intensive tasks can reduce operating time further. Even highly advanced humanoids remain significantly less energy efficient than humans. We estimate that a humanoid such as Tesla Optimus consumes roughly 500 watts while walking, whereas a human performing the same activity is 40-45% more energy efficient (Figure 144).

FIGURE 144. Humanoids' energy footprint

Source: AlphaSense, Douglas College Physics 1104 Custom Textbook, Barclays Research

From labour substitution to electricity substitution

Historically, economic growth has relied on expanding labour inputs. Robotics changes that equation. Our research finds that humanoids could eventually help offset demographic pressures in ageing economies by substituting for labour in manufacturing, logistics, healthcare and agriculture. **In China alone, an upper-bound scenario suggests humanoids could reach the equivalent of 3.8% of labour capacity by 2035**, helping offset a significant share of projected workforce declines.

However, replacing labour with robots does not eliminate resource consumption. Instead, it changes its composition. Physical work that was once powered by food and human labour increasingly becomes powered by electricity, batteries and semiconductors. In effect, robotics converts labour demand into energy demand.

Appendix

FIGURE 145. Federal Environmental Reviews and Authorizations

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Authorization for Liquefied Natural Gas Terminal Facilities, Onshore or in State Waters	FERC	Office of Energy Projects and Office of Energy Market Regulation	Liquefied Natural Gas Terminal Facilities (Onshore or in State Water), and associated Natural Gas Pipelines	Application for the siting, construction, expansion, or operation of an LNG terminal filed pursuant to section 3 of the Natural Gas Act	Applications for the siting, construction, expansion, or operation of an LNG Terminal must be submitted to FERC. An LNG Terminal includes all natural gas facilities located onshore or in State waters that are used to receive, unload, load, store, transport, gasify, liquefy, or process natural gas that is: (1) imported to the U.S. from a foreign country; (2) exported to a foreign country from the U.S.; or (3) transported in interstate commerce by waterborne vessel.	Section 3 of the Natural Gas Act [15 U.S.C. 717b]; 18 C.F.R. 153. Section 7(c) of the Natural Gas Act [15 U.S.C. 717f]; 18 C.F.R. 157.
Bald and Golden Eagle Protection Permit	DOI	FWS	All	Possible take/harm of eagles	The Bald and Golden Eagle Protection Act prohibits anyone from "taking" bald or golden eagles. Among other actions, "take" includes disturbance of eagles to the degree that it substantially interferes with breeding, feeding, or sheltering behavior or results in injury. The FWS can issue a permit for taking eagles when the take is associated with, but not the purpose of, an activity and cannot practicably be avoided. We refer to this type of take as "non- purposeful take." Authorization is subject to conditions to minimize impacts. The regulation authorizing non- purposeful take permits for bald and golden eagles is in 50 CFR 22.26.	Bald and Golden Eagle Protection Act
Business Resource Lease	DOI	BIA	Wind: Other than Federal Offshore Wind: Federal Offshore Solar Geothermal	Contract approved by the Secretary that authorizes possession of Indian land for a specific purpose and term.	Lease development activity to allow the possession of Indian land for wind or solar development or business purposes in exchange for pre- defined royalties.	25 CFR 162 - Leases and Permits. 25 CFR § 162.565 - 162.599 for Wind Energy Evaluation Leases (WEEL) and Wind Solar Resource (WSR) Leases. 25 USC 380, 393, 394, 397, 402, 403 and 415
Certificate of Public Convenience and Necessity for Interstate Natural Gas Pipelines	FERC	Office of Energy Projects and Office of Energy Market Regulation	Interstate Natural Gas Pipelines	Application for a certificate of public convenience and necessity to construct or extend interstate natural gas pipeline filed pursuant to section 7(c) of the Natural Gas Act	Applications must be submitted to FERC for a certificate of public convenience and necessity for the construction or extension of natural gas facilities used for the sale or transportation of natural gas in interstate commerce. No construction or extension of natural gas facilities that are subject to the jurisdiction of FERC can be undertaken unless there is a certification of public convenience and necessity issued by FERC authorizing such acts or operations. As provided by FERC regulations, environmental reviews are conducted for interstate natural gas pipelines and related jurisdictional facilities, including underground storage and LNG peak-shaving facilities.	Section 7(c) of the Natural Gas Act [15 U.S.C. 717f]; 18 C.F.R. 157. See also 18 C.F.R. 153.2, 380.5.

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Clean Water Act Section 402 Permit, National Pollutant Discharge Elimination System (EPA)	EPA or a State regulatory agency having delegated authority under CWA Section 404(b) State Permit Programs	EPA Region or a State regulatory agency having delegated authority under Section 404(b) State Permit Programs; NPDES permits are issued by states that have obtained EPA approval to issue permits or are issued by EPA Regions in states without such approval.	All	Proposed projects that would discharge pollutants regulated under the Clean Water Act through a point source of discharge into a water of the United States. Pollutant is defined as any type of industrial, municipal, or agricultural waste discharged into water.	Any person who discharges or proposes to discharge pollutants... and who does not have an effective permit, except persons covered by general permits under 40 CFR 122.28, excluded under 40 CFR 122.3, or a user of a privately owned treatment works unless the Director requires otherwise under 40 CFR 122.44(m), must submit a complete application to the EPA Region or delegated State Agency in accordance with 40 CFR 122 and 40 CFR 124, or in accordance with NPDES regulations of delegated State Agencies under 40 CFR 123.	Clean Water Act Section 404; 40 CFR 122.21 Application for a permit (also applicable to State programs, see 40 CFR 123.25).
Commercial Use Permit	DOI	BLM	Geothermal	Commercial use of federal lands for geothermal energy development	Prior to the commercial generation of electricity, an application for a Commercial Use Permit is submitted on a Sundry Notice form and approved by BLM prior to any commercial operations from a Federal geothermal lease, a Federal Unit, or a production facility. This document describes the location of all wells, allocation of that production, location of meters, allocation of schedules, power purchase agreement, and a number of other items. It generally takes about a month for BLM approval of a Commercial Use Permit.	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy and Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)
Conditional Letter of Map Revision	DHS	FEMA	All	Activity that modifies the floodway by causing any increase in the base flood discharge in regulatory floodways, or more than one foot increase in any portion of a Special Flood Hazard Area if no regulatory floodway is designated.	A Conditional Letter of Map Revision (CLOMR) is FEMA's comment on a proposed project that would, upon construction, affect the hydrologic or hydraulic characteristics of a flooding source and thus result in the modification of the existing regulatory floodway, the effective Base Flood Elevations (BFEs), or the Special Flood Hazard Area (SFHA). The letter does not revise an effective NFIP map, it indicates whether the project, if built as proposed, would be recognized by FEMA. FEMA charges a fee for processing a CLOMR to recover the costs associated with the review. Building permits cannot be issued based on a CLOMR, because a CLOMR does not change the NFIP map.	44 CFR 60.3; 44 CFR 65.7
Construction and Operations Plan	DOI	BOEM	Wind: Other than Federal Offshore Wind: Federal Offshore	Construction and operations of a commercial wind energy project on the OCS.	Prior to beginning construction activities on an OCS wind lease, a lessee must have an approved construction and operations plan (COP). The COP must describe all proposed activities including construction, commercial operations, and conceptual decommissioning of all facilities. BOEM regulations at 39 CFR 585.620 describe the information that must be in the COP.	Outer Continental Shelf Lands Act 43 U.S.C. 1337(p) and BOEM regulations at 30 CFR 585

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Development and Production Plan	DOI	BOEM	Offshore Oil & Gas	Development and production of oil and gas on the Outer Continental Shelf	Prior to developing outer continental shelf oil and gas resources, a lessee must have an approved development and production plan. The plan must describe the objectives and tentative schedule for start to completion. BOEM regulations at 30 CFR 550.242 describe the information that must accompany the plan. BOEM conducts both technical and environmental reviews of the plan. The plan must meet adequate provisions for safety, environmental protection, and conservation of natural resources and comply with the Outer Continental Shelf Lands Act and implementing regulations, and other Federal laws.	Outer Continental Shelf Lands Act 43 U.S.C. 1351 and BOEM regulations at 30 CFR 550
DOD Military Mission Impact Process	DOD	DOD	All	Developer files action with FAA or contacts DoD Siting Clearinghouse for a preliminary review	OE/AAA process protects national airspace from safety of flight hazards	
DOD Mission Compatibility Evaluation Process, Part 211 of Title 32 CFR	DOD; DOT	The DoD Siting Clearinghouse, established under the Deputy Under Secretary of Defense (Installations & Environment)	All	If the Secretary of Transportation decides that constructing or altering a structure may result in... after consultation with the Secretary of Defense, an adverse impact on military operations and readiness, the Secretary of Transportation shall conduct an aeronautical study to decide the extent of any adverse impact on the safe and efficient use of the airspace, facilities, or equipment.	A formal DOD review of a proposed project begins with the receipt from the Secretary of Transportation by the DOD Clearinghouse of a proper application filed with the Secretary of Transportation pursuant to 49 U.S.C. 44718... Not later than 30 days after receiving the application from the Secretary of Transportation, the Clearinghouse shall evaluate all comments and recommendations received and take one of three actions: (i) Determine that the proposed project will not have an adverse impact on military operations and readiness, in which case it shall notify the Secretary of Transportation of such determination. (ii) Determine that the proposed project will have an adverse impact on military operations and readiness but that the adverse impact involved is sufficiently attenuated that it does not require mitigation. When the Clearinghouse makes such a determination, it shall notify the Secretary of Transportation of such determination. (iii) Determine that the proposed project may have an adverse impact on military operations and readiness.	DOD Mission Compatibility Evaluation Process, 32 CFR § 211.6 Initiating a formal DoD review of a proposed project; 49 U.S.C 44718, Structures interfering with air commerce or national security

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Easement Administrative Action (USDA - NRCS)	USDA	NRCS	All	Identification of land use easements issued under the Agricultural Conservation Easement Program and the Healthy Forests Reserve Program during scoping by proponent, lead or cooperating agencies	Project lands may be held under easements through NRCS programs and identified such that any potential impacts may be mitigated during the Environmental Analysis. The Agricultural Conservation Easement Program (ACEP) helps landowners, land trusts, and other entities protect, restore, and enhance wetlands, grasslands, and working farms and ranches through conservation easements. Under the Agricultural Land Easements component, NRCS helps American Indian tribes, state and local governments and non-governmental organizations protect working agricultural lands and limit non-agricultural uses of the land. Under the Wetlands Reserve Easements component, NRCS helps to restore, protect and enhance enrolled wetlands.	2018 Farm Bill

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Endangered Species Act Consultation (DOI-FWS)	DOI	FWS	All	The proposed action may affect threatened/ endangered species and/ or their designated critical habitat	Under Section 7 of the ESA, Federal agencies must consult with NMFS and/or the USFWS when an action the agency carries out, funds, or authorizes (such as through a permit) may affect a listed endangered or threatened species and/or designated critical habitat. The purpose of the consultation is to assist the action agency in meeting its duty to insure that its action is not likely to jeopardize the continued existence of any listed species or result in the destruction or adverse modification of designated critical habitat. Upon completion of consultation the USFWS and/or NMFS provides the action agency with either a letter of concurrence that the proposed project is not likely to adversely affect any listed species or a biological opinion. If the biological opinion concludes that the project is not likely to jeopardize the continued existence of any listed species and is not likely to result in destruction or adverse modification of critical habitat for any listed species, the USFWS and/or NMFS also provides an incidental take statement including terms and conditions, which if complied with, provide an exemption from the ESA's prohibitions on incidental take of listed species. If the biological opinion concludes that the proposed action is likely to jeopardize the continued existence of any listed species (or result in the destruction or adverse modification of designated critical habitat), the USFWS and NMFS develop reasonable and prudent alternatives that would avoid jeopardy or adverse modification that are shared and discussed with the action agency prior to completion of the biological opinion. NOAA generally manages marine species. The USFWS generally manages freshwater and terrestrial species. NOAA and the USFWS share responsibility for sea turtles, Atlantic salmon, and gulf sturgeon.	Endangered Species Act (16 USC 1531- 1544) Focus on ESA Section 7(a)(2), ESA Section 9, and ESA Section 7(a)(1) Responsibilities of Federal Agencies To Protect Migratory Birds (EO 13186)

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Endangered Species Act Consultation (NOAA-NMFS)	DOC	NOAA - NMFS	All	The proposed action may affect threatened/ endangered species and/ or their designated critical habitat	Under Section 7 of the ESA, Federal agencies must consult with NMFS and/or the USFWS when an action the agency carries out, funds, or authorizes (such as through a permit) may affect a listed endangered or threatened species and/or designated critical habitat. The purpose of the consultation is to assist the action agency in meeting its duty to insure that its action is not likely to jeopardize the continued existence of any listed species or result in the destruction or adverse modification of designated critical habitat. Upon completion of consultation the USFWS and/or NMFS provides the action agency with either a letter of concurrence that the proposed project is not likely to adversely affect any listed species or a biological opinion. If the biological opinion concludes that the project is not likely to jeopardize the continued existence of any listed species and is not likely to result in destruction or adverse modification of critical habitat for any listed species, the USFWS and/or NMFS also provides an incidental take statement including terms and conditions, which if complied with, provide an exemption from the ESA's prohibitions on incidental take of listed species. If the biological opinion concludes that the proposed action is likely to jeopardize the continued existence of any listed species (or result in the destruction or adverse modification of designated critical habitat), the USFWS and NMFS develop reasonable and prudent alternatives that would avoid jeopardy or adverse modification that are shared and discussed with the action agency prior to completion of the biological opinion. NOAA generally manages marine species. The USFWS generally manages freshwater and terrestrial species. NOAA and the USFWS share responsibility for sea turtles, Atlantic salmon, and gulf sturgeon.	Endangered Species Act (16 USC 1531- 1544) Focus on ESA Section 7(a)(2), ESA Section 9, and ESA Section 7(a)(1) Responsibilities of Federal Agencies To Protect Migratory Birds (EO 13186)
Fish and Wildlife Coordination Act Review (DOI - FWS)	DOI	FWS	All	Coast Guard permits, Corps Section 10/404/103 permits, EPA Section 402 permits, FERC licenses, NRC power station licensing, BR water resource projects.	The amendments enacted in 1946 require consultation with the Fish and Wildlife Service and the fish and wildlife agencies of States where the "waters of any stream or other body of water are proposed or authorized, permitted or licensed to be impounded, diverted . . . or otherwise controlled or modified" by any agency under a Federal permit or license. Consultation is to be undertaken for the purpose of "preventing loss of and damage to wildlife resources."	Fish and Wildlife Coordination Act 16 U.S.C. 661- 667e

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Fish and Wildlife Coordination Act Review (NOAA)	DOC	NOAA - NMFS	All	Coast Guard permits, Corps Section 10/404/103 permits, EPA Section 402 permits, FERC licenses, NRC power station licensing, BR water resource projects, other actions dependent upon or resulting in the diversion, control or modification of a stream or other body of water.	The amendments enacted in 1946 require consultation with the National Oceanic and Atmospheric Administration (NOAA)*, the Fish and Wildlife Service, and the fish and wildlife agencies of States where the "waters of any stream or other body of water are proposed or authorized, permitted or licensed to be impounded, diverted . . . or otherwise controlled or modified" by any agency under a Federal permit or license. Consultation is to be undertaken for the purpose of "preventing loss of and damage to wildlife resources." * Congress has not amended the FWCA since 1958, and, therefore, the Act does not refer to NOAA or the National Marine Fisheries Service (NMFS) or expressly refer to the functions of the Secretary of Commerce. However, the Reorganization Plan of 1970 confers the transfer of functions, including the authorities under FWCA, to NOAA and the Secretary of Commerce.	Fish and Wildlife Coordination Act 16 U.S.C. 661- 667e
Floodplain Assessment	All	All	All	Potential to impact floodplain areas	Consideration of floodplain values	Section 404 of the Clean Water Act (water quality impacts), Endangered Species Act – Section 7 (habitat impacts), Executive Order (E.O.) 11988—Floodplain Management (May 24, 1977), Executive Order 11990—Protection of Wetlands (May 24, 1977)
Form 3200- 9, Notice of Intent to Conduct Geothermal Resource Exploration Operations	DOI	BLM	Geothermal	Geothermal exploration on federal lands	Form 3200- 9, Notice of Intent to Conduct Geothermal Resource Exploration Operations	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)
Geothermal Drilling Permit (GDP)	DOI	BLM	Geothermal	Drilling on federal lands (for geothermal resources)	"3261.11 - How do I apply for approval of drilling operations and well pad construction? (a) Send to BLM: (1) A completed and signed drilling permit application, Form 3260- 2; (2) A complete operations plan (3261.12); (3) A complete drilling program (3261.13); and (4) An acceptable bond (3261.18). (b) Do not start any drilling operations until after BLM approves the permit."	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)
Geothermal Exploration Bond	DOI	BLM	Geothermal	Geothermal exploration, drilling or utilization operations on federal lands	Form 3000- 004a, Geothermal Exploration Bond	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Geothermal Lease	DOI	BLM	Geothermal	Lease confers right to explore for, drill, produce and utilize geothermal resources on federal lands with limitations. Subsequent permits required prior to any surface disturbing activities. Lands must be made available for this use in BLM Land Use Plan.	Geothermal Lease Form (Form 3200- 24a). Nomination of Lands for Competitive Geothermal Leasing (Form 3203- 1). Conduct auction. Non-competitive leases available for two years if not purchased at auction (Form 3200- 24a).	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)
Geothermal Project Utilization Plan, Facility Construction Permit, DOI and Site License		BLM	Geothermal	Once a decision is made to construct a facility to utilize the geothermal resource and construct electric generation facilities. A site license is required of operators who are not the geothermal lease holder to operate the facility.	A Utilization Plan (no form, 43 CFR 3272) describes how the geothermal resource will be used, including all of the proposed structures and facilities and their locations necessary for plant operations, as well as plans for final reclamation and decommissioning. These documents are submitted as part of the NEPA process. The Utilization Plan also includes the construction permit for construction and operation of the electric generation facilities, direct- use steam plants, and related facility and well field operations, including well field production and injection. The site license is required if the operator is not party to the geothermal lease.	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)
Geothermal Sundry Notice	DOI	BLM	Geothermal	Surface disturbance activities prior to obtaining exploration or drilling permit *or* Change to surface use plan or drilling plan	To begin surface disturbance activities prior to obtaining a Geothermal Drilling Permit for the purpose of constructing roads, pads, etc.	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)
Lease of Power Privilege (DOI-BOR)	DOI	BOR	Federal Hydropower Non- Federal Hydropower - Leases	Authorization of hydropower development at certain facilities owned by the Bureau of Reclamation	Non- federal entities who develop hydropower resources at Reclamation facilities where Reclamation has authority to develop power, must execute a Lease of Power Privilege. The lease term may extend for up to 40 years. Reclamation has issued a directive and standard for Reclamation's Lease of Power Privilege that outlines the roles, responsibilities and process for obtaining a Lease of Power Privilege.	The Town Sites and Power Act of 1906; Reclamation Project Act of 1939 (43 USC 485h); Contributed Funds Act of 1921, 43 U.S.C. sec. 395
Loan Guarantee Program, Title XVII of EP Act 2005	DOE	Loan Program Office	Renewable Energy Production (all)	Entrance into loan guarantee underwriting due diligence	The Title XVII innovative clean energy projects loan program (Title XVII) provides loan guarantees to accelerate the deployment of innovative clean energy technology. The U.S. Department of Energy is authorized to issue loan guarantees pursuant to Title XVII of the Energy Policy Act of 2005. Loan guarantees are made to qualified projects and applicants who apply for funding in response to open technology- specific solicitations.	Title XVII of the Energy Policy Act of 2005 (42 USC Sec. 16511, et. Seq)/10 CFR Part 609

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Loan Program, Advanced Technology Vehicle Manufacturing	DOE	Loan Program Office	Manufacturing (all)	Entrance into ATVM loan underwriting due diligence	The Advanced Technology Vehicles Manufacturing (ATVM) direct loan program was established in Section 136 of the Energy Independence and Security Act of 2007 to support the production of fuel-efficient, advanced technology vehicles and qualifying components in the United States. The ATVM loan program provides direct loans to automotive or component manufacturers for reequipping, expanding, or establishing manufacturing facilities in the U.S. that produce fuel-efficient advanced technology vehicles or qualifying components, or for engineering integration performed in the U.S. for advanced technology vehicles or qualifying components.	Section 136 of the Energy Independence and Security Act of 2007, as amended (42 USC 17013)/10 CFR Part 611
Magnuson-Stevens Fishery Conservation and Management Act, Section 305 Essential Fish Habitat (EFH) Consultation	DOC	NOAA - NMFS	All	Projects in any sector with potential impact to essential fish habitat	Under the Magnuson Stevens Act (MSA) all Federal agencies must consult on any action they authorize, fund, or undertake, or propose to authorize, fund, or undertake if they determine their actions may adversely affect essential fish habitat (EFH). Permits are not issued under this authority. NOAA provides non-binding conservation recommendations to Federal action agencies to minimize the project's potential adverse affect on EFH.	Magnuson- Stevens Fishery Conservation and Management Act (16 USC 1801- 1891(d)) Section 305(b)
Marine Mammal Protection Act (MMPA) Incidental Take Authorization	DOC	NOAA - NMFS	All	Projects in any sector that may result in potential 'take' as defined (16 U.S.C. 1362 Sec.3(13) defines as harass, hunt, capture, or kill, or attempt to harass, hunt, capture, or kill; (18) further defines 'harassment' as any act of pursuit, torment, or annoyance which (i) has the potential to injure or has the potential to disturb by causing disruption of behavioral patterns...)	The Marine Mammal Protection Act (MMPA) prohibits the take of marine mammals. Federal agencies and individuals must seek authorization to incidentally take marine mammals when conducting otherwise lawful activities. Two types of incidental take permits can be issued: Incidental Harassment Authorizations (IHAs) and Letters of Authorization (LOAs). IHAs are issued for actions that do not have the potential to cause marine mammal mortality or serious injury. An LOA is required for actions that have the potential to cause mortality or serious injury.	Marine Mammal Protection Act (16 USC 1361- 1423)

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Migratory Bird Treaty Act permits	DOI	FWS	All	The proposed action may take birds.	The Migratory Bird Treaty Act makes it illegal for anyone to take any migratory bird or nests, or eggs of such a bird except under the terms of a valid permit issued pursuant to Federal regulations. The migratory bird species protected by the Act are listed in 50 CFR 10.13. Take means to pursue, hunt, shoot, wound, kill, trap, capture, or collect, or attempt to pursue, hunt, shoot, wound, kill, trap, capture, or collect. Take incidental to an action (non-purposeful take) is not permitted.	Migratory Bird Treaty Act of 1918 (16 U.S.C. 703- 712; Ch. 128; July 13, 1918; 40 Stat. 755)
National Marine Sanctuaries Act, Issuance of a General Permit or Authorization of a Permitted Activity	DOC	NOAA - National Ocean Service	All	National Marine Sanctuary General permit: Any activity that is prohibited by the national marine sanctuary regulations may be conducted in a national marine sanctuary if it is conducted in accordance with the scope, purpose, terms and conditions of a general permit. Authorization: A newly proposed activity that would be covered by a federal, state, or local permit, and would occur in a national marine sanctuary, but would be otherwise prohibited by the national marine sanctuary regulations and unable to qualify for an National Marine Sanctuaries Act general or special use permit.	National Marine Sanctuary General Permit: A person may conduct an activity prohibited by 15 CFR part 922, subparts F-O, if such activity is specifically authorized by and provided such activity is conducted in accordance with the scope, purpose, terms and conditions of, a National Marine Sanctuary General permit issued under 922.48. For the Florida Keys National Marine Sanctuary, the prohibited activity must be conducted in accordance with a general permit issued under 922.166. For Thunder Bay National Marine Sanctuary and Unwater Preserve, the prohibited activity must be conducted in accordance with a general permit issued under 922.195. The sanctuaries have three categories of general permits; they include: management, education, and research permits. Authorization: An authorization (15 C.F.R. 922.49) provides a sanctuary with the power to "authorize" or give legal or official approval to another applicable federal, state, or local permit to allow an activity otherwise prohibited by sanctuary regulations and unable to qualify for an ONMS general or special use permit. As of March 2016, six national marine sanctuaries—Florida Keys, Flower Garden Banks, Monterey Bay, Olympic Coast, Stellwagen Bank, and Thunder Bay—have the ability to issue authorizations. The ONMS considers the general permit regulatory review criteria at 15 C.F.R. pt. 922 when deciding whether to issue an authorization, but does not require that all criteria be met.	National Marine Sanctuary General Permit: 15 C.F.R. §§ 922.48, 922.166, and 922.195 Authorization: 15 C.F.R. § 922.49

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
National Marine Sanctuaries Act, Section 304(d) Consultation	DOC	NOAA - National Ocean Service	All	Federal actions that are likely to destroy, cause the loss of, or injure a national marine sanctuary resource	Under section 304(d) of the National Marine Sanctuaries Act (16 U.S.C. § 1434(d)), any Federal agency taking any action that is likely to destroy, cause the loss of, or injure a sanctuary resource is required to consult with NOAA. At the earliest practical time before final approval of the action, the Federal agency shall provide NOAA with a written statement describing the action and its potential effects. The term Federal action includes private activities authorized by licenses, leases, or permits. Sanctuary resources are defined as any living or nonliving resource that contributes to the conservation, recreational, ecological, historical, educational, cultural, archeological, scientific, or aesthetic value of the sanctuary. Upon review of the sanctuary resource statement, NOAA may issue recommendations to protect sanctuary resources by eliminating, reducing, or mitigating potential injury to sanctuary resources.	National Marine Sanctuaries Act, § 304(d) (16 U.S.C. §1434(d)); Oceans Act Section 2202(e) (for Stellwagen Bank National Marine Sanctuary Federal Agencies shall consult on activities that may affect sanctuary resources)
National Marine Sanctuaries Act, Special Use Permit, as defined in Section 310	DOC	NOAA - National Ocean Service	Offshore Broadband Infrastructure	The continued presence of commercial submarine cables on or within the submerged lands of any national marine sanctuary.	Under section 310 of the National Marine Sanctuaries Act (16 U.S.C. § 1441), the ONMS can issue a special use permit and collect fees for a limited number of activities that either “establish conditions of access to and use of any sanctuary resource” or “promote public use and understanding of a sanctuary resource.” Each of the seven categories for a special use permit must also meet four legislative “permit terms,” one of which is a finding that the project will not destroy, cause the loss of, or injure sanctuary resources. Sanctuary resources are defined as any living or nonliving resource that contributes to the conservation, recreational, ecological, historical, educational, cultural, archeological, scientific, or aesthetic value of the sanctuary. Here, the ONMS can issue a special use permit, "that establish[es] conditions of access to and use of any sanctuary resource," for the continued presence of commercial submarine cables on or within the submerged lands of any national marine sanctuary.	National Marine Sanctuaries Act, 16 U.S.C. § 1441; 78 FR 25957 (list of applicable special use permit categories of activities)

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Native American Graves Protection Act Compliance	All	All	All	Intentional excavation of Native American cultural items, including human remains and objects of cultural patrimony.	For activities on Federal lands, NAGPRA requires consultation with &; Indian tribes (including Alaska Native villages) or Native Hawaiian organizations prior to the intentional excavation, or removal after inadvertent discovery, of several kinds of cultural items, including human remains and objects of cultural patrimony. For activities on Native American or Native Hawaiian lands, which are defined in the statute, NAGPRA requires the consent of the Indian tribe or Native Hawaiian organization prior to the removal of cultural items.	Native American Graves Protection Act
Natural Gas Export Authorization	DOE	Office of Regulation and International Engagement	Liquefied Natural Gas Terminal Facilities - Onshore or in State Water Liquefied Natural Gas Terminal Facilities - Offshore	Application for export of LNG	Under Section 3 of the Natural Gas Act (NGA), DOE authorizes imports and exports of natural gas, including LNG and CNG. As needed, these imports and exports must be supported by natural gas commodity pipelines and gasification and liquefaction projects that involve large capital expenditures. FERC regulates most such projects under sections 3 and 7 of the NGA. But, if the projects are located in deepwater ports, MARAD is the principal regulatory authority. If the projects cross federal lands, BLM at the Department of Interior is the principal regulatory authority. Within the Department of Transportation, PHMSA is responsible for enforcing safety standards on both gas and oil pipelines.	Section 3 of the Natural Gas Act [15 U.S.C. 717b]
Non- Federal Hydropower Licenses	FERC	Office of Energy Projects	Non- Federal Hydropower - Licenses (including Non- Federal Marine and Hydrokinetic Projects)	License application submitted to FERC in accordance with Part I of the Federal Power Act	An application must be submitted to the Commission for a license to construct, operate, and maintain a non- federal hydropower project that would: (1) be located on navigable waters of the United States; (2) occupy lands or reservations of the United States; (3) use surplus water or water power from a Government dam; or (4) be located on non- navigable waters that are subject to the authority of Congress under the Commerce Clause, affect the interests of interstate or foreign commerce, and involve construction on or after August 26, 1935. FERC is authorized to issue original licenses, and new licenses following the expiration of an existing license (i.e., "relicenses").	Section 4(e), 15(a)(1), and 23(b) of the Federal Power Act [16 U.S.C. 797, 808, and 817, respectively]; 18 C.F.R. 4, 5, and 16.
Non- Impairment Determination (separate from NPS permit)	DOI	NPS	All	Any project crossing NPS jurisdictional lands for which a NPS permit would be required	NPS is required to by statute to preserve its lands unimpaired for the enjoyment of future generations. NPS is required to make a determination as to whether a project would impair park system resources before taking an agency action, including issuing permits.	National Park Service and Related Programs (commonly known as the NPS Organic Act), 54 USC 100101 et seq

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Notice of Proposed Construction - Form 7460	DOT	FAA	All	Construction/alteration of structures >200 ft or near airports, depending on distinction and length of nearby runway; siting within radar line- of- sight of air surveillance or communications facility	Any person/organization who intends to sponsor construction or alterations that fall under activities listed under CFR Title 14 Part 77.9 must notify the Administrator of the FAA via a Notice of Proposed Construction - Form 7460. This notice includes a plan for appropriate markings and lighting, after which FAA makes a Hazard Determination for the proposed construction.	Structures interfering with air commerce (49 USC 44718)
NPS Permit	DOI	NPS	All	Projects to be located on NPS lands	Permits for rights- of- way, easements or other non- park uses: Informational requirements are determined on a case- by- case basis, and applicants should consult with the Park Superintendent before making formal application. The applicant must provide sufficient information on the proposed non- park use, as well as park resources and resource- related values to be affected directly and indirectly by the proposed use in order to allow the Service to evaluate the application, assess the impact of the proposed use on the NPS unit and other environmental values, develop restrictions/stipulations to mitigate adverse impacts, and reach a decision on issuance of the instrument. Policies concerning regulation of special uses are described in the NPS Management Policies Notebook.	Rights- of- way through parks or reservations for power and communications facilities (16 USC 5), Rights- of- way for public utilities (16 USC 79) (NPS does not have authority, however, to issue rights- of- way for oil or gas pipelines.)
Nuclear Power Plant – Combined (construction and operating) License	NRC	Office of New Reactors	Nuclear Power Plant - Combined (construction and operating) license	Application for a Combined License by an applicant under 10 CFR Part 52.	Authorization under 10 CFR Part 52, for which review under the National Environmental Policy Act is required, allows construction and operation of a new nuclear power plant under the Atomic Energy Act.	Atomic Energy Act is Sections 103 and 185(b)
Nuclear Power Plant – Construction Permit	NRC	Office of Nuclear Reactor Regulation	Nuclear Power Plant - Construction Permit	Application for a Construction Permit by an applicant and then subsequent application for an Operating License under 10 CFR Part 50.	Authorization under 10 CFR Part 50, for which review under the National Environmental Policy Act is required, covers construction of a new nuclear power plant under the Atomic Energy Act. A construction permit does not allow operation but, if certain requirements are met, the applicant may convert the construction permit to an operating license as approved by the NRC.	Atomic Energy Act is Sections 103 and 185(b)
Oil and Gas Sundry Notice for Surface Disturbing Activity	DOI	BLM	Land- based Oil & Gas - Production/ Extraction	Proposal to build large oil and gas processing facility "on lease"	Surface disturbing activities "on lease" (i.e. located on the same lease, unit, or communitization agreement as the lease from which all processed fluids are produced) must be permitted on Sundry Notice Form 3160- 5. The proposal must include a Surface Use Plan.	Mineral Leasing Act (30 USC) 43 CFR3162.3- 2

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Oil Spill Response Plan (DOI - BSEE)	DOI	BSEE	Offshore Oil & Gas	You must submit, and BSEE must approve, an OSRP that covers each facility located seaward of the coast line before you may use that facility.	If you are the owner or operator of an oil handling, storage, or transportation facility, and it is located seaward of the coast line, you must submit an oil spill response plan to BSEE for approval.	30 CFR 254
Outer Continental Shelf (OCS) Air Permit	EPA	EPA Region	Any equipment, activity or facility which: (1) Emits or has the potential to emit any air pollutant; (2) Is regulated or authorized under the Outer Continental Shelf Lands Act ("OCSLA") (43 U.S.C. §1331 et seq.); and (3) Is located on the OCS or in or on waters above the OCS	Application for construction and operations of an offshore energy project on the OCS filed pursuant to section 328 of the CAA and 40 CFR part 55.	The Outer Continental Shelf (OCS) Air Regulations, found at 40 CFR part 55, establish the applicable air pollution control requirements, including provisions related to permitting, monitoring, reporting, fees, compliance, and enforcement, for facilities subject to the Clean Air Act (CAA) section 328. These regulations apply to OCS Sources that are located beyond state seaward boundaries up to 25 nautical miles, with the exception of OCS sources located in the Gulf of Mexico west of 87.5° longitude (i.e. offshore TX, LA, MS, AL) and areas offshore the North Slope of Alaska, which are under the authority of the Department of Interior's Bureau of Ocean Energy Management (DOI-BOEM). Applicants locating beyond 25 nautical miles from the state seaward boundary are also subject to these regulations and may need an OCS permit.	CAA Section 328 and 40 CFR Part 55
Outgrant Administrative Action	DOD, US Army	USAF, USN, USACE	All, with exception for Wind: Federal Offshore	Project proponent seeks to right to use DOD installation land or facilities (e.g. lease or easement).	An outgrant refers in general to a legal document that conveys or grants the use of DOD-controlled real property	Department of Defense Instruction 4165.70, Real Property Management
Operations Plan / Surface Use Plan	DOI	BLM	Geothermal	Surface disturbance activities prior to obtaining exploration or drilling permit, or start of NEPA approval process	The operator may submit the Operations Plan (a.k.a. surface use plan): (1) as part of the GDP application; or (2) by itself, prior to submitting a GDP application, by using a Sundry Notice (NEPA analysis and approval required before starting surface operations (see 43 CFR 3261 - Drilling Operations: Getting a Permit). This will allow the operator to begin the NEPA approval process earlier and begin surface preparations while the rest of the GDP application is prepared. Plan describes access to well location(s), size of drilling pad(s), environmental mitigation measures, etc.	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Right- of- way Authorization (DOI-- BIA)	DOI	BIA	Electricity Transmission "Rural Energy" Projects Surface Transportation (all) Broadband (all)	Application by prospective transportation developers, utilities and adjoining landowners to access, cross or provide services and utilities on Indian lands	Application by prospective transportation developers, utilities and adjoining landowners to access, cross or provide services and utilities on Indian lands	25 USC 323 - 328 and 25 USC 2218, where appropriate, 25 CFR 169 Rights- of- Way over Indian Lands
Right- of- Way Authorization (DOI- BLM)	DOI	BLM	- Wind energy : Other than Federal Offshore - Electricity Transmission (all) - Surface Transportation (all) - Broadband (all) - Water Resource Projects (all)	Filing a Right- of- Way application (SF- 299) to request a right- of- way grant for the use of public lands.	Applications for proposed rights- of- way over, upon, under, or through public lands, including, but not limited to, grants for wind energy site testing and monitoring, power distribution lines, trails, highways, communication site facilities, water retention basins, and water conveyance pipelines. The processing of right- of- way applications must comply with the BLM's regulatory requirements, including those for planning, environmental, and right- of- way. BLM may approve application, approve the application with modifications, or deny the application. • A BLM right- of- way grant is required for an oil or gas pipeline to cross Federal lands under BLM's jurisdiction or the jurisdiction of two or more Federal agencies (43 CFR 2881.11). • The BLM requires as a matter of policy that all prospective applicants schedule and participate in at least	Title V of the Federal Land Policy and Management Act of 1976, as amended (FLPMA) (43 USC 1763), 43 CFR 2800
Right- of- Way Authorization (DOI- FWS)	DOI	FWS	All	Use of national wildlife refuge lands	The National Wildlife Refuge System Improvement Act requires that any activity on Refuge lands be determined as compatible with the Refuge system mission and Refuge purpose(s). Compatibility determinations are made by the Service's Refuge Managers. The FWS will conduct a review of projects on lands managed by it and coordinate any responses with the reviewing/lead agency. NWR lands are mainly for environmental, wildlife, recreational, educational purposes.	Title V of the Federal Land Policy and Management Act of 1976, as amended (FLPMA) (43 USC 1763), 43 CFR 2800
Section 10 of the Rivers and Harbors Act of 1899	US Army	USACE	All	Placement of structures affecting course, location, condition, or capacity of navigable waters of U.S. (includes offshore wind within 3 miles of coast); exemptions exist)	Activities that may affect navigation on US waterways must be evaluated to ensure navigational capacity is maintained.	Rivers and Harbors Act of 1899 (33 U.S.C. 401 et seq.) Section 10 (33 USC 403)
Section 103 of the Marine Protection, Research, and Sanctuaries Act	US Army	USACE District Office -- Regulatory	All	All proposed transportation of dredged material for disposal in ocean waters	Ocean dumping requires a permit issued under Section 103 the MPRSA. In the case of dredged material, the decision to issue a permit is made by the U.S. Army Corps of Engineers, using EPA 's environmental criteria and subject to EPA 's concurrence. EPA is responsible for designating	Title I, Marine Protection Research and Sanctuaries Act (33 USC 1401 et seq.), Section 103; 33 CFR 325.1 Applications for permits.

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Section 106 Review	ACHP	ACHP	All	Activities that could involve Federal impacts on properties listed in or are eligible for National Register of Historic Places	recommended ocean dumping sites (through promulgation in the Code of Federal Regulations). Section 106 of the National Historic Preservation Act of 1966 (NHPA) requires Federal agencies to take into account the effects of their undertakings on historic properties, and afford the Advisory Council on Historic Preservation a reasonable opportunity to comment. Federal agencies must complete this process prior to moving forward with, or issuing a permit, license, or assistance for, an undertaking. While completion of the Section 106 process is not an "authorization" per se, a federal agency must be able to show evidence that it has properly concluded its review in accordance with the regulations.	Section 106 of the National Historic Preservation Act of 1966 (NHPA)
Section 1222 Project	DOE	DOE	Electric Transmission	Application/proposal	The Secretary of Energy, acting through the Southwestern Power Administration (Southwestern) or the Western Area Power Administration (Western), has the authority to design, develop, construct, operate, own, or participate with other entities in designing, developing, constructing, operating, maintaining, or owning two types of projects: (1) Electric power transmission facilities and related facilities needed to upgrade existing transmission facilities owned by Southwestern or Western (42 U.S.C 16421(a)), or (2) New electric power transmission facilities and related facilities located within any State in which Southwestern or Western operates (42 U.S.C. 16421(b)).	Section 1222 of the Energy Policy Act of 2005 (42 U.S.C. 16421)
Section 404 Clean Water Act	US Army	USACE -	Regulatory	All Discharge of dredged or fill materials into waters of U.S. (includes wetlands, defined as "areas that are inundated or saturated by surface or ground water at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions. Wetlands generally include swamps, marshes, bogs, and similar areas")	Clean Water Act Section 404 regulates the discharge of dredged and fill material into waters of the United States, including wetlands. In general, to obtain a Section 404 permit, applicants must demonstrate that the discharge of dredged or fill material would not significantly degrade the nation's waters and there are no practicable alternatives less damaging to the aquatic ecosystem, so long as the alternative does not have other significant adverse environmental consequences. Applicants should also describe steps taken to minimize impacts to water bodies and wetlands and provide appropriate and practicable mitigation, such as restoring or creating wetlands, for any remaining, unavoidable impacts.	Clean Water Act (33 USC 1251 et seq.) Section 404 (33 USC 1344)

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Section 408 Permit	US Army	USACE	All	Alteration/modification of a federal project	That the Secretary may, on the recommendation of the Chief of Engineers, grant permission for the alteration or permanent occupation or use of any of the aforementioned public works when in the judgment of the Secretary such occupation or use will not be injurious to the public interest and will not impair the usefulness of such work.	33 USC 408 (Chapter 9.1), Navigation and Navigable Waters: It is unlawful for any person(s) to build upon, alter, deface, destroy, move, injure, obstruct or... impair the usefulness of any levee or other work built by the U.S.
Service Line Agreement	DOI	BIA	All	Application by prospective transportation developers, utilities and adjoining landowners to access, cross or provide services and utilities on Indian lands	Authorized rights of way and easements for access to and across Indian lands and to transport a variety of commodities including water, oil, gas, broadband, telephone service and electricity across and to Indian lands.	25 U.S.C. 47, 323- 328, 450
Site License (DOI - BLM)	DOI	BLM	Geothermal	A site license is required of operators who are not the geothermal lease holder to operate the facility on the described land and determine rental fee.	Site License (no form)	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)
Special Use Permit (BLM)	DOI	BLM	Geothermal	Split estates where BLM is the surface managing agency but does not own the subsurface mineral resource (the mineral estate was not reserved to the Federal government)	Special Use Permit (no form). In split estate where BLM is the surface management (the mineral estate was not reserved to the Federal government), a special use permit, usually, a rights of way (ROW) is required to occupy the surface. The drilling permit will be approved by the appropriate State agency.	Geothermal Steam Act (30 USC 1001 et seq.); Federal Land Policy Management Act (43 USC 1701 et seq.); Energy Policy Act of 2005 (Pub. L. 109- 58)
Special Use Permit (FS)	USDA	FS	All	Proposed activities impact Forest Service owned lands	The Forest Service requires a special use permit for all uses and occupancies of National Forest System (NFS) lands with certain limited exceptions. The Agency's special- uses program authorizes uses on NFS land that provide a benefit to the general public and protect public and natural resource values. These uses cover a variety of activities ranging from individual private uses to large- scale commercial facilities and public services. Examples of authorized land uses include road rights- of- way accessing private residences and non- Federal lands, domestic water supplies and water conveyance systems, utility rights- of- way, communications uses, ski areas, resorts, marinas, outfitting and guiding services, and public parks and campgrounds. The Forest Service carefully reviews each application to determine how the request affects the public's use of NFS land.	Federal Land Policy and Management Act of 1976, the Organic Act of 1897, the Mineral Leasing Act and others

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
State, Local Tribal, or Other Non-Federal Action	All	All	All	Federally required decisions, authorizations, or permits that have been assumed by, or delegated to, State, Local, Tribal or other non-Federal agencies.	Federally required permits, environmental reviews, or authorizations that have been assumed by, or delegated to, State, Local, Tribal, or other non-Federal agencies may be tracked using this action. This decision or authorization may be added to the Permitting Dashboard if the State, Local, Tribal, or other non-Federal agency has opted in and the action is a prerequisite to the issuance of a decision or authorization by a Federal agency.	Various
USCG Bridge Permit	DHS	USCG - Office of Bridge Programs	Bridges	Construction of bridges or causeways over or in any navigable river or other navigable water, and international bridges regardless of the waterway's navigability.	Federal law prohibits the construction or modification of any bridge across navigable waters of the United States unless first authorized by the Coast Guard. The Coast Guard approves the location, plans and navigational clearances of bridges through the issuance of bridge permits or bridge permit amendments.	Section 9, Rivers and Harbors Appropriations Act of 1899, as amended (33 USC 401); the Act of March 23, 1906, amended (33 USC 491), the General Bridge Act of 1946, amended (33 USC 525); the International Bridge Act of 1972 (33 USC 535)
USCG Letter of Recommendation for Marine Operations	USCG	USCG Captain of the Port (COTG)	Liquefied Natural Gas Terminal Facilities	An owner or operator seeking approval from FERC to build and operate or expand a LNG facility, as defined in 33 CFR Part 127	An owner or operator intending to build a new facility handling LNG or LHG, or planning new construction to expand or modify marine terminal operations in an existing facility handling LNG or LHG, where construction, expansion, or modification would result in an increase in the size and/or frequency of LNG or LHG marine traffic on the waterway associated with a proposed facility or modification, must submit a Letter of Intent (LOI) to the Captain of the Port (COTP) of the zone in which the facility is or will be located, no later than the date that the owner or operator files a pre-filing request with FERC, but, in all cases, at least 1 year prior to the start of construction. The owner/operator must prepare or update a Waterway Suitability Assessment; after the COTP receives the LOI the COTP issues a Letter of Recommendation (LOR) as to the suitability of the waterway for LNG or LHG marine traffic to the Federal, State, or local government agencies having jurisdiction for siting, construction, and operation.	33 CFR 127.007 Letter of intent and waterway suitability assessment; 33 CFR 127.009 Letter of recommendation.

Permit/Review	Responsible Agency	Responsible Bureau/ Department /Office	Project Type	Activity/Trigger	Description	Statute/ Regulation
Use Authorization (DOI-BOR)	BOR	BOR	Federal Hydropower Non- Federal Hydropower - Leases Non- Federal Hydropower - Licenses (including Non- Federal Marine and Hydrokinetic Projects)	Through communication with the local Reclamation office and submission of an application using one of the following forms depending on the nature of the requested use:(a) Use SF 299 to request a use authorization for the placement, construction, and use of energy, transportation, water, or telecommunication systems and facilities on or across all Federal property including Reclamation land, facilities, or waterbodies. (b) Use Form 7-2540 to request any other type of use authorization.	Applications to obtain a use authorization for the placement, construction, and use of energy, transportation, water, or telecommunication systems and facilities on or across all Federal property including Reclamation land, facilities, or waterbodies. Reclamation uses SF 299, Application for Transportation and Utility Systems and Facilities on Federal Lands. The regulations addressing this are 43 CFR part 429, Use of Bureau of Reclamation Land, Facilities, and Waterbodies.	Title V of the Federal Land Policy and Management Act of 1976, as amended (FLPMA) (43 USC 1763), 43 CFR 2800
Wild and Scenic Rivers Act Determination/ Coordination	BLM	BLM, FWS, NPS, or USFS	Hydropower (all) Surface Transportation (all) Electricity Transmission (all) Water Resource Projects (all)	Projects on national wild and scenic rivers, congressionally authorized study rivers, or upstream, downstream or on a tributary of such rivers.	The Wild and Scenic Rivers Act prohibits a federal agency from issuing a permit, license, loan, grant, or other assistance for an activity that would adversely affect the free- flow, water quality, or outstandingly remarkable river values of a national wild and scenic river. The river administering agency (BLM, FWS, NPS, or USFS) makes the determination about effects, and coordinates with proponents to achieve compliant projects. FERC is prohibited from licensing construction of any hydroelectric dam or other project works regulated under the Federal Power Act on designated wild and scenic rivers and Congressionally authorized study areas.	Wild and Scenic Rivers Act, 16 U.S.C. 1271-1287: Sections 7, 10(a), and 12; • 36 CFR part 297 (USFS); • 43 CFR part 8350 (BLM).
Wind Energy Evaluation Lease - Indian Lands	DOI	DOI - BIA	Wind: Other than Federal Offshore	Contract approved by the Secretary that authorizes possession of Indian land for a specific purpose and term.	Lease allows the possession of Indian land for wind energy evaluation in exchange for pre- defined royalties.	

The Federal Environmental Review and Authorization Inventory identifies many of the most common federal environmental reviews, authorizations, licenses, permits, approvals, findings, determinations, and other administrative decisions involved in permitting.

Source: [US Government Permitting Dashboard](#)

Analyst(s) Certification(s):

Each research report excerpted herein was certified under SEC Regulation AC by the analyst primarily responsible for such report as follows: I hereby certify that: 1) the views expressed in this research report accurately reflect my personal views about any or all of the subject securities referred to in this report and; 2) no part of my compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this report.

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Materially Mentioned Stocks (Ticker, Date, Price)

APA Corporation (APA, 01-Sep-2026, USD 44.30), Equal Weight/Positive, CD/CE/J/K/M

Bloom Energy Corporation (BE, 01-Sep-2026, USD 213.63), Equal Weight/Positive, A/CE/D/E/FA/J/K/L/M/N

CNX Resources Corp. (CNX, 01-Sep-2026, USD 37.56), Underweight/Positive, CD/CE/J

Centrica (CNA.L, 01-Sep-2026, GBp 152), Overweight/Positive, CD/E/J/K/L/M/N

Cheniere Energy (LNG, 01-Sep-2026, USD 294.13), Overweight/Neutral, CD/CE/FA/J

Chevron Corporation (CVX, 01-Sep-2026, USD 211.05), Equal Weight/Positive, CD/CE/D/E/J/K/L/M/N

ConocoPhillips (COP, 01-Sep-2026, USD 136.19), Overweight/Positive, CD/CE/J/K/M/N

DT Midstream Inc. (DTM, 01-Sep-2026, USD 129.99), Equal Weight/Neutral, J/K/M

Diamondback Energy (FANG, 01-Sep-2026, USD 203.16), Overweight/Positive, CD/CE/D/E/J/K/L/M

Drax (DRX.L, 01-Sep-2026, GBp 800), Overweight/Positive, E/FA/J/K/L/N

EOG Resources, Inc. (EOG, 01-Sep-2026, USD 148.35), Equal Weight/Positive, A/CD/CE/D/E/J/K/L/M/N

EQT Corporation (EQT, 01-Sep-2026, USD 55.59), Overweight/Positive, CD/CE/D/E/J/K/L/M

Enbridge Inc. (ENB.TO, 01-Sep-2026, CAD 70.54), Equal Weight/Neutral, A/CD/CE/D/E/J/K/L/M

Enel (ENEI.MI, 01-Sep-2026, EUR 9.24), Overweight/Positive, CD/J/K/M/N

Energy Transfer, LP (ET, 01-Sep-2026, USD 21.42), Overweight/Neutral, A/CD/CE/D/E/J/K/L/M

Engie (ENGIE.PA, 01-Sep-2026, EUR 24.40), Overweight/Positive, A/CD/D/E/J/K/L/M/N

Eni (ENI.MI, 01-Sep-2026, EUR 23.67), Overweight/Positive, A/CD/CE/D/E/J/K/L/M/N

Entergy Corp. (ETR, 01-Sep-2026, USD 107.02), Overweight/Positive, A/CD/D/FB/J/L

Enterprise Products Ptns LP (EPD, 01-Sep-2026, USD 39.28), Overweight/Neutral, A/CE/D/E/J/K/L/M

ExxonMobil (XOM, 01-Sep-2026, USD 164.55), Overweight/Positive, CE/D/E/J/K/L/M/N

Fervo Energy (FRVO, 01-Sep-2026, USD 19.75), Overweight/Positive, A/CE/D/J/L

First Solar (FSLR, 01-Sep-2026, USD 199.65), Overweight/Positive, CE/J

Fortum (FORTUM.HE, 01-Sep-2026, EUR 20.94), Equal Weight/Positive, CD/D/J/K/L/M/N

Generac Holdings Inc (GNRC, 01-Sep-2026, USD 181.42), Equal Weight/Positive, CE/J

Halliburton Co. (HAL, 01-Sep-2026, USD 36.80), Overweight/Positive, CD/CE/D/FA/J/K/L/M/N

Kinder Morgan Inc. (KMI, 01-Sep-2026, USD 32.10), Overweight/Neutral, A/CD/CE/D/J/K/L/M

Kinetik Holdings Inc. (KNTK, 01-Sep-2026, USD 55.03), Equal Weight/Neutral, J

Kodiak Gas Services, Inc. (KGS, 01-Sep-2026, USD 59.55), Overweight/Neutral, A/CE/D/E/J/K/L/M

Marathon Petroleum Corp. (MPC, 01-Sep-2026, USD 383.00), Overweight/Neutral, A/CD/CE/D/E/J/L

NOV Inc. (NOV, 01-Sep-2026, USD 21.36), Underweight/Positive, CD/CE/E/J/K/L/M/N

Naturgy Energy (NTGY.MC, 01-Sep-2026, EUR 29.52), Equal Weight/Positive, E/J/K/L/M/N

NextEra Energy, Inc. (NEE, 01-Sep-2026, USD 82.93), Equal Weight/Positive, A/CE/D/E/J/K/L/M

OMV (OMVV.VI, 01-Sep-2026, EUR 69.90), Underweight/Positive, A/CD/D/E/J/K/L/M

Occidental Petroleum Corporation (OXY, 01-Sep-2026, USD 60.95), Overweight/Positive, A/CD/CE/D/J/K/L/M

Oklo Inc. (OKLO, 01-Sep-2026, USD 38.53), Overweight/Positive, D/J/K/L/M

Ormat Technologies, Inc. (ORA, 01-Sep-2026, USD 107.29), Overweight/Positive, CE/J

PPL Corp (PPL, 01-Sep-2026, USD 34.47), Overweight/Positive, A/CE/D/E/FA/FB/J/K/L/M/N

Pembina Pipeline Corp. (PPL.TO, 01-Sep-2026, CAD 68.16), Overweight/Neutral, J

Phillips 66 (PSX, 01-Sep-2026, USD 252.02), Equal Weight/Neutral, A/CD/CE/D/E/FA/J/K/L/M/N

Plains All American Pipeline (PAA, 01-Sep-2026, USD 25.86), Underweight/Neutral, A/CD/CE/D/J/K/L/M

RWE (RWE.GE, 01-Sep-2026, EUR 58.58), Overweight/Positive, A/CD/D/E/J/K/L/M/N

SLB (SLB, 01-Sep-2026, USD 57.15), Overweight/Positive, CE/J/K/N

SSE (SSE.L, 01-Sep-2026, GBp 2406), Overweight/Positive, A/CD/D/E/J/K/L/M/N

Shell Plc (SHEL.L, 01-Sep-2026, GBP 34.33), Overweight/Positive, D/E/J/K/L/M/N

Solaria Energia y Medio Ambiente (SLRS.MC, 01-Sep-2026, EUR 16.95), Overweight/Positive, A/D/E/J/L

Solaris Energy Infrastructure (SEI, 01-Sep-2026, USD 49.64), Overweight/Positive, A/CE/D/J/L

Sunoco LP (SUN, 01-Sep-2026, USD 76.99), Overweight/Neutral, A/CD/CE/D/FB/J/K/L/M

TC Energy (TRP.TO, 01-Sep-2026, CAD 87.81), Overweight/Neutral, A/CE/D/E/J/K/L/M

TGS (TGS.OL, 01-Sep-2026, NOK 137.70), Overweight/Positive, CD/J

Targa Resources Corp. (TRGP, 01-Sep-2026, USD 296.06), Overweight/Neutral, A/CD/CE/D/J/L

Technip Energies (TE.PA, 01-Sep-2026, EUR 31.34), Overweight/Positive, CD/J/K/N

TechnipFMC Plc (FTI, 01-Sep-2026, USD 78.31), Overweight/Positive, CD/CE/J/K/M/N

Tecnicas Reunidas (TRE.MC, 01-Sep-2026, EUR 26.34), Overweight/Positive, J/K/M/N

Tenaris (TENR.MI, 01-Sep-2026, EUR 24.39), Overweight/Positive, CE/J/K/N

TotalEnergies SE (TOTF.PA, 01-Sep-2026, EUR 77.61), Overweight/Positive, CD/D/E/J/K/L/M/N

United Utilities (UU.L, 01-Sep-2026, GBp 1402), Overweight/Positive, A/D/J/K/L/M/N

Valero Energy (VLO, 01-Sep-2026, USD 361.99), Overweight/Neutral, CD/CE/J

Vallourec (VLLP.PA, 01-Sep-2026, EUR 18.76), Overweight/Positive, CD/E/J/K/L/M/N

Vestis Corp. (VSTS, 01-Sep-2026, USD 12.13), Underweight/Neutral, CE/J

Weatherford (WFRD, 01-Sep-2026, USD 94.52), Overweight/Positive, A/CE/D/E/J/K/L/M/N

Xcel Energy Inc. (XEL, 01-Sep-2026, USD 75.89), Overweight/Positive, A/CD/CE/D/J/L

bp (BPL, 02-Sep-2026, GBP 5.43), Overweight/Positive, CD/CE/E/J/K/L/M/N

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Explanation of the U.S. Equity Strategy sector rating system:

Positive Outlook: We recommend clients increase their level of portfolio exposure to the sector (as defined below)

Neutral Outlook: We recommend clients maintain their current level of portfolio exposure to the sector (as defined below)

Negative Outlook: We recommend clients decrease their level of portfolio exposure to the sector (as defined below)

Sector Definitions:

Our custom sector definitions are derived from GICS sectors as follows:

No changes to GICS Level 1 for Energy, Consumer Staples, Industrials, Real Estate, Healthcare, Utilities, Materials, and Financials.

“Big Tech”: Seven stocks - AAPL, AMZN, GOOG, GOOGL, META, MSFT, NVDA - have been removed from their corresponding GICS Level 1 sectors.

“Rest of Tech” is GICS Level 1 Information Technology with AAPL, MSFT, and NVDA excluded.

“Communication Services” is GICS Level 1 Communication Services with GOOG, GOOGL, and META excluded.

“Discretionary” is GICS Level 1 Consumer Discretionary with AMZN excluded.

Explanation of the European Equity Strategy sector rating system:

Overweight (OW): The performance of the MSCI Europe sector is expected to outperform the MSCI Europe index in the next 3-6 months.

Marketweight (MW): The performance of the MSCI Europe sector is expected to perform in line with the MSCI Europe index in the next 3-6 months.

Underweight (UW): The performance of the MSCI Europe sector is expected to underperform the MSCI Europe index in the next 3-6 months.

Guide to the Barclays Fundamental Equity Research Rating System:

Our coverage analysts use a relative rating system in which they rate stocks as Overweight, Equal Weight or Underweight (see definitions below) relative to other companies covered by the analyst or a team of analysts that are deemed to be in the same industry (the “industry coverage universe”).

In addition to the stock rating, we provide industry views which rate the outlook for the industry coverage universe as Positive, Neutral or Negative (see definitions below). A rating system using terms such as buy, hold and sell is not the equivalent of our rating system. Investors should carefully read the entire research report including the definitions of all ratings and not infer its contents from ratings alone.

Stock Rating

Overweight - The stock is expected to outperform the unweighted expected total return of the industry coverage universe over a 12-month investment horizon.

Equal Weight - The stock is expected to perform in line with the unweighted expected total return of the industry coverage universe over a 12-month investment horizon.

Underweight - The stock is expected to underperform the unweighted expected total return of the industry coverage universe over a 12-month investment horizon.

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Industry View

Positive - industry coverage universe fundamentals/valuations are improving.

Neutral - industry coverage universe fundamentals/valuations are steady, neither improving nor deteriorating.

Negative - industry coverage universe fundamentals/valuations are deteriorating.

Below is the list of companies that constitute the “industry coverage universe”:

European Energy Services

ADNOC Drilling Company PJSC (ADNOCDRILL.AD)	Aker Solutions (AKSOL.OL)	Hunting (HTG.L)
Maire (MTCM.MI)	Saipem (SPMI.MI)	SBM Offshore (SBMO.AS)
Subsea 7 SA (SUBC.OL)	Technip Energies (TE.PA)	Tecnicas Reunidas (TRE.MC)
Tenaris (TENR.MI)	TGS (TGS.OL)	Vallourec (VLLP.PA)
Viridien (VIRI.PA)		

European Integrated Energy

ADNOC Distribution (ADNOCDIST.AD)	ADNOC Gas (ADNOCGAS.AD)	ADNOC Logistics & Services (ADNOCLS.AD)
Aker BP (AKERBP.OL)	bp (BP.L)	Eni (ENI.MI)
Equinor (EQNR.OL)	Fertiglobe Plc (FERTIGLB.AD)	Galp Energia (GALP.LS)
Harbour Energy (HBR.L)	Ithaca Energy (ITH.L)	Neste (NESTE.HE)
OMV (OMVV.VI)	Repsol (REP.MC)	Saudi Aramco (2222.SE)
Shell Plc (SHEL.L)	TotalEnergies SE (TOTF.PA)	Tullow Oil Plc. (TLW.L)
Vår Energi (VAR.OL)		

European Utilities & Clean Energy

Centrica (CNA.L)	CK Infrastructure Holdings (CKI.L)	Drax (DRX.L)
E.ON (EONGn.DE)	EDP (EDP.LS)	EDP Renovaveis (EDPR.LS)
Elia Group SA (ELI.BR)	Enagas (ENAG.MC)	Endesa (ELE.MC)
Enel (ENEL.MI)	Engie (ENGIE.PA)	EVN (EVNV.VI)

Fortum (FORTUM.HE)	Greenergy Renovables (GREG.MC)	Hidroelectrica S.A. (ROH2O.BX)
Iberdrola (IBE.MC)	Italgas (IG.MI)	National Grid (NG.L)
Naturgy Energy (NTGY.MC)	Orsted (ORSTED.CO)	Pennon Group (PNN.L)
Redeia (REDE.MC)	RWE (RWE.G.DE)	Severn Trent (SVT.L)
Snam (SRG.MI)	Solaria Energia y Medio Ambiente (SLRS.MC)	SSE (SSE.L)
Terna (TRN.MI)	Uniper (UN0k.DE)	United Utilities (UU.L)
Veolia (VIE.PA)	Verbund (VERB.VI)	

North America Clean Technology

Array Technologies, Inc. (ARRY)	Blink Charging (BLNK)	Bloom Energy Corporation (BE)
Brookfield Renewable Corporation (BEP)	Brookfield Renewable Partners LP (BEP)	Centrus Energy Corp. (LEU)
Enlight Renewable Energy Ltd. (ENLT)	Enphase Energy (ENPH)	ERock Inc. (EROC)
First Solar (FSLR)	Fluence Energy (FLNC)	GE Vernova Inc (GEV)
Generac Holdings Inc (GNRC)	Nextpower (NXT)	NuScale Power (SMR)
Oklo Inc. (OKLO)	Shoals Technologies (SHLS)	SolarEdge Technologies Inc. (SEDG)
Standard Nuclear, Inc. (STDN)	STEM (STEM)	Sunrun Inc. (RUN)
Wallbox NV (WBX)	XPLR Infrastructure LP (XIFR)	

North America Midstream and Refining

Cheniere Energy (LNG)	Cheniere Energy Partners LP (CQP)	Darling Ingredients (DAR)
DT Midstream Inc. (DTM)	EagleRock Land LLC (EROK)	Enbridge Inc. (ENB.TO)
Energy Transfer, LP (ET)	Enterprise Products Prtns LP (EPD)	Excelerate Energy Inc (EE)
HF Sinclair Corp. (DINO)	Keyera Corp. (KEY.TO)	Kinder Morgan Inc. (KMI)
Kinetik Holdings Inc. (KNTK)	Kodiak Gas Services, Inc. (KGS)	LandBridge (LB)
Marathon Petroleum Corp. (MPC)	MPLX LP (MPLX)	ONEOK Inc. (OKE)
Pembina Pipeline Corp. (PPL.TO)	Phillips 66 (PSX)	Plains All American Pipeline (PAA)
Plains GP Holdings LP (PAGP)	South Bow Corp. (SOBO)	South Bow Corp. (SOBO.TO)
Sunoco LP (SUN)	SunocoCorp (SUNC)	Targa Resources Corp. (TRGP)
TC Energy (TRP.TO)	Valero Energy (VLO)	WaterBridge Infrastructure (WBI)
Williams Cos. (WMB)		

North America Power & Utilities

AES Corp. (AES)	Algonquin Power & Utilities (AQN)	Algonquin Power & Utilities (AQN.TO)
Alliant Energy Corporation (LNT)	Ameren Corp. (AEE)	American Electric Power Company, Inc. (AEP)
American Water Works Company, Inc. (AWK)	Atmos Energy (ATO)	Avista Corp. (AVA)
CenterPoint Energy, Inc. (CNP)	Chesapeake Utilities Corp (CPK)	CMS Energy Corporation (CMS)
Consolidated Edison, Inc. (ED)	Constellation Energy (CEG)	Dominion Energy (D)
DTE Energy (DTE)	Duke Energy Corporation (DUK)	Edison International (EIX)
Emera Inc. (EMA)	Emera Inc. (EMA.TO)	Entergy Corp. (ETR)
Essential Utilities, Inc. (WTRG)	Eversource Energy (ES)	Eversource Energy (ES)
Exelon Corporation (EXC)	FirstEnergy Corp. (FE)	Fortis Inc. (FTS)
Fortis Inc. (FTS.TO)	H2O America (HTO)	Hawaiian Electric Inds (HE)
Hydro One Ltd. (H.TO)	IDACORP INC (IDA)	NextEra Energy, Inc. (NEE)
NiSource, Inc. (NI)	NorthWestern Corporation (NWE)	NRG Energy Inc (NRG)
OGE Energy Corp. (OGE)	PG&E Corporation (PCG)	Pinnacle West Capital Corporation (PNW)
Portland General Electric Company (POR)	PPL Corp (PPL)	Public Service Enterprise Group (PEG)
Sempra (SRE)	Talen Energy Corporation (TLN)	The Southern Company (SO)
TXNM Energy (TXNM)	WEC Energy Group, Inc. (WEC)	Xcel Energy Inc. (XEL)

U.S. Business & Professional Services

ADT Inc. (ADT)	Bright Horizons (BFAM)	Cintas Corp. (CTAS)
Clarivate Analytics PLC (CLVT)	Ecolab Inc. (ECL)	Equifax Inc. (EFX)
FactSet Research Systems (FDS)	Fair Isaac Corp. (FICO)	First Advantage (FA)

Gartner Inc. (IT)	KinderCare (KLC)	ManpowerGroup (MAN)
McGrath RentCorp (MGRC)	Mobility Global (MBGL)	Moody's Corp. (MCO)
MSCI Inc. (MSCI)	NIQ (NIQ)	Robert Half International (RHI)
Rollins, Inc. (ROL)	S&P Global Inc. (SPGI)	Thomson Reuters Corp. (TRI)
TransUnion (TRU)	UniFirst (UNF)	Verisk Analytics (VRSK)
Vestis Corp. (VSTS)	WillScot (WSC)	

U.S. Energy Services & Technology

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Cactus, Inc. (WHD)	Expro Group (XPRO)	Fervo Energy (FRVO)
Halliburton Co. (HAL)	Helmerich & Payne Inc. (HP)	Innovex International Inc. (INNVX)
Liberty Energy (LBRT)	Nabors Industries (NBR)	National Energy Services Reunited Corp (NESR)
Noble Corp. (NE)	NOV Inc. (NOV)	Oceaneering International (OII)
Ormat Technologies, Inc. (ORA)	Patterson-UTI Energy (PTEN)	ProPetro Services (PUMP)
Seadrill Limited (SDRL)	SLB (SLB)	Solaris Energy Infrastructure (SEI)
TechnipFMC Plc (FTI)	Tenaris S.A. (TS)	Tidewater Inc. (TDW)
Transocean Ltd. (RIG)	Valaris Limited (VAL)	Weatherford (WFRD)

U.S. Integrated Oil & E&P

Antero Resources (AR)	APA Corporation (APA)	BKV Corporation (BKV)
California Resources Corp. (CRC)	Chevron Corporation (CVX)	CNX Resources Corp. (CNX)
ConocoPhillips (COP)	Devon Energy Corporation (DVN)	Diamondback Energy (FANG)
EOG Resources, Inc. (EOG)	EQT Corporation (EQT)	Expand Energy (EXE)
ExxonMobil (XOM)	Murphy Oil Corporation (MUR)	NET Power (NPWR)
Occidental Petroleum Corporation (OXY)	Ovintiv Inc. (OVV)	Range Resources Corp. (RRC)
Viper Energy Partners LP (VNOM)		

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Primary Issuers/Bonds

BP PLC, Overweight, CD/CE/E/J/K/L/M/N

Valuation Methodology: Our Overweight rating on BP reflects improving rating momentum following the strategic reset, with management prioritising debt reduction and disciplined capital allocation. We view BP's diversified integrated business model, strong cash-generation capability and commitment to maintaining a strong IG profile as supportive of further credit improvement. We also see scope for spread tightening as leverage trends towards management's targeted levels and credit metrics move closer to higher-rated peers.

Risks that May Impede Achievement of the Rating: Weaker commodity prices, delays in executing asset disposals or a shift towards more shareholder-friendly capital allocation could slow deleveraging and reduce scope for further spread tightening. Execution of the strategic reset and delivery against management's debt-reduction targets remain key to the credit story.

Representative Bond: BPLN 1.594 07/03/28 (EUR 96.98, 01-Sep-2026)

LG ENERGY SOLUTION LTD, Overweight, CD/J

Valuation Methodology: The LGES curve is the widest in the Korea HG Corporate sector, and is also wide of the BBB curve for global machinery and equipment peers. Although ratings remain vulnerable, we think current valuations are already indicative of a low-BBB credit profile despite its mid-BBB ratings. At the same time, we think LGES's efforts to boost profitability and liquidity are starting to bear fruit, and leverage metrics could start falling in 2H26.

Risks that May Impede Achievement of the Rating: The possibility that LGES misses its revenue targets and earnings remain weak. LGES is LG Chem's main earnings contributor. Hence, weak LGES performance could exert pressure on LG Chem's credit profile; and

Continued weak performance at LG Chem's petrochemical operations putting its, and LGES's ratings at risk.

Representative Bond: LGENSO 5 1/2 07/02/34 (USD 97.95, 01-Sep-2026)

Representative Bond: LGENSO 5 3/4 09/25/28 (USD 101.13, 01-Sep-2026)

Representative Bond: LGENSO 5 3/8 07/02/27 (USD 100.42, 01-Sep-2026)

Representative Bond: LGENSO 5 3/8 07/02/29 (USD 100.39, 01-Sep-2026)

Representative Bond: LGENSO 5 5/8 09/25/26 (USD 100.11, 01-Sep-2026)

SANTOS FINANCE LTD, Overweight, CD/J

Valuation Methodology: Our OW rating reflects the the Santos bonds' wide valuations against the BBB E&P curve, and our expectations that earnings will rise in 2026 due to higher energy prices and volumes, with rating upgrades likely in 2027.

Risks that May Impede Achievement of the Rating: Key risk to our view is a sharp decline in energy prices, especially if LNG supply normalises in the coming months.

Representative Bond: STOAU 6 7/8 09/19/33 (USD 106.37, 01-Sep-2026)

TOTALENERGIES SE, Overweight, CD/D/E/J/K/L/M/N

Valuation Methodology: We are OW on TTEFP as we expect the company to maintain a robust credit profile over the cycle and remains well positioned to be the highest-rated issuer within the sector. TotalEnergies' spreads may outperform sector peers on rating upgrade potential. In any case, we expect TotalEnergies' RV to be supported by the group's strong fundamental position, being one of the highest rated and most robust corporate names in Europe.

Risks that May Impede Achievement of the Rating: Weaker commodity prices, a shift towards more aggressive shareholder distributions or a material increase in investment spending could weaken credit metrics. Delivery of the group's disciplined capital allocation framework remains key to the credit story.

Representative Bond: TTEFP 0 3/4 07/12/28 (EUR 95.57, 01-Sep-2026)

Materially Mentioned Issuers/Bonds

AYDEM YENILENEBILIR ENERJI AS, Market Weight, CD/J

Representative Bond: AYDEMT 9 7/8 09/30/30 (USD 99.00, 01-Sep-2026)

AYDEMT 9 7/8 09/30/30 (USD 99.00, 01-Sep-2026)

AZULE ENERGY FINANCE PLC, Overweight, CD/J

Representative Bond: AZURGY 8 1/8 01/23/30 (USD 102.00, 01-Sep-2026)

AZURGY 8 1/8 01/23/30 (USD 102.00, 01-Sep-2026)

CENTRICA PLC, Underweight, CD/E/J/K/L/M/N

Representative Bond: CNALN 7 09/19/33 (GBP 106.29, 01-Sep-2026)

ELECTRICITE DE FRANCE SA, Overweight, A/CD/D/E/J/K/L/M/N

Representative Bond: EDF 4 1/2 11/12/40 (EUR 97.86, 01-Sep-2026)

FLNG LIQUEFACTION 2 LLC, CD/J/K/M

FLIQII 5.92 12/31/39 (USD 97.98, 01-Sep-2026, Barclays)

FLNG LIQUEFACTION 3 LLC, CD/J/K/M

FLQIII 5.55 03/31/39 (USD 94.36, 01-Sep-2026)

INDIKA ENERGY TBK PT, CD/J

INDYIJ 8 3/4 05/07/29, Overweight (USD 102.38, 01-Sep-2026)

KOREA ELECTRIC POWER CORP, Underweight, CD/CE/J

Representative Bond: KORELE 4 06/14/27 (USD 99.50, 01-Sep-2026)

Representative Bond: KORELE 4 1/8 11/12/30 (USD 97.09, 01-Sep-2026)

Representative Bond: KORELE 4 3/4 02/13/28 (USD 100.00, 01-Sep-2026)

Representative Bond: KORELE 4 7/8 01/31/27 (USD 100.12, 01-Sep-2026)

Representative Bond: KORELE 5 1/2 04/06/28 (USD 101.24, 01-Sep-2026)

Representative Bond: KORELE 5 1/8 04/23/34 (USD 99.79, 01-Sep-2026)

Representative Bond: KORELE 7 02/01/27 (USD 100.97, 01-Sep-2026)

KOREA GAS CORP, Underweight, CD/J

Representative Bond: KORGAS 2 07/13/31 (USD 87.45, 01-Sep-2026)

Representative Bond: KORGAS 2 7/8 07/16/29 (USD 95.11, 01-Sep-2026)

Representative Bond: KORGAS 3 1/8 07/20/27 (USD 98.76, 01-Sep-2026)

Representative Bond: KORGAS 3 7/8 07/13/27 (USD 99.50, 01-Sep-2026)

Representative Bond: KORGAS 4 1/4 07/10/30 (USD 98.00, 01-Sep-2026)
 Representative Bond: KORGAS 4 7/8 07/05/28 (USD 100.28, 01-Sep-2026)
 Representative Bond: KORGAS 5 07/08/29 (USD 100.76, 01-Sep-2026)
 Representative Bond: KORGAS 6 1/4 01/20/42 (USD 107.27, 01-Sep-2026)
 Representative Bond: KORGAS FLOAT 07/10/28 (USD 100.26, 01-Sep-2026)

KOSMOS ENERGY GTA HOLDINGS, CD/D/J/L

KOS 11 1/4 01/29/31 (USD 106.05, 01-Sep-2026)

KOSMOS ENERGY LTD, Market Weight, A/CD/CE/D/FA/J/L

Representative Bond: KOS 8 3/4 10/01/31 (USD 89.88, 01-Sep-2026)

LG CHEM LTD, Market Weight, CD/FA/J

Representative Bond: LGCHM 2 3/8 07/07/31 (USD 86.63, 01-Sep-2026)

NATURGY CAPITAL MARKETS SA, Underweight, E/J/K/L/M/N

Representative Bond: NTGYSM 3 7/8 05/21/35 (EUR 96.88, 01-Sep-2026)

NEXTERA ENERGY CAPITAL HOLDINGS INC, Underweight, A/CD/D/E/J/K/L/M

Representative Bond: NEE 2.44 01/15/32 (USD 86.89, 01-Sep-2026)

NICKEL INDUSTRIES LTD, CD/J

NICAU 9 09/30/30, Overweight (USD 103.13, 01-Sep-2026)

PERUSAHAAN PERSEROAN PERSERO PT PERUSAHAAN LISTRIK NEGARA, CD/J

PLNIJ 5.45 02/03/36 (USD 94.19, 01-Sep-2026)

PETRONAS CAPITAL LTD, CD/J

PETMK 4.55 04/21/50 (USD 82.14, 01-Sep-2026)

PHILIPPINE GOVERNMENT INTERNATIONAL BOND, Underweight, CD/J/K/M

Representative Bond: PHILIP 2.457 05/05/30 (USD 91.95, 19-Aug-2026)

Representative Bond: PHILIP 2.95 05/05/45 (USD 65.03, 19-Aug-2026)

PHILIP 3.229 03/29/27 (USD 99.29, 01-Sep-2026)

POSCO HOLDINGS INC, Market Weight, CD/CE/J

Representative Bond: PKX 5 1/8 05/07/30 (USD 99.50, 01-Sep-2026)

Representative Bond: PKX 5 3/4 05/07/35 (USD 99.84, 01-Sep-2026)

RELIANCE INDUSTRIES LTD, Market Weight, CD/D/E/J/K/L/M/N

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Representative Bond: RILIN 4 7/8 02/10/45 (USD 87.58, 01-Sep-2026)

ROP SUKUK TRUST, Underweight, CD/J

Representative Bond: PHILIP 5.045 06/06/29 (USD 100.41, 19-Aug-2026)

SEPLAT ENERGY PLC, Market Weight, CD/J

Representative Bond: SEPLLN 9 1/8 03/21/30 (USD 105.26, 01-Sep-2026)

TENNET GMBH & CO KG, Market Weight, CD/J

Representative Bond: TENNGR 4 5/8 07/15/46 (EUR 96.94, 01-Sep-2026)

TENNET HOLDING BV, Overweight, CD/D/J/K/L/M/N

Representative Bond: TENNNL 1 3/4 06/04/27 (EUR 99.18, 01-Sep-2026)

TRANSOCEAN INTERNATIONAL LTD, CD/J/K/N

RIG 6.8 03/15/38, Overweight (USD 94.75, 01-Sep-2026)

RIG 8 1/2 05/15/31, Overweight (USD 104.88, 01-Sep-2026)

VENTURE GLOBAL LNG INC, A/CD/D/J/L

VENLNG 6 3/8 12/15/34, Market Weight (USD 99.13, 01-Sep-2026)

VENLNG 6 5/8 06/15/36, Market Weight (USD 100.00, 01-Sep-2026)

VENLNG 7 01/15/30, Market Weight (USD 102.75, 01-Sep-2026)

VENLNG 8 3/8 06/01/31, Market Weight (USD 104.00, 01-Sep-2026)

VENLNG 9 1/2 02/01/29, Market Weight (USD 107.48, 01-Sep-2026)

VENLNG 9 7/8 02/01/32, Market Weight (USD 106.38, 01-Sep-2026)

VISTRA OPERATIONS CO LLC, Overweight, A/CD/D/J/K/L/M

Representative Bond: VST 5.35 01/31/36 (USD 95.32, 01-Sep-2026)

WOODSIDE FINANCE LTD, Market Weight, CD/D/J/K/L/M/N

Representative Bond: WDSAU 3.7 03/15/28 (USD 98.32, 01-Sep-2026)

Representative Bond: WDSAU 3.7 09/15/26 (USD 99.96, 01-Sep-2026)

Representative Bond: WDSAU 4 1/2 03/04/29 (USD 98.81, 01-Sep-2026)

Representative Bond: WDSAU 5.1 09/12/34 (USD 96.49, 01-Sep-2026)

Representative Bond: WDSAU 5.7 09/12/54 (USD 91.72, 01-Sep-2026)

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Overweight (OW):

For sectors rated against the Bloomberg U.S. Credit Index, the Bloomberg Pan-European Credit Index, the Bloomberg EM Asia USD High Grade Credit Index or the Bloomberg EM USD Corporate and Quasi-Sovereign Index, the analyst expects the six-month excess return of the sector to exceed the six-month excess return of the relevant index.

For sectors rated against the Bloomberg U.S. High Yield 2% Issuer Capped Credit Index, the Bloomberg Pan-European High Yield 3% Issuer Capped Credit Index excluding Financials, the Bloomberg Pan-European High Yield Finance Index or the Bloomberg EM Asia USD High Yield Corporate Credit Index, the analyst expects the six-month total return of the sector to exceed the six-month total return of the relevant index.

Market Weight (MW):

For sectors rated against the Bloomberg U.S. Credit Index, the Bloomberg Pan-European Credit Index, the Bloomberg EM Asia USD High Grade Credit Index or the Bloomberg EM USD Corporate and Quasi-Sovereign Index, the analyst expects the six-month excess return of the sector to be in line with the six-month excess return of the relevant index.

For sectors rated against the Bloomberg U.S. High Yield 2% Issuer Capped Credit Index, the Bloomberg Pan-European High Yield 3% Issuer Capped Credit Index excluding Financials, the Bloomberg Pan-European High Yield Finance Index or the Bloomberg EM Asia USD High Yield Corporate Credit Index, the analyst expects the six-month total return of the sector to be in line with the six-month total return of the relevant index.

Underweight (UW):

For sectors rated against the Bloomberg U.S. Credit Index, the Bloomberg Pan-European Credit Index, the Bloomberg EM Asia USD High Grade Credit Index or the Bloomberg EM USD Corporate and Quasi-Sovereign Index, the analyst expects the six-month excess return of the sector to be less than the six-month excess return of the relevant index.

For sectors rated against the Bloomberg U.S. High Yield 2% Issuer Capped Credit Index, the Bloomberg Pan-European High Yield 3% Issuer Capped Credit Index excluding Financials, the Bloomberg Pan-European High Yield Finance Index or the Bloomberg EM Asia USD High Yield Corporate Credit Index, the analyst expects the six-month total return of the sector to be less than the six-month total return of the relevant index.

Sector definitions:

Sectors in U.S. High Grade Research are defined using the sector definitions of the Bloomberg U.S. Credit Index and are rated against the Bloomberg U.S. Credit Index.

Sectors in U.S. High Yield Research are defined using the sector definitions of the Bloomberg U.S. High Yield 2% Issuer Capped Credit Index and are rated against the Bloomberg U.S. High Yield 2% Issuer Capped Credit Index.

Sectors in European High Grade Research are defined using the sector definitions of the Bloomberg Pan-European Credit Index and are rated against the Bloomberg Pan-European Credit Index.

Sectors in Industrials and Utilities in European High Yield Research are defined using the sector definitions of the Bloomberg Pan-European High Yield 3% Issuer Capped Credit Index excluding Financials and are rated against the Bloomberg Pan-European High Yield 3% Issuer Capped Credit Index excluding Financials.

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Sectors in Asia High Grade Research are defined on Barclays Live and are rated against the Bloomberg EM Asia USD High Grade Credit Index.

Sectors in Asia High Yield Research are defined on Barclays Live and are rated against the Bloomberg EM Asia USD High Yield Corporate Credit Index.

Sectors in EEMEA and Latin America Research are defined on Barclays Live and are rated against the Bloomberg EM USD Corporate and Quasi Sovereign Index. These sectors may contain both High Grade and High Yield issuers.

To view sector definitions and monthly sector returns for Asia, EEMEA and Latin America Research, go to <https://live.barcap.com/go/research/EMSectorReturns> on Barclays Live.

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Explanation of the Barclays Research Corporate Credit Rating System

For all High Grade issuers covered in the US, Europe or Asia, and for all issuers in Latin America and EEMEA, the credit rating system is based on the analyst's view of the expected excess return over a six-month period of the issuer's index-eligible corporate debt securities* relative to the expected excess return of the relevant sector, as specified on the report.

Overweight (OW): The analyst expects the six-month excess return of the issuer's index-eligible corporate debt securities to exceed the six-month expected excess return of the relevant sector.

Market Weight (MW): The analyst expects the six-month excess return of the issuer's index-eligible corporate debt securities to be in line with the six-month expected excess return of the relevant sector.

Underweight (UW): The analyst expects the six-month excess return of the issuer's index-eligible corporate debt securities to be less than the six-month expected excess return of the relevant sector.

Rating Suspended (RS): The rating has been suspended temporarily due to market events that make coverage impracticable or to comply with applicable regulations and/or firm policies in certain circumstances including where the Investment Bank of Barclays Bank PLC is acting in an advisory capacity in a merger or strategic transaction involving the company.

Coverage Suspended (CS): Coverage of this issuer has been temporarily suspended.

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For all High Yield issuers (excluding those covered in EEMEA or Latin America), the credit rating system is based on the analyst's view of the expected total returns over a six-month period of the rated debt security relative to the expected total return of the relevant sector, as specified on the report.

Overweight (OW): The analyst expects the six-month total return of the debt security subject to this rating to exceed the six-month expected total return of the relevant sector.

Market Weight (MW): The analyst expects the six-month total return of the debt security subject to this rating to be in line with the six-month expected total return of the relevant sector.

Underweight (UW): The analyst expects the six-month total return of the rated debt security subject to this rating to be less than the six-month expected total return of the relevant sector.

Rating Suspended (RS): The rating has been suspended temporarily due to market events that make coverage impracticable or to comply with applicable regulations and/or firm policies in certain circumstances including where the Investment Bank of Barclays Bank PLC is acting in an advisory capacity in a merger or strategic transaction involving the company.

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Where a recommendation is made at the issuer level, it does not apply to any sanctioned securities, where trading in such securities would be prohibited under applicable law, including sanctions laws and regulations.

*In EEMEA and Latin America (and in certain other limited instances in other regions), analysts may occasionally rate issuers that are not part of the Bloomberg U.S. Credit Index, the Bloomberg Pan-European Credit Index, the Bloomberg EM Asia USD High Grade Credit Index or Bloomberg EM USD Corporate and Quasi Sovereign Index. In such cases the rating will reflect the analyst's view of the expected excess return over a six-month period of the issuer's corporate debt securities relative to the expected excess return of the relevant sector, as specified on the report.

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27% have been assigned an Overweight rating which, for purposes of mandatory regulatory disclosures, is classified as a Buy rating; 57% of issuers with this rating category are investment banking clients of the Firm; 70% of the issuers with this rating have received financial services from the Firm.

53% have been assigned Market Weight rating which, for purposes of mandatory regulatory disclosures, is classified as a Hold rating; 63% of issuers with this rating category are investment banking clients of the Firm; 77% of the issuers with this rating have received financial services from the Firm.

20% have been assigned an Underweight rating which, for purposes of mandatory regulatory disclosures, is classified as a Sell rating; 57% of issuers with this rating category are investment banking clients of the Firm; 71% of the issuers with this rating have received financial services from the Firm.

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24% have been assigned an Overweight rating which, for purposes of mandatory regulatory disclosures, is classified as a Buy rating; 40% of bonds with this rating category are investment banking clients of the Firm; 63% of the issuers with this rating have received financial services from the Firm.

55% have been assigned Market Weight rating which, for purposes of mandatory regulatory disclosures, is classified as a Hold rating; 43% of bonds with this rating category are investment banking clients of the Firm; 64% of the issuers with this rating have received financial services from the Firm.

21% have been assigned an Underweight rating which, for purposes of mandatory regulatory disclosures, is classified as a Sell rating; 40% of bonds with this rating category are investment banking clients of the Firm; 56% of the issuers with this rating have received financial services from the Firm.

Explanation of the Barclays EM Sovereign Credit Issuer Rating System

Overweight (OW):

The analyst expects the six-month excess return of the country's index eligible bonds to exceed the six-month excess return of the Bloomberg EM USD Sovereign Index*.

Market Weight (MW):

The analyst expects the six-month excess return of the country's index eligible bonds to be in line with the six-month excess return of the Bloomberg EM USD Sovereign Index*.

Underweight (UW):

The analyst expects the six-month excess return of the country's index eligible bonds to be less than the six-month excess return of the Bloomberg EM USD Sovereign Index*.

Rating Suspended (RS):

The rating has been suspended temporarily due to market events that make coverage impracticable or to comply with applicable regulations and/or firm policies in certain circumstances including where the Investment Bank of Barclays Bank PLC is acting in an advisory capacity.

*Analysts may occasionally rate sovereign issuers that are not part of the Bloomberg EM USD Index. In such cases, the rating will still reflect the analyst's view of the expected excess return over a six-month period of the sovereign debt securities relative to the expected excess return of the Bloomberg EM USD Index.

Distribution of ratings assigned by Barclays Emerging Markets Sovereign Research at the issuer level:

32% have been assigned an Overweight rating which, for purposes of mandatory regulatory disclosures, is classified as a Buy rating; 25% of issuers with this rating category are investment banking clients of the Firm; 83% of the issuers with this rating have received financial services from the Firm.

41% have been assigned Market Weight rating which, for purposes of mandatory regulatory disclosures, is classified as a Hold rating; 7% of issuers with this rating category are investment banking clients of the Firm; 67% of the issuers with this rating have received financial services from the Firm.

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